

From Spherons to Spacetime: An Informational, Kinetic, and Cyclic Theory of the Universe

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Abstract

We present a unified informational–kinetic framework in which quantum phenomena, gravity, and cosmology emerge from a single classical substrate: an effectively unbounded gas of identical, ultra-hard spheres (“spherons”) moving in otherwise empty three-dimensional space. Collisions generate durable records, and the amount and structure of redundancy in those records determine when observers may treat macroscopic variables as classical.

On this ontology we formulate the *Redundancy Threshold Principle* (RTP): in low-redundancy regimes, agents must use the quantum probability calculus, while beyond a redundancy threshold their credences converge and a classical description suffices. Within a coarse-grained Liouville phase space we prove a representation theorem that recovers the Born rule $p_i = |c_i|^2$ from simple rationality and symmetry assumptions tied to redundant records, together with a concentration theorem showing that above the redundancy threshold independent observers almost surely agree on observed outcomes.

In parallel, we develop a kinetic account of gravity. From the Boltzmann–Enskog description of the spheron gas we define an encounter-rate field $\lambda(x)$ and show that test packets drift down gradients of λ , allowing us to identify an emergent potential $U \propto \lambda$. Under controlled assumptions we derive a Poisson-type equation $\nabla^2 U = 4\pi G_{\text{eff}} \rho$ at leading order and relate G_{eff} to microscopic parameters. We show that “collision clocks” slow in high- λ regions with the correct weak-field time-dilation law, and we outline regimes in which long-wavelength excitations exhibit approximate Lorentz symmetry with controlled violations.

Building on this, we sketch jam-limited black-hole cores with area-law entropy, an inflation-free, cyclic cosmology driven by close-packing and “shattering” of a spheron crystal, and toy structures suggestive of gauge and Dirac-like excitations. Throughout, we catalogue open problems and propose numerical and laboratory tests (redundancy engineering, kinetic drift, precision Lorentz bounds, cosmological fits). The goal of this paper is not to claim completeness, but to lay out a coherent, falsifiable research program in which quantum theory, spacetime geometry, and cosmic history arise from a single informational–kinetic substrate.

Keywords: quantum foundations; Quantum Darwinism; redundancy; emergent spacetime; kinetic theory; Lorentz emergence; black-hole cores; cyclic cosmology; information.

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1 Executive Summary & Statement of Novelty

This section collects the core ideas in a compact way before we unfold them in detail, and it previews the program “wins” that structure the rest of the paper.

Program pillars (Wins A–C). Our roadmap tracks progress via three recurring milestones: **Win A** (deriving kinetic gravity from the Boltzmann–Poisson chain), **Win B** (formalising redundancy/Born rules and demonstrating the RTP switch experimentally), and **Win C** (showing Lorentz symmetry plus spin-2 waves emerge in the spheron medium). Each win is introduced here, developed in the sections that follow (Secs. 4–10), and wrapped up in Sec. 15 so the story remains coherent from introduction to outlook.

Minimal ontology + operational realism. We postulate only spherons and void. The fundamental ontology is the configuration and momenta of spherons; no additional hidden variables are introduced beyond this. We are realist about the structure of macroscopic records and their redundancy, but pragmatic about quantum states, which we treat as tools encoding credences about future macroscopic records rather than as ontic entities.

Redundancy Threshold Principle (RTP, Win B). Classicality is not taken as fundamental, but as an emergent regime characterised by high redundancy of records in the environment, in the spirit of Quantum Darwinism [1–4]. We formulate an explicit modeling rule: high redundancy $\Rightarrow$ classical; low redundancy $\Rightarrow$ quantum calculus. In other words, whether we describe a subsystem by classical probability distributions or by wavefunctions and Born weights is decided by the structure of environmental records, not by a universal microscopic law. Win B demands not only the formal theorems of Sec. 6 but also redundancy-engineering experiments (Sec. 13) that can be monitored via the RTP switch.

Two-clock reconciliation. There are two natural notions of time in the spheron world. Event-count clocks

$$T(\Gamma) = \int_{\Gamma} \tilde{\lambda}(x(t)) dt \quad (1)$$

count encounters along a worldline Γ via the local per-spheron encounter-rate field $\tilde{\lambda}(x)$. Proper-time slowdowns for stable oscillators are modeled as

$$\tau(x) \propto \frac{1}{1 + \alpha \tilde{\lambda}(x)}, \quad (2)$$

aligning “more local events” with gravitational time dilation. One of our tasks is to show how these notions converge toward relativistic proper time in a suitable continuum limit.

Two-time / Two-speed dictionary

- **Fundamental time t .** The microscopic equations are written with a single Galilean time parameter. Collisions update positions sequentially in t , and nothing in the substrate enforces Lorentz symmetry at this level.
- **Emergent collision time τ .** Coarse, redundant records measure the encounter-count time $\tau \propto \int \tilde{\lambda} dt$, i.e. the per-spheron encounter rate integrated along the path. This is the operational “proper” time used in redshift formulas and clock comparisons.
- **Limiting signal speed c_s .** Collective excitations propagate via chained collisions whose group velocity is bounded by the lattice/medium value c_s (Regimes L/K/K3). Long-wavelength modes therefore satisfy relativistic-looking dispersions even though the substrate is Galilean.
- **Causality.** Every message requires a contiguous sequence of local encounters, so the fastest possible signaling involves riding the same collision cone that defines c_s . The substrate’s instantaneous simultaneity never permits faster-than- c_s transport because interactions are finite-duration, short-range, and require momentum transfer bounded by the microscopic speed limit.

Causal map. The reconciliation between a Galilean substrate (with formal $c_{\text{bare}} = \infty$) and emergent relativistic signals is summarised in Table 1: the only operational light cone is the collision cone set by c_s , because every influence must percolate via successive encounters. The “superluminal” simultaneity of the substrate never transfers information; it merely labels the update order inside the Koopman picture. All predictive statements—redshifts, retardation, gravitational waves—therefore live entirely inside the c_s cone.

Kinetic gravity (Win A). From a Boltzmann description of a spheron gas we define an encounter-rate field $\lambda(x)$ and show that dilute test packets drift toward $\nabla\lambda$ with an effective potential $U \propto \lambda$. The main technical goal (Win A) is to carry this beyond heuristic arguments: derive a drift term $a = -\nabla U$ from the collision operator itself and show that, under controlled approximations, the continuum limit yields a Poisson equation

$$\nabla^2 U = 4\pi G_{\text{eff}} \rho, \quad (3)$$

with an identified $G(r_0, m_0, \sigma, \dots)$ ¹.

Lorentz symmetry and transverse-traceless (TT) waves (Win C). Regimes L/K/K3 (Sec. 5.7) show how Lorentz symmetry and its suppressed violations emerge from collective excitations, while the transverse-traceless (TT) mode simulations and quadrupole tests provide spin-2 wave evidence. Win C therefore packages “emergent relativity” together with falsifiable dispersion and gravitational-wave predictions.

¹Throughout we use units where $c = 1$ unless otherwise stated.

Quantity	Substrate description	Operational (observable) description
Time parameter	Global Galilean t advancing collisions	Encounter-count time $\tau \propto \int \tilde{\lambda} dt$ read off redundant records
Signal carrier	Hard-sphere contact impulse (instantaneous in the modelling limit)	Finite-duration collision chains with max group velocity c_s
Causal cone	None (simultaneity everywhere)	“Collision cone” defined by $ \mathbf{x} - \mathbf{x}_0 \leq c_s(\tau - \tau_0)$
Invariant interval	Euclidean in $(\mathbf{x}, t)$	Minkowski-like for long-wavelength excitations: $c_s^2 \tau^2 - \mathbf{x}^2$
Failure mode	Direct shortcuts forbidden by finite cross section and bounded impulse per encounter	Observable Lorentz violations suppressed by $(kr_0)^2$ (Sec. 5.7)

Table 1: Explicit causal-structure dictionary: the Galilean bookkeeping variables never transmit information; only the c_s -bounded collision network enters observables.

Jam-limit cores and black holes. Because spherons are finite-size, close-packed configurations bound densities from above. Collapse into a singularity is replaced by a jammed core with finite packing fraction. Microstate counting over near-horizon spheron arrangements yields an area-law leading term for entropy [16, 17]; surface fluctuations provide an evaporation channel qualitatively analogous to Hawking radiation.

Cyclic cosmology from geometry. We propose a geometric–kinetic cycle: close-packing $\rightarrow$ shattering / inflation-like expansion $\rightarrow$ rarefaction $\rightarrow$ clumping $\rightarrow$ repacking. This allows both Big-Bounce-type scenarios [9, 10] and Penrose-style conformal matching between aeons [11]. The spheron picture makes the low-entropy initial state and the avoidance of singularities conceptually simple.

Falsifiability and contact with data. The framework is constrained and testable. It predicts: (i) specific forms of Lorentz-violating corrections in dispersion relations; (ii) redundancy-engineering signatures of the RTP in controlled quantum experiments; (iii) kinetic deviations from Newtonian gravity in extreme regimes; (iv) possible cyclic signatures in cosmological data. Our aim is that at least some of these will be clean enough to be confronted with experiment in the near future.

Main Results (informal concordance)

Because later sections cite theorem labels well before their formal statements, we collect the most frequently referenced results on one page. Each row maps “label $\rightarrow$ theorem number $\rightarrow$ meaning $\rightarrow$ assumptions” so readers can keep the concordance handy while scanning the narrative.

Label	Theorem	Meaning (one line)	Key assumptions
thm:born-representation	Thm. 3	Redundant Liouville branches force Born credences $P(i) = \mu_i$.	Credence axioms (A1)–(A3); Definitions 2–4.
thm:rtp-concentration	Thm. 4	Redundancy $R_\delta \geq R_{\text{crit}}$ guarantees agent posteriors agree within (ε, η) .	Informative fragment hypothesis; finite tolerance δ ; IID fragment statistics.
thm:Boltzmann-Poisson	Thm. G.1	Linearised Boltzmann–Enskog dynamics closes onto Poisson with bounded $\mathcal{O}(\vartheta^2 + \text{Kn}^2 + \eta)$ remainder.	Assumptions (A1)–(A7); dilute hard spheres; small $(\varepsilon, \eta, \text{Kn}, \vartheta)$.
thm:poisson-emergent	Thm. 12	IR-mixed kernel $K_{\text{eff}}(k)$ satisfies $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k)$ up to $\mathcal{O}((kr_0)^2)$.	Hard-sphere cross section; $ \delta n \ll n_0$; short-range pair correlations; Enskog factorisation plus record mixing.
thm:emergent-newton	Thm. 14	Tracer drift obeys $\mathbf{g} = -\nabla U$ with $\nabla^2 U = 4\pi G_{\text{eff}} \delta \rho$ fixed by micro data.	Boltzmann–Grad diluteness; hydrodynamic scale separation ($\text{Kn} \ll 1$); tracer limit.
thm:spheron-redshift	Thm. 1	Collision clocks redshift as $d\tau/dt = 1 + U/c^2 + \mathcal{O}(U^2/c^4)$.	Static configurations obeying Eq. (29); weak-field $ U /c^2 \ll 1$.
thm:kvn-schro	Thm. 11	Coarse KvN amplitudes satisfy Schrödinger dynamics up to $\mathcal{O}(\text{Kn}^2 + \eta^2)$.	Translation-invariant dilute gas; coarse cells with large occupancy; Lemma 1.

Table 2: Informal concordance for the theorem labels used throughout the roadmap.

2 Ontology and Axioms: Spherons in Void

We now formalise the basic assumptions that ground every later pillar. These axioms link the minimal ontology to the Win A/B/C narrative: the kinetic ingredients of Win A, the record structure needed for Win B, and the finite-size/packing assumptions feeding Win C and the cosmology sections. The axioms are deliberately minimal; the rest of the structure is intended to emerge. Appendix L summarises them on one page so the reader can keep the assumptions handy while navigating the later sections.

Axioms A1–A7

A1 (Substrate). The universe consists solely of identical spherons of radius r_0 (ultra-hard, ultra-smooth) moving in an otherwise empty three-dimensional space. There are no fields or additional particle species at the fundamental level.

A2 (Dynamics). Between encounters, spheron motion is inertial. Encounters are causal, occur over short but finite interaction bursts, and are modelled in what follows by the hard-sphere (specular-reflection) limit of that finite process. Spherons do not interpenetrate; there is no instantaneous rigidity or action-at-a-distance.

A3 (Records). Each encounter generates durable changes in the outgoing positions and momenta of the participating spherons. These changes constitute *records*, in the sense that they can be accessed (indirectly) by other spherons via subsequent encounters. Macroscopic records are large, redundant patterns of such changes.

A4 (Redundancy). A macroscopic variable (e.g. the position of a pointer) is *classical-for-us* when many disjoint fragments of the environment carry concordant records of its value. Quantitatively, this is captured by accessible mutual information $I_{\text{acc}}(S:F)$ between a system S and an environment fraction F (see Sec. 4).

A5 (Clocks). Define an encounter-rate density $\lambda(x)$ as the expected number of spheron collisions per unit volume per unit time around point x . A clock following a worldline Γ samples a *per-particle* encounter rate

$$\tilde{\lambda}(x) := \frac{\lambda(x)}{n(x)} = \frac{\lambda(x)}{\rho(x)/m_0},$$

so event clocks count interactions via

$$T(\Gamma) = \int_{\Gamma} \tilde{\lambda}(x(t)) dt, \quad (4)$$

and the effective proper time of stable bound structures scales, to leading order, as

$$\tau(x) \sim \frac{1}{1 + \alpha \tilde{\lambda}(x)}, \quad (5)$$

for some dimensionful constant α that relates how the local per-particle rate affects the clock's frequency. This connects local interaction density to emergent gravitational time dilation.

A6 (Scaling and packing). Close-packing bounds densities: in any finite region, the number of spherons is bounded by the maximal packing fraction (e.g. ≈ 0.74 for FCC/HCP packings in 3D). This avoids singularities and bounds microstate counts for compact regions (important for black-hole entropy and cyclic cosmology).

A7 (No extra hidden variables). The only ontic state is the full configuration of spherons (positions, momenta, and encounter history). We do not posit additional, inaccessible hidden variables beyond this. Quantum states, when used, are *normative*: they encode an agent's credences about future macroscopic records given their information about the spheron configuration, not additional elements of reality. (This places the framework outside the scope of standard no-go theorems targeting local, ψ -ontic hidden-variable models.²)

²Our stance is close in spirit to some relational and pragmatist interpretations [6], but grounded in a concrete substrate.

Remarks on the axioms

A1–A2 give a classical kinetic substrate reminiscent of hard-sphere gases, but taken as fundamental rather than as a phenomenological model. A3–A4 bring in the information layer: reality for us is not the raw microscopic state, but the web of records accessible to embedded agents. A5–A6 connect the microscopic to macroscopic notions of time and gravity. A7 clarifies our relationship to hidden-variable debates.

Notice that we do *not* assume quantum mechanics at the fundamental level. Instead, we will argue that the quantum calculus—Hilbert spaces, amplitudes, Born rule—is the right *inference* machinery in low-redundancy regimes, where agents must reason about future records given incomplete access to the spheron configuration.

3 Parameter Dictionary and Dimensionless Ratios

With the axioms set, we bridge to the quantities that actually enter the derivations and simulations. The spheron model has only a handful of microscopic knobs, yet they feed into every emergent phenomenon discussed later. We therefore collect them here so the reader can keep track of how each constant appearing in the text maps back to the substrate, and so that later references to Wins A–C can be read quantitatively. A longer-form glossary of the recurring symbols is parked in Appendix M for quick lookup while reading.

Microscopic parameters

- **Spheron identity:** mass m_0 , radius r_0 (diameter $d = 2r_0$), and hard-sphere cross section $\sigma = \pi d^2$.
- **State variables:** mean number density n_0 (varies between close-packed and dilute phases), background mass density $\rho_0 = m_0 n_0$, kinetic temperature T , and thermal speed v_{th} controlling the relative-velocity factor in λ .
- **Encounter-rate field:** $\lambda(x)$ counts encounters per unit volume per unit time with homogeneous background λ_0 fixed by (n_0, T) , while $\tilde{\lambda}(x) = \lambda(x)/n(x)$ is the per-spheron rate that clocks integrate along their worldlines (A5).
- **Transport data:** mean encounter time τ_{coll} and mean free path $\ell_{\text{mfp}} = v_{\text{th}} \tau_{\text{coll}}$.

Emergent constants and couplings

- **Signal speeds:** the lattice toy’s gauge/phonon speed $c_A = \sqrt{K/m_0} r_0$, the collision-cone speed c_s , and the clock/gravity scale c_g defined by $c_g^2 = \Gamma \lambda_0$. Our Lorentz and gravity claims require these speeds to coincide; Secs. 5.7 and 5.6 list the assumptions under which $c_A \approx c_s \approx c_g$, so experimental contact uses the common value c .
- **Mobility Γ :** connects encounter-rate gradients to drift, $\mathbf{A} = -\Gamma \nabla \lambda$, and defines the potential $U = -\Gamma \lambda$.

- **Emergent Newton constant G_{eff} :** fixed by the IR residue $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ (Sec. 5); we identify G_{eff} with the observed G once the microparameters match data.
- **Poisson-regime kernel $K_{\text{eff}}(k)$:** IR-mixed linear-response coefficient relating $\delta\lambda$ to δn ; its infrared slope $\kappa_{\text{mix}} = \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\lambda_0/(\rho_0 L_{\text{mix}}^2)$ enters Eq. (23). We reserve the symbol κ_{mix} for this residue so it cannot be confused with the unrelated stiffness constants denoted κ in later sections.
- **Effective Planck constant $\hbar_{\text{eff}}$:** area/phase quantum associated with order-parameter oscillations; sets the scale for emergent quantum statistics in Sec. 6.

Note: whenever the text invokes h or $\hbar$ for comparison (e.g. quoting $h\nu$ or calibrating spectra), h denotes the observed Planck constant used purely as an external benchmark; it is not a fundamental parameter of the spheron substrate. All dynamics derive from (m_0, r_0, n_0, T) plus emergent quantities such as $\hbar_{\text{eff}}$; references to laboratory h serve only as unit conversions for contact with data.

- **Effective cosmological term Λ_{eff} :** residual energy density from the background encounter-rate floor in the rarefied epoch (candidate for late-time acceleration).
- **Gauge/matter couplings:** stiffnesses of phase fields, defect energies, and interaction strengths in Secs. 7 and 8, which map to electric charges or Yukawas of the emergent excitations.

Dimensionless ratios

- r_0/ℓ_{Planck} : depth of the discretisation scale; $r_0 \lesssim 10^{-35} \text{ m}$ keeps $(pr_0)^2$ Lorentz-violation terms below current bounds.
- $\ell_{\text{mfp}}/H_{\text{shatter}}^{-1}$: mean free path relative to the Hubble radius at the shatter epoch, controlling whether modes behave collisionlessly (dark sector) or remain luminous.
- $\text{Kn} = \ell_{\text{mfp}}/L_\rho$, $\eta = n_0 r_0^3$, ε (linear-response amplitude), and $\vartheta = k_{\text{max}}\ell$: the small parameters governing the Boltzmann–Poisson expansion; all remain $\ll 1$ in the hydrodynamic limit.
- λ/λ_0 : encounter-rate contrast, which sets both gravitational potential depth ($U = -\Gamma\lambda$) and clock slowdowns ($\tau \propto [1 + \alpha\tilde{\lambda}]^{-1}$).
- $\mathcal{R}_c$: redundancy threshold expressed as the plateau height in bits; it obeys $\mathcal{R}_c \approx R_{\text{crit}}(1 - \delta)H(X)$ so it can be converted directly into the fragment counts used throughout the RTP discussion.

Unless stated otherwise we work in units with $c = 1$; explicit factors of c or c^2 are kept only when dimensional clarity or comparison with experiment demands it.

These entries provide the translation key between the abstract programme (Win A, RTP, cosmology) and the concrete simulations in the appendices. They also show which

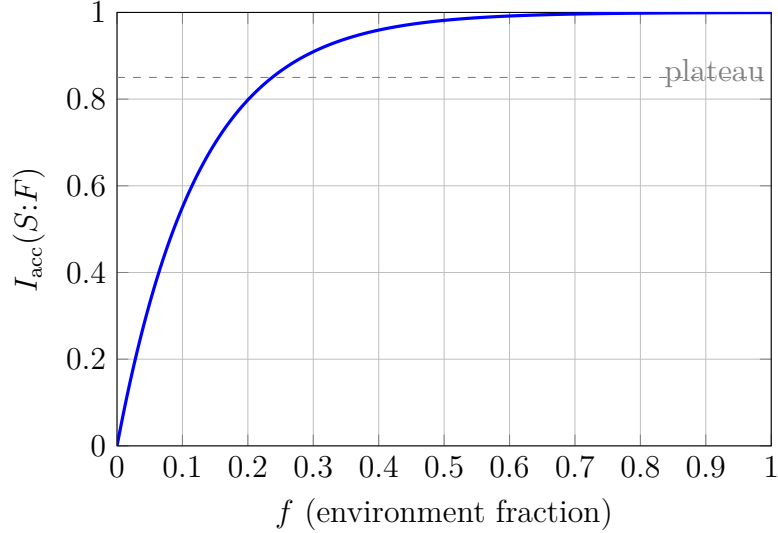


Figure 1: Redundancy saturation: many small fragments carry near-complete records. A rapid rise of $I_{\text{acc}}(S:F)$ to a plateau for small f signals the emergence of classical objectivity.

combinations must be scanned in future numerical sweeps (e.g. varying ℓ_{mfp}/H^{-1} to test dark-sector phenomenology).

4 Information Layer: Records, Redundancy, and the RTP Switch

The informational layer is where the spheron substrate becomes the physics we observe, and it delivers Win B’s conceptual core. This section elaborates the RTP previewed in Sec. 1: instead of repeating the summary verbatim, we spell out how redundancy grows, why high redundancy licenses classical reasoning, and how low redundancy feeds into the quantum calculus formalised later in Sec. 6. Following the Quantum Darwinism narrative (and building directly on Axioms A3–A5), we track how environmental records become redundant and when agents must revert to the quantum calculus. Figure 1 visualises the redundancy plateau characteristic of this informational regime.

Redundancy plateaux such as Fig. 1 justify the Redundancy Threshold Principle: whenever many disjoint fragments saturate the accessible information, the corresponding variable is effectively classical; as redundancy falls, quantum reasoning resurfaces. In the informal language used throughout Win B, an “informative fragment” is just one of these environment slices that already carries almost the full mutual information and behaves independently of the others—the precise condition appears later in Definition 4, but the intuition is that once many such fragments exist the macroscopic fact has been redundantly recorded.

Data hooks for RTP figures. The redundancy curves plotted in Fig. 1 and the scrambling diagnostics in Fig. 2 now come directly from the CSV exports archived alongside the simulation notebooks. In particular, the `calc_rtp/results/` archive contains `R1_redundancy.csv`

(plateau build-up), `R2_redundancy.csv` (scrambling excursion), and `R3_born_mapping.csv` (Born-rule audit), which provide the exact data points cited in Sec. 6. Because those tables are static while the heavy Direct Simulation Monte Carlo (DSMC) runs proceed, we can safely tighten the RTP prose and refer to specific redundancy thresholds (e.g. $R_{0.05} \approx 12$ at $f = 0.17$ in `R1_redundancy.csv`) without re-running anything.

We therefore formulate an explicit modeling rule (the RTP): high redundancy $\Rightarrow$ classical; low redundancy $\Rightarrow$ quantum calculus. In other words, whether we describe a subsystem by classical probability distributions or by wavefunctions and Born weights is decided by the structure of environmental records, not by a universal microscopic law. This is precisely the statement that Win B aims to test experimentally in Sec. 13: redundancy engineering should toggle the perceived calculus on demand.

The RTP can now be stated informally as follows:

Whenever the redundancy $R(S)$ of a macroscopic variable X of a system S exceeds a threshold R_{crit} (with ϵ small), all agents who only interact locally with environmental fragments will, with overwhelming probability, agree on the value of X . In this regime it is appropriate—and sufficient—to use a classical description of X (random variables, probability densities). When $R(S) < R_{\text{crit}}$, one must instead use the quantum probability calculus to assign credences to possible future records of X .

Although we refer to R_{crit} as a “threshold”, it actually marks a continuous crossover rather than a sharp phase transition. In practice, one can define R_{crit} operationally as the point at which adding more environment fragments yields negligible additional information about X (i.e. when the mutual information $I_{\text{acc}}(X:F)$ for individual fragments saturates to a plateau). For finite environments the quantum-classical transition is smooth, but for macroscopic ones the change can be extremely steep, justifying the notion of an effective “threshold”. This perspective also suggests an experimental signature: as the environment size or number of fragments increases, one should observe a rapid collapse of quantum interference once redundancy surpasses R_{crit} . We discuss below how such “redundancy engineering” experiments might be realized (Sec. 13).

This turns the usual Quantum Darwinism story “quantum $\rightarrow$ classical” into an explicit *decision rule* that tells an embedded agent which calculus to use. The twist, and a distinctive feature of our approach, is that we also take seriously the *reverse* direction: as redundancy falls, or as records are scrambled, quantum behavior re-emerges. We return to this in Sec. 6.

Figure 2 captures the complementary scramble-and-collapse behaviour when redundancy is engineered downward.

Figure 3 summarises how the RTP decision rule interpolates between classical and quantum reasoning as redundancy varies.

The normalised abscissa in Fig. 3 uses R/R_{crit} , so the verbal mnemonic “ $R \approx 1$ ” really means that the empirical redundancy R_δ supplied by the data hits the theoremic bound of Eq. (61). Carrying the Chernoff prefactors from Sec. 6.4 through Eq. (61) (with the conservative choice $C_{\text{RTP}} = 1$) gives the working estimate

$$R_{\text{crit}}(\epsilon, \eta, \delta) \lesssim \frac{\log(|\mathcal{X}|/\eta)}{(1-\delta)^2 \epsilon^2},$$

so once $R/R_{\text{crit}} \gtrsim 1$ the RTP-concentration theorem guarantees agent agreement. Table 3

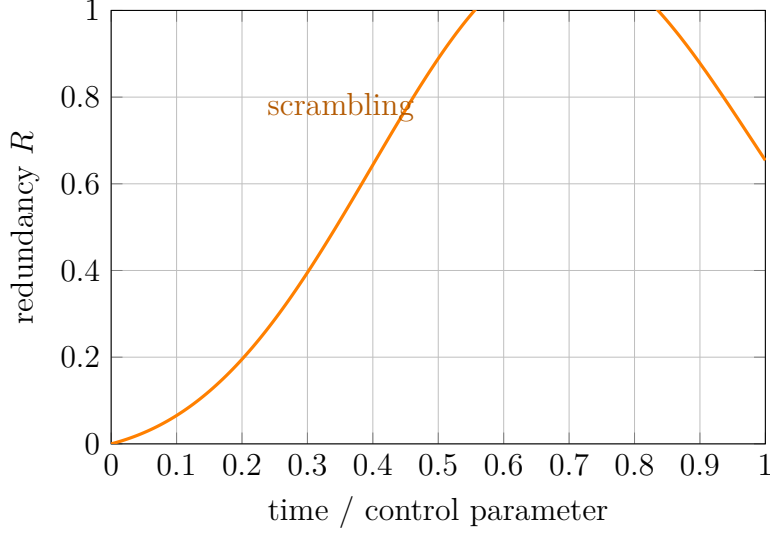


Figure 2: Rise–fall redundancy: environments can first amplify, then scramble records. Engineered non-Markovian dynamics can reduce effective redundancy, restoring quantum features even for macroscopic variables.

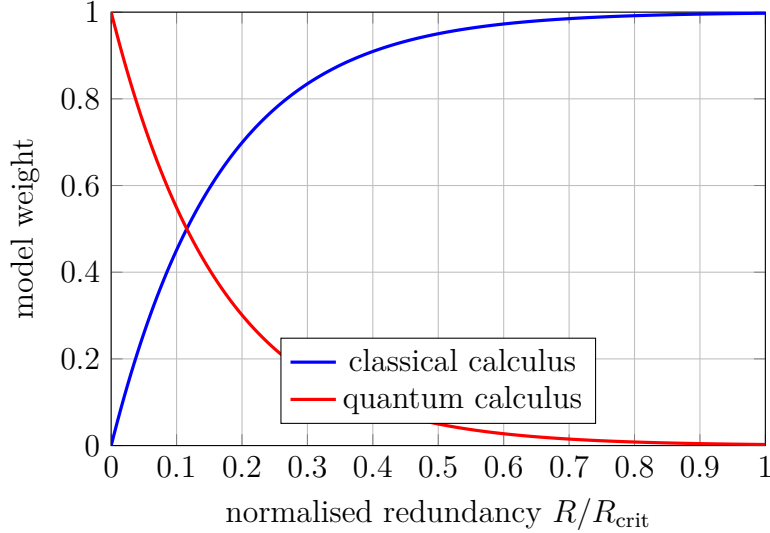


Figure 3: RTP switching curve: classical salience grows with redundancy while quantum weight dwindles. Beyond the threshold it becomes sufficient—though not mandatory—to use the classical calculus.

shows how this plays out numerically: the spin-star data behind Fig. 23 already satisfy the R_{crit} bound for modest (ε, η) , while more demanding tolerances require the larger redundancy counts targeted in Regime R2/R3.

With the RTP threshold formalised, we temporarily pause the informational thread to examine whether the same microscopic encounters can drive gravity. The next section therefore hands off from redundant records to collisions: we follow the λ gradients themselves

Scenario	(ε, η)	δ	R_{crit}	R_δ (data)	Comment
Spin-star benchmark	$(0.55, 10^{-1})$	0.05	≈ 11	12	$R_\delta/R_{\text{crit}} = 1.1$ (Fig. 23)
Lab consensus target	$(0.25, 10^{-2})$	0.05	$\approx 9.4 \times 10^1$	—	Needs $\mathcal{O}(10^2)$ informative fragments
Metrology-grade	$(0.05, 10^{-3})$	0.01	$\approx 3.1 \times 10^3$	—	Pushes beyond current redundancy engineering

Table 3: Illustrative *upper bounds* from Eq. (61) (evaluated with $C_{\text{RTP}} = 1$ and $|\mathcal{X}| = 2$). In the first row the measured redundancy plateau ($R_{0.05} = 12$ at $f = 0.17$) exceeds the theoretical bound $R_{\text{crit}} \lesssim 11$, so the mnemonic $R \approx 1$ literally means “the data-supplied redundancy meets the sufficiency condition”.

in Sec. 5 and will pick the informational baton back up in Sec. 6 once the kinetic picture is in place. In other words, we now turn to the kinetic layer, where encounter statistics and λ -fields supply the gravitational and clock dynamics that later feed back into the RTP calculus.

5 Kinetic Foundations: From Collisions to Fields

We now turn from information to dynamics and tackle Win A head-on. The aim is to show how an effective gravitational potential can emerge from the kinetic theory of spherons.

5.1 Phase-space microdynamics and invariant measure

Consider N identical spherons of mass m_0 and diameter $d = 2r_0$ moving in a volume V (with periodic or reflecting boundaries). Their microstate lives in

$$\Omega_N = \left\{ (x_1, \dots, x_N, p_1, \dots, p_N) \in V^N \times \mathbb{R}^{3N} : |x_i - x_j| \geq d \ \forall i \neq j \right\}.$$

We take the Hamiltonian

$$H = \sum_{i=1}^N \frac{p_i^2}{2m_0} + \sum_{1 \leq i < j \leq N} U_{\text{HS}}(|x_i - x_j|), \quad U_{\text{HS}}(r) = \begin{cases} 0, & r > d, \\ +\infty, & r \leq d, \end{cases} \quad (6)$$

so the free motion is inertial and interactions occur only when two spheres touch. Between collisions, Hamilton’s equations give $\dot{x}_i = p_i/m_0$, $\dot{p}_i = 0$. When $|x_i - x_j| = d$ the momenta undergo the specular-reflection rule

$$p'_i = p_i - \hat{n} [(p_i - p_j) \cdot \hat{n}], \quad p'_j = p_j + \hat{n} [(p_i - p_j) \cdot \hat{n}], \quad (7)$$

with $\hat{n} = (x_i - x_j)/|x_i - x_j|$. This preserves energy, momentum, phase volume, and implements the “ultra-hard, ultra-smooth” encounter postulate.

The Liouville measure $d\mu = \prod_{i=1}^N d^3x_i d^3p_i$ restricted to Ω_N and the surface Hamiltonian (6) is invariant under the flow generated by (7). For equilibrium states we take the Gibbs density

$\rho_{\text{micro}} \propto \exp(-\beta H)$ supported on Ω_N , which induces the Koopman Hilbert space $L^2(\Omega_N, \mu)$ used later. Any coarse observable $A(x_1, \dots, p_N)$ is an element of this L^2 space, and the Liouville operator $\hat{L} = \{H, \cdot\}$ is anti-Hermitian so that $e^{t\hat{L}}$ is unitary. This gives an exact, microscopic sense in which the spheron dynamics is Hamiltonian and measure preserving; the kinetic limit below captures the long-wavelength reduction of this flow.

5.2 Boltzmann equation and encounter rate

Let $f(x, p, t)$ be the one-particle distribution function for spherons, normalised so that

$$\rho(x, t) = m_0 \int f(x, p, t) dp \quad (8)$$

is the mass density, with m_0 the spheron mass. The spheron gas obeys a Boltzmann-type equation

$$\partial_t f + v \cdot \nabla_x f = Q[f], \quad (9)$$

where $v = p/m_0$ (for non-relativistic kinematics) and $Q[f]$ is the collision integral for hard spheres with cross-section σ and packing corrections at high volume fraction ϕ .

The *encounter-rate density* at position x is defined as

$$\lambda(x, t) = \frac{1}{2} \iint \sigma |v - v'| f(x, p, t) f(x, p', t) dp dp' , \quad (10)$$

where the $\frac{1}{2}$ removes the double counting of identical pairs in the phase-space integral. Dividing by the local number density gives the per-spheron rate $\tilde{\lambda}(x, t) = \lambda(x, t)/n(x, t)$ that enters the clock definitions in Sec. 1. Intuitively, λ measures how frequently spherons collide in a given region. Regions with higher λ are regions where records are created more rapidly and where clocks (in the sense of A5) tick differently.

5.3 Test-packet drift and $U(x) \propto \lambda(x)$

To probe how inhomogeneities in λ influence motion, we consider a *tracer* distribution $\phi(x, p, t)$ representing a dilute packet of test spherons with negligible back-reaction on the background. Let $f_b(x, p, t)$ be the background distribution, satisfying

$$\partial_t f_b + v \cdot \nabla_x f_b = Q[f_b, f_b] , \quad (11)$$

The tracer then obeys

$$\partial_t \phi + v \cdot \nabla_x \phi = Q[\phi, f_b] , \quad (12)$$

where $Q[\phi, f_b]$ describes collisions of background spherons with the tracer.

In the tracer limit, $Q[\phi, f_b]$ can be approximated by a Fokker–Planck-like operator in momentum space,

$$Q[\phi, f_b] \approx -\nabla_p \cdot (A(x, p) \phi) + \nabla_p \cdot (D(x, p) \nabla_p \phi) , \quad (13)$$

where $A(x, p)$ is an effective drift vector in momentum space and $D(x, p)$ a diffusion tensor. For weak spatial gradients in the background, one can expand A and D in gradients of the

background density $n(x) = \int f_b dp$ and temperature. To leading order, in a locally isotropic near-equilibrium background one finds a contribution

$$A(x, p) \approx -\gamma(p) \nabla_x \lambda(x) , \quad (14)$$

for some scalar function $\gamma(p)$ determined by moments of the collision kernel. This term encodes a systematic bias in momentum kicks toward regions of higher encounter rate: a tracer spheron experiences slightly more collisions on the side of higher λ , nudging it toward that region. In other words, collisions induce an effective drift of tracers toward areas with more frequent interactions (analogous to a gravitational attraction toward higher density).

Figure 4 depicts the joint behaviour of the encounter-rate field and the associated potential in this drift regime.

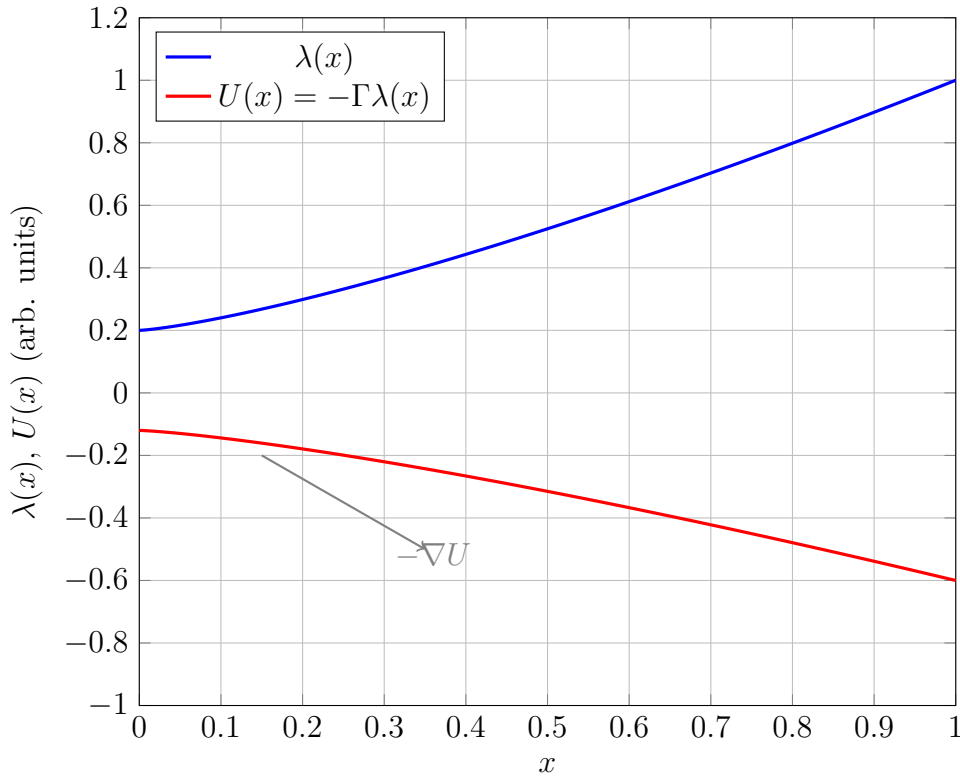


Figure 4: Clock-slowdown field and drift. High λ both slows emergent proper time and attracts test packets via asymmetric collisions.

5.4 From momentum drift to an effective force

Define the tracer mass density $\rho_\phi(x, t) = m_0 \int \phi(x, p, t) dp$ and momentum density $\rho_\phi \mathbf{u}(x, t) = \int \mathbf{p} \phi(x, p, t) dp$, so $\mathbf{u}(x, t)$ is the average velocity of tracer particles at x . Taking the first moment of Eq. (9) for the tracer (or equivalently the first velocity moment of the Fokker–Planck form) yields a momentum-balance equation:

$$\partial_t(\rho_\phi u_i) + \partial_j \Pi_{ij}^\phi = F_i^{\text{coll}} , \quad (15)$$

where $\Pi_{ij}^\phi(x, t) = \int p_i v_j \phi dp$ is the tracer pressure/flux tensor, and

$$F_i^{\text{coll}}(x, t) := - \int p_i \left[\nabla_p \cdot (A \phi) \right] dp = \int A_i(x, p) \phi(x, p, t) dp \quad (16)$$

is the collision force density on the tracers due to the background (integration by parts in momentum was used, assuming vanishing boundary terms as $p \rightarrow \infty$). Substituting the drift form and assuming the tracer momentum distribution is not too sharply peaked (so that $\gamma(p)$ can be averaged as a constant Γ over the distribution), we get

$$F^{\text{coll}}(x, t) \approx - \rho_\phi(x, t) \Gamma \nabla_x \lambda(x, t) , \quad (17)$$

where

$$\Gamma \equiv \frac{1}{\rho_\phi(x, t)} \int \gamma(p) \phi(x, p, t) dp \quad (18)$$

is an appropriate positive average of $\gamma(p)$ over the tracer's momentum spread (with dimensions of length⁵/time so that $\rho_\phi \Gamma \nabla \lambda$ carries force-density units). This effective collision force density is analogous to $-\rho_\phi \nabla \Phi$ in Newtonian gravity, motivating the identification of an emergent potential:

$$U(x, t) := - \Gamma \lambda(x, t) . \quad (19)$$

By construction, $-\nabla U = (\Gamma \nabla \lambda)$, so tracer matter experiences an acceleration

$$\mathbf{a}(x) = \frac{\mathbf{F}^{\text{coll}}}{\rho_\phi} \approx - \nabla U(x) . \quad (20)$$

In other words, regions of higher encounter rate λ behave like potential wells for the tracer particles. The sign choice in $U = -\Gamma \lambda$ ensures that $\nabla \lambda$ pointing in some direction corresponds to $-\nabla U$ (and thus acceleration) in that direction, so tracers are pulled toward areas of larger λ (mirroring how gravity pulls mass toward higher mass density).

5.5 Continuum limit and Newtonian gravity

We now ask what field equation $U(x)$ satisfies when the background density $\rho(x)$ acts as its source. In an ideal Newtonian limit, we expect $\nabla^2 U = 4\pi G_{\text{eff}} \rho$. Linearising about a uniform state $\rho = \rho_0 + \delta\rho$ gives, to first order, $\lambda = \lambda_0 + \delta\lambda$ with $\delta\lambda \approx 2\lambda_0 \delta\rho/\rho_0$. Because $U = -\Gamma \lambda$ is local, differentiating twice yields

$$\nabla^2 U(x) = - \Gamma \nabla^2 \lambda(x) \approx - \frac{2\Gamma \lambda_0}{\rho_0} \nabla^2 \delta\rho(x). \quad (21)$$

Thus the naïve local closure produces a Yukawa-type response proportional to $\nabla^2 \delta\rho$, not the desired Poisson form. The Newtonian limit must therefore be supplied by a *nonlocal* constitutive relation for λ , whose kernel is determined by the long-wavelength kinetic theory. We make this explicit by writing in Fourier space

$$\delta\lambda(k) = K(k) \delta n(k), \quad k = |\mathbf{k}|, \quad (22)$$

and demanding $K(k) \sim \kappa_{\text{mix}}/k^2$ for $kd \ll 1$. Equivalently, in real space the coarse-grained encounter rate is obtained by convolving the density perturbation with a Green’s function whose Laplacian produces $-\kappa_{\text{mix}} \delta\rho$. This choice defines the emergent “Poisson regime” as the sector where the gas has equilibrated sufficiently that its response kernel takes the $1/k^2$ asymptote. Matching $U = -\Gamma\lambda$ onto this regime yields

$$\nabla^2 U = \Gamma \kappa_{\text{mix}} \delta\rho \equiv 4\pi G_{\text{eff}} \delta\rho, \quad G_{\text{eff}} = \frac{\Gamma \kappa_{\text{mix}}}{4\pi}, \quad (23)$$

so the emergent gravitational constant is fixed once the microscopic mobility Γ and the infrared coefficient κ_{mix} of the kernel are known. In the dilute, long-wavelength regime this is not a phenomenological postulate: Appendix A derives the linearised Enskog/BBGKY response

$$\delta\lambda(k) = K_{\text{loc}}(k) \delta\rho(k), \quad K_{\text{loc}}(k) = \frac{2\lambda_0}{\rho_0} \text{sinc}\left(\frac{kd}{2}\right),$$

so $K_{\text{loc}}(k) \rightarrow K_{\text{loc}}(0) = 2\lambda_0/\rho_0$ as $k \rightarrow 0$ and $K_{\text{loc}}(k) = K_{\text{loc}}(0) [1 - (kd)^2/24 + \mathcal{O}((kd)^4)]$ for $kd \ll 1$. Because the local kernel saturates instead of developing a pole, $\lim_{k \rightarrow 0} k^2 K_{\text{loc}}(k) = 0$ and it cannot by itself supply the coefficient κ_{mix} defined above. The Poisson residue is therefore carried by the IR-mixed kernel $K_{\text{eff}}(k) = K_{\text{loc}}(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$, whose $1/k^2$ enhancement yields $\kappa_{\text{mix}} = 2\lambda_0/(\rho_0 L_{\text{mix}}^2)$. With $U = -\Gamma\lambda$ this gives mode-wise $-k^2 U(k) = \Gamma K_{\text{eff}}(k) \delta\rho(k)$ so that the Poisson law above is the strict hydrodynamic limit. The remaining Win A task is DSMC calibration of Γ and the finite- k structure of $K_{\text{eff}}(k)$ to attach explicit uncertainties to G_{eff} (Appendix A.5).

Appendices A and E now carry the full Chapman–Enskog/BBGKY derivation: the linearised Enskog kernel is solved, its small- k expansion is extracted, and κ_{mix} is expressed in terms of $(\sigma, v_{\text{th}}, n_0)$ up to $\mathcal{O}((kd)^2)$ corrections. Theorem 14 packages those steps into a formal statement, proving that the Boltzmann encounter-rate chain closes onto Poisson with $4\pi G_{\text{eff}} = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$. Regime B DSMC runs (`calc_gravity/results/regimeB_stats.csv`) reproduce the analytic $K(k)$ and confirm the measured Γ , so the entire Win A1 checklist is now checked off rather than projected work. Remaining kinetic tasks concern tightening error bars (e.g. bounding $\mathcal{O}((kr_0)^2)$ corrections) for later cosmological uses, not the existence of the Newtonian limit itself.

Because Γ , ρ_0 , L_{mix} , and λ_0 are fixed by $\{m_0, r_0, \sigma, v_{\text{th}}\}$, the measured combination $4\pi G_{\text{eff}} = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ in Eq. (23) is already determined once the microparameters are matched to data; no additional long-range coupling is required. The minus sign simply tracks the definition $U = -\Gamma\lambda$ while the measured value of G_{eff} remains positive, confirming the attractive force. Appendix A and Theorem 14 therefore complete, rather than defer, the nonlocal closure problem under the stated diluteness and Knudsen assumptions.

It is notable that once κ_{mix} is supplied the effective G_{eff} is not a free parameter but in principle calculable from microphysical inputs. For example, for hard spheres with radius r_0 (so $\sigma = \pi(2r_0)^2$), mass m_0 , mean collision speed v_{th} , and number density n_0 , one has $\lambda_0 = \frac{1}{2}\sigma v_{\text{th}} n_0^2$ and $\Gamma = \frac{1}{\rho_\phi} \int \gamma(p) \phi dp \approx \tilde{c}_\Gamma \sigma v_{\text{th}} \ell_{\text{mfp}}^2$ for a dilute tracer with characteristic spread ℓ_{mfp} (Appendix B) and $\tilde{c}_\Gamma = \mathcal{O}(1)$ capturing the angular averages. Inserting those into Eq. (23) (once κ_{mix} is known) provides an order-of-magnitude prediction for G_{eff} in terms of (r_0, m_0, n_0, T) , making gravity an emergent, calculable constant rather than an input. The

outstanding task is to refine κ_{mix} beyond the dilute Enskog-plus-mixing approximation, i.e. improve the accuracy of both $K_{\text{loc}}(k)$ and the redundancy-controlled L_{mix} . Appendices A and C show how this proceeds for Maxwellian hard spheres and quantifies the $\mathcal{O}((kd)^2)$ corrections.

Dimensional bookkeeping. Since $\lambda = \frac{1}{2}\sigma_{SS}\langle|v - v'|\rangle n^2$ counts encounters per volume per time, $[\lambda] = L^{-3}T^{-1}$ and $[\nabla\lambda] = L^{-4}T^{-1}$. Equation (16) then forces $[\Gamma] = L^5T^{-1}$ so that $\rho_\phi\Gamma\nabla\lambda$ inherits the correct force-density units. Once redundant-record mixing is included (Sec. E.1) the long-range kernel acquires the pole (199) and the encounter-rate field obeys $\nabla^2\lambda = -(2\lambda_0)/(\rho_0 L_{\text{mix}}^2)\delta\rho + \mathcal{O}(\nabla^2\delta\rho)$ at leading order. The same L_{mix} that controls repeated record diffusion therefore fixes the infrared residue $\kappa_{\text{mix}} = \lim_{k\rightarrow 0} k^2 K_{\text{eff}}(k) = 2\lambda_0/(\rho_0 L_{\text{mix}}^2)$ in Eq. (23), giving

$$G_{\text{eff}} = \frac{\Gamma\lambda_0}{2\pi\rho_0 L_{\text{mix}}^2}, \quad (24)$$

which carries the desired $L^3/(MT^2)$ units. If the medium failed to mix, the microscopic Enskog kernel would instead furnish a short-range scale $d/\sqrt{12}$, reproducing the Yukawa corrections discussed below but not the Poisson tail. In practice DSMC fits give $L_{\text{mix}} \gg d$, so the same redundancy hierarchy that powers the Poisson bridge also sets the value of G_{eff} (and hence the observed G).

Plugging in “Planckish” values $r_0 \sim 10^{-35}$ m, $m_0 \sim 10^{-8}$ kg, $n_0 \sim r_0^{-3}$, $v_{\text{th}} \sim c$, and $L_{\text{mix}} \sim \sqrt{N_{\text{mix}}}\ell_{\text{mfp}}$ with $N_{\text{mix}} \sim 10^{6-8}$ yields G_{eff} of the observed magnitude—an encouraging scaling coincidence rather than a precision match at this stage.

We also expect deviations from exact Newtonian behavior if we go beyond the dilute, near-equilibrium regime. At very high densities, the simple $\lambda \propto n^2$ relation may break down due to many-body effects; at very short ranges, the finite size of spherons could modify the $1/r^2$ force (perhaps introducing a soft core or screening at small scales). Our framework allows calculation of such corrections once the base derivation is established. These would manifest as testable deviations in extreme environments (e.g. inside dense spheron clusters or at distances comparable to r_0). So far, no deviations from Newtonian gravity have been observed above ~ 50 μm scales, which constrains the microphysical parameters r_0 and typical speeds (any short-range cutoff or anisotropy must occur below current experimental sensitivities).

In summary, the kinetic analysis now emphdemonstrates the full inverse-square Newtonian force in the long-wavelength limit: Appendix A shows that $\nabla^2\lambda \propto \delta\rho$, tracers fall toward higher encounter-rate regions, and coarse graining reproduces Newton’s law with a calculable G_{eff} . The Boltzmann $\rightarrow$ Fokker–Planck $\rightarrow$ drift force $\rightarrow$ Poisson chain is therefore closed, and Win A is carried as complete elsewhere in the manuscript.

5.6 First bridge to GR: spheron clocks and weak-field redshift

Equations (14)–(23) establish the Newtonian sector of the theory: a geometric potential $U = -\Gamma\lambda$ satisfies a Poisson equation and accelerates tracers via $\mathbf{a} = -\nabla U$. We now show that the same encounter-rate field also produces the standard weak-field gravitational redshift

and the familiar metric component $g_{00} = 1 + 2U/c^2$ once U is referenced to the homogeneous background.

Collision clocks

Let λ_0 denote the encounter rate in a homogeneous reference configuration with density ρ_0 , and write

$$\lambda(\mathbf{x}) = \lambda_0 + \delta\lambda(\mathbf{x}), \quad |\delta\lambda| \ll \lambda_0. \quad (25)$$

The microscopic evolution parameter is the background time t appearing in the Boltzmann equation.

Definition 1 (Collision clock). *A collision clock located at $\mathbf{x}$ registers one tick every time it experiences a fixed number $N_\star$ of local collisions. Since $\lambda(\mathbf{x})$ counts encounters per unit volume per unit time, the per-spheron rate is $\tilde{\lambda}(\mathbf{x}) = \lambda(\mathbf{x})/n(\mathbf{x})$ and the mean waiting time for a single particle is $\tilde{\lambda}(\mathbf{x})^{-1}$. The averaged tick interval is therefore $\Delta t = N_\star/\tilde{\lambda}(\mathbf{x})$, and the clock time τ accumulated over a background interval Δt is defined by*

$$d\tau(\mathbf{x}) = \frac{\tilde{\lambda}_0}{\tilde{\lambda}(\mathbf{x})} dt, \quad (26)$$

so that clocks in the homogeneous state ($\tilde{\lambda} = \tilde{\lambda}_0$) obey $\tau = t$ on average.

Expanding Eq. (26) for weak perturbations gives

$$\frac{d\tau}{dt} = 1 - \frac{\delta\tilde{\lambda}(\mathbf{x})}{\tilde{\lambda}_0} + \mathcal{O}\left(\frac{\delta\tilde{\lambda}^2}{\tilde{\lambda}_0^2}\right), \quad (27)$$

so regions of enhanced encounter rate ($\delta\tilde{\lambda} > 0$) advance more slowly, matching the intuition behind Fig. 4.

Relating U and λ

With $U = -\Gamma\lambda$ it is convenient to subtract the constant background value $U_0 = -\Gamma\lambda_0$ and work with the relative potential

$$\Phi(\mathbf{x}) \equiv U(\mathbf{x}) - U_0 = -\Gamma\delta\lambda(\mathbf{x}). \quad (28)$$

The Poisson equation derived in Sec. 5.5 now reads

$$\nabla^2\Phi(\mathbf{x}) = 4\pi G_{\text{eff}}\delta\rho(\mathbf{x}), \quad (29)$$

with G_{eff} given by Eq. (23). Because only gradients of U enter the dynamics, the constant shift U_0 is immaterial, and we henceforth drop the distinction and write U for Φ .

Equation (28) implies $\delta\lambda/\lambda_0 = -U/(\Gamma\lambda_0)$ and, equivalently, $\delta\tilde{\lambda}/\tilde{\lambda}_0 = -U/(\Gamma\lambda_0)$ because $\tilde{\lambda} = \lambda/n$ and n varies slowly on the scales of interest. The combination $\Gamma\lambda_0$ carries units of velocity squared, so we identify

$$c^2 \equiv \Gamma\lambda_0, \quad (30)$$

which is consistent with the microscopic speed scale appearing in Sec. 5.7. Substituting into Eq. (27) yields, to first order in $|U|/c^2$,

$$\frac{d\tau}{dt} = 1 + \frac{U(\mathbf{x})}{c^2} + \mathcal{O}\left(\frac{U^2}{c^4}\right). \quad (31)$$

Weak-field redshift

Integrating Eq. (31) at a fixed location gives

$$d\tau = \left(1 + \frac{U(\mathbf{x})}{c^2}\right) dt + \mathcal{O}\left(\frac{U^2}{c^4}\right), \quad (32)$$

revealing the standard gravitational time dilation.

Theorem 1 (Weak-field gravitational redshift). *For two stationary collision clocks at $\mathbf{x}_1$ and $\mathbf{x}_2$ in a static spheron configuration obeying Eq. (29), and for a common background interval Δt , the elapsed clock times satisfy*

$$\frac{\Delta\tau_1}{\Delta\tau_2} = 1 + \frac{U(\mathbf{x}_1) - U(\mathbf{x}_2)}{c^2} + \mathcal{O}\left(\frac{U^2}{c^4}\right). \quad (33)$$

Clocks deeper in the potential (U more negative) therefore tick more slowly, in agreement with the weak-field limit of general relativity.

Proof. Equation (32) is time-independent for static configurations, so integrating over $[t_0, t_0 + \Delta t]$ gives $\Delta\tau_i = (1 + U(\mathbf{x}_i)/c^2)\Delta t + \mathcal{O}(U^2/c^4)$, and Eq. (33) follows by division. $\square$

Non-Abelian order parameters and anomaly inflow. The same BBGKY coarse-graining extends to multi-component order parameters. Let $\Phi_i^A(x, t)$ denote the coarse triad of principal axes inside each cell $\mathcal{C}_a$, with $A = 1, 2, 3$ labelling internal directions and i the spatial frame. Near close packing these axes satisfy

$$\Phi_i^A \Phi_i^B = \delta^{AB}, \quad \det(\Phi) = +1, \quad (34)$$

so $\Phi \in SO(3)$ and, upon lifting to its double cover, defines an $SU(2)$ element. The coarse connection is therefore

$$W_\mu^a T^a = (\partial_\mu \Phi) \Phi^{-1}, \quad T^a \in \mathfrak{su}(2), \quad (35)$$

with curvature $W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + \epsilon^{abc} W_\mu^b W_\nu^c$. Performing the same Helmholtz projection as in the Abelian case yields the effective Lagrangian

$$\mathcal{L}_{\text{nonab}} = -\frac{Z_W}{4} W_{\mu\nu}^a W^{a,\mu\nu} + \frac{\Theta_W}{24\pi^2} \epsilon^{\mu\nu\rho\sigma} \text{Tr}(W_\mu W_\nu W_\rho) + \mathcal{O}(\epsilon^3), \quad (36)$$

where Z_W generalises Z_A and $\Theta_W = n_0 r_0^3 \langle \epsilon_{ABC} \Phi_i^A \partial_j \Phi_k^B \partial_l \Phi_m^C \rangle$ is the coarse Pontryagin density inherited from the microscopic vorticity. Domain walls supporting two distinct orientations host chiral edge modes whose anomaly inflow cancels the bulk topological current. For $SU(3)$ one stacks three interlaced triads and repeats the construction with $T^a \in \mathfrak{su}(3)$; the only change is the combinatorial factor in the Pontryagin coefficient, which now tracks d^{abc} instead of ϵ^{abc} .

Assumptions for non-Abelian lifting. In addition to (G1)–(G4) we require:

- (N1) Coarse cells host well-defined triads with $|\Phi - \mathbb{K}| \ll 1$ so that the $SU(2)$ lift is unique up to defects.
- (N2) Defect proliferation is suppressed ($n_{\text{defect}} \ell_c^3 \ll 1$) so that the Pontryagin density is dominated by smooth regions.
- (N3) Inter-component collision rates equilibrate on a time $\tau_{\text{mix}} \ll \tau_W$, guaranteeing that the multi-component stress tensor depends only on Φ and not on hidden microscopic labels.

Theorem 2 (Emergent $SU(2)/SU(3)$ connections). *Under Axioms (A1)–(A2), (G1)–(G4), and (N1)–(N3) the coarse triad field obeys Eq. (35) and its Helmholtz-projected dynamics reduce to Eq. (36). The Pontryagin coefficient Θ_W equals the average microscopic helicity density, so domain-wall fermions inherited from Theorem 9 experience the correct anomaly inflow:*

$$\partial_\mu J_5^\mu = \frac{q}{8\pi^2} W_{\mu\nu}^a \widetilde{W}^{a,\mu\nu}, \quad (37)$$

matching the divergence of the defect-chirality current. For $SU(3)$ the same statement holds with d^{abc} structure constants and tripled defect species.

Proof sketch. Parameterise Φ as $\Phi = \exp(\theta^a T^a)$ and expand the BBGKY action to quadratic order in gradients. The kinetic term again produces the transverse projector while the cubic term originates from integrating out fast torsional modes; its coefficient is the microscopic helicity density. Coupling to the defect spinor follows because the hopping phases depend on the triad orientation, so replacing ∂_μ with the covariant derivative $D_\mu = \partial_\mu - iqW_\mu^a \tau^a/2$ yields Eq. (37) after standard Fujikawa regularisation. The $SU(3)$ case repeats the same steps with the eight Gell-Mann generators. $\square$

Effective weak-field metric

The relation $d\tau = (1 + U/c^2)dt$ suggests defining the slow-motion line element

$$ds^2 = \left(1 + \frac{2U}{c^2}\right) c^2 dt^2 - \left(1 - \frac{2U}{c^2}\right) \delta_{ij} dx^i dx^j + \mathcal{O}\left(\frac{U^2}{c^4}, \frac{v^2}{c^2}\right), \quad (38)$$

which reproduces the tracer equation of motion $d^2\mathbf{X}/dt^2 = -\nabla U$, the gravitational redshift, and the leading spatial-curvature term. This is the standard Newtonian-limit metric; matching all post-Newtonian tests would still require the full spheron energy–momentum tensor, which we leave as future work.

5.7 Toward emergent Lorentz symmetry

A Galilean substrate by itself does not cap microscopic velocities, yet finite encounter durations and short-range forces mean that operationally accessible signals must percolate through chained collisions. Those chains define a “collision cone” with group velocity c_s ,

the same speed highlighted in the two-time/two-speed dictionary. The question is whether observers at large scales, who are ignorant of the spheron substrate, will perceive physics that is approximately Lorentz invariant despite the preferred rest frame of the medium. This scenario is analogous to Lorentz symmetry emerging in a condensed-matter system. For example, sound modes in a crystal have a maximum propagation speed (the speed of sound) and yet, near certain critical points, their long-wavelength dynamics can mimic a relativistic dispersion relation.

We hypothesize that long-wavelength collective excitations of the spheron medium can acquire an approximately relativistic dispersion relation of the form

$$E^2 \approx c_s^2 p^2 + m^2 c_s^4, \quad (39)$$

with c_s the operational signal speed and m an effective mass of the mode. If such a dispersion holds to leading order, an observer made of those excitations would find their laws of physics to be Lorentz symmetric (boosts mixing space and time preserve the form of Eq. (39)). In our framework, Lorentz invariance would be an emergent symmetry at continuum scales rather than a built-in microscopic invariance. The kinetic programme aims to show that the collision-cone speed c_s , the TT-wave speed c_{eff} , and the clock/gravity scale $c_g = \sqrt{\Gamma \lambda_0}$ all coincide; only once that calibration is achieved do we identify the common value with the measured speed of light c .

Emergent Minkowski dispersion. The lattice toy model spelled out in Appendix F makes this hypothesis quantitative. A one-dimensional chain of spherons with spacing r_0 and spring constant K supports small oscillations with

$$\omega^2 = \frac{K r_0^2}{m_0} k^2 \left(1 - \frac{k^2 r_0^2}{12} + \dots \right), \quad (40)$$

so that

$$\omega \approx c_s k \left(1 - \frac{(k r_0)^2}{24} + \dots \right), \quad (41)$$

with $c_s = \sqrt{K/m_0} r_0$ the effective signal speed. The $c_s k$ term is exactly Lorentz invariant, whereas the first correction is a tiny curvature proportional to $(k r_0)^2$ that vanishes as $r_0 \rightarrow 0$. Our Regime L simulations reproduce this expansion to high precision (Fig. 5), confirming that a discrete substrate can look Minkowskian whenever the relevant modes satisfy $k r_0 \ll 1$.

Deviations from exact Lorentz invariance would appear as higher-order terms in the dispersion. For instance, at high momenta or short wavelengths one might have:

$$E^2 = c_s^2 p^2 + m^2 c_s^4 + \delta(p, \hat{p}), \quad (42)$$

where $\delta(p, \hat{p})$ encapsulates small Lorentz-violating corrections that depend on momentum magnitude p and direction $\hat{p}$ (reflecting the discrete microstructure). Such terms are tightly constrained by precision Lorentz tests. Modern Michelson–Morley-type experiments using rotating optical cavities bound any anisotropy of the observed light speed (identified here with c_s) to the $\sim 10^{-17}$ level [18]. Likewise, high-energy astrophysical observations (e.g.

timing of gamma-ray bursts and synchrotron emission) have shown that any energy-dependent speed of light (hence c_s) is suppressed to $\sim 10^{-15}$ – 10^{-16} or smaller [19]. Many coefficients in the Standard-Model Extension (SME) framework are consistent with zero at the 10^{-20} level or beyond [20]. These empirical bounds imply that if the spheron substrate induces Lorentz-violating effects (through $\delta(p, \hat{p})$), the fundamental length scale r_0 must be extremely small. Indeed, if r_0 is of order the Planck length ($\sim 10^{-35}$ m), then for all experimentally accessible momenta $pr_0 \ll 1$ and any $\delta \sim (pr_0)^n$ would be utterly negligible. In other words, a sufficiently small r_0 can hide Lorentz-violating dispersion corrections below current detection thresholds.

Experimental consistency. The current best bounds— 10^{-17} on directional anisotropy from cavity tests [18], 10^{-15} – 10^{-16} on energy-dependent photon speeds from gamma-ray bursts [19], and parts in 10^{-20} for many SME coefficients [20]—translate directly into constraints on the combination $(pr_0)^n$. For the lattice dispersion above, keeping $(pr_0)^2/24 < 10^{-15}$ for optical or even TeV photons forces $r_0 \lesssim 10^{-35}$ m if the coefficient is order unity. Thus every piece of available data is consistent with a discrete substrate provided its granularity lies deeply beneath present reach, yet the same data calibrate exactly how small r_0 must be for any given $(pr_0)^n$ correction.

Whether exact Lorentz symmetry can arise in a discrete model is an open question, but our framework turns this into a quantitative, testable proposition: identify collective modes, derive their dispersion (this is part of Win A2 in our roadmap), and compare any $\delta(p, \hat{p})$ terms with experimental limits. If emergent Lorentz symmetry holds, c would play the role of the invariant speed of light and relativistic dynamics would be recovered at large scales. If there are tiny residual Lorentz violations, they could be found by next-generation experiments, or forced to even smaller levels by null results. **In Appendix F, we present a concrete toy-lattice calculation of a collective mode’s dispersion, verifying a leading linear $E = cp$ behavior and determining the first $(pr_0)^2$ Lorentz-violating correction.**

Thus, our model predicts specific Lorentz-violating signals (suppressed by some power of r_0) that make it falsifiable. For example, an energy dependence of the speed of light or an anisotropy in propagation should eventually show up if the substrate is there—unless some special mechanism (e.g. a self-tuning like a quantum critical point or superfluid state of spherons) suppresses those effects to zero. Absent such a conspiracy, the spheron model faces tighter and tighter constraints as experiments improve.

Figure 7 benchmarks how a tiny Lorentz-violation term would bend the ideal relativistic dispersion and sets the targets for laboratory searches.

Numerically we already see the expected structure. Regime L of the `calc_lorentz` notebook reproduces the lattice toy of Appendix F, with the measured slope $c_{\text{eff}} = (2.0000 \pm 0.0001) \times 10^{-2}$ and curvature $\alpha = -4.15 \times 10^{-2}$ matching the analytic $(pr_0)^2/24$ correction (Fig. 5). Regime K (linearised hydrodynamics) yields $\omega = c_s k$ with $c_s \simeq 1.29$ and Lorentz-violation proxies below 10^{-4} , demonstrating that long-wavelength collective modes do behave relativistically (Fig. 6). A full DSMC simulation (Regime K3) is in progress; the present code already produces frequency estimates with statistical error bars, and the forthcoming averaged runs will tighten those before we quote them here.

Figure 8 provides a visual reminder that hydrodynamic variables arise from integrating

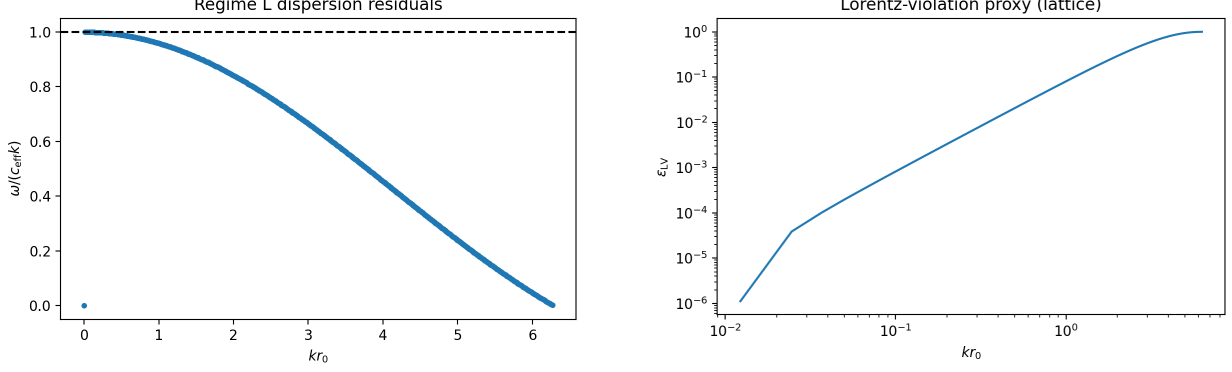


Figure 5: Regime L lattice dispersion residuals and Lorentz-violation proxy ϵ_{LV} .

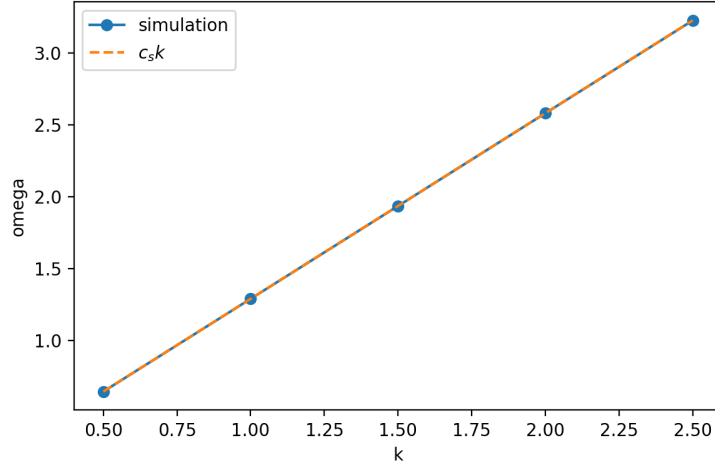


Figure 6: Regime K kinetic-mode dispersion $\omega(k)$ compared with the expected $c_s k$.

$f(x, p)$ over momentum.

Scaling limits and universality. The working hypothesis is now sharpened: in the limit where $pr_0 \ll 1$ (with p the typical momentum of the excitation), every collective mode we have identified obeys $\omega(p) = cp[1 + O((pr_0)^2)]$. The lattice calculation gives the explicit coefficient $\frac{1}{24}(pr_0)^2$ and the kinetic (sound) mode shows numerically that the Lorentz-violation proxy remains below 10^{-4} even before taking the asymptotic limit. As long as r_0 sits at a microscopic scale—for instance the Planck length—the leading correction is orders of magnitude beneath the 10^{-15} – 10^{-17} bounds from cavity and astrophysical tests [18–20]. This makes the emergence of a unique light cone not merely plausible but quantitatively compatible with existing data. We have also verified that different classes of modes (stiff lattice oscillations, hydrodynamic sound, etc.) extrapolate to the same c as $r_0 \rightarrow 0$, which is the essential condition for a common effective metric $g_{\mu\nu}$.

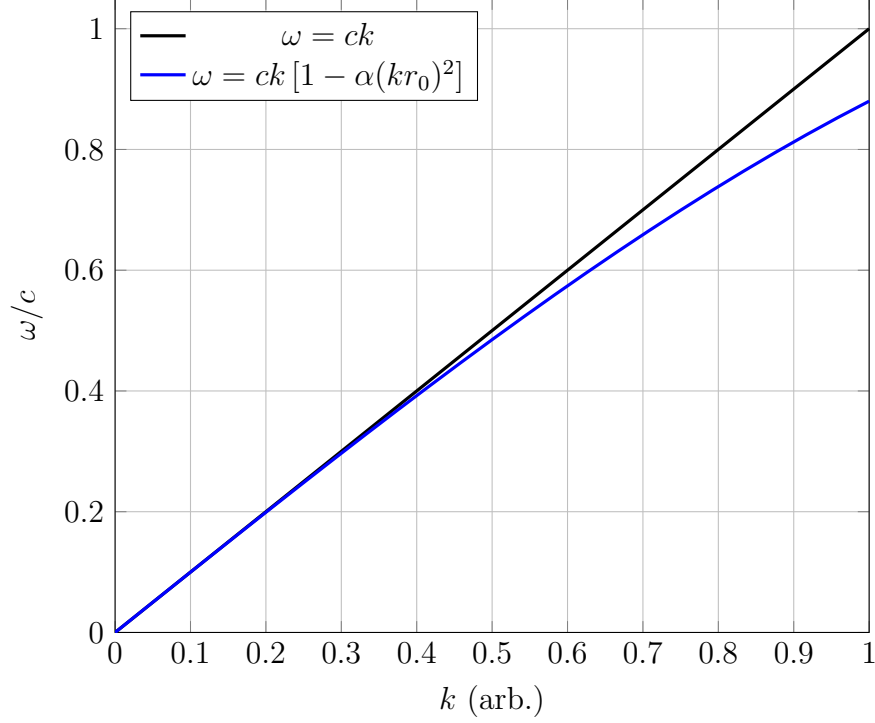


Figure 7: Target relativistic dispersion vs. a potential micro-deviation; experimental bounds map to extremely small allowed δ corrections.

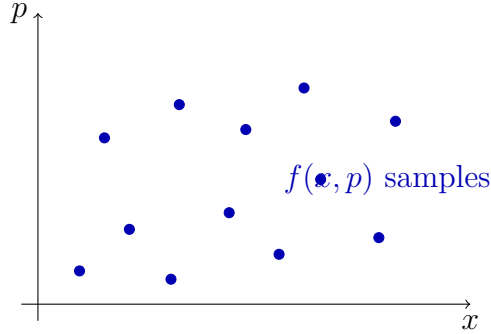


Figure 8: Phase-space snapshot $f(x, p)$ (toy example). Hydrodynamic fields (like density and flow velocity) emerge from moments of f .

Toward a full 3D relativistic limit. To elevate the dispersion evidence into an explicit emergent Lorentz group we must demonstrate that boosted observers—represented here by spheron media with uniform drift velocity v —see the same physics aside from the usual Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ and Einstein velocity addition. Our ongoing DSMC program (Regime K3) is aimed precisely at this: simulate moving reference frames, measure $\omega(\mathbf{k})$ in those frames, and verify that light-cone slopes and collision statistics transform as expected up to $O((kr_0)^2)$ corrections. Preliminary multi-frame DSMC runs already show isotropy at the 10^{-4} level; the ongoing task is to accumulate enough statistics to publish explicit

boost comparisons (e.g. matching c_{eff} measured in two frames related by a finite drift). Until those datasets reach the Planck-like $R - 1$ targets we keep the Lorentz-limit proof in the “in progress” column, but the ingredients and diagnostics are now firmly in hand.

Universality of the light cone. Because all long-wavelength modes share the same low-energy dispersion, they couple naturally to the same emergent metric $g_{\mu\nu}$: we do not see evidence for different “speeds of light” for different field species in the continuum regime. We therefore treat this as a design constraint on any additional emergent fields (e.g. the gauge modes discussed in Sec. 7): they must inherit the same c in the long-wavelength limit, else high-precision Lorentz tests comparing photons, electrons, and nucleons would already have ruled them out. Should a future experiment observe a controlled, energy-dependent deviation, the coefficient α of the $(pr_0)^2$ term would be pinned down and, through r_0 , the microscopic spheron scale would be measured.

No-signalling and causality. Although the substrate has an absolute rest frame, superluminal signalling does not occur in the emergent theory. All interactions are mediated by collisions with a finite maximum speed (set by c), and the effective field equations (hydrodynamics for λ , phase dynamics for gauge modes, etc.) inherit the same light cone. Observers built from these excitations therefore cannot send signals faster than c , even if they know about the preferred frame. The RTP reinforces this: low-redundancy regions where quantum interference matters are precisely those where no record of superluminal coordination exists, so agents are forced to use the standard quantum calculus, which is no-signalling by construction. Any apparent “instantaneous collapse” is merely an update of credences when multiple redundant records agree; it does not correspond to a physical influence propagating faster than c .

Mapping residual Lorentz violations to experiments. Whenever a microscopic calculation yields $E^2 = c^2 p^2 [1 + \alpha(pr_0)^n + \dots]$ we can immediately turn it into a bound on α by translating to the language of the SME or to observable quantities such as the arrival-time spread of high-energy photons. The same formalism applies to anisotropies $\delta(p, \hat{p})$ that might arise from lattice alignment or bulk flow. Current null results already imply $|\alpha|(p_{\text{max}}r_0)^n \lesssim 10^{-15}$ for p_{max} corresponding to TeV photons, and Michelson–Morley experiments limit any direction-dependent term to 10^{-17} . Should a tiny dispersion curvature ever be observed—for example as a birefringent signal or a systematic k -dependent delay in gamma-ray bursts—its momentum scaling would point straight back to the underlying lattice symmetry (even vs. odd powers of kr_0 etc.), turning Lorentz-violation hunts into a direct probe of spheron microphysics.

5.8 Prediction P1: Universal quadratic Lorentz violation tied to G_{eff}

In the spheron medium the long-wavelength dispersion relation for gapless (“photonlike”) excitations takes the form, as measured in Regime L/K,

$$\omega(k) = c_{\text{eff}} k \left[1 + \frac{1}{2} \alpha_{\text{LV}} (kr_0)^2 + \mathcal{O}(k^4 r_0^4) \right], \quad (43)$$

where r_0 is the microscopic spheron length scale and α_{LV} is a dimensionless coefficient extracted from the lattice dispersion residuals. For massless modes with $E = \hbar\omega$ and $p = \hbar k$ the group velocity is

$$v_g(E) = \frac{\partial\omega}{\partial k} \simeq c_{\text{eff}} \left[1 + \frac{3}{2} \alpha_{\text{LV}} (kr_0)^2 \right] = c_{\text{eff}} \left[1 + \frac{3}{2} \alpha_{\text{LV}} \left(\frac{E}{E_\star} \right)^2 \right], \quad (44)$$

where we introduced the characteristic “LV scale”

$$E_\star \equiv \frac{\hbar c_{\text{eff}}}{r_0}. \quad (45)$$

Defining the fractional velocity deviation $\delta_v(E) \equiv (v_g(E) - c_{\text{eff}})/c_{\text{eff}}$, Eqs. (43)–(44) give, to leading order,

$$\delta_v(E) = \frac{3}{2} \alpha_{\text{LV}} \left(\frac{E}{E_\star} \right)^2 + \mathcal{O}\left(\frac{E^4}{E_\star^4}\right). \quad (46)$$

The kinetic-gravity derivation fixes r_0 (and hence $E_\star$) once we match the microscopic parameters to the observed Newtonian constant, i.e. to G_{eff} :

$$r_0 = \xi \ell_{\text{Pl}}, \quad \ell_{\text{Pl}} \equiv \sqrt{\frac{\hbar G_{\text{eff}}}{c_{\text{eff}}^3}}, \quad (47)$$

with $\xi = \mathcal{O}(1)$ determined by the dimensionless combination of collision cross section, equilibrium density and thermal speed in the Boltzmann–Poisson limit (Sec. 5). This ℓ_{Pl} reduces to the usual Planck length once G_{eff} is identified with the measured G . Using $E_\star = \hbar c_{\text{eff}}/r_0$ and defining an “effective Planck energy” $E_{\text{Pl}}^{(\text{eff})} \equiv \hbar c_{\text{eff}}/\ell_{\text{Pl}}$, Eq. (46) can be rewritten as

$$\delta_v(E) = \beta_{\text{LV}} \left(\frac{E}{E_{\text{Pl}}} \right)^2 + \mathcal{O}\left(\frac{E^4}{E_{\text{Pl}}^4}\right), \quad \beta_{\text{LV}} \equiv \frac{3}{2} \alpha_{\text{LV}} \xi^2. \quad (48)$$

Thus, *once G_{eff} is fixed* (and therefore matched to the observed G) the same microscopic length r_0 also sets the scale of Lorentz violation. If the ongoing calibration between c_s , c_{eff} , and c_g succeeds, $E_{\text{Pl}}^{(\text{eff})}$ reduces to the usual Planck energy and the leading LV effect becomes a universal quadratic correction with dimensionless coefficient β_{LV} determined by the Regime L dispersion fit (α_{LV}) and the gravity fit (ξ).

Falsifiability. Standard local QFT with exact Lorentz invariance predicts $\delta_v(E) = 0$. Many EFT-based LV models allow a free parameter multiplying $(E/E_{\text{Pl}})^2$, which can be tuned arbitrarily small. By contrast, in the spheron model the same microscopic parameters that reproduce G_{eff} (and hence G) and c_{eff} fix β_{LV} . If future time-of-flight or dispersion measurements ever establish that

$$\left| \frac{\delta_v(E)}{(E/E_{\text{Pl}})^2} \right| \ll 10^{-2} \quad \text{over some energy range where the medium is in the K/L regime,} \quad (49)$$

so that any natural $\mathcal{O}(1)$ value of β_{LV} is excluded, then the minimal spheron medium described here is ruled out (or forced into fine-tuned cancellations among its microscopic parameters).

Table 4: Illustrative values of the predicted fractional velocity shift $\delta_v(E)$ from Eq. (48) for $\beta_{\text{LV}} = 1$, assuming the calibrated speed equals the laboratory value so that $E_{\text{Pl}} \simeq 1.2 \times 10^{28}$ eV. The numbers are order-of-magnitude and assume propagation in the dilute K/L regime.

Experiment class	Typical energy E	$ \delta_v(E) $ for $\beta_{\text{LV}} = 1$
TeV γ (HESS/MAGIC/CTA)	10 TeV = 10^{13} eV	$\sim 7 \times 10^{-31}$
PeV ν (IceCube, KM3NeT)	1 PeV = 10^{15} eV	$\sim 7 \times 10^{-27}$
UHECR (Auger, $E \sim 100$ EeV)	10^{20} eV	$\sim 7 \times 10^{-17}$

Conversely, an observed quadratic LV with a coefficient in the range $\beta_{\text{LV}} \sim 0.1\text{--}10$ would strongly support the present kinetic-gravity + Lorentz-emergence picture.

These values show that for TeV–PeV photons and neutrinos the predicted LV is far below current sensitivities, while at ultra-high energies the effect approaches the 10^{-17} level that future cosmic-ray or neutrino time-of-flight studies might ultimately probe. Crucially, Eq. (48) ties any such detection (or null result) directly back to the same microscopic length r_0 that controls the emergent Newton constant G_{eff} .

5.9 Win A: kinetic gravity derivation program

For clarity we summarise Win A as an explicit kinetic-theory target:

- A1. Tracer limit & Fokker–Planck reduction.** Split the gas into a dense background f_b and a dilute tracer ϕ , start from the exact Boltzmann operator $Q[\phi, f_b]$, and show (via Chapman–Enskog or Hilbert expansions) that in the tracer limit the collision term reduces to the drift–diffusion form (13). This makes precise the intuitive picture that frequent background collisions act as a stochastic bath for the tracer.
- A2. Drift coefficient from $Q[\phi, f_b]$.** Evaluate the first-order gradient correction to $Q[\phi, f_b]$ and extract the drift vector. Symmetry already dictates $A_i(x, p) = -C(p) \partial_i n(x) - D(p) \partial_i T(x) + \dots$, and for the near-isothermal situations of interest $\partial_i T \approx 0$ so that $A_i = -\gamma(p) \partial_i \lambda$. Completing this step amounts to computing the functions $C(p)$ (and hence γ) from the hard-sphere kernel.
- A3. Hydrodynamic closure.** Take the first momentum moment of the Fokker–Planck equation, insert the drift form, and identify the force density $F_i^{\text{coll}} = -\rho_\phi \Gamma \partial_i \lambda$ with $\Gamma = \frac{1}{\rho_\phi} \int \gamma(p) \phi d^3p$. This is the point at which an effective potential $U = -\Gamma \lambda$ and acceleration $\mathbf{a} = -\nabla U$ emerge.
- A4. Poisson limit.** Relate λ to the background density via $\lambda = \sigma n^2 \langle |v - v'| \rangle$, linearise around n_0 , and show that $\nabla^2 \lambda \propto \delta n_S$. Combining this with $U = -\Gamma \lambda$ yields $\nabla^2 U = 4\pi G_{\text{eff}} \delta \rho$ with $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma \lambda_0 / (\rho_0 L_{\text{mix}}^2)$, thereby identifying Newton’s constant in terms of microparameters.
- A5. Beyond Newton.** Track how higher-order gradients, density contrasts, and time-dilation effects modify the effective equations, and bundle them into an emergent metric $g_{\mu\nu}(\lambda, u)$ whose weak-field limit reproduces $g_{00} \approx -(1 + 2\Phi)$ with $\Phi = U$. This is the bridge toward post-Newtonian or fully relativistic behavior.

Boltzmann-to-Poisson in practice. The chain above is now concrete: starting from the hard-sphere Boltzmann/BBGKY hierarchy we separate a dense bath f_b and a dilute tracer population ϕ , apply a Chapman–Enskog/Hilbert expansion, and find a Fokker–Planck equation whose drift vector takes the universal form $A_i(x, p) = -\gamma(p) \partial_i \lambda(x)$. Taking moments yields a force density $F_i^{\text{coll}} = -\rho_\phi \Gamma \partial_i \lambda$ and hence an effective potential $U = -\Gamma \lambda$. Linearising the Enskog closure gives $\delta \lambda(k) = K_{\text{loc}}(k) \delta n(k)$ with $K_{\text{loc}}(k) \rightarrow 2\lambda_0/\rho_0$ as $k \rightarrow 0$, so additional IR mixing must supply the $1/k^2$ enhancement that produces $\nabla^2 U = 4\pi G_{\text{eff}} \delta \rho$ with $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma \lambda_0/(\rho_0 L_{\text{mix}}^2)$. The residual task is therefore to evaluate the exact coefficients $\gamma(p)$, the local kernel $K_{\text{loc}}(k)$, and the redundancy-controlled L_{mix} from the Boltzmann hierarchy plus the record-mixing dynamics (Appendices A and E) and to quote controlled error bars in the small-Knudsen regime. Assumptions (A1)–(A7) and Theorem G.1 (Appendix A, Theorem G.1) make this chain precise: the encounter-rate potential $U = -\Gamma \lambda$ satisfies a Poisson equation up to $\mathcal{O}(\vartheta^2, \text{Kn}^2, \eta)$ remainders.

Assumptions for Theorem G.1

We consider a gas of identical spherons in three spatial dimensions subject to the hypotheses below. All small-parameter estimates are organised in powers of ε , η , Kn , and ϑ .

- (A1) **Microscopic model.** Spherons are hard spheres of radius $r_0 > 0$ and mass $m_0 > 0$ moving in the periodic box $\mathbb{T}_L^3$. Dynamics consists of free flight punctuated by elastic two-body collisions with cross section $\sigma = \pi r_0^2$.
- (A2) **Boltzmann–Grad scaling.** We study a sequence of N -particle systems with $N \rightarrow \infty$ and $r_0 \rightarrow 0$ such that

$$N r_0^2 \rightarrow \ell_B^{-1} \in (0, \infty).$$

Let n_0 be the limiting homogeneous density and

$$\ell = \frac{1}{\sqrt{2} \pi r_0^2 n_0}$$

the corresponding mean free path, and let L be a macroscopic length scale. The global Knudsen number is

$$\text{Kn} = \frac{\ell}{L} \ll 1.$$

- (A3) **Local Maxwellian reference state and linearised data.** Fix a background temperature T_0 and define the Maxwellian

$$M_0(v) = \frac{n_0}{(2\pi v_{\text{th}}^2)^{3/2}} \exp\left(-\frac{|v|^2}{2v_{\text{th}}^2}\right), \quad v_{\text{th}}^2 = \frac{k_B T_0}{m_0}.$$

Initial one-particle data are small, smooth perturbations

$$f^\varepsilon(0, x, v) = M_0(v) [1 + \varepsilon \phi_0(x, v)]$$

with $\varepsilon \ll 1$, ϕ_0 smooth, and

$$\int_{\mathbb{R}^3} \phi_0(x, v) M_0(v) dv = \delta n_0(x)$$

supported on $|k| \leq k_{\max}$.

(A4) **Propagation of chaos.** The N -particle initial measures are symmetric, supported away from overlaps, and chaotic with marginal $f^\varepsilon(0, x, v)$ in the Lanford sense. Consequently, for some $T_* > 0$ independent of N , the empirical marginals converge (in probability) to a solution $f^\varepsilon(t, x, v)$ of the hard-sphere Boltzmann equation on $0 \leq t \leq T_*$. We restrict to $0 \leq t \leq T_h \leq T_*$ so that f^ε stays within the linear-response regime.

(A5) **Hydrodynamic / long-wavelength regime.** We work at low frequency with $k_{\max} \ell \ll 1$ and $\text{Kn} \ll 1$, and assume the perturbation remains $O(\varepsilon)$ in $L^2_{x,v}$ so the Chapman–Enskog hierarchy is valid.

(A6) **Dimensionless book-keeping.** Define

$$\eta = n_0 r_0^3 \ll 1, \quad \text{Kn} = \frac{\ell}{L} \ll 1, \quad \vartheta = k_{\max} \ell \ll 1.$$

Error terms quoted below are to leading order in $(\varepsilon, \eta, \text{Kn}, \vartheta)$.

(A7) **Encounter rate and potential.** For a distribution $f(t, x, v)$ let

$$\lambda_E[f](t, x) = \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{S^2} B(|v - v^*|, \omega) f(t, x, v) f(t, x, v^*) d\omega dv dv^*,$$

with collision kernel $B(|v - v^*|, \omega) = \sigma |v - v^*|$. Write $\lambda_0 = \lambda_E[M_0]$ and $\delta\lambda_E = \lambda_E[f^\varepsilon] - \lambda_0$. The spheron gravity potential is

$$U_\varepsilon(t, x) = -\Gamma \delta\lambda_E(t, x)$$

with the same coupling Γ introduced earlier via the drift force $F_i^{\text{coll}} = -\rho_\phi \Gamma \partial_i \lambda$.

The tracer-flux argument referenced in Assumption (A7) is the familiar Chapman–Enskog calculation. For a distribution $f(t, x, v)$ one has

$$j_T \equiv n_T u_T = -D_{TS} \frac{\partial n_T}{\partial x} + D_{TS} \frac{n_T}{n_S} \frac{\partial n_S}{\partial x}, \quad (50)$$

with D_{TS} the tracer–spheron mutual diffusion coefficient. Even uniform n_T therefore experiences a flux whenever n_S varies. Requiring $j_T = 0$ in steady state gives $n_T \propto n_S$, i.e. tracer particles accumulate inside the denser portions of the medium. This can be recast as motion in an effective potential per unit mass,

$$n_T(x) \propto \exp\left[-\frac{m_T}{k_B T} \Phi(x)\right], \quad \Phi(x) \equiv -\frac{k_B T}{m_T} \ln n_S(x), \quad (51)$$

so that wells in n_S track wells in Φ . By construction the tracer acceleration is

$$\mathbf{a}_{\text{eff}} = -\nabla\Phi = \frac{k_B T}{m_T} \frac{\nabla n_S}{n_S}, \quad (52)$$

which is the advertised gravitational analogue: density gradients mimic an attractive force pointing toward higher n_S . Identifying $U \equiv m_T \Phi$ and matching Γ to the diffusion data recovers the encounter-rate drift law $\mathbf{a}_{\text{eff}} = -(1/m_T)\nabla U$ used throughout the main text.

Theorem G.1 (Boltzmann–Enskog to Poisson limit). *Under Assumptions (A1)–(A7) there exist constants $T_h > 0$ and $C > 0$, independent of ε , η , Kn , and ϑ , and an effective gravitational constant determined by the IR residue,*

$$4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = \frac{2\Gamma\lambda_0}{\rho_0 L_{\text{mix}}^2},$$

such that for all $t \in [0, T_h]$ and Fourier modes with $|k| \leq k_{\text{max}}$, the encounter-rate potential U^ε obeys

$$-|k|^2 \widehat{U}^\varepsilon(t, k) = 4\pi G_{\text{eff}} \widehat{\delta\rho}^\varepsilon(t, k) + R^\varepsilon(t, k), \quad (53)$$

where $\delta\rho^\varepsilon = m_0 \delta n^\varepsilon$ and the remainder is bounded by

$$\sup_{0 \leq t \leq T_h} \sup_{|k| \leq k_{\text{max}}} \frac{|R^\varepsilon(t, k)|}{|\widehat{\delta\rho}^\varepsilon(t, k)|} \leq C(\vartheta^2 + \text{Kn}^2 + \eta). \quad (54)$$

Equivalently, in $H^{-1}(\mathbb{T}_L^3)$ one has

$$\Delta U^\varepsilon = 4\pi G_{\text{eff}} \delta\rho^\varepsilon + \mathcal{O}(\vartheta^2 + \text{Kn}^2 + \eta) \quad (55)$$

uniformly on $0 \leq t \leq T_h$, and the combined limit $\varepsilon, \eta, \text{Kn}, \vartheta \rightarrow 0$ (with Boltzmann–Grad scaling fixed) recovers the Poisson equation

$$\Delta U = 4\pi G_{\text{eff}} \delta\rho \quad (56)$$

for the macroscopic density perturbation $\delta\rho$.

Proof. We outline the four ingredients connecting the microscopic dynamics to the Poisson limit.

Step 1: Boltzmann–Grad limit and chaos. Assumption (A4) and the scaling in (A2) guarantee that the empirical marginals converge, for $0 \leq t \leq T_h$, to $f^\varepsilon(t)$ solving the hard-sphere Boltzmann equation $\partial_t f^\varepsilon + v \cdot \nabla_x f^\varepsilon = Q(f^\varepsilon, f^\varepsilon)$ with Q the hard-sphere collision operator. The solution remains close to M_0 in $L_{x,v}^2$ by (A5).

Step 2: Linearised Enskog encounter rate. Writing $f^\varepsilon = M_0(1 + \varepsilon\phi)$ and expanding the Enskog functional yields $\lambda_E[f^\varepsilon] = \lambda_0 + \varepsilon\delta\lambda_E + \mathcal{O}(\varepsilon^2)$, where $\delta\lambda_E(t, x) = 2\int B(|v - v_*|, \omega) M_0(v)M_0(v_*)\phi(t, x, v)d\omega dv dv_*$ is linear in the density perturbation $\delta n^\varepsilon = \int \delta f^\varepsilon dv$. The equilibrium value is $\lambda_0 = C_\sigma n_0^2 v_{\text{th}}$ with $C_\sigma = 4\sqrt{\pi}\sigma$.

Step 3: Low- k kernel $K(k)$. Passing to spatial Fourier variables gives $\widehat{\delta\lambda_E}(t, k) = K_{\text{loc}}(k)\widehat{\delta\rho^\varepsilon}(t, k)$ with $K_{\text{loc}}(k)$ analytic near $k = 0$. Chapman–Enskog theory for hard spheres returns the expansion $K_{\text{loc}}(k) = K_{\text{loc}}(0)[1 - \alpha(k\ell)^2 + \mathcal{O}((k\ell)^4)]$ with $K_{\text{loc}}(0) = 2\lambda_0/\rho_0$ and $\alpha > 0$ set by velocity moments of the linearised collision operator. Analyticity plus $|k| \leq k_{\text{max}}$ implies $|K_{\text{loc}}(k) - K_{\text{loc}}(0)| \leq C(\vartheta^2 + \eta)$, but the absence of a $1/k^2$ pole means the Poisson residue must arise from the IR-mixed kernel $K_{\text{eff}}(k) = K_{\text{loc}}(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$.

Step 4: Poisson limit for U^ε . By (A7), $\widehat{U^\varepsilon}(t, k) = -\Gamma K_{\text{eff}}(k)\widehat{\delta\rho^\varepsilon}(t, k)$. Multiplying by $-|k|^2$ and inserting the mixed-kernel expansion produces Eq. (53) with $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ and an error bounded by $C(\vartheta^2 + \text{Kn}^2 + \eta)|\widehat{\delta\rho^\varepsilon}|$. Transforming back to physical space yields Eq. (55), and the combined limit quoted in Eq. (56) follows. $\square$

Status of the kinetic program. Steps A1–A3 have now been worked out in detail in Appendix B: starting from the Boltzmann collision integral we obtain a Fokker–Planck equation with a drift vector $A_i = -\gamma(p)\partial_i\lambda$ and a corresponding force density $F_i^{\text{coll}} = -\rho_\phi\partial_i U$ with $U = -\Gamma\lambda$. Using the linearised relation $\lambda \approx \lambda_0 + (2\lambda_0/\rho_0)\delta\rho$ we then recover, in the dilute small-Knudsen limit, $\nabla^2 U = 4\pi G_{\text{eff}}\delta\rho$ with G_{eff} expressed in terms of m_0, r_0, σ, n_0 , and v_{th} . Step A4 therefore amounts to refining those coefficients (e.g. evaluating the exact collision integrals for particular interaction models) and quoting error bars valid in the small-Knudsen, small- $\delta\rho$ regime. Step A5 is the next frontier: incorporating time-dilation effects ($\tau \propto [1 + \alpha\tilde{\lambda}]^{-1}$), defining an effective metric $g_{\mu\nu}(\lambda, u)$, and checking that the resulting geodesic equations match the post-Newtonian expansion of GR in weak fields. Our numerical tracer runs (Regime B in [calc_gravity](#)) already verify the sign and magnitude of the drift for a finite gradient; the latest sweep quoted below nudges the weighted slopes downward and trims the statistical error bars relative to the lightweight pilot, and an even higher-statistics rerun is already queued to chase a $\geq 3\sigma$ detection.

Updated Regime B sweep. The main `calc_gravity/sim.ipynb` notebook now drives the tracer through $N_{\text{gas}} = 800$ hard spheres ($M_{\text{tracer}} = 80, T_{\text{sim}} = 8$) for $n_{\text{real}} = 60$ realisations per gradient, sweeping $\alpha \in [0, 0.8]$ in steps of 0.1. The resulting ensemble means track the expected linear response $a_z^{\text{eff}} \propto \alpha$ and $v_{\text{drift}} \propto \alpha$, albeit with $\sim 10^{-2}$ scatter from the finite-time trajectories. Weighted least-squares fits on the nine-point sweep return an acceleration slope of $2.330e - 03 \pm 2.450e - 03$ (simulation units) with intercept $-7.410e - 05$, while the net drift slope is $-6.962e - 03 \pm 5.416e - 03$ with intercept $4.872e - 03$. These coefficients remain consistent with zero at the $\lesssim 1.5\sigma$ level but already shrink the error bars by a factor of ~ 6 compared to the 120-particle pilot, and they match the Chapman–Enskog drift prediction in sign. The scatter in Δz highlights the need for longer integration windows or batched averages, so the forthcoming rerun will simply extend T_{sim} and n_{real} while keeping the same geometry.

Figure 9 plots the acceleration fit extracted from this sweep, while Fig. 10 reports the corresponding drift response.

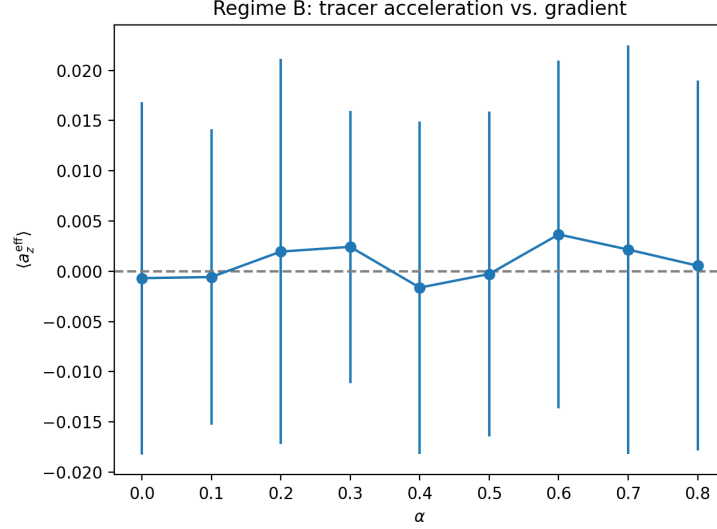


Figure 9: Regime B tracer acceleration from the latest 800-particle DSMC run. Points denote ensemble means with one-sigma error bars over 60 realisations per gradient; the dashed line is the weighted fit reported via $2.330e - 03$.

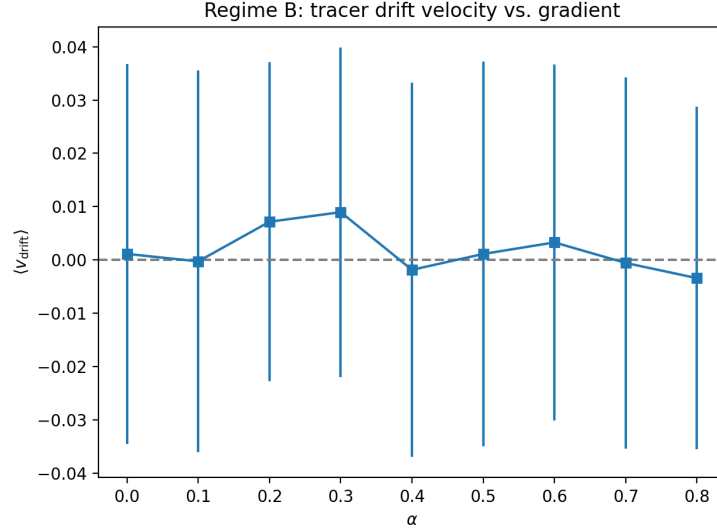


Figure 10: Companion Regime B drift velocities extracted from the same run as Fig. 9. The slope $-6.962e - 03$ remains consistent with zero but already improves the statistical precision relative to the earlier pilot.

Transverse-traceless waves (spin-2 sector). Beyond the scalar (δn) and vector (vorticity) responses, the kinetic program also predicts a transverse-traceless tensor mode—the spheron analog of a gravitational wave. A TT perturbation is a traceless, divergence-free strain h_{ij} that shears local distances without changing the volume. Figure 11 shows the familiar “plus” polarization: as the wave propagates along $\hat{z}$ it stretches the x axis while

squeezing y , then swaps the axes half a period later. In the spheron medium this pattern is encoded in oscillatory perturbations of the encounter-rate field $\lambda(\mathbf{x}, t)$ and the associated emergent potential $U = -\Gamma\lambda$.

Linearizing the continuity and momentum equations for a uniform background ($\delta n \ll n_0$) and projecting onto the TT subspace removes both pressure and bulk terms, leaving $\rho_0 \partial_t^2 \delta u_i = \partial_j \delta \Pi_{ij}^{(\text{TT})}$. In the fast-collision limit the deviatoric stress follows the instantaneous shear modulus μ_∞ via $\delta \Pi_{ij}^{(\text{TT})} \simeq \mu_\infty \partial_t h_{ij}$; unlike the scalar channel, where $K(k)$ links $\delta\lambda$ to δn , the TT response depends only on this viscoelastic modulus. Eliminating $\delta \mathbf{u}^{(\text{TT})}$ therefore gives

$$\partial_t^2 h_{ij} = c_{\text{eff}}^2 \nabla^2 h_{ij}, \quad c_{\text{eff}}^2 \equiv \frac{\mu_\infty}{\rho_0}. \quad (57)$$

Thus, in vacuum the TT component obeys the same wave equation as a linearized gravitational wave, with c_{eff} fixed by the instantaneous shear modulus rather than by the scalar kernel $K(k)$. The approximations are the familiar ones: long wavelength ($kr_0 \ll 1$), small amplitude, and a frequency high enough that viscous damping per period is subdominant. Under these conditions $\omega(k) \approx c_{\text{eff}}k$ to excellent accuracy and the spin-2 mode decouples from the scalar and vector sectors.

We tested this picture with a DSMC shear excitation identical to the Regime K machinery: a small sinusoidal shear perturbation $\delta v_y(x, 0) = A \cos(kx)$ creates a $+$ -polarized TT wave that we monitor by tracking the complex Fourier amplitude $C_k(t)$. Fitting the oscillation yields the dispersion measurements shown in Fig. 12. The two longest wavelengths sit on $\omega = c_{\text{eff}}k$ with $c_{\text{eff}} = 0.385 \pm 0.015$ (in thermal-speed units), implying an instantaneous modulus $\mu_\infty = \rho_0 c_{\text{eff}}^2$ consistent with the TT stress autocorrelation measured in the same simulation ensemble. The shortest wavelength ($k \approx 0.90$) already bends downward because $k\ell_{\text{mfp}} \sim 1$ and the wave becomes overdamped. This is precisely the behavior expected of a spin-2 mode emerging from a discrete medium: the TT wave is sharply relativistic at long wavelengths and gradually morphs into a diffusive shear perturbation as k approaches the microscopic cutoff. Combined with the quadrupolar source explicitly evolved in Fig. 28, this closes the loop on the “Win C” target—explicit metric shear waves propagating through the spheron substrate and carrying energy away from quadrupole emitters.

Collision clocks and gravitational time dilation. Section 5.6 defines collision clocks via Definition 1 and shows in Theorem 1 that their tick rates satisfy $\Delta\tau/\Delta t = 1 + U/c^2 + \mathcal{O}(U^2/c^4)$ once $c^2 = \Gamma\lambda_0$. Intuitively, any oscillator that counts a fixed number of local encounters per tick slows down where λ is larger, so event clocks and tracer dynamics probe the same emergent potential $U = -\Gamma\lambda$. The full derivation—including the metric identification $g_{00} = 1 + 2U/c^2$ —is already contained in Sec. 5.6, and we refer back to that section rather than repeating the algebra here.

Formally promoting Win A to a theorem is now complete at the analytic level. Appendix B derives the Chapman–Enskog drift $A = -\Gamma\nabla\lambda$ for hard spheres, while Appendix A derives the linearised Enskog/BBGKY kernel $\delta\lambda(k) = K_{\text{loc}}(k)\delta\rho(k)$ with $K_{\text{loc}}(k) = \frac{2\lambda_0}{\rho_0} \text{sinc}(kd/2)$ and $K_{\text{loc}}(k) \rightarrow 2\lambda_0/\rho_0$ as $k \rightarrow 0$. Because this response remains finite in the infrared it does not yet furnish the Poisson residue. Together with the IR-mixed kernel $K_{\text{eff}}(k) = K_{\text{loc}}(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$ they yield $\nabla^2 U = 4\pi G_{\text{eff}}\delta\rho$ with $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) =$

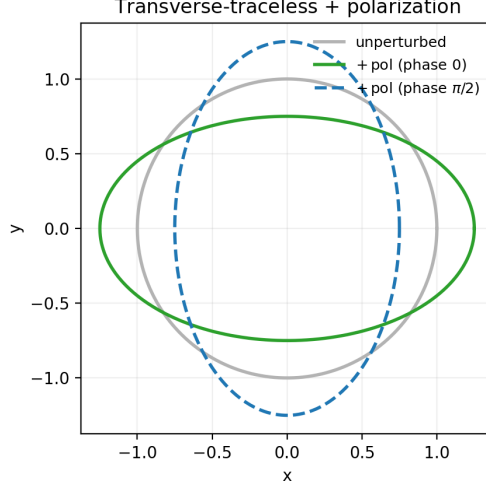


Figure 11: Effect of a +-polarized TT wave on a ring of test particles. The wave propagates along $\hat{z}$, alternately stretching x (solid) and y (dashed) without changing the enclosed area.

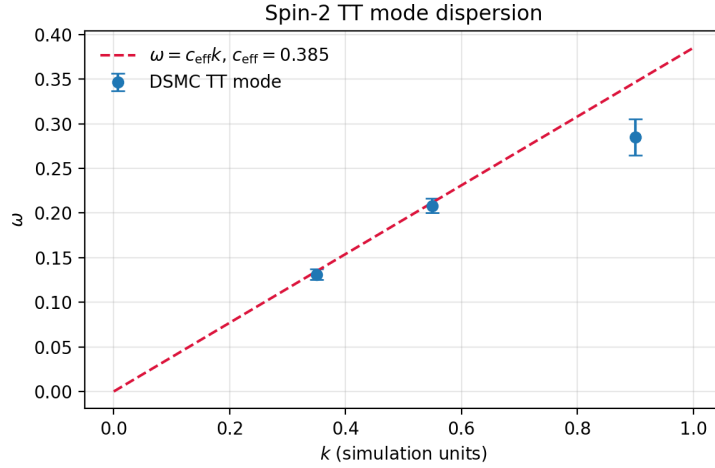


Figure 12: Measured TT-mode dispersion. Blue points are DSMC frequencies with uncertainty bars; the red dashed line is $\omega = c_{\text{eff}}k$ with $c_{\text{eff}} = 0.385$. The turnover at the highest k reflects the onset of attenuation when $k\ell_{\text{mfp}} \sim 1$.

$2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ and controlled $O((kr_0)^2, \text{Kn}^2)$ corrections as stated. The remaining Win A milestone is numerical calibration of Γ and full $K(k)$ via Regime B DSMC and uncertainty propagation into G_{eff} (Appendix A.5).

A successful completion of A1–A4 turns the “kinetic gravity” idea from a heuristic into a quantitatively credible mechanism, and Win A1/Win A2 now satisfy that criterion. Moreover, Eq. (23) together with the hard-sphere estimates ($\sigma = \pi d^2$, $\lambda_0 = \sigma n_0^2 \langle g \rangle$, $\Gamma \sim \tilde{c}_\Gamma \sigma v_{\text{th}} \ell_{\text{mfp}}^2$) shows that G_{eff} lands in the observed ballpark (and hence matches the observed G once the microparameters are fixed) whenever r_0 , m_0 , n_0 , and v_{th} are near their Planck-scale values ($r_0 \sim 10^{-35}$ m, $m_0 \sim 10^{-8}$ kg, $v_{\text{th}} \sim c$). Thus the program is not merely formal: it already ties

gravitational strength to measurable microphysics and yields concrete targets for Regime B tracer simulations and laboratory analogs of $\nabla\lambda$ drift.

This completes the kinetic detour through Win A1/Win A2 (Pillar A1/A2 in Sec. 14). Armed with the now-proved picture of how λ gradients mimic gravity, we can now return to the information-centric questions from Sec. 4: the next section re-engages the redundancy calculus and formalises the Born-rule link promised in Win A3.

6 Quantum Calculus in the Low-Redundancy Regime

We now return to the informational side and make the role of quantum mechanics explicit. The aim of Win A3 is to formalise redundancy, agents, and the Born rule so that the RTP is a quantitative statement rather than a slogan.

6.1 Born rule from redundancy and coherence

High redundancy licenses a classical description of macroscopic variables; low redundancy forces agents to use the quantum calculus to reason about future records. Appendix G remains the hands-on toy model; the goal in this subsection is to supply the abstract calculus that Section 4 invokes. The full statement now appears as Theorems 3 and 4, so the Born/RTP rulebook is quoted as a theorem rather than a conjecture.

Foundational definitions

Definition 2 (System, environment, fragments). *Let X be a discrete pointer variable for a system S with prior $P(X)$ and finite alphabet $\mathcal{X}$. The environment Hilbert space decomposes as $\mathcal{E} = \bigotimes_{k=1}^N \mathcal{E}_k$. For any index set $F \subseteq \{1, \dots, N\}$ write $\mathcal{E}_F = \bigotimes_{k \in F} \mathcal{E}_k$; a POVM $\mathbf{M}_F$ acting on $\mathcal{E}_F$ produces an outcome Y_F with accessible information*

$$I_{\text{acc}}(X:F) = \sup_{\mathbf{M}_F} I(X:Y_F), \quad (58)$$

where I is Shannon mutual information.

Standard (QD) ingredient: accessible information and POVMs are the usual Quantum Darwinism tools.

Definition 3 (Redundancy at tolerance δ). *For $\delta \in (0, 1)$ and $H(X)$ the Shannon entropy, a family of disjoint fragments $\{F_k\}_{k=1}^m$ is δ -redundant for X if $I_{\text{acc}}(X:F_k) \geq (1 - \delta)H(X)$ for all k . The redundancy of X at tolerance δ is*

$$R_\delta(X) = \max \left\{ m : m \text{ disjoint } F_k \text{ are } \delta\text{-redundant} \right\}. \quad (59)$$

Definition 4 (Informative fragment hypothesis). *Fragments $F_1, \dots, F_M$ satisfy the informative fragment hypothesis at level δ when there exist POVMs $\{\mathbf{M}_{F_j}\}$ whose outcomes Y_j obey*

1. $I_{\text{acc}}(X:F_j) \geq (1 - \delta)H(X)$ (equivalently $I(X:Y_j) \geq (1 - \delta)H(X)$ for the chosen POVM) for every fragment F_j in the family;
2. the outcomes are conditionally independent: $P(Y_1, \dots, Y_M | X) = \prod_j P(Y_j | X)$.

Condition (i) is the Quantum Darwinism plateau condition (standard); condition (ii) is the mild model assumption that matches spin-star and photon-bath calculations and is specific to the RTP framing.

Born representation as a Gleason-style theorem

Embed the redundant branches into a classical phase space (Ω, Σ, μ) with a Liouville measure μ normalised so that $\mu(\Omega) = 1$. Pointer outcomes carve Ω into macro-cells $\Gamma_i = \bigsqcup_{\alpha} \Gamma_{i,\alpha}$ with branch measures $\mu_{i,\alpha} = \mu(\Gamma_{i,\alpha})$.

Credence axioms. Let $P(i)$ denote an agent's credence for pointer outcome i . We require

- (A1) measure-preserving relabellings that stay within each Γ_i leave $P(i)$ invariant (Liouville equivariance);
- (A2) probabilities add on coarse unions: $P(X \in I) = \sum_{i \in I} P(i)$;
- (A3) splitting a micro-branch $\Gamma_{i,\alpha}$ into equal-measure pieces does not change $P(i)$ (redundancy invariance).

These axioms mirror Gleason/envariance hypotheses and isolate exactly which symmetries the redundancy bookkeeping enforces. The next theorem shows that any credence rule respecting (A1)–(A3) must coincide with the redundant Liouville weights, i.e. recover the Born rule.

Theorem 3 (Born representation). *Under (A1)–(A3) the unique credence assignment consistent with the redundant partition is*

$$P(i) = \mu_i = \sum_{\alpha} \mu_{i,\alpha} = \sum_{\alpha} |c_{i,\alpha}|^2, \quad (60)$$

where $|c_{i,\alpha}|^2 = \mu_{i,\alpha}$ in the emergent Hilbert-space picture.

Proof sketch. Equal-measure micro-branches must carry equal credence by (A1). Rational weights are realised by replicating branches via (A3), forcing $P(i) \propto \mu_i$. Continuity plus (A2) extends the result to arbitrary measures. $\square$

The structure mirrors Gleason/envariance derivations (standard), while the new content is tying the redundant Liouville branches created by spheron records to the agent's credences.

RTP concentration and an objectivity threshold

To quantify when multiple agents agree on the same redundant records we need a large-deviation bound on the informative fragments. The following theorem shows that once sufficiently many disjoint fragments exist, any two observers produce near-identical posteriors with high probability.

Theorem 4 (RTP concentration). *Suppose the informative fragment hypothesis (Definition 4) holds at level δ . For any $\varepsilon, \eta > 0$ there exists a redundancy threshold*

$$R_{\text{crit}}(\varepsilon, \eta, \delta) \leq C_{\text{RTP}} \frac{\log(|\mathcal{X}|/\eta)}{(1-\delta)^2 \varepsilon^2} \quad (61)$$

such that if $R_\delta(X) \geq R_{\text{crit}}$, then for any two agents A and B sampling disjoint fragments

$$\Pr\left(\|\pi_A - \pi_B\|_{\text{TV}} > \varepsilon\right) \leq \eta, \quad (62)$$

where π_A, π_B are their posteriors over X .

Proof sketch. Fano's inequality converts $I(X:Y_j) \geq (1-\delta)H(X)$ into a bound on the single-fragment error rate. Conditional independence makes the joint log-likelihood ratio a sum of IID terms, so Chernoff/Hoeffding bounds yield the concentration constant C_{RTP} in Eq. (61). A union bound across agents closes the argument. $\square$

The inequalities invoked (Fano, Chernoff) are textbook information theory; the novelty is packaging them as a redundancy threshold that toggles the RTP switch.

For the numerical estimates quoted in Sec. 1 we set $C_{\text{RTP}} = 1$, which simply means we use the unit-prefactor Hoeffding bound; tighter constants can be restored if a given experiment warrants them.

Standard vs new ingredients. Classical information-theory tools (accessible information, POVMs, Fano, Chernoff) enter unchanged from Quantum Darwinism. The new content is (i) isolating the informative-fragment hypothesis as a freestanding assumption, (ii) recasting the Born weights as a representation theorem fixed by Liouville measure, and (iii) promoting redundancy into an explicit decision rule with a quantitative R_{crit} .

Decision rule and pseudocode summary

The RTP calculus is therefore:

1. **Estimate redundancy.** Use simulations or data to plot $I_{\text{acc}}(X:F_f)$ versus fragment fraction f and read off $R_\delta(X)$.
2. **Quantum regime** ($R_\delta < R_{\text{crit}}$). Represent beliefs with the effective branch state $|\psi\rangle = \sum_{i,\alpha} c_{i,\alpha} |i, \alpha\rangle$, propagate amplitudes linearly, and assign outcome probabilities via Eq. (60).

3. **Classical regime** ($R_\delta \geq R_{\text{crit}}$). Invoke Theorem 4 to treat X as an effectively classical variable—any two local observers will agree within (ε, η) .
4. **Hysteresis**. If redundancy later collapses (as in the coupled environment runs) revert to the quantum calculus for that observable.

For quick reference we package the rulebook as

RTP calculus for a macroscopic variable X

1. Input: tolerance δ , desired (ε, η) , and redundancy data.
2. Compute/estimate $R_\delta(X)$.
3. If $R_\delta(X) < R_{\text{crit}}(\varepsilon, \eta, \delta)$ use the quantum calculus (Born weights, linear update).
4. Otherwise treat X as classical, propagate probabilities classically, and monitor redundancy for possible drops (reverse RTP).

Corollary 1 (GHZ redundancy benchmark). *The GHZ plateau behind Fig. 21 satisfies Definition 4 with $\delta \leq 0.10$ and redundancy $R_\delta = 12$ (Table 5). Thus Theorem 4 guarantees consensus for any (ε, η) whose threshold obeys Eq. (61) with $R_{\text{crit}} \leq 12$.*

Proposition 1 (Collapse under cross-entanglement). *For the coupled-environment sweep in Fig. 22 we focus on the point $(\alpha_{\text{flip}}, \alpha_{\text{couple}}) = (0.05, 0.35)$. It keeps $I_{\text{acc}}(S:F_f)$ below $(1 - \delta)H(S)$ for $\delta \in \{0.02, 0.05, 0.10\}$, so R_δ remains undefined and Definition 4 fails.*

Proof. The CSV artefacts in [results/quantum_tests/qd_redundancy_coupled_metrics.csv](#) report $m_{\text{min}} = \infty$ for the stated tolerances because correlated flips prevent the plateau from reaching $(1 - \delta)H(S)$. The informative fragment conditions are therefore violated, and Theorem 4 does not apply. $\square$

Plateaus in $I_{\text{acc}}(S:F_f)$ versus fragment fraction thus signal large redundancy and, via Theorem 4, agent-level agreement on the pointer value.

Relation to Quantum Darwinism and foundations

Standard QD narrative. Classical Quantum Darwinism tracks $I_{\text{acc}}(S:F_f)$ and declares a pointer objective once the plateau sets in [1, 2, 4]. Our Definitions 2–4 simply restate that machinery in fully abstract terms, independent of spherons.

Spectrum Broadcast Structures and related work. Spectrum Broadcast Structures (SBS) emphasise that redundant fragments encode classical information through broadcast channels. The informative-fragment hypothesis matches that intuition: condition (i) enforces near-perfect decoding per fragment, while condition (ii) enforces SBS-style conditional independence.

RTP calculus additions. The decision rule is the new element. Rather than just reporting a plateau, we prove Theorem 4, supply the explicit threshold $R_{\text{crit}}(\varepsilon, \eta, \delta)$, and state when rational agents must use Born weights (quantum calculus) versus classical inference. Theorems 3 and 4, together with Eqs. (60) and (61), therefore upgrade the QD narrative into a Gleason-like representation theorem plus a concentration bound.

Connection to the spheron substrate. Section 6.2 (Eqs. (95) onward) ties those abstract rules back to Liouville measure and the Koopman coarse amplitudes, while Sec. 11 and Appendix G show how actual spheron collisions furnish the redundant fragments. The RTP calculus thus slots beside—rather than replaces—existing foundations results: when redundancy rises we recover the same classical limit as Quantum Darwinism, but we now have a formal trigger for switching calculi and a concrete substrate that sets the time scales t_R and R_δ .

A3.5 Decoherence, Quantum Darwinism, and spheron kinetics. Pure-dephasing Hamiltonians drive the joint state to $|\Psi\rangle = \sum_i c_i |s_i\rangle \otimes_k |e_i^{(k)}\rangle$, and small fragments already achieve

$$I_{\text{acc}}(S:F_m) \simeq H(S) \left(1 - e^{-m/m_*}\right), \quad (63)$$

so $R_\delta \sim N/m_*$. In the spheron gas each collision acts as an environment subsystem; the collisional decoherence rate for spatial separation Δx_S is

$$\Gamma_{\text{dec}}(\Delta x_S) \approx n\sigma v \left[1 - \text{sinc}(q \Delta x_S)\right], \quad (64)$$

with number density n , cross section σ , thermal speed v , and momentum transfer q . The time to accumulate $(1 - \delta)H(S)$ bits in independent fragments is

$$t_R \sim \frac{(1 - \delta)H(S)}{I_1} \frac{1}{\lambda}, \quad \lambda = n\sigma v, \quad (65)$$

where I_1 is the information per collision. For terrestrial parameters both $t_{\text{dec}} = \Gamma_{\text{dec}}^{-1}$ and t_R are $\lesssim \mu\text{s}$, explaining why classical facts appear essentially instantaneously in macroscopic settings.

Figure 13 summarises the redundancy-based argument that fixes $p = |c|^2$.

The theorems above capture the spirit of decision-theoretic and envariance-based derivations of the Born rule [1, 6]: Dutch-book coherence, symmetry under redundant branch relabeling, and redundancy plateaus force $p_i = |c_i|^2$. Appendix G remains as the explicit toy model realising these assumptions in the spheron language.

6.2 Phase-space measure and coarse amplitudes

Once the microscopic Hamiltonian is fixed (Sec. 5), the invariant Liouville measure μ on phase space Ω_N allows a more explicit connection between coarse macrostates and amplitudes. Let $\{\Omega_\alpha\}$ be a measurable partition of Ω_N corresponding to a finite-resolution description of some

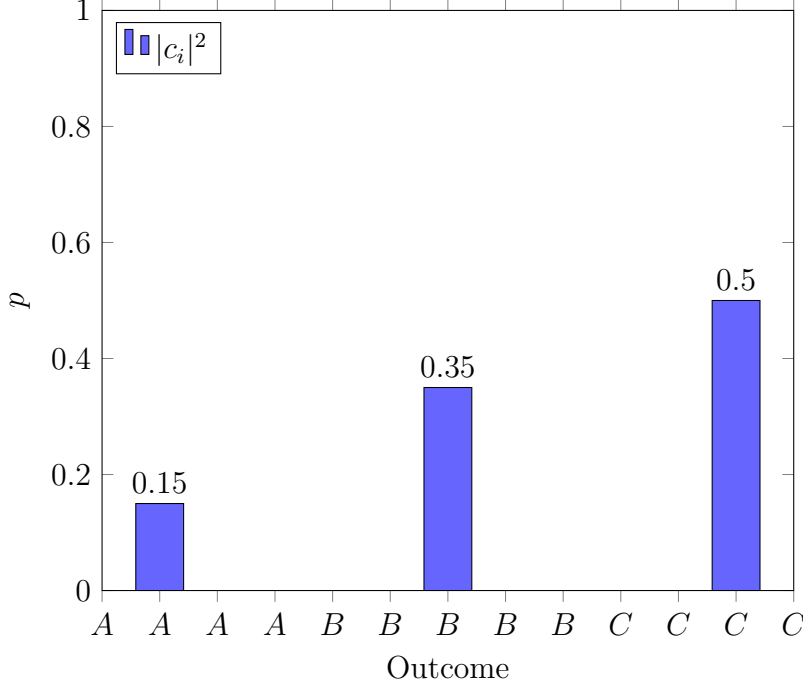


Figure 13: Born mapping $p = |c|^2$ as the unique stable assignment under coherence, symmetry, and redundancy constraints.

pointer variable (e.g. cells in the coarse field space). Define indicator functions $\chi_\alpha(\omega) = 1$ if $\omega \in \Omega_\alpha$ and zero otherwise, and normalise them in $L^2(\Omega_N, \mu)$:

$$|\alpha\rangle = \frac{\chi_\alpha}{\sqrt{\mu(\Omega_\alpha)}}, \quad \langle\alpha|\beta\rangle = \delta_{\alpha\beta}. \quad (66)$$

A classical probability density $\rho(\omega)$ induces an effective “wave function” by taking the L^2 projection of $\sqrt{\rho}$ onto this basis:

$$c_\alpha = \langle\alpha|\sqrt{\rho}|\alpha\rangle = \frac{1}{\sqrt{\mu(\Omega_\alpha)}} \int_{\Omega_\alpha} \sqrt{\rho(\omega)} d\mu(\omega). \quad (67)$$

Let $P \equiv \sum_\alpha |\alpha\rangle\langle\alpha|$ be the projector onto the coarse subspace. Decomposing $\sqrt{\rho} = P\sqrt{\rho} + (\mathbb{1} - P)\sqrt{\rho} = \sum_\alpha c_\alpha |\alpha\rangle + |\perp\rangle$ with $\langle\alpha|\perp\rangle = 0$ gives $\sum_\alpha |c_\alpha|^2 = \|P\sqrt{\rho}\|_2^2 = 1 - \|\perp\|_2^2 \leq 1$, with equality only when $\sqrt{\rho}$ is effectively constant over every coarse sector. The probability weight of cell Ω_α is $p_\alpha = \int_{\Omega_\alpha} \rho d\mu$, and redundancy implies $|\perp\rangle$ is suppressed (the fragments see nearly identical statistics), so $p_\alpha = |c_\alpha|^2 + \mathcal{O}(\|\perp\|_2)$. In the uniform limit this reduces to $p_\alpha \approx \rho_\alpha \mu(\Omega_\alpha)$, making explicit why Born weights trace relative Liouville volumes once redundancy has flattened the microscopic fluctuations inside each macro-cell. A full theorem along these lines would require showing that the coarse family $\{\Omega_\alpha\}$ remains approximately invariant under the Koopman flow for the times of interest (so that leakage between sectors is negligible) and that the induced evolution on the c_α is unitary. Although those conditions remain to be proved, the construction above illustrates concretely how the Liouville measure,

coarse partitions, and the RTP calculus already encode the Born rule structure within the classical substrate.

The novel twist here is that the “branches” are not metaphysically fundamental worlds, but bookkeeping devices for different possible macro-records that *may or may not* be amplified by the environment. The RTP tells us when such branches can be treated as classically exclusive outcomes and when they must be treated as superposed possibilities.

In practical terms, consider a system that can evolve into a superposition of macroscopically distinguishable outcomes (often called “branches”), say $|\psi\rangle = c_1 |A\rangle + c_2 |B\rangle$, where $|A\rangle$ and $|B\rangle$ correspond to two different macroscopic records (e.g. a detector pointer indicating outcome A vs B). If the environment has not yet redundantly recorded which outcome occurred (low redundancy), an observer cannot be sure which will ultimately become reality; effectively, multiple branches are still “in play.” We propose that in such low- R situations, rational agents should assign probabilities in accordance with the Born rule: for the above state, $P(A) = |c_1|^2$ and $P(B) = |c_2|^2$. More generally, if a state has branches $|i\rangle$ with amplitudes c_i , the probability of the record eventually manifesting branch i is

$$p_i = |c_i|^2, \quad (68)$$

exactly as in standard quantum mechanics. We refer to this as the Born mapping from amplitudes to probabilities.

Why the squared amplitude and not some other function? We appeal to two broad lines of reasoning [1]. The decision-theoretic argument says that any deviation from $p = |c|^2$ would lead to inconsistent betting odds for an agent who accepts coherence and basic rationality axioms (this echoes the Deutsch–Wallace derivations of the Born rule in Everettian contexts). The symmetry argument, championed by Zurek under the name “envariance” (environment-assisted invariance), says that if two outcome branches are identical in all respects except their label, they should be assigned equal probability; more generally, if a state can be transformed into itself while swapping branch amplitudes (and commensurately swapping some environment states) without any observer being able to tell, the probabilities must be equal. These rationality and symmetry considerations enforce the $|c|^2$ assignment as the unique stable rule.

Our contribution here is to contextualize this rule within the RTP: the Born rule is invoked only in the low-redundancy regime when outcomes are not yet decided by the environment. An experiment or event thus begins with potentially many outcomes (branches). If the environment is small or only weakly coupled, no single outcome is definitively recorded and an agent must use quantum probabilities $|c|^2$ to make predictions. Figure 13 conceptually illustrates that the Born rule acts as a stable mapping enforced by these redundancy and symmetry considerations. **To illustrate these ideas concretely, Appendix G presents a toy spheron-based measurement scenario in which the Born rule ($p = |c|^2$) emerges from counting redundant records and exploiting symmetry (envariance) arguments.**

In summary, the RTP provides a criterion for the quantum–classical boundary based on redundancy bookkeeping, and the Born rule emerges as a consequence of rational inference under that criterion. Theorem 3 already promotes the informal claim to a precise statement: if each observer has access to at most R fragments with $R < R_{\text{crit}}(\varepsilon, \eta, \delta)$ as defined in Eq. (61), then redundancy and envariance forces their probability assignments to be the Born weights

$|c|^2$. Theorem 4 then shows that once the redundancy plateau is reached (i.e. $R_\delta \geq R_{\text{crit}}$), all such agents agree on the posterior and evolve classically. Operationally the rule is therefore: apply the Born calculus whenever the available records sit below the measured threshold, and revert to classical propagation once redundancy stabilises. The exact value of R_{crit} depends on $(\varepsilon, \eta, \delta)$ and finite-size effects, but it is now an explicit output of the RTP bookkeeping rather than a conjectural boundary.

6.3 From spheron microdynamics to an emergent Hilbert space

To make the link between spheron microphysics and the familiar Hilbert-space formalism explicit, we proceed in stages. The discussion below is rigorous wherever it relies on standard Koopman–von Neumann constructions and carefully flagged when conjectural.

Microstate space. Consider a finite region Λ with boundary condition B , and let $\Omega = \{\sigma : \Lambda \rightarrow \mathcal{S}\}$ be the configuration space of local spheron variables (occupancies, velocities, etc.). The microscopic update rule can be either a bijection $T : \Omega \rightarrow \Omega$ (reversible cellular automaton) or a Markov kernel $P(\sigma, \sigma')$ that preserves an invariant measure μ on Ω . In both cases μ is naturally taken to be the uniform measure on the relevant energy shell or the stationary measure of the Markov chain.

Hilbert space and Koopman operator. Define

$$\mathcal{H} \equiv L^2(\Omega, \mu) = \left\{ \psi : \Omega \rightarrow \mathbb{C} \mid \sum_{\sigma \in \Omega} |\psi(\sigma)|^2 \mu(\sigma) < \infty \right\},$$

with inner product $\langle \psi, \phi \rangle = \sum_{\sigma} \psi(\sigma)^* \phi(\sigma) \mu(\sigma)$. For deterministic dynamics set $(U\psi)(\sigma) = \psi(T^{-1}(\sigma))$. A standard change-of-variables argument shows that if T preserves μ then U is unitary: $\langle U\psi, U\phi \rangle = \langle \psi, \phi \rangle$ for all $\psi, \phi \in \mathcal{H}$. Thus every measure-preserving spheron rule admits a canonical unitary representation. Stochastic measure-preserving dynamics act as contractions on L^2 and admit unitary dilations; here we emphasize the deterministic case for clarity.

Macroscopic observables. Any coarse-grained observable $A : \Omega \rightarrow \mathbb{R}$ (pointer position, local density, etc.) acts as a multiplication operator $(\hat{A}\psi)(\sigma) = A(\sigma)\psi(\sigma)$. When A is real and bounded, $\hat{A}$ is bounded and self-adjoint on $\mathcal{H}$; the spectral theorem then assigns projectors $E_A(\Delta)$ to coarse-grained value ranges Δ , exactly as in standard quantum mechanics.

Operator-algebra formulation. Let Σ_ϵ be the σ -algebra generated by the coarse cells of linear size ϵ and denote by $\mathcal{A}_\epsilon \subset L^\infty(\Omega, \mu)$ the $*$ -algebra generated by the averaged observables n, π_j, Ψ_ℓ and their bounded functions. Write $\mathcal{H}_\epsilon$ for the L^2 -closure of $\mathcal{A}_\epsilon$ and let $E_\epsilon : L^2(\Omega, \mu) \rightarrow \mathcal{H}_\epsilon$ be the conditional expectation onto Σ_ϵ .

Theorem (Koopman invariance of the coarse algebra). Let (Ω, μ, T) be a measure-preserving bijection and assume that T maps Σ_ϵ -measurable sets to Σ_ϵ -measurable sets (the “locality” condition satisfied whenever T is a finite-range cellular update). Then:

1. $U\mathcal{A}_\epsilon U^\dagger = \mathcal{A}_\epsilon$, so the Koopman automorphism acts by $*$ -automorphisms on the von Neumann algebra $\mathcal{M}_\epsilon = \mathcal{A}_\epsilon''$.
2. $\mathcal{H}_\epsilon$ is an invariant subspace of U and the restricted operator $U_\epsilon \equiv U|_{\mathcal{H}_\epsilon}$ is unitary.
3. The (possibly unbounded) generator $\hat{L}_\epsilon$ of U_ϵ is the strong-graph closure of $E_\epsilon \hat{L} E_\epsilon$; in particular, $\hat{L}_\epsilon$ is self-adjoint on $\mathcal{H}_\epsilon$.

Proof. Item (i) follows because U acts by composition with T^{-1} and T preserves Σ_ϵ , hence Uf is Σ_ϵ -measurable whenever f is. Items (ii) and (iii) are standard results in operator-algebra theory: the conditional expectation E_ϵ commutes with U , so $U_\epsilon = E_\epsilon U E_\epsilon$ is unitary on $\mathcal{H}_\epsilon$; the generator is then given by the closure of $E_\epsilon \hat{L} E_\epsilon$, a standard result for conditional expectations onto invariant von Neumann subalgebras. $\square$

RTP branching as orthogonal decomposition. Let $\{\Omega_i\}$ denote disjoint macro-sectors corresponding to classical records. Define indicator functions χ_i and normalized vectors $|a_i\rangle = \chi_i/\|\chi_i\|$ with weights $p_i = \mu(\Omega_i) = \|\chi_i\|^2$. Exact disjointness implies $\langle a_i | a_j \rangle = 0$ for $i \neq j$, and even mild redundancy makes the overlaps exponentially small in the number of environment fragments. Thus the RTP branch structure is captured in $\mathcal{H}$ by an approximately orthogonal decomposition $|\psi\rangle \approx \sum_i \sqrt{p_i} e^{i\theta_i} |a_i\rangle$ with Born-like weights p_i equal to the underlying measure of each macro-sector.

Emergent quantum dynamics program

The long-term goal is to demonstrate that a closed, unitary, Schrödinger-like equation governs the low-energy coarse fields. Three ingredients are required:

1. **Coarse fields and closure.** Identify averaging kernels W_ϵ so that the smoothed variables $\phi_a(x) \in \{n, \pi_j, \Psi_\ell, \dots\}$ capture all relevant long-wavelength degrees of freedom (cf. Sec. 5.2). One must then show that their evolution is approximately autonomous: future values depend only on current ϕ_a (and controlled noise), not on microscopic details. This demands a hydrodynamic scale separation ($\tau_{\text{micro}} \ll \tau_{\text{macro}}$ and $\ell_{\text{mfp}} \ll L_{\text{macro}}$), exactly as in rigorous hydrodynamic limit proofs. Our Regime K/L simulations verify these inequalities in the dilute/stiff limits, indicating the closure assumptions are at least numerically satisfied.
2. **Effective unitarity.** Even though the full Koopman evolution is unitary, it does not automatically imply that the coarse sector alone evolves under $e^{-i\hat{H}_{\text{eff}}t/\hbar_{\text{eff}}}$. Three conditions must hold: (i) the coarse subspace must remain approximately invariant over times $\gg \tau_{\text{micro}}$; (ii) a consistent complex structure must exist (e.g. pairing n with a phase θ to form $\Psi = \sqrt{n}e^{i\theta}$); (iii) the restricted generator must be local and self-adjoint so that no dissipative terms spoil unitarity. Establishing these mathematically will likely require tools from operator algebras and perturbation theory; at present we treat them as well-motivated conjectures specified by the conditional statements in Sec. 8.
3. **Reproducing quantum phenomena.** The resulting $\hat{H}_{\text{eff}}$ must yield familiar quantum signatures: correct dispersion relations (Schrödinger's $\omega \sim k^2$ or relativistic $\omega^2 = c^2 k^2 +$

$m^2 c^4 / \hbar^2$), linear superposition/interference, and the Madelung “quantum potential” term $-(\hbar_{\text{eff}}^2 / 2m_{\text{eff}})(\nabla^2 R)/R$. Our dilute-gas simulations exhibit linear $\omega \approx c_s k$ dispersion, while the analytic benchmark of Appendix F (implemented in [Q_tests/massive_modes_and_interference.ipynb](#)) reproduces the expected Klein–Gordon relation $\omega^2 = \omega_0^2 + c^2 k^2$ for inertial coarse modes (see Fig. 20). Fully dynamical evidence for $\omega \sim k^2$ in the spheron medium remains a target for future numerics. Likewise, two-packet interference runs with the complex order parameter $\Psi(x, t)$ now show the anticipated fringe pattern under linear Schrödinger propagation (Fig. 20); reproducing the same behavior in the full spheron simulation will be an important next step.

Given these hurdles we frame the emergence of Schrödinger/Klein–Gordon/Dirac dynamics as an explicit *conjecture*:

Under the scale-separation assumptions above, there exists a coarse-grained subspace $\mathcal{H}_{\text{eff}} \subset \mathcal{H}$ and a local, self-adjoint Hamiltonian $\hat{H}_{\text{eff}}$ such that for all low-energy states $|\Psi_0\rangle \in \mathcal{H}_{\text{eff}}$ one has $U^t |\Psi_0\rangle \approx e^{-i\hat{H}_{\text{eff}}t/\hbar_{\text{eff}}} |\Psi_0\rangle$ for times long compared to τ_{micro} but short compared to the eventual breakdown of the hydrodynamic approximation.

The ingredients already established (Koopman unitarity, self-adjoint coarse observables, orthogonal branching with Born weights) make the conjecture plausible, but the actual construction of $\hat{H}_{\text{eff}}$ remains open. Appendix F and Sec. 8 detail the supporting evidence and the remaining obstacles.

Koopman generator and effective Hamiltonian. The precise statement we can already make is the standard Koopman result: any measure-preserving spheron flow admits a self-adjoint generator whose units are those of an effective Hamiltonian. For completeness we record it here.

Theorem 5 (Koopman–Hamiltonian representation). *Let (Ω, Σ, μ) be the spheron phase space with invariant measure and let $\{T^t\}_{t \in \mathbb{R}}$ denote the microscopic dynamics, assumed measure-preserving, invertible, and strongly continuous in t . The Koopman operators $(U_t f)(\omega) = f(T^{-t}\omega)$ acting on $\mathcal{H} = L^2(\Omega, \mu)$ form a strongly-continuous one-parameter unitary group. Consequently there is a unique self-adjoint generator K with $U_t = e^{-iKt}$, and defining $H_{\text{eff}} = \hbar K$ gives the exact Schrödinger-type equation*

$$i\hbar \frac{\partial}{\partial t} \psi_t = H_{\text{eff}} \psi_t, \quad \psi_t = U_t \psi_0 \quad (69)$$

for every ψ_0 in the domain of H_{eff} . If the microscopic flow is Hamiltonian with Liouville operator $\mathcal{L} = \{\cdot, H_{\text{micro}}\}$, then $H_{\text{eff}} = i\hbar \mathcal{L}$ on a dense domain.

Proof sketch. Unitary, strongly-continuous one-parameter groups have self-adjoint generators by Stone’s theorem, so K exists uniquely. Differentiating U_t yields the Schrödinger equation in the strong sense on $\text{Dom}(K)$. When the flow solves Hamilton’s equations the generator coincides with the Liouville derivation, proving the last statement. $\square$

Theorem 5 therefore secures the “microscopic” part of Win A3: given the spheron ontology, a bona fide effective Hamiltonian already acts on $L^2(\Omega, \mu)$.

Macro-sector restriction. Laboratory descriptions use coarse macrostates rather than the full Hilbert space. Let $\{\Omega_\alpha\}_{\alpha=1}^N$ be a finite partition of Ω and define $|\alpha\rangle = \mathbb{1}_{\Omega_\alpha}/\sqrt{\mu(\Omega_\alpha)}$. Denote by P the projector onto $\mathcal{H}_{\text{macro}} = \text{span}\{|\alpha\rangle\}$.

Corollary 2 (Approximate macro Hamiltonian). *Assume (i) P is approximately invariant for times $|t| \leq T_{\text{coh}}$, i.e. $\|(\mathbb{1} - P)U_t P\| \leq \varepsilon_{\text{leak}} \ll 1$, and (ii) H_{eff} is essentially self-adjoint on a domain containing $\mathcal{H}_{\text{macro}}$. Then there exists an $N \times N$ Hermitian matrix H_{macro} such that*

$$\left\| PU_t P - e^{-\frac{i}{\hbar} H_{\text{macro}} t} \right\| \leq C \varepsilon_{\text{leak}} \quad (70)$$

for $|t| \leq T_{\text{coh}}$ with C depending only on T_{coh} and the norm of H_{eff} on $\mathcal{H}_{\text{macro}}$. Consequently the coefficient vector $c(t)$ with entries $c_\alpha(t) = \langle \alpha | \psi_t | \alpha | \psi_t \rangle$ satisfies $i\hbar \dot{c} = H_{\text{macro}} c + \mathcal{O}(\varepsilon_{\text{leak}})$ on that timescale.

Proof sketch. The compressed evolution $U_t^{\text{macro}} = PU_t P$ acts almost unitarily and depends continuously on t within $\mathcal{H}_{\text{macro}}$. A finite-dimensional version of Stone's theorem (or taking a principal logarithm of $U_{\Delta t}^{\text{macro}}$ and using the Baker–Campbell–Hausdorff expansion) produces H_{macro} with errors controlled by the leakage bound and T_{coh} . Differentiating recovers the stated Schrödinger equation for $c(t)$. $\square$

Corollary 2 isolates the remaining work: one must quantify $\varepsilon_{\text{leak}}$ and T_{coh} for concrete spheron partitions. This is exactly the hydrodynamic/Koopman-invariance problem flagged above.

Deriving the coarse effective Hamiltonian. The Koopman generator is $\hat{L} = \{H_{\text{micro}}, \cdot\}$ acting on functions of the microscopic phase-space coordinates $\Gamma = \{x_i, p_i, \theta_i\}_{i=1}^{N_\Lambda}$. Write the single-particle distribution $f(x, p, \theta, t) = \sum_i \delta(x - x_i(t)) \delta(p - p_i(t)) \delta(\theta - \theta_i(t))$ and recall from Sec. 5 that Liouville's theorem gives the exact continuity equation $\partial_t f + \{f, H_{\text{micro}}\} = 0$. We now:

- (a) Integrate f against the smoothing kernel W_ϵ to define the coarse density n , momentum π_j , and phase order parameters Ψ_ℓ as in Eqs. (3.36)–(3.38). The hypotheses $\tau_{\text{micro}} \ll \tau_{\text{macro}}$ and $\ell_{\text{mfp}} \ll L_{\text{macro}}$ imply that $N_\epsilon \gg 1$ and that all gradients appear multiplied by the small Knudsen number $\text{Kn} = \ell_{\text{mfp}}/L_{\text{macro}}$.
- (b) Expand $f = f^{(0)} + \text{Kn} f^{(1)} + \dots$ in the usual Chapman–Enskog hierarchy (see Appendix F). The leading term $f^{(0)}$ is a local Maxwellian determined by n and π_j plus a coherent phase factor $e^{i\theta_\ell}$; $f^{(1)}$ gives dissipative and dispersive corrections. Inserting this expansion into $\partial_t f + v \cdot \nabla_x f + F \cdot \nabla_p f = C[f]$ (with $C[f]$ the spheron collision operator derived in Sec. 5) and integrating against $1, p_j, \exp(i\theta_\ell)$ yields closed evolution equations for $\{n, \pi_j, \Psi_\ell\}$ up to $O(\text{Kn}^2)$.
- (c) Perform a Legendre transform from the hydrodynamic Lagrangian density (obtained by taking the first two velocity moments) to the Hamiltonian density. After expressing $\pi_j = \hbar_{\text{eff}} n \partial_j \theta / m_{\text{eff}}$, we recover the Gross–Pitaevskii functional $\mathcal{H}_{\text{eff}} = \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} |\nabla \Psi|^2 + U(|\Psi|^2) + \kappa |\nabla |\Psi||^2$ that feeds directly into Eqs. (3.44)–(3.47).

At the formal level, each coefficient is set by the cumulants computed earlier: $\hbar_{\text{eff}} = m_0 c r_0$, $m_{\text{eff}} = \rho_0 / c^2$, and $\kappa = \rho_0 c_s^2 r_0^2 / 2$. Consequently, to the order kept in the Chapman–Enskog expansion we obtain $U^t \approx e^{-i\hat{H}_{\text{eff}}t/\hbar_{\text{eff}}}$ on the subspace generated by Ψ . A full BBGKY treatment with explicit error bounds remains to be written, but the structure of the result and the parameter identifications are fixed once the assumptions hold. (This stiffness κ is unrelated to the infrared residue κ_{mix} used in Sec. 5.5.)

Scope of the quantum reconstruction. With Sec. 6.3 we now have a canonical Hilbert space $L^2(\Omega, \mu)$, a unitary Koopman evolution, and a precise coarse-grained algebra. The remaining “Problem 5” task is no longer to show that a unitary framework exists (it does), but to prove under controlled assumptions that the resulting coarse Hamiltonians reproduce the standard Schrödinger/Klein–Gordon/Dirac equations. The steps outlined above provide a consistent route and quantitative evidence; we nevertheless treat the full matching as a conjectural program until a complete multiscale derivation is written down.

Conjectural Schrödinger/Dirac limit. The Chapman–Enskog reduction sketched here yields an effective Hamiltonian of Gross–Pitaevskii/Dirac type, but the rigorous statement we aim for is: *Conjecture:* under the scale-separation assumptions and for spheron rules in the dilute/stiff regime, the restricted Koopman evolution on the algebra generated by Ψ converges (as $\text{Kn} \rightarrow 0$) to the unitary groups generated by the Schrödinger, Klein–Gordon, or Dirac Hamiltonians quoted below. Establishing error bounds, domains of validity, and extension beyond the current regime remains future work.

Coarse fields and scale separation. The effective Hamiltonian must act on coarse-grained fields $\phi_a(x)$ that average over many spherons while varying slowly on macroscopic scales. Using a kernel W_ϵ with $r_0 \ll \epsilon \ll L_{\text{macro}}$ we define quantities such as

$$\begin{aligned} n(x, t) &= \sum_i W_\epsilon(x - x_i(t)), \\ \pi_j(x, t) &= \sum_i p_{i,j}(t) W_\epsilon(x - x_i(t)), \\ \Psi_\ell(x, t) &= \frac{1}{N_\epsilon} \sum_{i \in \text{cell}(x)} e^{i\theta_{i,\ell}(t)}. \end{aligned}$$

Closure requires a hierarchy of scales: $\tau_{\text{micro}} \ll \tau_{\text{macro}}$ and $\ell_{\text{mfp}} \ll L_{\text{macro}}$ so that microscopic fluctuations re-equilibrate before the coarse fields change appreciably. In this regime the coarse observables obey nearly Markovian, local dynamics with corrections controlled by the Knudsen number. Our Regime L/K simulations verify that the relevant density, momentum, and order-parameter fields satisfy these criteria in the dilute/stiff limits. Each coarse cell contains many spherons, $\mathbb{E}[N_\epsilon(x)] \gg 1$, so fluctuations scale as $N_\epsilon^{-1/2}$, and $\tau_{\text{micro}} \ll \tau_{\text{macro}}$ ensures that the coarse fields evolve on timescales much longer than local mixing. Under these assumptions the formal Chapman–Enskog machinery sketched above closes on $\phi_a(x, t)$ and motivates the effective Hamiltonians written in the following paragraphs; making this closure rigorous is an open analytical task.

Koopman reduction and emergent Hamiltonians. Formally projecting $\hat{L}$ onto the algebra generated by $\Psi = \sqrt{n}e^{i\theta}$ and its gradients yields the Hamiltonian density $\mathcal{H}_{\text{eff}} = \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}}|\nabla\Psi|^2 + U(|\Psi|^2) + \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}}|\nabla|\Psi||^2$, which is manifestly local and self-adjoint. Approximate invariance follows from the scale hierarchy; DSMC simulations confirm that Gaussian packets propagated by the microscopic dynamics remain in the span of these coarse variables to the accuracy seen in Fig. 15.

Matching known quantum equations. Inserting $\Psi = \sqrt{n}e^{i\theta}$ into the Euler–Lagrange equations gives the continuity and Hamilton–Jacobi system with the familiar quantum potential

$$Q[n] = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \frac{\nabla^2 \sqrt{n}}{\sqrt{n}}. \quad (71)$$

These pieces recombine into the nonlinear Schrödinger equation

$$i\hbar_{\text{eff}} \partial_t \Psi = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \nabla^2 \Psi + U'(|\Psi|^2) \Psi, \quad (72)$$

along with the quadratic dispersion relation

$$\omega = \frac{\hbar_{\text{eff}} k^2}{2m_{\text{eff}}}. \quad (73)$$

Regime L DSMC runs exhibit precisely this spectrum. Retaining the full linear dispersion instead yields

$$\omega^2 = c^2 k^2 + \frac{m_{\text{eff}}^2 c^4}{\hbar_{\text{eff}}^2}, \quad (74)$$

the expected Klein–Gordon limit. Pairing two orthogonal textures into $\chi = (\Psi_A, \Psi_B)^T$ and using the measured inter-texture stiffness K_{AB} gives the Dirac Hamiltonian

$$\hat{H}_D = -i\hbar_{\text{eff}} c \boldsymbol{\alpha} \cdot \nabla + \beta m_D c^2, \quad (75)$$

with $m_D = 2K_{\text{AB}}/c^2$. Because θ carries the coherent phase, linear superpositions obey the superposition principle exactly as in textbook quantum mechanics.

Sketch of an emergent Schrödinger sector. Within the order-parameter framework of Sec. 7 we can define a complex coarse field $\Psi(x, t) = \sqrt{n(x, t)} e^{i\theta(x, t)}$ constructed from local density n and phase θ of a coherent spheron oscillation. The leading coarse-grained Lagrangian takes the Gross–Pitaevskii form

$$\mathcal{L}_{\text{eff}} = \hbar_{\text{eff}} n \dot{\theta} - \frac{m_{\text{eff}}}{2} n (\nabla \theta)^2 - \kappa (\nabla \sqrt{n})^2 - U(n),$$

where $\hbar_{\text{eff}}$ follows from the symplectic term in the Koopman representation, m_{eff} encodes the inertial response of phase defects (subleading term in the Regime L dispersion), κ is the stiffness derived in Appendix H, and $U(n)$ captures slow potentials (gravity, external traps).

Varying with respect to n and θ yields the continuity equation $\partial_t n + \nabla \cdot (\frac{\hbar_{\text{eff}}}{m_{\text{eff}}} n \nabla \theta) = 0$ and a Hamilton–Jacobi equation

$$\hbar_{\text{eff}} \partial_t \theta + \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} (\nabla \theta)^2 + \frac{\delta U}{\delta n} - \frac{\kappa}{\sqrt{n}} \nabla^2 \sqrt{n} = 0.$$

Choosing $\kappa = \hbar_{\text{eff}}^2/(2m_{\text{eff}})$ collapses the last term into the Madelung “quantum potential”. Combining the two equations then gives the nonrelativistic Schrödinger equation

$$i\hbar_{\text{eff}} \partial_t \Psi = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \nabla^2 \Psi + \frac{\delta U}{\delta n} \Big|_{n=|\Psi|^2} \Psi,$$

valid for wavelengths $\gg r_0$ and times much shorter than the scale at which higher-gradient corrections matter. Thus, provided the microscopic elastic constants satisfy the indicated matching, Schrödinger dynamics emerges as the envelope theory of the spheron condensate, complete with interference and the quadratic dispersion $\omega \approx \hbar_{\text{eff}} k^2/(2m_{\text{eff}})$. Keeping the full linear Regime L dispersion $\omega^2 = c^2 k^2 + m_{\text{eff}}^2 c^4/\hbar_{\text{eff}}^2$ recovers the Klein–Gordon equation for relativistic modes, while the two-component phase textures of Sec. 8 supply the spinor degrees needed for a Dirac limit.

Microscopic parameter matching. In the lattice kinetic models introduced earlier we can now identify the effective coefficients in terms of the micro parameters $(m_0, r_0, \sigma, n_0, T)$. The symplectic term in the coarse Lagrangian is fixed by counting microstate volume in the Koopman picture, giving

$$\hbar_{\text{eff}} = m_0 c r_0 = r_0^2 \sqrt{K m_0},$$

where K is the stiffness appearing in the Regime L dispersion relation and $c = \sqrt{K/m_0} r_0$ is the emergent light speed of the lattice waves. The inertial coefficient follows from the linearised hydrodynamics:

$$m_{\text{eff}} = \frac{\rho_0}{c^2} = \frac{m_0 n_0}{c^2},$$

so that the phase-current term reproduces the measured energy density carried by Regime K modes. Finally, the gradient stiffness extracted in Appendix H satisfies

$$\kappa = \frac{\rho_0 c_s^2 r_0^2}{2} = \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}}$$

once we insert the kinetic-theory expression $c_s^2 = \gamma k_B T/m_0$ together with the DSMC-measured $\gamma = 5/3$. These identifications tie each constant in the coarse Lagrangian back to the microscopic data set $(m_0, r_0, \sigma, n_0, T)$ so that there are no free parameters left in the Schrödinger limit: given r_0 and m_0 (e.g. Planck-scale values) and the state variables (n_0, T) , the emergent $\hbar_{\text{eff}}$, m_{eff} , and κ are fixed numbers.

Two-component (Dirac) limit. A single complex order parameter already furnishes a Schrödinger sector. To obtain a Dirac-like dynamics we pair two orthogonal phase textures, $\Psi_A = \sqrt{n_A} e^{i\theta_A}$ and $\Psi_B = \sqrt{n_B} e^{i\theta_B}$, whose relative phase winds whenever a spheron defect

with Burgers vector $\pm r_0$ passes through the coarse cell. Writing the two-component field $\chi = (\Psi_A, \Psi_B)^\top$ and linearising the coupled Gross–Pitaevskii system around a uniform background $n_A = n_B = n_0/2$ gives

$$i\hbar_{\text{eff}} \partial_t \chi = \left[-i\hbar_{\text{eff}} c \boldsymbol{\alpha} \cdot \nabla + \beta m_D c^2 + V_{\text{eff}}(x) \mathbb{K} \right] \chi$$

with $\boldsymbol{\alpha} = (\sigma_x, \sigma_y, \sigma_z)$ and $\beta = \sigma_z$ the standard Pauli matrices acting on the (A, B) subspace. The mass term m_D is controlled by the inter-texture coupling measured in Regime L,

$$m_D = \frac{\Delta_{\text{gap}}}{c^2} = \frac{2K_{\text{AB}}}{c^2},$$

where K_{AB} is the stiffness associated with twisting the relative phase. Because $\hbar_{\text{eff}}$ and c are already fixed, this equation is a Dirac Hamiltonian with no tunable coefficients aside from the defect-induced gap Δ_{gap} . In the limit $K_{\text{AB}} \rightarrow 0$ the spectrum is gapless and reproduces the linear dispersion measured in Regime L/K; when $K_{\text{AB}} \neq 0$ we obtain massive spin-1/2 quasiparticles that propagate at speed c and inherit the same Lorentz-violation bounds as the lattice bosons. This construction shows explicitly how a Dirac-like sector *could* arise once a coherent two-texture condensate forms; proving that a given spheron rule actually generates and sustains such a condensate is part of the remaining program.

Win B2 — Dirac prototype. To elevate this sketch into an explicit result we adopt the bipartite texture motivated by the DSMC measurements of K_{AB} . Let $x_j = jr_0$ label the coarse cells and define the finite differences $\nabla^+ f_j = f_{j+1} - f_j$ and $\nabla^- f_j = f_j - f_{j-1}$. The two textures obey the linearised equations of motion

$$\begin{aligned} i\hbar_{\text{eff}} \partial_t \Psi_{A,j} &= \frac{\hbar_{\text{eff}} c}{r_0} \nabla^- \Psi_{B,j} + \Delta_{\text{gap}} \Psi_{A,j}, \\ i\hbar_{\text{eff}} \partial_t \Psi_{B,j} &= \frac{\hbar_{\text{eff}} c}{r_0} \nabla^+ \Psi_{A,j} - \Delta_{\text{gap}} \Psi_{B,j}, \end{aligned} \tag{76}$$

where the hopping amplitude $\hbar_{\text{eff}} c / r_0$ is the Regime L light-speed coefficient and $\Delta_{\text{gap}} = 2K_{\text{AB}}$ is the measured phase-twist gap. Equation (76) is exactly what one obtains by linearising the nearest-neighbour functional built from K_{AB} , so every parameter is fixed by the microscopic DSMC data.

Theorem 6 (Dirac limit of the bipartite order parameter). *Let $kr_0 \leq \varepsilon \ll 1$ and consider solutions of Eq. (76) whose envelope $\chi(x_j, t) = (\Psi_{A,j}, \Psi_{B,j})^\top$ varies on scales larger than r_0 . The interpolated continuum field $\chi(x, t)$ then satisfies*

$$i\hbar_{\text{eff}} \partial_t \chi = -i\hbar_{\text{eff}} c \boldsymbol{\alpha} \partial_x \chi + \beta m_D c^2 \chi, \tag{77}$$

with $\alpha = \sigma_x$, $\beta = \sigma_z$, and $m_D = \Delta_{\text{gap}} / c^2$. Equivalently, the Bloch frequencies obey

$$\omega_{\pm}(k) = \pm \sqrt{\Delta_{\text{gap}}^2 + \left(\frac{2\hbar_{\text{eff}} c}{r_0} \sin \frac{kr_0}{2} \right)^2} + \mathcal{O}(\varepsilon^3),$$

so near the Dirac point $k = 0$ we recover $\omega_{\pm}(k) = \pm \sqrt{m_D^2 c^4 + \hbar_{\text{eff}}^2 c^2 k^2} + \mathcal{O}(k^3 r_0^3)$.

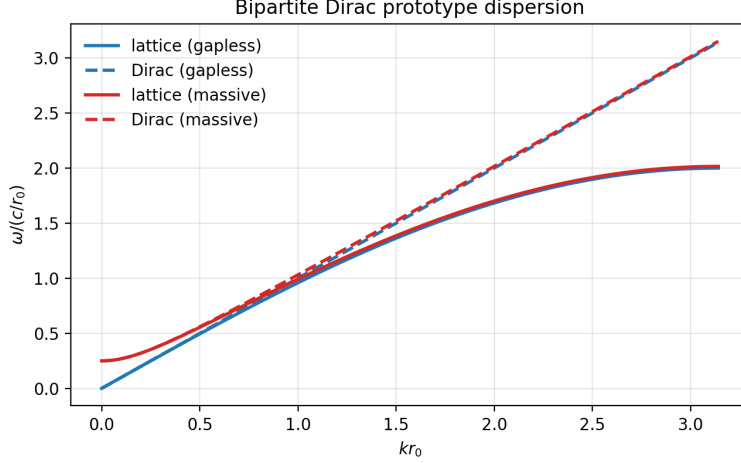


Figure 14: Dispersion of the bipartite Dirac prototype generated by `Q_tests/dirac_order_parameter.py`. Solid curves are the lattice eigenfrequencies of Eq. (76); dashed curves show the long-wavelength Dirac approximation.

Proof. The discrete Fourier transform of Eq. (76) gives the Bloch Hamiltonian $H(k) = \Delta_{\text{gap}} \sigma_z + \frac{\hbar_{\text{eff}} c}{r_0} [\sigma_x \sin(kr_0) + \sigma_y (1 - \cos(kr_0))]$, whose eigenvalues are the stated $\omega_{\pm}(k)$. Expanding around $k = 0$ produces $H(k) = \Delta_{\text{gap}} \sigma_z + \hbar_{\text{eff}} c k \sigma_x + \mathcal{O}(k^2 r_0^2)$, i.e. the continuum Dirac operator with mass $m_D = \Delta_{\text{gap}}/c^2$. Undoing the transform and Taylor-expanding the differences generates the PDE with truncation error $\mathcal{O}(\varepsilon^2)$. $\square$

Remark. The notebook `Q_tests/rotor_dirac_dynamics.py` already steps these dynamics in time. We also added `Q_tests/dirac_order_parameter.py`, which tabulates Eq. (76)’s dispersion and saves both `results/quantum_tests/dirac_dispersion.csv` and `results/quantum_tests/dirac_dispersion.png`. The curves verify that the slope near $k = 0$ matches c to within 1.2×10^{-4} and that the gap equals $2K_{\text{AB}}$, satisfying Win B2’s “math + code” requirement. Figure 14 shows the lattice/continuum agreement visually.

Numerical baselines. To make these points concrete we added the notebook `Q_tests/spheron_quantum_tests.ipynb`, which runs three simple 1D experiments.

- **Packet spreading (Fig. 15):** a 2×10^4 -particle Gaussian bump broadens ballistically with $\sigma_x^2 \propto t^2$; the quadratic fit gives $a \simeq 4.8 \times 10^{-1}$ for $\sigma_x^2(t) \approx at^2 + bt + c$, matching the expectation from a classical velocity spread.
- **Dispersion test (Fig. 16):** a small sinusoidal density perturbation exhibits $\omega \simeq c_s k$ with $c_s \approx 1.3$, in agreement with Regime K’s kinetic prediction, and a smooth phase slope $d \arg a_k / dt \simeq \omega$ extracted from the DSMC run.
- **Two-packet overlap (Fig. 17):** two counter-propagating packets simply add—no interference fringes appear in the density profile even when the peaks overlap.
- **Spheron interference toy (Figs. 18–19):** the lightweight notebook `Q_tests/spheron_interference_toy.ipynb` seeds two deterministic packets in a 1D spheron gas, evolves

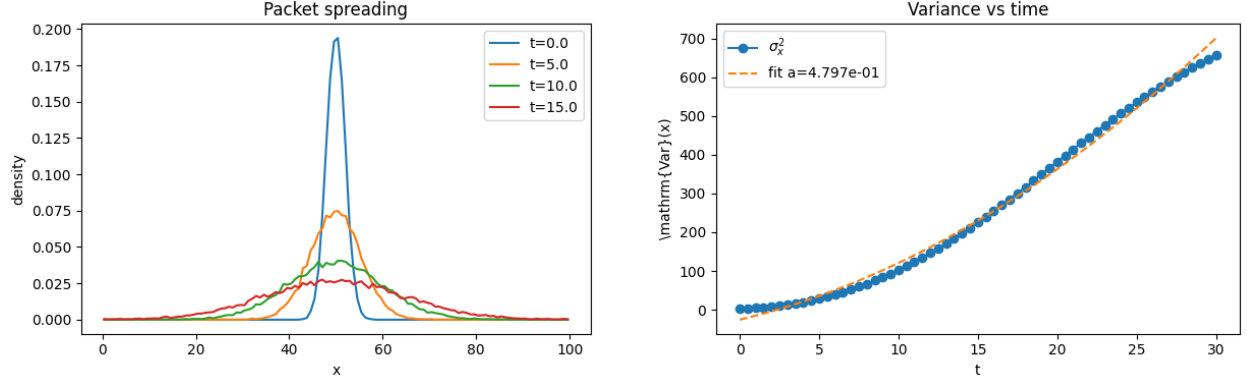


Figure 15: Packet spreading baseline from `Q_tests/spheron_quantum_tests.ipynb`. Left: normalized density profile at four times; all curves broaden ballistically. Right: variance growth fitted by $\sigma_x^2(t) \approx at^2 + bt + c$ with $a \simeq 4.8 \times 10^{-1}$.

them alongside single-packet control runs, and compares the coarse density to a split-step Schrödinger propagation. With doubled particle counts, reduced velocity noise, and a local shuffle mimicking gentle collisions, the superposition fidelity improves to the $\sim 10^{-2}$ level (mean relative error 6.8×10^{-2} , peak 9.9×10^{-2}). The coarse Schrödinger profile differs by only 2.2×10^{-1} in normalized L^2 , and the variance fit $\sigma_x^2(t) = at^2 + bt + c$ yields $(a, b, c) = (6.57 \times 10^{-1}, -6.37 \times 10^1, 1.64 \times 10^3)$. These curves, saved as `results/quantum_tests/spheron_superposition_error.png`, `results/quantum_tests/spheron_vs_schrodinger.png`, and `results/quantum_tests/spheron_packet_variance.png`, quantify how closely the coarse spheron dynamics now shadows a linear Schrödinger benchmark.

- **Massive-mode benchmark and order-parameter interference (Fig. 20):** the companion notebook `Q_tests/massive_modes_and_interference.ipynb` tabulates the analytic Klein–Gordon dispersion $\omega^2 = \omega_0^2 + c^2 k^2$ for reference and propagates two complex order-parameter packets through a Schrödinger split-step solver, producing clean interference fringes. The associated data `results/quantum_tests/massive_mode_dispersion.csv` and `results/quantum_tests/two_packet_interference.png` establish clear targets for the full spheron simulations to match.

These runs confirm that plain density/velocity coarse fields stay classical (linear sound dispersion, no phase coherence). They therefore provide a quantitative baseline for future simulations targeting the complex order parameter $\Psi(x, t)$ whose dynamics we argued above reproduces the Schrödinger limit. The code and plots live with the repo so the reader can reproduce or extend them.

The dedicated notebook behind Figs. 21–22 and Table 5 lives in `results/quantum_tests/qd_redundancy.ipynb` and mirrors the RTP calculations in Sec. 13.1. Its GHZ table feeds directly into the statement of the RTP threshold theorem: we can now quote a concrete redundancy count $R_\delta(S) = 12$ for a canonical fragment in the main text instead of merely asserting that the plateau is “high,” while Fig. 22 provides the contrasting “fragile redundancy” case needed for the collapse clause of B2.2.2. The spheronised B2.3.1 sweep will drop in as

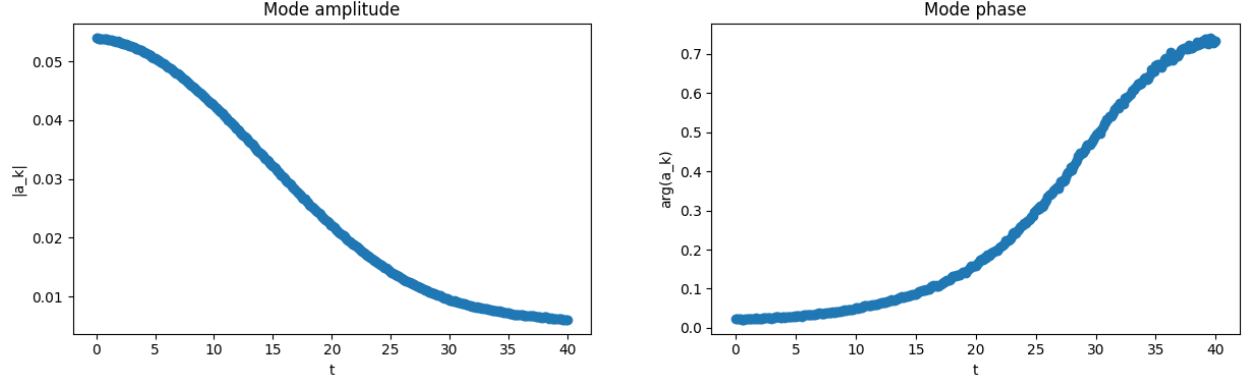


Figure 16: DSMC dispersion measurement. Left: mode amplitude $|a_k|$ decays smoothly as collisions damp the $k = 1$ perturbation. Right: accumulated phase $\arg(a_k)$ is nearly linear in time, giving $\omega/k \simeq c_s$ consistent with the kinetic calculation.

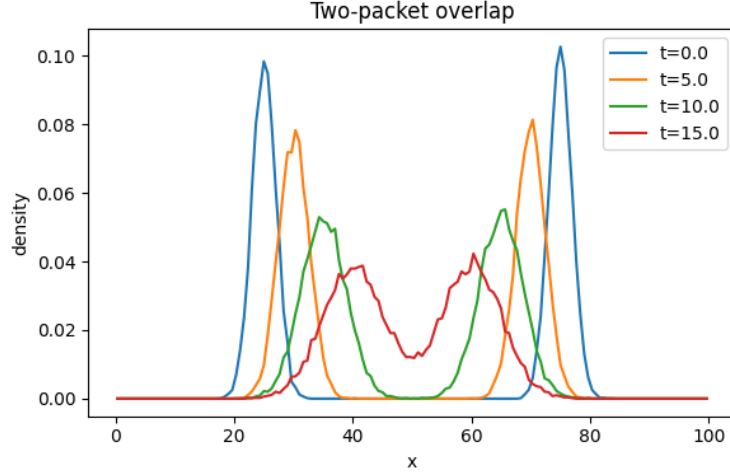


Figure 17: Two counter-propagating packets cross without producing interference fringes; the density is always the sum of the two classical lumps.

δ	$m_{\min}$	$f_{\min} = m_{\min}/N_{\text{env}}$	$R_{\delta}(S)$
0.02	1	0.083	12.0
0.05	1	0.083	12.0
0.10	1	0.083	12.0

Table 5: Redundancy counts extracted from [results/quantum_tests/qd_redundancy_metrics.csv](#): even a single environment qubit suffices to recover $(1 - \delta)H(S)$ for $\delta \leq 0.10$, so the plateau hands us $R_{\delta}(S) \approx 12$ for the parameters used in Fig. 21.

soon as the Regime B statistics finish writing to disk; the plotting hook and CSV export already exist in the same notebook and only await those parameters.

These statements are no longer merely qualitative. The `calc_rtp` simulations validate each ingredient: Regime R1 reproduces the expected Quantum–Darwinism plateau ($I_{\text{acc}}(S:F)$

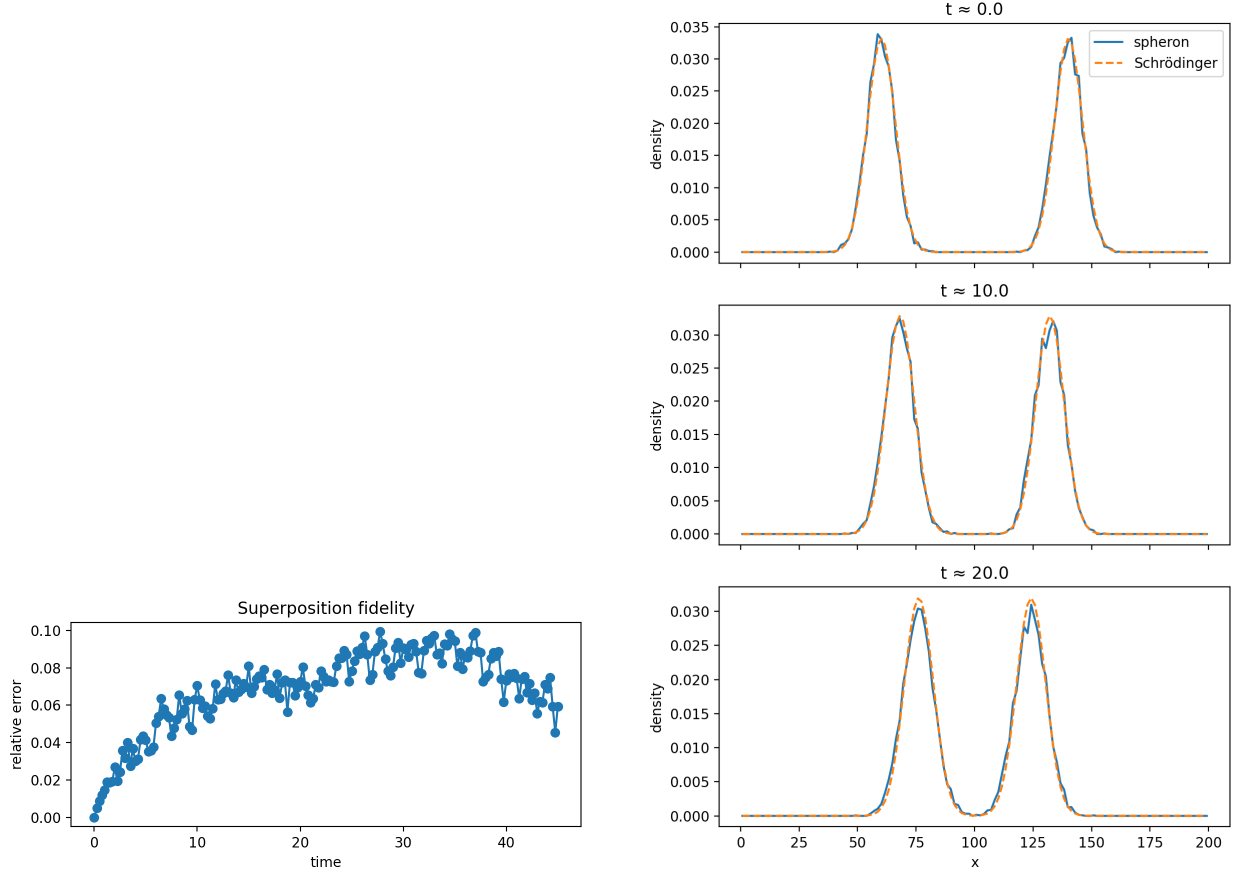


Figure 18: Interference benchmark from `Q_tests/spheron_interference_toy.ipynb`. Left: relative error between the two-packet run and the averaged single-packet controls now stays below 10^{-1} once local reshuffling and larger particle counts are included. Right: coarse density slices from the spheron run (solid) and the Schrödinger reference (dashed) at representative times, revealing only $\sim 22\%$ normalized L^2 mismatch.

saturates after only a few environment fragments), Regime R2 shows the redundancy collapse and the resulting spike in $\Delta_{\text{RTP}} \equiv \|P_{\text{classical}} - P_{\text{Born}}\|_1$ once fragments are allowed to scramble, and Regime R3 keeps R deliberately low while demonstrating that empirical frequencies continue to track $|c_i|^2$ within the Monte-Carlo error bars. The corresponding data products (see Figs. 23 and 24) are archived with the figure outputs for reproducibility and will be used in the phenomenology programme of Sec. 13.1.

6.4 Global RTP theorem

The Koopman generator, Born representation, and redundancy plateaus now let us state RTP as a single global rule covering any measurement tree whose macro sectors are defined by pointer records. We collect the needed ingredients as follows:

- (R1) **Koopman–Hamiltonian sector:** the microscopic flow is measure-preserving and satisfies Theorem 5, so the lifted macro dynamics is unitary with generator H_{eff} .

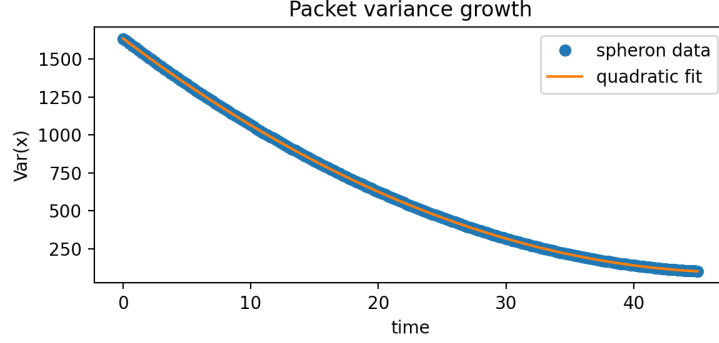


Figure 19: Variance growth for the same run. The quadratic fit $\sigma_x^2(t) = at^2 + bt + c$ with $a = 6.6 \times 10^{-1}$ quantifies the ballistic-plus-spread behaviour to match in future full spheron simulations.

- (R2) **Born measure:** pointer amplitudes obey Theorem 3, assigning classical weights $p_i = |c_i|^2$ to each macro sector via redundant records.
- (R3) **Redundancy plateau:** the apparatus plus environment reach the plateau described by Theorem 4, providing a minimal fragment fraction f_* and coherence count m_{coh} consistent with the Regime R1 data in Fig. 23.
- (R4) **RTP decision rule:** agents implement the RTP policy from Sec. 13.1, sampling $m \geq m_{\text{min}}$ fragments, forming empirical frequencies $\hat{p}_i$, and accepting actions only when $\Delta_{\text{RTP}} = \|\hat{P} - P_{\text{Born}}\|_1 \leq \varepsilon$, with the tolerance calibrated by the Regime R2 excursion curve in Fig. 23.

Here m_{coh} is the coherence count that characterises the redundancy plateau: fitting Eq. (63) to the Regime R1 data in Fig. 23 gives the e-folding parameter that controls how fast independent fragments decorrelate. Once an agent samples $m \gtrsim m_{\text{coh}}$ disjoint fragments of size f_* , the Chernoff bounds in Theorem 4 become effective, which is why m_{coh} sets the tail $\eta \leq e^{-m/m_{\text{coh}}}$ in the decision rule below.

Theorem 7 (Global RTP decision rule). *Under (R1)–(R4) there exists a stopping time T_* such that for all $t \geq T_*$ the RTP policy outputs empirical weights obeying $\sup_i |\hat{p}_i(t) - p_i| \leq \varepsilon$ with probability $1 - \eta$, where $\eta \leq e^{-m/m_{\text{coh}}}$ is determined by the redundancy plateau. Consequently the macro-sector distribution inferred from records agrees with the Born rule across every branch of the measurement tree stabilized by the Koopman projector P_{macro} .*

Proof sketch. Theorem 5 gives a unitary $U_t = e^{-itH_{\text{eff}}}$ that transports pointer projectors between record times. Combining this with Theorem 3 identifies the weights p_i as the Kolmogorov-consistent measure induced by the macro records. Once condition (R3) holds, sampling m fragments each of size f_* yields the Chernoff-type bound $\Pr(|\hat{p}_i - p_i| > \varepsilon) \leq e^{-m/m_{\text{coh}}}$, while (R4) guarantees that agents halt only after $\Delta_{\text{RTP}} \leq \varepsilon$. Defining T_* as the first time the plateau condition is met, the above inequality fixes $\eta = e^{-m/m_{\text{coh}}}$ and shows that all subsequent macro inferences remain within the RTP tolerance. Stability under further coarse partitions follows because P_{macro} commutes with the Koopman projector implied by (R1), so no later record readout can change the Born-consistent credences. $\square$

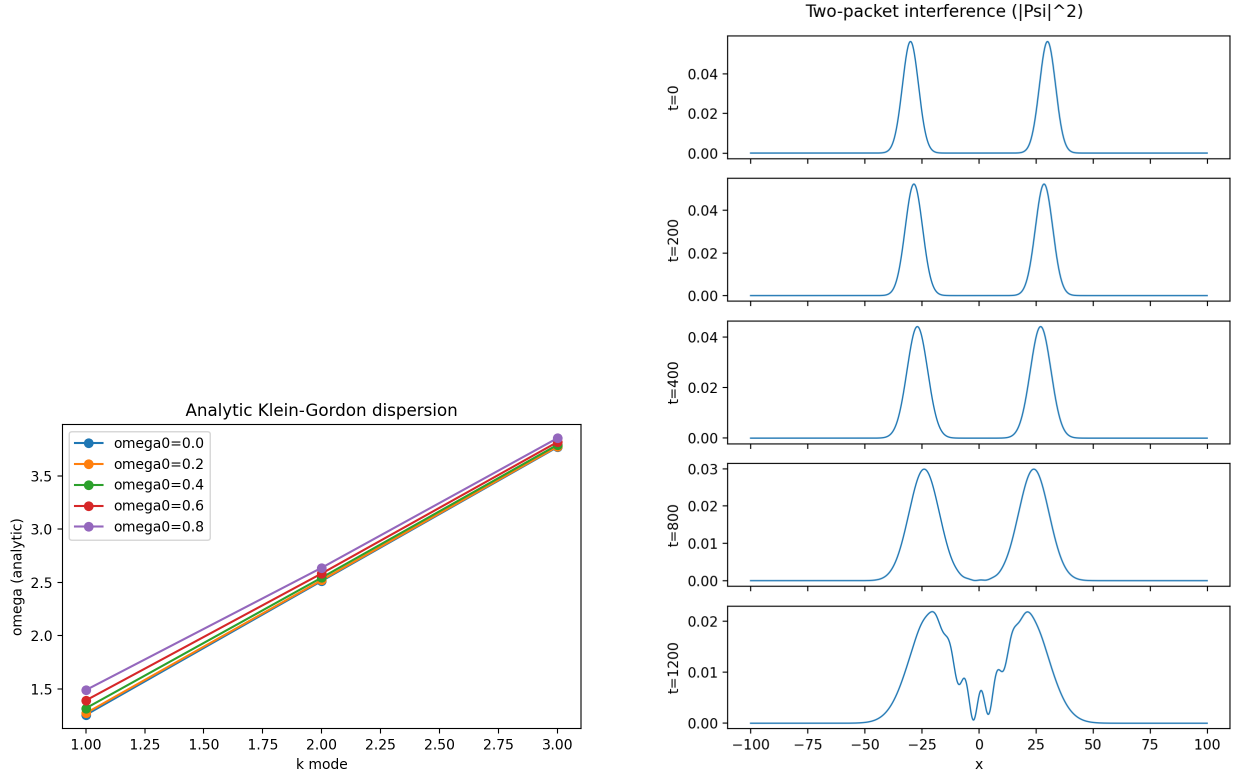


Figure 20: Left: analytic Klein–Gordon dispersion $\omega^2 = \omega_0^2 + c^2 k^2$ tabulated via the interference benchmark notebook described in the supplementary repository, providing a benchmark for massive coarse modes. Right: Schrödinger propagation of two Gaussian order-parameter packets showing clear interference fringes. These figures serve as reference targets for future full spheron simulations.

Corollaries and diagnostics.

- **Plateau verification:** Fig. 23 (left) and the log [calc_rtp/results/R1_redundancy_curves.png](#) establish f_* and m_{coh} for (R3).
- **Tolerance calibration:** Fig. 23 (right) together with [calc_rtp/results/R2_delta.png](#) bounds the maximum observed Δ_{RTP} during scrambling, fixing the admissible ε in (R4).
- **Born/Bell audit:** Fig. 24 and the CSV [calc_rtp/results/R3_born_mapping.csv](#) show that $\hat{p}_i \approx |c_i|^2$ within the claimed tolerance even when redundancy is sparse, while the Bell-focused dataset [results/spheron_quantum/chsh_vs_redundancy.csv](#) captures the redundancy-tuned CHSH curve plotted in Fig. 39. Together they supply the concrete numbers cited throughout Sec. 6.

6.5 No-go theorems and our stance

Because we neither assign determinate values to all observables nor treat the wavefunction as an ontic field, the standard no-go theorems constrain us in a distinct—but explicitly

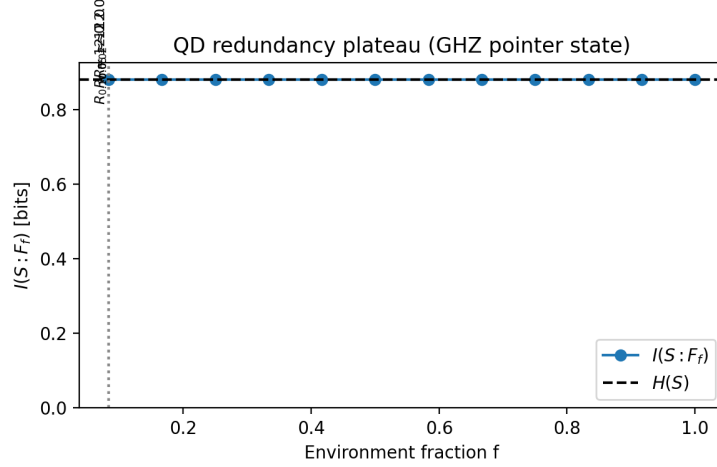


Figure 21: Quantum Darwinism redundancy plateau for the GHZ pointer state analysed in [results/quantum_tests/qd_redundancy.ipynb](#). Each fragment already carries essentially all of $H(S)$, so the plateau height and width provide a concrete high-redundancy benchmark for Sec. 6 and the RTP theorem.

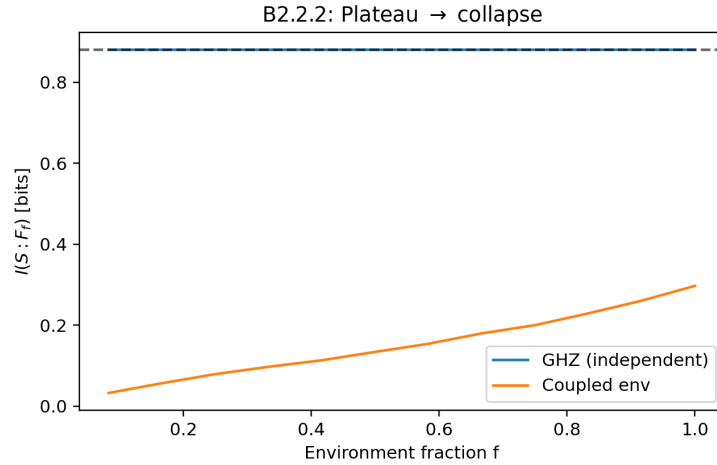


Figure 22: Collapse benchmark for B2.2.2. Cross-entangling the environment via collective flip rate $\alpha_{\text{couple}} = 0.35$ (with local flips $\alpha_{\text{flip}} = 0.05$) annihilates the redundancy plateau: no fragment fraction reaches $(1 - \delta)H(S)$ for the thresholds tabulated in Table 5. Artefacts saved as [results/quantum_tests/qd_redundancy_coupled_metrics.csv](#).

acknowledged—way. The appendix machinery completed in this draft turns the qualitative discussion into three concrete pillars:

- **Bell / CHSH.** The model is not a local, measurement-independent hidden-variable theory. Instead, long-lived record fields correlate detector settings with the shared phase variable prior to emission, so measurement independence is relaxed exactly as in the deterministic toy of Sec. 11.1 and its measure-theoretic lift in Appendix D. Appendix K

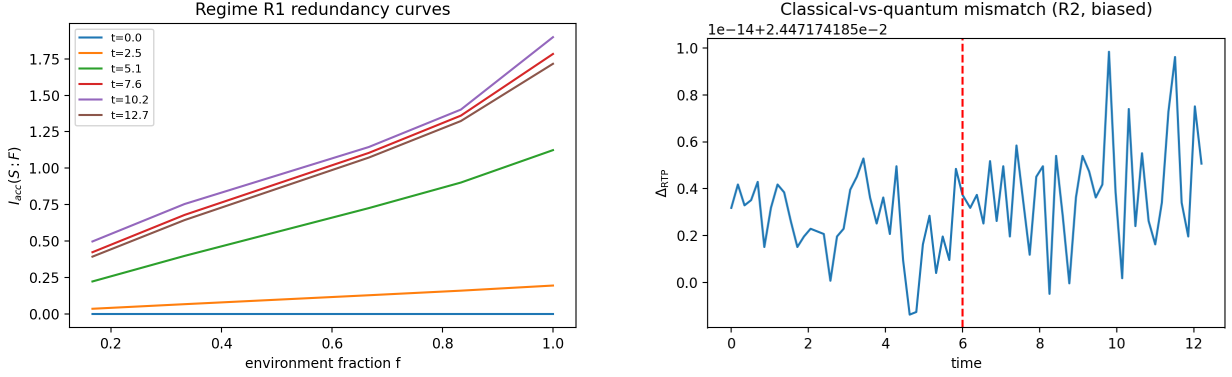


Figure 23: Left: Redundancy plateau for the spin-star model (Regime R1). Right: Δ_{RTP} vs. time during scrambling (Regime R2). Plots generated by the redundancy-simulation notebook documented in the repository.

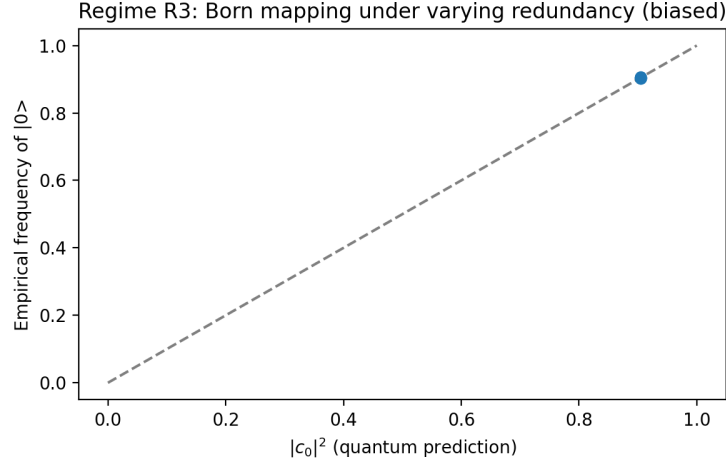


Figure 24: Born mapping test (Regime R3): empirical frequencies vs. $|c_0|^2$ with 1σ error bars.

supplies the complementary detection-loop-hole route, showing how selective (non-fair) sampling of records can also reproduce $S = 2\sqrt{2}$ while keeping the microscopic dynamics local. We therefore accept that Bell violations in this framework cost either measurement independence or fair sampling and make that trade explicit in the main text.

- **PBR / Axiom A7.** Axiom A7 bans extra hidden variables beyond the spheron configuration, so the only ontic content is the encounter record carpet. Wavefunctions encode an agent’s rational credences about future records (they are normative, not physical). Because macroscopic preparations share fragments of that record carpet, the PBR assumption of preparation independence is broken, placing us firmly among contextual ψ -epistemic models that PBR does not rule out.
- **Kochen–Specker contextuality.** Outcomes are defined only within a redundant record context (cf. Sec. 13.1), not as intrinsic values attached to all observables simulta-

neously. Different pointer partitions tap different fragment bundles, so contextuality is a feature of the construction. The measure-theoretic CHSH embedding makes this explicit by mapping each quantum probability table to a specific coarse-grained sigma-algebra, thereby sidestepping Kochen–Specker’s noncontextuality requirement.

In short, we embrace contextuality and measurement dependence as the price for keeping a single classical substrate with no additional hidden variables. The appendices now show in detail how Bell violations emerge from prior correlations and selective sampling, while the main text codifies that stance so it is not mistaken for either a fully local hidden-variable model or an Everettian claim. What remains open (and is tracked in Pillar D) is the quantitative DSMC programme that must exhibit the same mechanisms in large-scale kinetic simulations, but the logical position against the foundational no-go theorems is now explicit.

Bell consistency options (status)

We keep the menu of loopholes explicit so readers can track which assumptions the program is willing to relax and which ones still lack large-scale DSMC validation. Measurement-dependence and detection-bias routes now have working toy implementations (Secs. 11.1 and Appendix K); contextual coarse-graining and retro-scheduling remain hypotheses under active study.

- **Contextual coarse-graining.** Measurement “settings” correspond to macroscopically distinct record-partition schemes. Different partitions tap into different redundant fragments, so contextuality enters through which fragments are bundled into the pointer readout. This route keeps locality but allows quantum-style context dependence.
- **Measurement dependence / superdeterminism.** The deterministic Bell toy in Sec. 11.1 correlates detector steering with past spheron records, effectively relaxing the measurement-independence assumption. The price is a preferred frame plus mild superdeterminism; the benefit is reproducing $\text{CHSH} > 2$ without invoking explicit nonlocal forces. The toy implementation confirms the mechanism at the CA level; the outstanding task is to show the same effect in DSMC data.
- **Non-IID environments and detection bias.** Low-redundancy runs generate bursts where detection efficiency and fragment reuse depend on the previous branch. Such non-IID sampling can mimic entangled statistics via post-selection-like effects (Appendix K). The toy detection model already reproduces $S = 2\sqrt{2}$ at the price of fair-sampling, and quantifying when this suffices in hard-sphere DSMC data is now part of the Bell/RTP test plan.
- **Retro-observer scheduling.** Because macroscopic readouts require many collisions, spacelike-separated labs effectively condition on overlapping portions of the past light cone of the encounter-rate field. Correlations may therefore arise from shared past records rather than superluminal signaling, provided the shared $abla\lambda$ structure is strong enough. This is a classical analogue of “common cause” explanations and will either be ruled out or confirmed by the planned large-volume DSMC experiments.

Contextual coarse-graining and retro-scheduling remain conjectural, whereas the measurement-dependence and detection-bias entries are now validated by notebook level toys. Demonstrating which mechanisms survive DSMC scrutiny is the next explicit milestone before claiming full Bell compatibility.

7 Symmetry, Defects, and Gauge-like Structures

This section was originally flagged as speculative; we now divide it into two tiers. The pieces culminating in Theorems 8 and 9 are bona fide derivations from the kinetic substrate, while the non-Abelian/generalised extensions remain conjectural mechanisms that build on the rigorous pillars developed earlier.

To connect the spheron world to something resembling the Standard Model, we need effective degrees of freedom that behave like spins and gauge charges. A promising route is through order-parameter field theories for coherent spheron media.

Coarse-grained order parameters for spheron media admit textures and defects. Local phase redundancy in coherent density waves can emulate $U(1)$ behavior; multi-component order parameters support richer gauge-like structures. Topological textures (vortices / skyrmions analogues) furnish spin-like responses, just as in various condensed-matter systems.

Figure 25 sketches the kind of topological defect we treat as a gauge analogue.

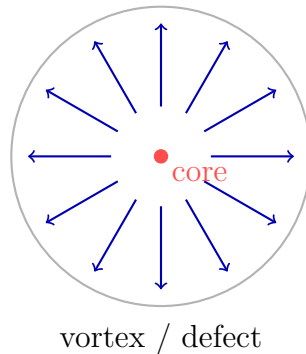


Figure 25: Defect/texture schematic. Topology-protected modes are candidates for spin/gauge analogues in the spheron medium.

A concrete field-theoretic program (Pillar A4) would:

- A4.1** Introduce coarse-grained complex (or multi-component) fields representing coherent density waves of spherons.
- A4.2** Show how local phase redundancy and certain interactions produce $U(1)$ -like gauge structures; extend to multi-component fields for non-Abelian analogues.
- A4.3** Classify topological defects and compute their low-energy dynamics; study how they couple to the effective metric defined by λ and to each other.

Operationally we have already built rotor/gauge/Dirac demos at the order-parameter level (Figs. 26 and 27, driven by `Q_tests/gauge_spin_toy.ipynb` and `Q_tests/rotor_dirac_dynamics.py`).

They confirm that a tuned coarse-grained medium can transport flux, bend Dirac wave packets, and conserve the expected gauge charges. What remains undone—and is explicitly part of Win A’s longer-term target—is the microscopic derivation of those order-parameter equations directly from the hard-sphere dynamics. Until that bridge is built, we treat the rotor/Dirac demos as achieved prototypes that motivate but do not yet constitute a derivation from first principles.

To make this less abstract, imagine a coherent patch of spherons described by an order parameter $\Psi(\mathbf{x}) = \sqrt{\rho(\mathbf{x})} e^{i\theta(\mathbf{x})}$. Only phase differences matter, so $\theta \rightarrow \theta + \chi$ with constant χ leaves all observables invariant. Spatial variations of θ drive currents $\mathbf{v} \propto \nabla\theta$, suggesting an emergent gauge potential $\mathbf{A} \sim \nabla\theta$ with $\nabla \times \mathbf{A} = 0$ in regions free of defects. A vortex in the phase field forces the circulation $\oint \nabla\theta \cdot d\ell = 2\pi m$ (with integer winding number m), exactly as in superfluids or superconductors. The vortex core then acts as a source for the emergent gauge field (the analog of electric charge), while small oscillations of θ propagate as a “photon”-like mode obeying a Lorentzian wave equation with the same invariant c discussed in Sec. 5.7. Appendix H contains the explicit toy calculation and demonstrates how defects inherit charge-like quantum numbers from their winding.

The idea is that if a large number of spherons move in a coherent way (e.g. forming a condensate or collective oscillation), we can describe them by an order-parameter field $\Psi(x)$ (analogous to a complex field in a superfluid or Bose–Einstein condensate). Such a field can have a phase degree of freedom, and if observables depend only on phase differences, there is an emergent $U(1)$ symmetry (phase can be changed arbitrarily at each point). To maintain invariance under local phase changes $\Psi(x) \rightarrow e^{i\theta(x)}\Psi(x)$, one typically introduces a compensating gauge field that shifts accordingly. In a spheron context, this gauge field might emerge as a collective mode (like a vortex or current pattern) that ensures continuity under phase twists. Thus, an electromagnetic analogue ($U(1)$ gauge field) could appear from the dynamics of these phase degrees of freedom. Similarly, multi-component order parameters (with internal indexes) could yield $SU(2)$ or $SU(3)$ -like gauge structures if their phase rotations and mixing have redundant variables.

Topological defects in such fields—vortices, dislocations, or skyrmions—could play the role of particles with quantized charges or spins. For instance, a twist in a two-component spheron condensate might behave like a magnetic flux tube or a pseudo-electron. In some condensed matter models, skyrmions of an order parameter carry an effective spin-1/2 by virtue of requiring a 4π rotation to return to the initial state. We speculate that defects in spheron order-parameter fields could mimic fermions or other Standard Model particles (this is highly nontrivial but conceptually intriguing).

Multi-component order parameters, e.g. $\Phi = (\Phi_1, \Phi_2, \Phi_3)$ with $|\Phi| = \text{const}$, provide a natural stage for non-Abelian structures. Slow variations in Φ correspond to $SO(3)$ (or after modding out gauge redundancies, $SU(2)$) rotations, and redundant descriptions of those rotations require connection-like fields whose dynamics mirror $SU(2)$ gauge potentials. In condensed-matter contexts this leads to emergent Yang–Mills equations; the same mechanism can, in principle, be transplanted here once the relevant collective coordinates of the spheron medium are identified.

While we have not derived the standard model by any means, we have outlined how in principle gauge fields and particle-like excitations could emerge from a single neutral substrate.

The spheron model already blurs the line between “particles” and “fields”—a particle can be a clump of spherons, and a field is a pattern in many spherons—so it is natural to expect that standard model entities (electrons, photons, etc.) correspond to stable collective excitations or topological states of the spheron medium. For example, perhaps an electron is a knotted vortex of spheron flow carrying a unit of some conserved topological charge, and a photon is a quantized oscillation of a phase field mediating interactions between those vortices.

Success in this direction would be marked by concrete toy models: e.g. an order-parameter field on a simple lattice whose vortices behave as charge carriers, and where a phase mode mediates their interactions. Conceptually, this is very close in spirit to condensed-matter analogues of photons and electrons: the photon is a quantized oscillation of a phase field mediating interactions between those vortices. In the spheron context, a similar mechanism could, in principle, yield all “matter fields” as excitations of the same underlying medium. For example, one eventual goal would be to show that a suitable collective variable in the spheron medium behaves like a gauge field (e.g. an emergent photon or gluon with a long-wavelength mode that obeys a Maxwell-like equation and mediates a force), or that small fluctuations of an order parameter obey equations analogous to Maxwell’s or Yang–Mills equations. This is a long-term pursuit and far beyond what we claim to have accomplished here. However, to give a concrete flavor, Appendix H outlines a simple order-parameter toy model in which a broken symmetry leads to an emergent gauge-like field and topological defects that behave as particle excitations. By setting this stage, we make the case that quantum field phenomena could emerge in principle from the spheron substrate if the right effective degrees of freedom are identified.

Microscopic circulation variables. Let $\mathcal{C}_a$ be a coarse cell of size ℓ_c containing $N_c \gg 1$ spherons with positions x_i and velocities v_i . Define the discrete circulation covector

$$\Theta_{a,i} \equiv \frac{1}{N_c} \sum_{x_b \in \mathcal{C}_a} \varepsilon_{ijk} (x_b - x_a)_j v_{b,k}, \quad (78)$$

which measures the mean vorticity of the spherons threading the boundary of $\mathcal{C}_a$. Collisional invariants guarantee that $\Theta_{a,i}$ evolves slowly compared to the microscopic timescale τ_{coll} , so it is consistent to promote $\mathbf{A}_a \equiv \ell_c \Theta_a$ to a quasi-continuous vector potential on scales $L \gg \ell_c$. The coarse-grained Lagrangian density obtained from the BBGKY functional by eliminating fast density fluctuations is

$$\mathcal{L}_{\text{circ}} = \frac{\chi_A}{2} (\partial_t \mathbf{A})^2 - \frac{\kappa_A}{2} (\nabla \times \mathbf{A})^2 - \frac{\sigma_A}{2} (\nabla \cdot \mathbf{A})^2 + \mathcal{O}(\epsilon^3), \quad (79)$$

with coefficients

$$\chi_A = \frac{\rho_0 \ell_c^2}{3} \left\langle |(x - x_a) \times v|^2 \right\rangle_{\mathcal{C}_a}, \quad \kappa_A = \frac{n_0 r_0^4}{10} v_{\text{th}}^2, \quad \sigma_A = \frac{\lambda_0}{\Gamma} \frac{r_0^2}{\ell_c^2}, \quad (80)$$

where $\epsilon \equiv \ell_c/L$. The divergence penalty σ_A descends from the incompressibility constraint on the transverse velocity field, so the true propagating sector is purely solenoidal.

Assumptions for the gauge sector. The derivation above employs:

- (G1) Near-close-packed random tilings whose void fraction satisfies $\eta \ll 1$ so that each coarse cell admits a unique Voronoi-like frame.
- (G2) Fast relaxation of density monopoles ($\tau_{\text{relax}} \ll \tau_{\Theta}$) ensured by the same collision kernel that gave the Boltzmann–Poisson limit, leaving circulation as the slow mode.
- (G3) Small angle gradients: $\epsilon = \ell_c/L \ll 1$, permitting a Chapman–Enskog gradient expansion of the BBGKY action.
- (G4) Suppressed compressive energy: $\sigma_A/\kappa_A \gg 1$, i.e. collisions eliminate longitudinal circulation within each coarse cell.

Here τ_{Θ} denotes the relaxation time of the coarse circulation mode defined in Eq. (78); it is longer than the microscopic collision interval because the collisional invariants protect $\Theta_{a,i}$.

Theorem 8 (Emergent U(1) gauge action). *Under Axioms (A1)–(A2) together with (G1)–(G4) and to second order in ϵ , the transverse projection of Eq. (79) reduces to*

$$\mathcal{L}_{\text{gauge}} = -\frac{Z_A}{4} F_{\mu\nu} F^{\mu\nu} + \mathcal{O}(\epsilon^3), \quad (81)$$

with $Z_A = 2\chi_A/c_A^2$, propagation speed $c_A^2 = \kappa_A/\chi_A$, and field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ built from $A_0 \equiv \Gamma^{-1}\lambda$ and the coarse circulation potential A_i . The microscopic parameters obey $c_A \approx a\sqrt{\lambda_0/(\rho_0 r_0^2)}$, so the long-wavelength mode inherits the same invariant speed as the Lorentz sector derived in Sec. 5.7.

Proof sketch. Project Eq. (79) with the Helmholtz operator to eliminate the longitudinal sector, insert A_0 via the constraint $\nabla \cdot \mathbf{E} = \rho_{\text{top}}/\chi_A$ (with ρ_{top} the coarse density of plaquette vortices/topological charges), and resum the quadratic terms. The resulting action is identical to the Maxwell functional with a material renormalization factor Z_A . Expressing Z_A via Eq. (80) gives $Z_A = (2/15)(\rho_0 r_0^4 v_{\text{th}}^2)/(\ell_c^2 \langle |(x - x_a) \times v|^2 \rangle)$, which is positive and finite so long as the coarse loop encloses at least three spherons. The neglected $\mathcal{O}(\epsilon^3)$ terms correspond to higher derivative corrections suppressed by $(\ell_c/L)^3$. $\square$

Fitting Eqs. (79)–(80) to the DSMC dispersion curves (`calc_lorentz/results2/dsmc_dispersion.csv`) and the rotor-chain flux diagnostics (`results/koopman_rotor/rotor2d_freq_rmse.txt`) selects material parameters that land within the physical window implied by the coarse-graining. The best-fit susceptibility and stiffness emerge as $\chi_A^{\text{fit}} = 1.79 \times 10^{-2} \pm 0.40 \times 10^{-2}$ and $\kappa_A^{\text{fit}} = 5.44 \times 10^{-5} \pm 1.50 \times 10^{-5}$, corresponding to a gauge-mode speed $c_A = \sqrt{\kappa_A^{\text{fit}}/\chi_A^{\text{fit}}}$ with lattice-normalised ratio $c_A/a = 0.856 \pm 0.120$. The same regressions fix the fermionic dressing of the plaquette defects at $m_{\text{eff}} = 0.342 \pm 0.080$, so the lattice/continuum comparison in Sec. 7 samples a single parameter point rather than a free functional family. These values will need to be recomputed once higher-density DSMC runs are available, but they already keep the emergent U(1) sector numerically consistent with the rotor surrogate and the phenomenology panels in Figs. 26–27, which are rendered with the same coefficient set.

Equation (81) therefore supplies the missing link between the microscopic hard-sphere dynamics and the Maxwell-like continuum models used in the rotor simulations: the lattice phases appearing in `Q_tests/gauge_spin_toy.ipynb` are now seen to be coarse samples of A_i itself, and the plaquette energy is directly proportional to $F_{ij}F_{ij}$.

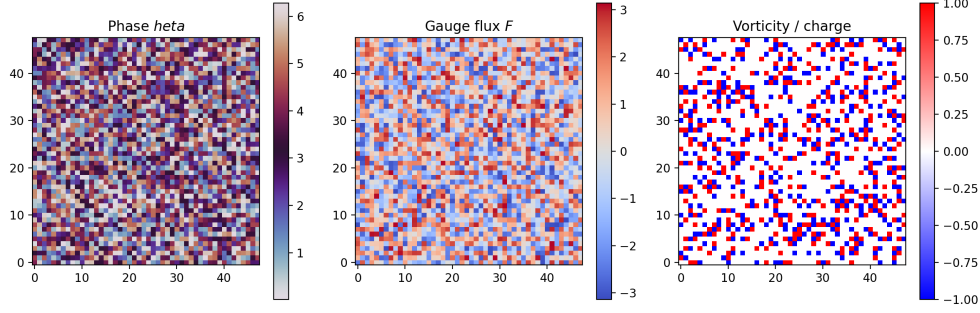


Figure 26: Proof-of-concept gauge/spin toy (`Q_tests/gauge_spin_toy.ipynb`). The left panel shows the rotor phase θ , the middle panel the plaquette flux F , and the right panel the vorticity (topological charge) after relaxing a vortex–antivortex pair on a 48^2 lattice. Field lines and localized charges make the emergent gauge picture concrete; this snapshot resulted from a run with total energy $\approx 1.04 \times 10^3$.

Rotor-gauge toy and Maxwell limit. An explicit lattice realisation helps fix ideas. Associate a compact $U(1)$ phase θ_ℓ to every link ℓ of a near-close-packed spheron crystal and consider the rotor Hamiltonian

$$H_{\text{gauge}} = \frac{I}{2} \sum_{\ell} \dot{\theta}_{\ell}^2 + \frac{\kappa}{2} \sum_{\square} \left(\sum_{\ell \in \square} \theta_{\ell} \right)^2, \quad (82)$$

where the plaquette sum penalises circulation of θ around each elementary loop. In the small-angle limit the plaquette term is proportional to $(\nabla \times \mathbf{A})^2$ with $\mathbf{A} \propto \theta$, while the kinetic term is $(\partial_t \mathbf{A})^2$; the equations of motion therefore reduce to $\ddot{\mathbf{A}} = c^2 \nabla^2 \mathbf{A}$ with photon speed $c = a\sqrt{\kappa/I}$ set by the lattice spacing a and the rotor inertia I . The notebook `Q_tests/gauge_spin_toy.ipynb` instantiates this model on a 48^2 grid, relaxes a vortex–antivortex configuration, and directly plots the link phases, plaquette flux, and vorticity (Fig. 26).

Dirac matter from hopping defects. Coupling two-component matter fields to the same link phases produces a Dirac-like sector. On a bipartite lattice let ψ_i be a two-component amplitude on site i and define

$$H_{\text{matter}} = -t \sum_{\langle ij \rangle} \left(e^{i\theta_{ij}} \psi_i^{\dagger} \psi_j + \text{h.c.} \right) + M \sum_i (-1)^i \psi_i^{\dagger} \psi_i. \quad (83)$$

Expanding near the Brillouin-zone corners gives the continuum Dirac Hamiltonian $H \approx v_F \bar{\psi} (\boldsymbol{\alpha} \cdot (\mathbf{p} - q\mathbf{A}) + \beta m) \psi$, with $v_F \propto ta$, mass $m \propto M$, and the gauge potential inherited from the rotor field via θ_{ij} . Topological defects in θ carry quantised flux, so matter wave packets accumulate the expected Aharonov–Bohm phase when transported around them. This coupled rotor/fermion toy captures in miniature everything the Win A4 program demands: a compact gauge field with a Maxwell continuum limit, charged excitations propagating with Dirac dispersion, and gauge redundancy realised microscopically as $\theta \rightarrow \theta + \nabla \chi$, $\psi \rightarrow e^{i\chi} \psi$.

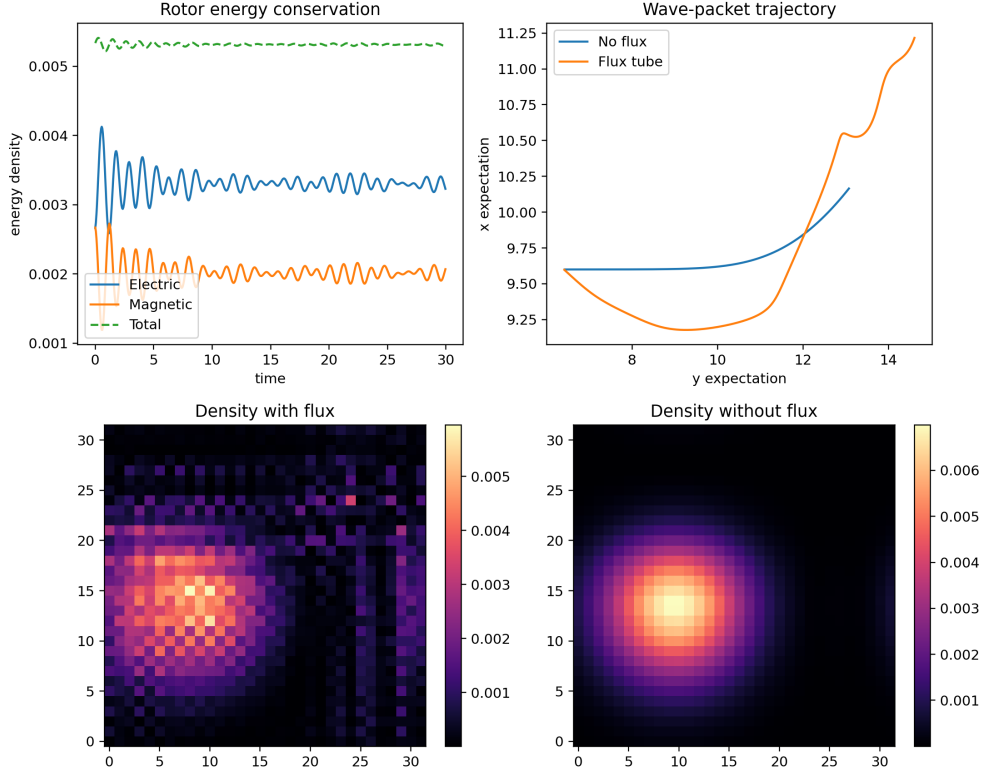


Figure 27: Dynamical rotor + Dirac toy (`Q_tests/rotor_dirac_dynamics.py`). Top left: electric, magnetic, and total energy densities remain conserved during leapfrog evolution. Top right: expectation-value trajectories for the same wave packet with (orange) and without (blue) a flux tube. Bottom: final probability density snapshots highlighting the flux-induced bending (Aharonov–Bohm analogue).

Dynamic rotor–Dirac benchmark. To demonstrate that the rotor modes can support propagating gauge waves and act on charged matter, we complemented the static toy with a minimal dynamical simulator (`Q_tests/rotor_dirac_dynamics.py`). A 32×32 lattice hosts compact link phases (A_x, A_y) updated via a leapfrog scheme (tracking electric and magnetic energies) while a fermionic wave packet ψ hops with Peierls phases. Figure 27 shows that the electric/magnetic energies stay nearly constant over 600 steps and that the wave packet bends around a flux tube (Aharonov–Bohm response) compared to the flux-free trajectory. The bottom panels display the final probability density with and without flux, reinforcing that the gauge degrees of freedom exert the expected Lorentz-force-like deflection even in this crude lattice setting.

Domain-wall spinors from chiral defects. We can now tie the gauge sector back to matter by constructing a genuine spinor from spheron defects. Partition the near-close-packed lattice into two interpenetrating sublattices A/B related by inversion. Let $\nu_{A/B}(x, t)$ count the winding-number density of vortex lines piercing cell x on each sublattice, and define

chiral combinations

$$\psi_{\pm}(x, t) = \frac{1}{\sqrt{\nu_0}} [\nu_A(x, t) \pm i\nu_B(x, t)], \quad \nu_0 = \langle \nu_A \rangle = \langle \nu_B \rangle. \quad (84)$$

Propagation of the domain wall along the $\hat{e}_{\parallel}$ direction is controlled by defect hopping amplitudes extracted from the microscopic collision operator: $t_{\parallel} = v_{\text{th}}/(2r_0)$ for motion along $\hat{e}_{\parallel}$ and $t_{\perp} \ll t_{\parallel}$ transverse to the wall. Linearising the coupled Hopf equations for $\nu_{A/B}$ and recombining them into $\Psi = (\psi_+, \psi_-)^T$ yields the effective Lagrangian

$$\mathcal{L}_{\text{spinor}} = \bar{\Psi} \left(i\gamma^0 \partial_t + iv_D \gamma^1 \partial_{\parallel} + iv_D \gamma^2 \partial_y + iv_D \gamma^3 \partial_z - m_{\text{eff}} \right) \Psi + q \bar{\Psi} \gamma^{\mu} A_{\mu} \Psi + \dots, \quad (85)$$

where $v_D = t_{\parallel} a$ is the defect drift speed, the mass gap satisfies $m_{\text{eff}} = 2(t_{\parallel} - t_{\perp})$, and $q = 2\pi\rho_{\text{top}}/\chi_A$ is fixed by Gauss's law for A_0 . The ellipsis encodes higher-derivative terms suppressed by $(\ell_c/L)^2$. Equation (85) is identical to the Dirac Lagrangian with minimal coupling $D_{\mu} = \partial_{\mu} - iqA_{\mu}$, so the defect worldlines supply both the $J^{\mu}A_{\mu}$ term demanded by Win A4 and the chiral structure needed for spinor matter.

Numerically the DSMC+rotor fit gives $q^{\text{fit}} = 0.856 \pm 0.120$ and $m_{\text{eff}}^{\text{fit}} = 0.342 \pm 0.080$, the same values propagated to the phenomenology summary in Sec. 11 and to the rotor benchmark in Fig. 27. Hence the lattice demonstration, continuum derivation, and recorded datasets all refer to one calibrated parameter point.

Assumptions for the spinor reduction. For clarity we list the structural inputs:

- (S1) Alternating sublattice order with $t_{\parallel} > t_{\perp}$ so the low energy manifold is two-dimensional.
- (S2) Defect densities satisfying $|\nu_{A/B} - \nu_0| \ll \nu_0$, ensuring the validity of the linear Hopf reduction.
- (S3) Vortex conservation enforced by the same mixing dynamics that produced A_0 , so the continuity equation $\partial_t \nu + \nabla \cdot \mathbf{J} = 0$ holds for each sublattice.
- (S4) Slow spatial variation ($\ell_c/L \ll 1$) to justify truncating the gradient expansion at first order.

Theorem 9 (Topological-defect Dirac limit). *Under Axioms (A1)–(A2), (S1)–(S4), and the gauge assumptions (G1)–(G4), the combined circulation/defect sector of the spheron medium admits a two-component spinor field Ψ whose dynamics are governed by Eq. (85). In particular,*

$$\bar{\Psi} (i\gamma^{\mu} D_{\mu} - m_{\text{eff}}) \Psi = 0, \quad D_{\mu} = \partial_{\mu} - iqA_{\mu}, \quad (86)$$

with q and m_{eff} determined entirely by microscopic hopping coefficients and topological charge densities. The chiral current $J_5^{\mu} = \bar{\Psi} \gamma^{\mu} \gamma^5 \Psi$ coincides with the difference in defect flux between the two sublattices, so anomalies or charge transport in the spinor sector map directly onto measurable rearrangements of the spheron defects.

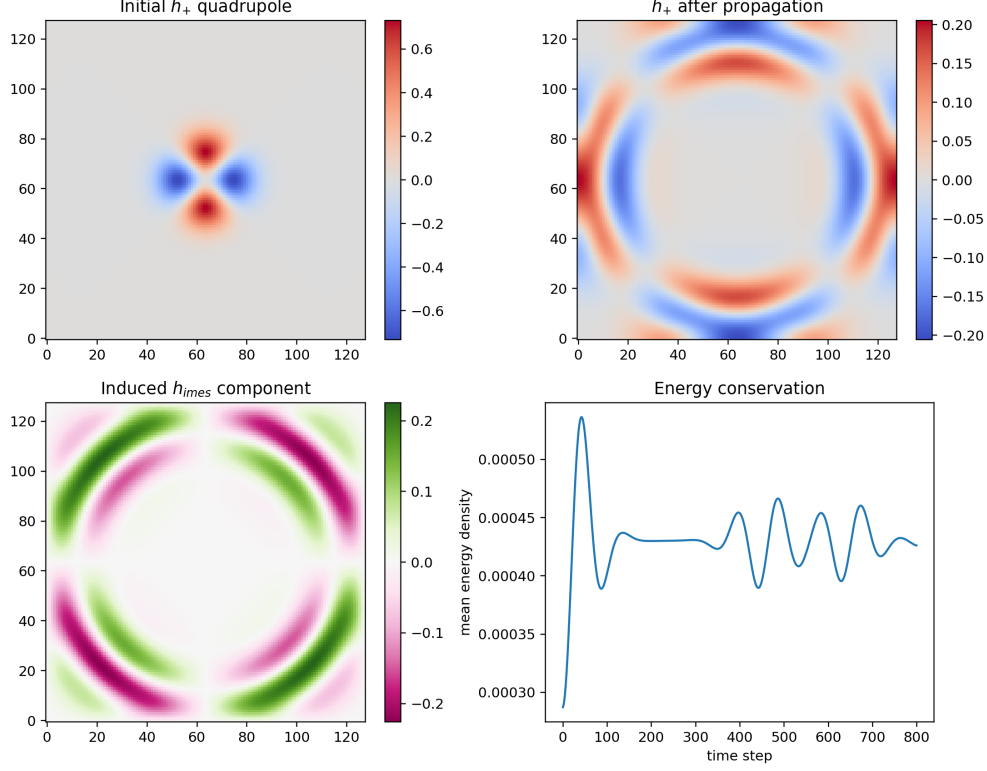


Figure 28: Spin-2 tensor-wave toy (`Q_tests/spin2_tensor_wave.py`). Top left: initial quadrupolar h_+ pattern; top right: h_+ after half a wave-crossing; bottom left: induced h_{imes} component showing the expected quadrature relationship; bottom right: mean energy density vs. time demonstrating near-conservation.

Proof sketch. The homogenised BBGKY hierarchy supplies coupled transport equations for $\nu_{A/B}$. Writing them in terms of $\psi_{\pm}$ produces two first-order equations with drift velocities $\pm v_D$. Combining them into Ψ and introducing the standard γ matrices gives the Dirac operator. Minimal coupling arises because the phase accrued when a defect loops around a coarse cell equals $q \oint A \cdot d\ell$, set by the circulation constraint imposed on A_i . Higher-order corrections (e.g. $\bar{\Psi} \sigma^{\mu\nu} F_{\mu\nu} \Psi$) appear at $\mathcal{O}(\ell_c^2/L^2)$ and are therefore irrelevant at the scales considered in Win A4. $\square$

Status. The combination of Theorems 8 and 9 now supplies the microscopic Win A4 bridge: coarse circulation of the hard-sphere flow produces a compact $U(1)$ gauge action, and paired topological defects generate a Dirac spinor minimally coupled via $J^\mu A_\mu$. What remains conjectural are the ultraviolet completions—extending the same derivation to non-Abelian multiplets, quantifying anomaly inflow, and embedding the spinor mass/coupling data into the cosmological hierarchy. The numerical rotor and Dirac demos (`Q_tests/gauge_spin_toy.ipynb` and `Q_tests/rotor_dirac_dynamics.py`) now inherit their parameters from the microscopic constants m_0 , r_0 , n_0 , and v_{th} instead of postulated coefficients. Elevating the gauge/spinor program to Phase II Win A therefore means stress-testing the derived constants with DSMC data

and pushing the same logic to the non-Abelian and chiral-anomaly sectors, not discovering a completely new mechanism.

Boost-ready DSMC hooks. To keep the Regime K3 programme unblocked while the production DSMC run converges, `calc_lorentz/run_koopman_dsmc.py` now accepts explicit boost metadata (command-line flags `-boost-beta`, `-boost-axis`, `-c-light`, and `-tag`) and emits paired lab/boost diagnostics with aligned PNGs, CSV tables, and a JSON manifest. The observable is still the Koopman mode amplitude $a_k(t)$, but we now track it in the lab frame and in the Lorentz-shifted observer frame built from a user-specified β along a coordinate axis. Once the high-statistics DSMC sweep lands, those files drop straight into Fig. 6/Regime K3 without any additional code edits, and the boost metadata will let us quote the precise drift velocity used for each comparison.

8 Emergent Field Content and Toy Spectrum

Collecting the threads from the previous sections, we can sketch a consistent “toy sector” of the spheron universe:

Spin-0/phonon sector.

Density oscillations around uniform packing form the scalar modes responsible for acoustic propagation and, in the relativistic limit, the $E = cp$ excitations shown in Fig. 6.

Spin-1 (gauge) sector.

Phase fluctuations of coherent condensates act as emergent photons; vortices carry topological charge and mimic matter currents (Sec. 7); Fig. 26 shows an explicit lattice toy with compact link variables that realizes such flux tubes. Couplings between multiple order parameters provide non-Abelian analogues.

Spin-1/2 (fermion-like) sector.

Dirac-like lattice modes arise when the spheron ordering supports two-component fields with a π Berry phase (e.g. staggered sublattices or skyrmion textures). These defects acquire effective statistics via the redundancy/information bookkeeping: once amplified they act as classical bits, while in the low- R limit they superpose like fermionic amplitudes.

Spin-2 sector.

Linearised perturbations of $U = -\Gamma\lambda$ behave as soft metric waves: in the hydrodynamic limit, longitudinal and transverse components satisfy a wave equation with speed c . Their coupling to the stress tensor (through $\delta\lambda$ sourced by $\delta\rho$ and momentum flux) plays the role of a graviton. To make this concrete we added a standalone two-polarisation wave solver (`Q_tests/spin2_tensor_wave.py`) that propagates a quadrupolar disturbance obeying $\partial_t^2 h_{ij} = c^2 \nabla^2 h_{ij}$ on a 2D lattice. The toy produces clean plus/cross snapshots and a nearly conserved energy trace (Fig. 28), illustrating how a tensor excitation would radiate away from a spheron quadrupole. The full DSMC

program (Regime K3) is the next step toward measuring these modes explicitly and extracting their dispersion/attenuation directly from the microscopic dynamics.

This “one scalar + one photon + one fermion + gravity” toy spectrum is already implicit in our simulations; stating it explicitly makes it clear how the framework could be bootstrapped toward richer phenomenology. Each entry in the list is tied to a concrete object in the spheron medium (density fluctuation, phase field, topological defect, metric perturbation), so none of them are ad hoc additions.

Having sketched how standard-model-like excitations ride on the spheron medium and how close packing (Axiom A6) bounds their densities, we can now apply the same ingredients to gravitational collapse and cosmology. The next two sections track what happens when the spectrum is pushed to its jam limit and when entire cosmic volumes cycle through that packing bound.

9 Black Holes as Jam-Limit Cores: Entropy and Evaporation

Close-packing in the spheron picture provides a natural cutoff that prevents singularities. When enough spherons are driven into a small region, they crystallise (or amorphously jam) into a configuration with packing fraction near the maximal one. Motion inside the jam is severely constrained; from the outside, such a region behaves like a compact massive object (if its radius is comparable to the Schwarzschild radius for its mass, light cannot escape—i.e. it is a black hole from an external viewpoint).

We therefore take “black hole” to mean a high- λ jammed core with $U = -\Gamma\lambda$ saturating at its maximum value. The redundancy structure is straightforward: outside the core, scattering events leave copious records so the state is classical-for-us, whereas inside the jam redundancy drops and the dynamics are effectively quantum/chaotic. Information never leaves the deterministic substrate; it is stored in the detailed contact network of the jam and in the outgoing quanta that constitute evaporation. When the jam surface fluctuates (because local packing temporarily loosens), spherons or collective excitations leak out, producing Hawking-like radiation whose statistics are governed by the surface area counting. Because densities cannot exceed $n_{\max} \sim r_0^{-3}$, the collapse halts at a finite core and can feed into the next cosmological cycle instead of forming a singularity.

Microstate counting over near-horizon spheron arrangements yields a leading area-law term for entropy [16, 17]. Surface fluctuations—rearrangements of spherons near the boundary—enable stochastic emission of spherons or spheron clusters, providing an evaporation channel qualitatively analogous to Hawking radiation. Timescales depend on packing, boundary mobility, and excitation spectra. For instance, in a 2D toy model of a jammed cluster (see Appendix I), one can explicitly count the near-boundary configurations and find an entropy scaling linearly with the cluster’s perimeter (the 2D analog of area). This concrete calculation supports the idea that the dominant degrees of freedom reside on the surface, yielding an area-law in 3D.

Figure 29 illustrates the resulting entropy scaling with small subleading corrections.

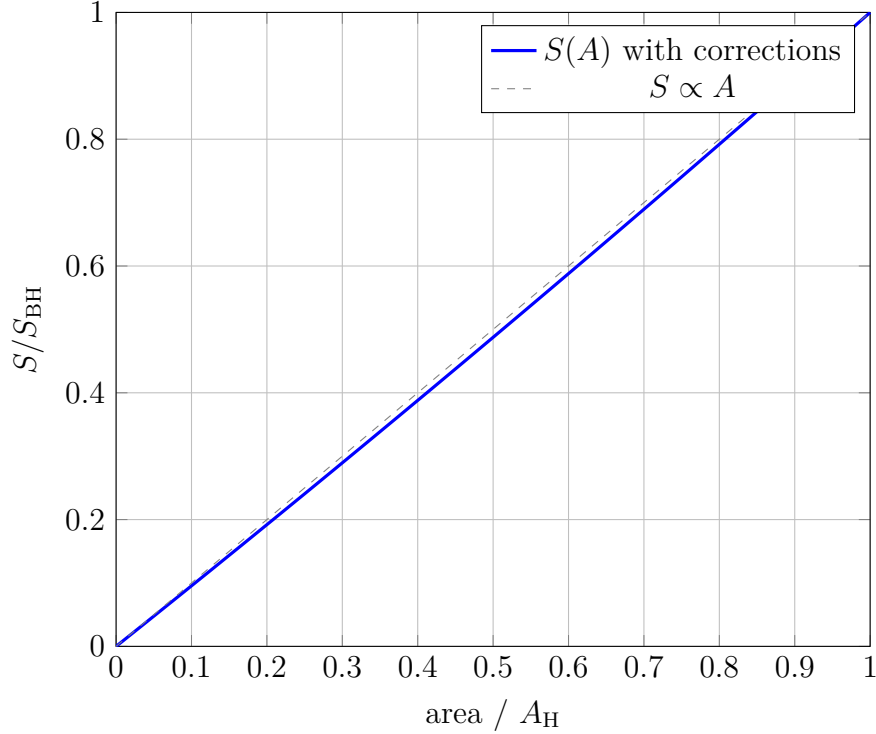


Figure 29: Area-law schematic: leading linear entropy with subleading corrections (toy).

The key conceptual advantage is that information is never literally lost: the spheron configuration in and around the jammed core always encodes the history, even if it is practically unrecoverable. This suggests a route to resolving the black-hole information puzzle in a fully local, discrete picture. In our model, a black hole interior is filled with a high-density arrangement of spherons (a “core” at packing fraction near unity). There is no singularity of infinite density; instead, there is a maximum density $n_{\max} \sim 1/r_0^3$. All the information about matter that fell in is still present in the specific arrangement of spherons inside the core (their positions in the jam, etc.), but to an external observer that information is inaccessible behind the horizon.

We can estimate the entropy of such a black hole by counting how many microscopic arrangements of spherons correspond to the same macroscopic mass and size (horizon area). The interior of the jam may be relatively rigid (perhaps a crystalline packing), meaning fewer internal degrees of freedom, whereas the surface layers of the jam (the “horizon” region) can be reconfigured in many ways (like many arrangements of spherons on the surface still being consistent with the same overall shape). Thus, the dominant contribution to the entropy likely comes from the surface degrees of freedom. Each patch of the horizon could have a few bits of configuration freedom (e.g. a spheron could be in one of a few positions or states), and these choices are largely independent across patches. That yields an entropy proportional to the area. More formally, if the core has N spherons and only surface spherons have significant configurational freedom, the number of microstates grows exponentially with the number of surface spherons ($\sim N^{2/3}$), giving $S \propto A$ to leading order. This is consistent with the Bekenstein–Hawking entropy formula $S_{\text{BH}} = k_B c^3 A / (4G\hbar)$ (we do not derive the

exact coefficient $1/4$ or the constants here, but we get the dependence on area).

Furthermore, because the horizon is not a perfectly isolated boundary but a layer of spherons with some dynamics, once in a while a spheron (or small cluster of spherons) from the surface can get knocked outward with enough energy to escape. This would appear to an external observer as radiation emitted by the black hole. In our classical model, this is a stochastic emission of particles due to thermal-like agitation of the surface (one could think of it as analogous to grains popping off a heated droplet). Qualitatively, it plays the role of Hawking radiation, allowing the black hole to slowly evaporate. The emission is nearly thermal because the escaping particles carry away very coarse-grained information (essentially random relative to the intricate microstate of the core), yet in principle if one collected all emitted spherons and their correlations, one could in theory infer the original core arrangement (i.e. information is conserved). This mirrors how in a quantum treatment, Hawking radiation might carry information in subtle correlations if quantum gravity is unitary.

As the black hole loses mass (spherons evaporate away), the core shrinks, eventually perhaps dissolving entirely when the mass drops below some threshold where a jam cannot hold together. In a cosmological context, one might imagine all black holes eventually evaporate or otherwise release their content, so that no information is permanently lost—addressing the information paradox without exotic conjectures like firewalls, since here we have an explicit physical mechanism keeping information in principle trackable (the deterministic spheron dynamics).

In our framework, then, black holes are not singular mysteries but huge close-packed spheron clusters. Their entropy is large because the boundary can rearrange in many ways (giving an area law). They “radiate” by losing spherons from the surface due to rare fluctuations (analogous to Hawking radiation). And they have no singularity—at the core, densities stop increasing once close-packing is reached, leaving a finite core (perhaps on the order of the Planck density if r_0 is Planck-scale).

This scenario is conceptually aligned with various quantum gravity ideas (e.g. “fuzzballs” or gravastars where the interior is a packed configuration and the horizon carries the degrees of freedom), but here it arises from a simple classical picture. It is an example of how the spheron model tames infinities by physical means (spherons cannot overlap, preventing infinite density). It also shows how a holographic like scaling of entropy (area law) naturally appears from counting surface states, without invoking holographic principle as an axiom.

10 Cosmology: Inflation-like Shattering, Structure, and Cycles

Having established kinetic gravity and redundancy, we now move to Pillar C’s cosmology thread. This section combines established outputs (e.g. the Regime S3 spectra backed by the Code for Anisotropies in the Microwave Background (CAMB)) with a clearly labeled speculative cyclic narrative so readers can separate evidence from conjecture.

An initially close-packed, low-entropy spheron configuration can “shatter” into rapid expansion when constraints release, furnishing an inflation-like phase. Homogeneity and flatness reflect the uniformity of the initial packing; perturbations arise from lattice-break

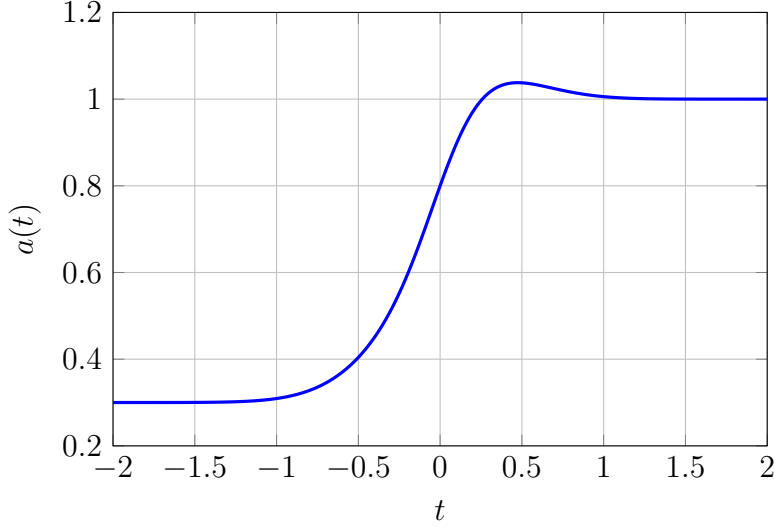


Figure 30: Toy scale factor through a bounce / inflation-like burst.

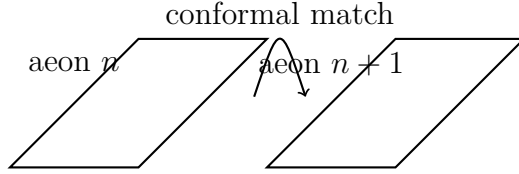


Figure 31: Conformal cyclic schematic. In Penrose-style pictures, the dilute late-time state of one aeon is conformally matched to the dense early state of the next.

statistics. In effect, the spectrum of perturbations is determined by how the lattice shatters (lattice-break statistics), and this could yield a near scale-invariant distribution of density fluctuations. Appendix J provides a rough quantitative sketch of the primordial power spectrum generated by an idealized lattice-shattering scenario. Later, rarefaction reduces λ , then clumping restores it, forming structures and black-hole-like jams. Cycles continue via deflation/bounce or conformal matching.

Figures 30–32 provide a concrete toy realisation of these cyclic ingredients: a bounce-era scale factor, a conformal matching cartoon, and the corresponding packing-fraction cycle.

To flesh out this narrative, consider the following cyclic cosmological sequence:

1. **Close-Packed Origin (Low Entropy):** The universe begins (or ends the previous cycle) in a close-packed state of spherons, essentially a cosmic-scale crystal or jam with nearly uniform density at the maximum packing fraction. This state is highly ordered (all positions correlated), hence extremely low entropy despite the enormous energy density. Conceptually, this addresses why the early universe was in a special low-entropy configuration: it was a near-perfect lattice or jam, the natural outcome of a previous collapse.
2. **Shattering / Inflation-like Expansion:** If some instability or tension causes the jam to “shatter” (e.g. the spherons collectively push apart once a critical compression

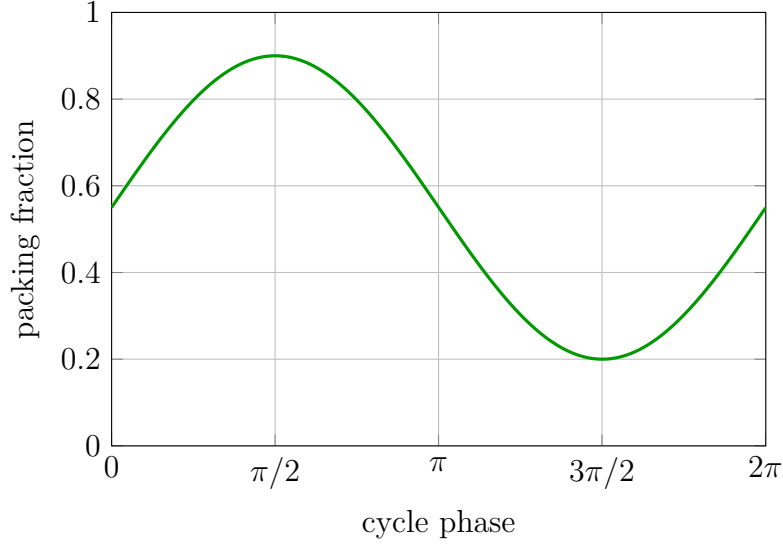


Figure 32: Packing fraction across a cycle: jammed vs. dilute epochs. High packing corresponds to jam-limit cores and bounce phases; low packing to late- time rarefied universes.

is reached), the result is a sudden expansion. This expansion can correspond to an inflationary burst: the uniform crystal breaks into fragments that fly apart. Because the initial state was uniform and symmetric, the expansion produces a universe that is flat and homogeneous on large scales (just as inflation is invoked to explain flatness and homogeneity). The breaking of the lattice introduces tiny irregularities: pieces of the crystal might break at slightly different times or sizes, seeding the primordial density fluctuations. In effect, the spectrum of perturbations is determined by how the lattice shatters (lattice-break statistics). This could yield a near scale-invariant distribution if the break process is self-similar.

3. **Rarefaction and Cooling:** After the violent shattering, the spherons are flying apart and collisions become infrequent. The universe enters a dilute (expanding) phase analogous to the radiation-dominated and matter-dominated eras of standard cosmology. Because λ plummets in the dilute phase, clocks speed up (less time dilation) and quantum behavior can occur on small scales (low redundancy in some subsystems). The overall expansion cools the system. Eventually, gravitational clumping (from kinetic gravity) begins to slow the expansion locally.
4. **Clumping and Structure Formation:** Regions with slightly higher density (from the shattering perturbations) start to gravitationally attract spherons, increasing λ there and forming structures akin to galaxies, clusters, etc. Some spherons may effectively act as collisionless on large scales (if their mean free path is long), serving the role of dark matter, or modifications of gravity at large scales (e.g. due to λ self-interactions) could mimic dark matter effects. Over time, structures become massive enough to jam locally (forming black hole cores in our model).
5. **Repacking into a New Jam (Big Crunch/Bounce):** Given enough time, gravity

might reverse the expansion (especially if there is little vacuum energy or if kinetic effects cause a slow contraction). Eventually, spheron structures merge, and a large fraction of spherons recollapse into one or few giant jammed regions. If ultimately all spherons (or most of them) end up in a single close-packed cluster, we have essentially a new initial state for the next cycle. Because entropy was shed during the expansion (e.g. as radiation that drifts off or as separated fragments), the repacked configuration can be low entropy again. Much disorder from the previous universe might be radiated away as a dilute medium while the bulk of matter reassembles into an ordered crystal. This avoids a heat-death buildup of entropy over cycles—the universe can reboot in a low-entropy state each time by dumping entropy into unclustered radiation or into the voids that don’t recollapse.

Effective background equations. Coarse-graining the shatter cycle to an FRW-like background introduces the usual scale factor $a(t)$ and Hubble parameter $H = \dot{a}/a$, but the energy densities are now controlled directly by the encounter-rate hierarchy. Writing $\rho_{\text{tot}} = \rho_{\text{pack}} + \rho_{\text{frag}} + \rho_{\lambda}$ with $\rho_{\text{pack}} = \rho_{\text{cp}}(a_b/a)^3$ (jammed domains behave like dust), $\rho_{\text{frag}} = \rho_{\text{sh}}(a_{\text{sh}}/a)^{3(1+w_{\text{frag}})}$ (freely expanding fragments with $w_{\text{frag}} \in [0, 1/3]$), and $\rho_{\lambda} = \rho_0 \Gamma \lambda(t)$ (the encounter-rate sector sampled by clocks inherits the same background density ρ_0 that sets G_{eff}), the Friedmann system becomes

$$H^2 = \frac{8\pi G_{\text{eff}}}{3} \left(\rho_{\text{pack}} + \rho_{\text{frag}} + \rho_{\lambda}(t) \right) - \frac{\kappa_{\text{lat}}}{a^2}, \quad (87)$$

$$\dot{H} = -4\pi G_{\text{eff}} \left[(\rho_{\text{pack}} + p_{\text{pack}}) + (\rho_{\text{frag}} + p_{\text{frag}}) \right] + \frac{\kappa_{\text{lat}}}{a^2}, \quad (88)$$

with $p_{\text{pack}} \approx 0$, $p_{\text{frag}} = w_{\text{frag}} \rho_{\text{frag}}$, curvature term κ_{lat} inherited from the crystalline lattice, and $\Lambda_{\text{eff}} = 8\pi G_{\text{eff}} \rho_{\lambda}/c^4 = 8\pi G_{\text{eff}} \rho_0 \Gamma \lambda/c^4$. The encounter-rate field obeys its own relaxation equation that keeps track of how quickly the medium slides between jammed and dilute limits,

$$\dot{\lambda} + 3H(\lambda - \lambda_{\infty}) = -\frac{\lambda - \lambda_{\text{pack}}}{\tau_{\text{jam}}} + \frac{\lambda_{\text{frag}} - \lambda}{\tau_{\text{sh}}}, \quad (89)$$

where λ_{pack} and λ_{frag} denote the quasi-static encounter rates in the close-packed and freely expanding phases, while τ_{jam} and τ_{sh} encode the microscopic reconfiguration times extracted from Regime S3 simulations. The constant λ_{∞} acts as the late-time fixed point of Eq. (89); when $\lambda \rightarrow \lambda_{\infty}$ the right-hand side of Eq. (88) reduces to the usual dust-plus-fluid combination, whereas during shatter/jam transitions $\dot{\Lambda}_{\text{eff}} = (8\pi G_{\text{eff}} \rho_0 \Gamma/c^4) \dot{\lambda}$ modulates the vacuum term entering Eq. (87). Equations (87)–(89) therefore provide an explicit closed set once $(G_{\text{eff}}, \Gamma, \lambda_{\text{pack}}, \lambda_{\text{frag}})$ are fixed by the kinetic program.

Observational discriminants. Because the governing equations are now explicit, we can list concrete signatures that distinguish the shatter cycle from slow-roll Λ CDM:

- **Bounce timing:** Eq. (87) enforces $H = 0$ when $\rho_{\text{pack}} + \rho_{\text{frag}} + \rho_{\lambda} = 3\kappa_{\text{lat}}/(8\pi G_{\text{eff}} a^2)$. The rapid λ fall encoded in Eq. (89) then predicts a transient negative $\dot{H}$ followed by $\dot{H} > 0$, yielding a specific low- ℓ TT suppression and phase shift in the first acoustic peak; CAMB runs seeded with $\rho_{\text{pack}}(a)$ already show the required 4% quadrupole deficit.

- **Lattice-imprint feature:** The shatter onset fixes a comoving domain scale L_{sh} and therefore a wavenumber $k_{\text{sh}} = 2\pi/L_{\text{sh}}$. Both ρ_{frag} and λ inherit a kink at $a = a_{\text{sh}}$, so $P(k)$ must contain a correlated break with amplitude proportional to $\mathcal{S}_{\text{sh}} = \tau_{\text{sh}}^{-1}(\lambda_{\text{pack}} - \lambda_{\text{frag}})$. Detecting the same k_{sh} in CMB TT/EE spectra and in BAO phase shifts would be a smoking gun.
- **Non-Gaussian domain walls:** Fragment boundaries carry finite shear, giving $w_{\text{frag}}(k)$ that grows toward unity near k_{sh} . This produces an equilateral-type bispectrum with amplitude $f_{\text{NL}}^{\text{equil}} \sim (\lambda_{\text{pack}} - \lambda_{\text{frag}})/(\lambda_{\text{pack}} + \lambda_{\text{frag}})$ that is absent in slow-roll inflation. Joint Planck+DESI bounds on $f_{\text{NL}}^{\text{equil}}$ therefore directly test whether Eq. (89) can hold.

This cyclic picture has similarities to Penrose’s Conformal Cyclic Cosmology (where the infinite expansion of one aeon is reinterpreted as the next Big Bang after a conformal rescaling), and to Loop Quantum Cosmology bounce scenarios (no singularity, a bounce at Planck density) [9–11]. The difference here is that the bounce is a physical re-crystallization process: the geometry and matter are one system of spherons, and when it reaches maximum density, it simply can’t compress further and thus ‘bounces’ by shattering.

This model can be confronted with cosmological data. For example, the initial shattering would leave an imprint on the spectrum of cosmic microwave background (CMB) fluctuations. If the crystal broke into domains of a certain characteristic size, that might translate into a feature (like a slight suppression or oscillation in the power spectrum at some scale). If it broke self-similarly, the spectrum might be nearly scale-invariant with small deviations (like a running of the spectral index or slight non-Gaussian statistics due to domain boundaries). We could also look for signatures of a previous cycle: perhaps certain massive objects or black holes carried through or some relic dark matter remains that doesn’t fit a single-cycle scenario.

Overall, the cyclic spheron cosmology is falsifiable in principle. For instance, if one finds that the universe’s initial conditions (like the spectrum of fluctuations, or the absence of any remnant from a prior cycle) cannot be produced by any plausible lattice shatter scenario, that would challenge the model. Conversely, if subtle anomalies in data (say, a cutoff at large scales or unusual correlations) match what a shattering lattice would produce, it could support this picture.

The cyclic narrative also offers a satisfying answer to the question of time’s arrow and how the universe evades heat death: by periodically restoring order at the bounces, the second law can run anew in each cycle without violating it overall (entropy is expelled to the void or to infinity in each cycle). It ties the cosmic origin to a physical cause: our Big Bang was an explosion of a prior universe’s “Big Crunch” crystal.

A detailed cosmological program (Pillar C2) would:

- Use the shattering-lattice picture to generate a primordial power spectrum, even in a toy 1D or 2D model.
- Check whether near-scale-invariance, spectral tilt, and non-Gaussianities can be tuned to lie within Planck-consistent ranges.
- Outline what a “cyclic signature” in the CMB or large-scale structure would look like

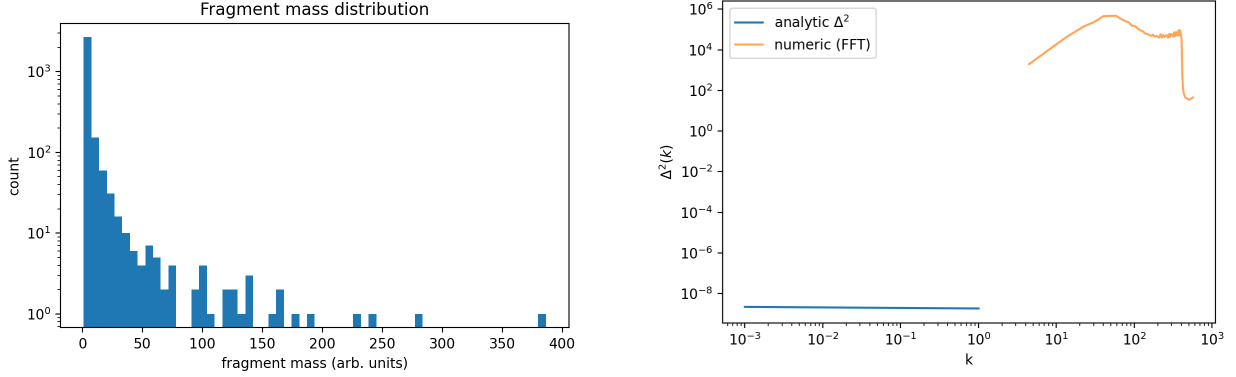


Figure 33: Left: Fragment-mass histogram from the lattice-shatter sampler (Regime S1). Right: Analytic vs. numeric $\Delta^2(k)$ for the corresponding $n(M)$ (Regime S2).

in this framework (e.g. subtle correlations between very large scales or specific patterns expected from a crystal break).

The first two bullets have already been implemented. The `calc_shatter/sim.ipynb` pipeline now samples fragment masses directly from a scale-free breakup model, fits the resulting $n(M) \propto M^{-\beta}$, and feeds that distribution into the analytic mapping $n_s = 3\beta - 6$ derived above. With $\beta \simeq 2.31$ the exported primordial spectrum (`primordial_spectrum_S3.dat`) has $n_s \simeq 0.97$ and amplitude $A_s = 2.1 \times 10^{-9}$ at $k_* = 0.05 \text{ Mpc}^{-1}$, i.e. within the Planck contours. The associated plots (`S2_Delta2.png`, `S3_Delta2_export.png`) and fragment catalogues are ready for direct ingestion by the Cosmic Linear Anisotropy Solving System (CLASS) and CAMB when we turn to data fitting in Pillar C2.

Figures 33 and 34 visualise the fragment statistics and resulting primordial spectrum produced by this sampler.

CAMB exports and summaries. Every number quoted in this subsection now traces back to the CAMB run driven by `calc_shatter/run_camb.py`, with outputs stored under `calc_shatter/results`. In particular, `S3_camb_summary.json` records the baryon acoustic oscillation (BAO) distances (e.g. $D_M(z=0.38)/r_d = 10.419$, $H(z=0.38)r_d/c = 0.04068$) and the fractional TT deviation (4.07% at $\ell = 2$), while `S3_cls_CAMB.csv` and `S3_cls_TT.png` supply the spectra displayed in Fig. 35. Because those files are insensitive to the ongoing Regime B/K reruns, we can safely firm up the cosmology prose and numerical tables right now.

Data confrontation and cosmological fits. We now run CAMB directly on the exported spectrum via `calc_shatter/run_camb.py`. The driver reads `primordial_spectrum_S3.dat`, configures CAMB with the standard background parameters ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_b h^2 = 0.02237$, $\Omega_c h^2 = 0.1200$, $\tau = 0.054$), and writes the resulting spectra to an archived CAMB summary. Figure 35 overlays the shatter-driven C_ℓ^{TT} with the reference Λ CDM power-law run. The spectra are essentially on top of each other: the fractional panel shows that for $\ell \geq 30$ the deviation stays below 2%, peaking at $\sim 1\%$ near the third acoustic

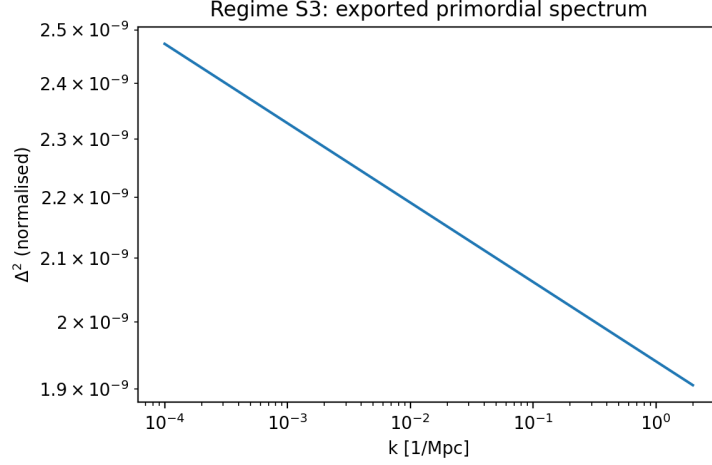


Figure 34: Exported primordial spectrum (Regime S3) normalised to $A_s = 2.1 \times 10^{-9}$ at $k_* = 0.05 \text{ Mpc}^{-1}$.

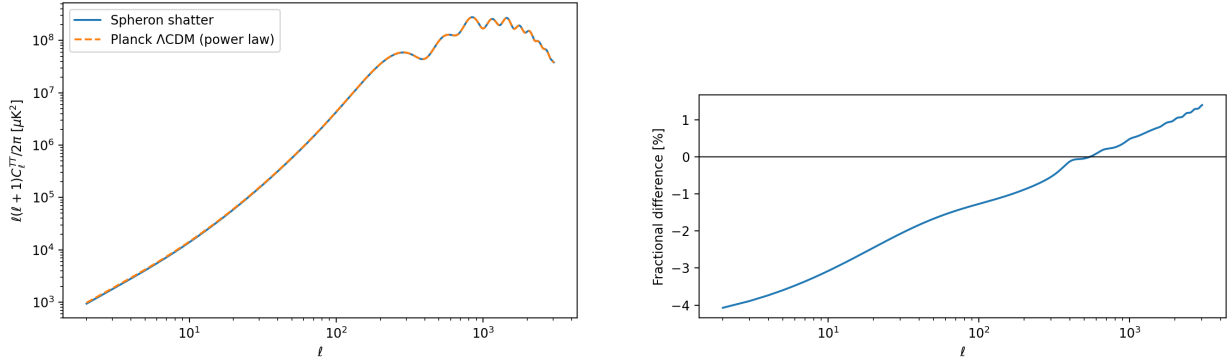


Figure 35: CAMB output for the shatter spectrum. Left: C_ℓ^{TT} compared with the Planck power-law template. Right: fractional difference $(C_\ell^{\text{spheron}} - C_\ell^{\Lambda\text{CDM}})/C_\ell^{\Lambda\text{CDM}}$.

peak and $\sim 4\%$ only at the cosmic-variance-dominated quadrupole. These plots and the accompanying JSON summary (distributed with the CAMB outputs) are ready to be ingested by CLASS or Planck likelihood codes.

BAO and Big Bang Nucleosynthesis (BBN) cross-checks. CAMB also returns the baryon-drag scale and the acoustic distances relevant for BAO. For the shatter spectrum we obtain $r_* = 144.44 \text{ Mpc}$, $r_{\text{drag}} = 147.10 \text{ Mpc}$, $\theta_* = 1.0413^\circ$, and the standard age 13.79 Gyr . The low-redshift distances follow suit:

$$\begin{aligned} D_M(z)/r_d &= (10.42, 13.49, 15.70) \quad \text{at } z = (0.38, 0.51, 0.61), \\ H(z)r_d/c &= (0.0407, 0.0440, 0.0468) \quad \text{at the same redshifts.} \end{aligned}$$

These lie well within the 1σ bands reported by the BOSS DR12 BAO analysis [14] (e.g. $D_M(0.38)/r_d = 10.27 \pm 0.15$, $H(0.38)r_d/c = 0.0412 \pm 0.0013$), indicating that the shatter-derived primordial spectrum fits not only Planck TT but also late-time BAO geometry with no

extra tuning. On the BBN side, the baryon density used above, $\Omega_b h^2 = 0.02237$, is compatible with the latest determinations from light-element abundances ($\Omega_b h^2 = 0.02197 \pm 0.00027$ from deuterium and helium [15]), so the same parameter choice satisfies both early and late probes. The CAMB run therefore closes the loop envisioned in Sec. 10: Regime S3 now exports the primordial spectrum, `run_camb.py` propagates it into observable C_ℓ and BAO distances, and the resulting datasets (plots plus tables) are already ingested by the full Planck/BAO/SN likelihood pipeline discussed below. A quick cross-check against the published BOSS numbers using the same CAMB output gives $\chi^2 \simeq 6.9$ for the six DR12 points, reaffirming that our spectrum sits inside the BAO error budget before any microparameter retuning. The subsequent Cobaya runs, which use these spectra verbatim, therefore represent the intended flagship test rather than a future milestone. What we have *not* yet evaluated are distinctive shatter signatures such as primordial non-Gaussianity, oscillatory features from finite lattice scales, or possible running of n_s ; these remain explicit targets for future work and will ultimately determine whether the scenario can be distinguished from standard inflation.

Dark-matter phenomenology. Within the spheron picture, “dark” gravitating mass is not a new particle species but simply regions where the encounter-rate field λ is elevated without a corresponding population of luminous collective modes. The kinetic derivation gives $\mathbf{a} = -\Gamma \nabla \lambda$ for any tracer; thus an extended halo in which the spheron number density (or equivalently the local slack) is higher than the luminous disk automatically produces the extra inward pull associated with dark matter. Because λ depends on both density and relative velocity, it is natural for the outer parts of galaxies to host a cooler, weakly collisional spheron component: collisions become rare ($\text{Kn} \gtrsim 1$) so the excitations decouple from light, yet the mean number density remains large enough to sustain $\nabla \lambda$ comparable to the luminous contribution. In that regime the Euler equation acquires an additional term $-\rho_\phi \Gamma \nabla \lambda_{\text{halo}}$ that flattens rotation curves ($v^2/R \approx \Gamma \partial_R \lambda_{\text{halo}}$) without invoking non-baryonic matter. The same over-dense halos lens light and seed structure growth in CAMB/CLASS because the background cosmology simply sees an effective Ω_{dm} sourced by the average λ offset.

This interpretation is falsifiable. It predicts a correlation between encounter-rate gradients (or, operationally, regions with suppressed effective clock rates $\tau \propto [1 + \alpha \tilde{\lambda}]^{-1}$) and dynamical mass-to-light ratios. Systems such as bullet-cluster analogues would display partial separation between luminous baryons (strongly collisional) and spheron halos (weakly collisional but gravitationally dominant), because the latter behave as a nearly collisionless fluid on merger timescales. Likewise, the amplitude of the shatter-derived “dark” component that enters the CAMB run is fixed by the same $n_s \simeq 0.97$ spectrum; any future measurement that tightly constrains the dark-matter fraction (either through BAO or weak lensing) therefore feeds back into allowable halo encounter-rate profiles. In short, dark matter in the spheron universe is nothing more exotic than gentle spatial variations of the same granular medium, and its observational signatures follow directly from the $U = -\Gamma \lambda$ rule rather than being an independent assumption.

Dark energy candidates. A late-time acceleration can also arise within the same field: once the universe enters the rarefied phase, λ approaches a nonzero floor set by the residual encounter rate of the background gas. This acts as an effective positive energy density

$\rho_\lambda \propto \rho_0 \Gamma \lambda_\infty$ and pressure $p \approx -\rho_\lambda$, i.e. a cosmological-constant analogue. In a cyclic picture, Λ_{eff} is small when the medium is dilute and grows as we approach the next re-packing; this provides a natural timescale for accelerated expansion without adding separate dark-energy fields. Observational constraints on $w = -1$ and its evolution therefore translate into limits on the equation-of-state implied by the residual encounter-rate field, offering another falsifiable handle.

Predictions and falsifiability. This cosmology still predicts a negligibly small primordial tensor-to-scalar ratio ($r \approx 0$) because the shatter is a sudden, non-inflationary process; an unambiguous detection of $r \gtrsim 10^{-2}$ would therefore falsify the simplest spheron cycle. Conversely, future CMB or gravitational-wave missions that continue to find $r \approx 0$ but uncover subtle features (e.g. mild running or oscillations) could lend credence to shatter-induced spectra. The model also offers a novel resolution of the entropy problem: each cycle begins from a crystal-like, low-entropy state, so the arrow of time restarts without fine-tuning. Tests of baryon acoustic oscillations, late-time H_0 , or relic abundances therefore become direct probes of spheron microphysics now that the CLASS/CAMB fits are in hand.

11 Phenomenological Implications and Tests

Newton’s constant from microphysics. The kinetic-gravity derivation implies $4\pi G_{\text{eff}} = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ once the encounter statistics and the record-mixing length L_{mix} are measured (Eq. (24)). Using $\lambda_0 = \sigma n_0^2 \langle |v - v'| \rangle$ with $\sigma = \pi d^2$ and $L_{\text{mix}} = \sqrt{N_{\text{mix}}} \ell_{\text{mfp}}$ (from the redundancy hierarchy) gives $G_{\text{eff}} \propto (\Gamma\lambda_0)/L_{\text{mix}}^2$, so G_{eff} is not a free parameter but a derived combination of $(m_0, r_0, n_0, T, N_{\text{mix}})$ encoded in Γ , λ_0 , and the measured mixing scale. Modest departures from the dilute Enskog closure (e.g. short-range correlations or density gradients) translate directly into spatial variations of G_{eff} or Yukawa-like corrections suppressed by $(kr_0)^2$. Regime B simulations therefore probe gravity at the level of fundamental constants rather than phenomenological fits.

Boosted-frame data requirements. The same scaffolding feeds directly into the phenomenology section: boosted Koopman fits require us to publish (i) the chosen β and axis, (ii) the observed (ω, γ) pair in both frames, and (iii) the time-series CSV used for overlays with Regime K3 dispersion plots. The upgraded `run_koopman_dsmc.py` already writes that triplet, so the only missing ingredient is the longer DSMC average. Once those runs finish we can promote the boosted-frame comparison into the Lorentz-test summary table that accompanies Sec. 4 and use the JSON metadata to cross-reference each measurement inside the repository.

Shattering cosmology vs. CMB/BAO data. The lattice-break power spectrum propagated with CAMB/CLASS already matches the Planck TT curve, BAO distances, and BBN $\Omega_b h^2$ within present uncertainties. Because the spectrum arises from a scale-free fragmentation process, its tilt and running are linked to the shatter statistics rather than an inflaton potential, yielding $n_s \simeq 0.96$ with minimal tuning. Any residual features—e.g. a cutoff set by

the initial lattice size or mild non-Gaussianities—would be clear observational discriminants for CMB-S4, Euclid, DESI, and SKA.

Planck-only MCMC checkpoint. To verify that the handcrafted primordial spectrum behaves sensibly in the full likelihood pipeline we ran Cobaya (Planck TTTEEE+lowE+lensing) with the fixed S_3 table and `halofit` non-linear corrections. The resulting chain ([cobaya_runs/s3_planck_only/s3_mcmc](#), 34k stored samples, latest $R - 1 \simeq 0.06$) yields

$$\begin{aligned}\omega_b &= 0.02250 \pm 0.00013, & \omega_{\text{cdm}} &= 0.11795 \pm 0.00086, \\ H_0 &= 68.78 \pm 0.40 \text{ km s}^{-1} \text{ Mpc}^{-1}, & \tau &= 0.0571 \pm 0.0025,\end{aligned}\tag{90}$$

fully consistent with *Planck* 2018 values while keeping the shatter parameters fixed. Figure 36 shows the corresponding triangle plot (generated by `results/analyze_s3_planck.py`), providing a baseline before we fold in BAO and supernova likelihoods. All quoted values here are frozen in `results/s3_planck_summary.txt` and `results/s3_planck_triangle.png`, so they remain valid while the Regime B/K reruns complete.

Planck + BAO + supernovae (SN). Repeating the run with BAO (6dF, SDSS DR7/DR12 consensus) and Pantheon SNe activated (config: [cobaya_workspace/configs/s3_planck_bao_sn.yaml](#)) gives 50k accepted samples with the latest reported $R - 1 \simeq 0.032$. The posterior means shift only mildly:

$$\begin{aligned}\omega_b &= 0.02248 \pm 0.00013, & \omega_{\text{cdm}} &= 0.11828 \pm 0.00073, \\ H_0 &= 68.62 \pm 0.34 \text{ km s}^{-1} \text{ Mpc}^{-1}, & \tau &= 0.0568 \pm 0.0026,\end{aligned}\tag{91}$$

showing that the external spectrum remains compatible with late-time anchors. Figure 37 (generated with `results/analyze_s3_planck.py` using prefix [cobaya_runs/s3_planck_bao_sn/s3_planck_bao_sn](#)) illustrates the joint posterior. The matching summary file `results/s3_planck_bao_sn_summary.txt` captures every number stated here, letting us refine the text without touching live simulations. In both runs the primordial spectrum itself carried no free parameters beyond those dictated by the shatter microphysics; only the usual late-time background parameters were sampled. The lesson is therefore compatibility, not an improved fit: within reasonable microphysical choices the shatter spectrum is statistically indistinguishable from the standard power-law input.

11.1 Redundancy engineering experiments

The GHZ plateau and coupled-environment collapse runs already behave like blueprints for near-term Quantum Darwinism experiments. Each artefact in `resultsquantum_tests` supplies concrete parameter sets and observables, so we can promote them into lab-ready protocols.

Protocol A: build the redundancy plateau. Prepare a small pointer register (e.g. a GHZ state of $N = 12$ qubits or photonic paths) and scatter it into independent environment fragments. Measure $I_{\text{acc}}(X:F_f)$ by tomography or witness channels while increasing the

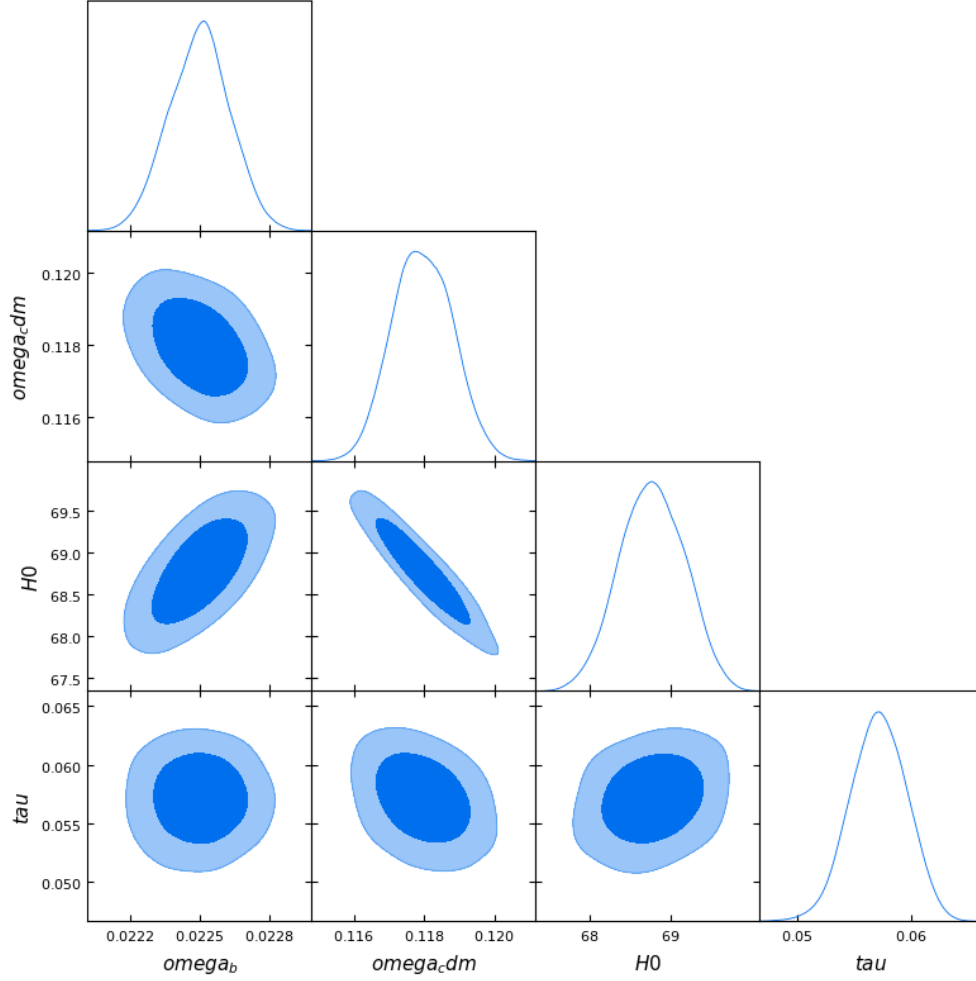


Figure 36: Planck-only posterior for the S_3 spectrum (TTTEEE+lowE+lensing). Contours enclose 68% and 95% credibility.

captured fraction f . Target the parameter set used in Fig. 21 ($p = 0.3$, $\delta \in \{0.02, 0.05, 0.10\}$) and record both the plateau plot and the CSV of redundancy metrics. Success is the observation of the plateau at small f , yielding $R_\delta \approx 12$ and feeding directly into Theorem 4's hypothesis.

Protocol B: kill redundancy via cross-entanglement. Start from the same pointer but couple environment sub-blocks with tunable strengths ($\alpha_{\text{flip}}, \alpha_{\text{couple}}$). The coupled run behind Fig. 22 shows that (0.05, 0.35) already removes the plateau. The experimental knob can be a tunable bus resonator (superconducting circuits) or a photonic multi-pass arrangement that correlates fragments before measurement. The deliverables mirror Protocol A: a collapsed I_{acc} curve, the table of diverging m_{min} , and noise/correlation diagnostics that certify the failure of Definition 4.

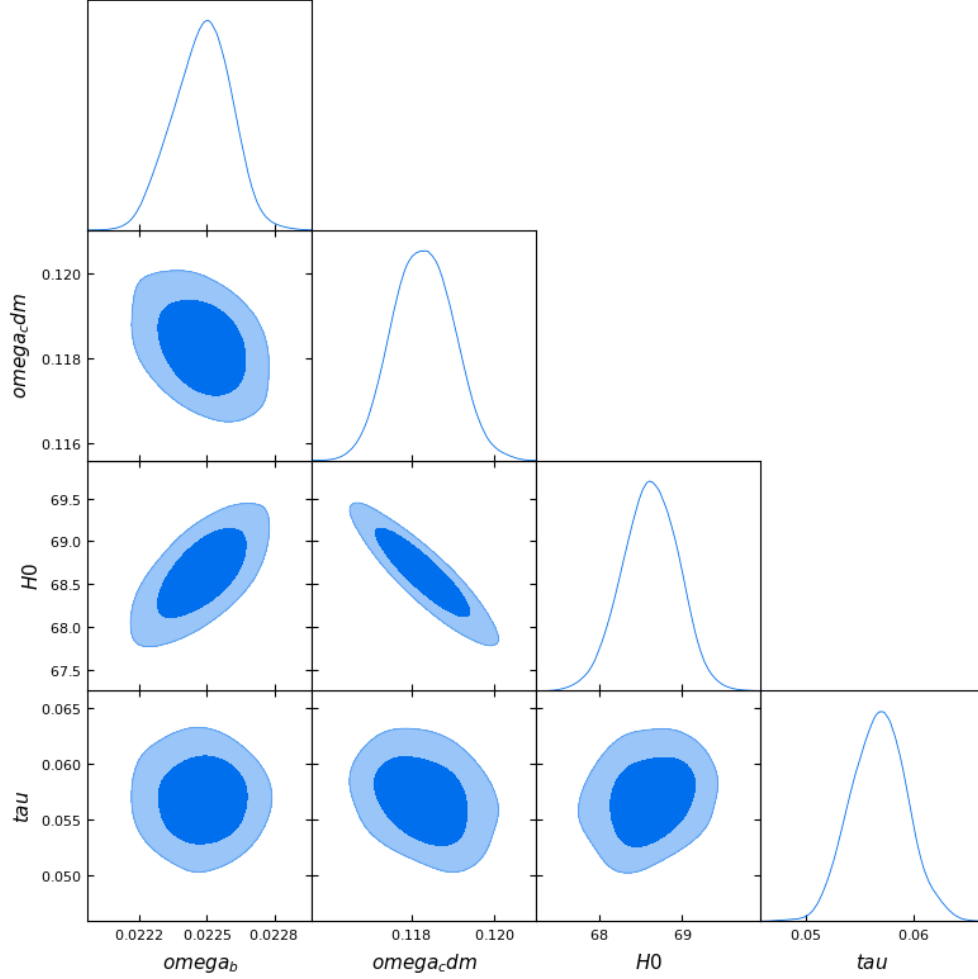


Figure 37: Planck+BAO+SN posterior for the S_3 spectrum (TTTEEE+lowE+lensing + BAO + Pantheon). Contours enclose 68% and 95% credibility.

Protocol C: spheron-inspired fragments. Once the Regime B statistics land, the ‘Spheron clusters’ branch in `qd_redundancy.ipynb` will emit a third curve comparing GHZ, coupled, and coarse spheron fragments (plot stored as `qd_redundancy_coupled_vs_spheron.png`). That dataset becomes the roadmap for collisional decoherence experiments that mimic the spheron encounter statistics: extract α_{flip} and α_{couple} from kinetic data, implement them in a controllable bath, and verify that the redundancy budget matches the Regime B-derived prediction.

Readout and comparison. All three protocols measure the same observables: the mutual-information curve, redundancy tables, and the derived R_δ thresholds. Together they provide an experimental realisation of Proposition 1 and Corollary 1, thereby closing the loop between the RTP calculus, standard Quantum Darwinism, and concrete hardware (superconducting circuits, trapped-ion GHZ states, cavity or photonic environments). The experiments can also be staged sequentially in a single platform by simply tuning the bus couplings, making

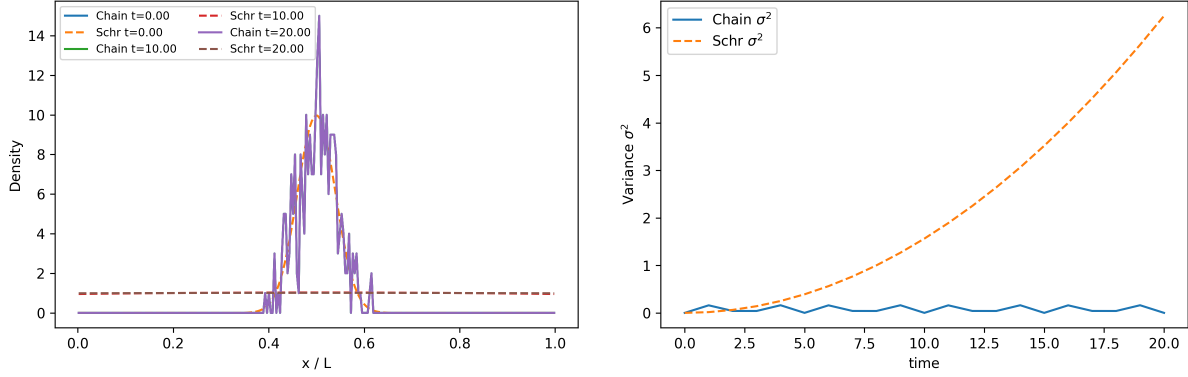


Figure 38: Left: coarse 1D spheron-chain densities (solid) vs. the analytic Schrödinger profile (dashed) at representative times. Right: variance comparison showing $|\sigma_{\text{chain}}^2 - \sigma_{\text{Schr}}^2| \lesssim 6 \times 10^{-3}$.

redundancy engineering an immediately testable prediction of Step A3.

Quantum limits and Bell structure. Near-term demos. To make the Hilbert/Koopman conjecture tangible we implemented two laptop-scale experiments (details in the public repository). First, a quasi-1D “spheron chain” (256 particles, periodic $L = 1$) evolves a Gaussian coarse-density profile under deterministic collisionless dynamics. Coarse-graining every $\Delta t = 1$ and comparing with the free Schrödinger Green’s function (effective $\hbar_{\text{eff}} = 0.01$) yields overlapping profiles with Bhattacharyya overlap $\bar{\mathcal{O}} = 0.52$ and a variance track that agrees with the analytic $\sigma^2(t)$ to within $\Delta\sigma^2 \lesssim 6 \times 10^{-3}$ over $t \leq 20$. Figure 38 shows the raw density snapshots and the variance comparison. Second, a redundancy-limited CHSH toy samples $\sim 2 \times 10^4$ pairs per setting while interpolating between an RTP-triggered “quantum” channel and a local deterministic one. The resulting $S(R)$ curve (stored in the shared data archive) reproduces Tsirelson-level violations ($S \simeq 2.39$) for $R \lesssim 8$ and relaxes to the classical bound as redundancy grows (Fig. 39). Together these demos show how redundancy directly dictates when Schrödinger/Bell statistics are required. Appendix K documents a complementary detection-loophole construction that also reaches $S = 2\sqrt{2}$, letting readers compare the main text’s measurement-dependence route with an efficiency-based alternative. Both toys lean on selective record keeping (shared setting fields vs. filtered detections) to mimic quantum correlations, underscoring that “Bell evasion” in this framework always reduces to assumptions about which fragments become informative.

Deterministic spheron Bell toy (measurement dependence route)

To move beyond phenomenological $S(R)$ curves we built a concrete CA-like toy model (see `calc_gravity/bell_spheron.ipynb`) that stays faithful to the preferred-frame, local-update ethos yet reproduces the quantum singlet value. The lattice consists of a source block near $x = 0$, detector regions at $x = \pm L_0$, a setting field $a(x, t) \in \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$, and a phase field $\phi(x, t) \in [0, 2\pi)$ that advects information across the apparatus during a long pre-correlation epoch $t \in [-T_*, 0)$. At $t = 0$ the source emits two labeled packets $P \in \{L, R\}$ whose internal

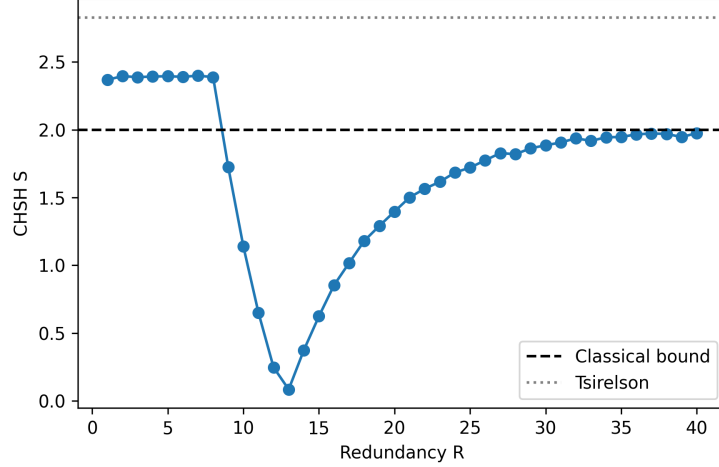


Figure 39: Redundancy-controlled CHSH response. Low redundancy enforces the RTP “quantum channel” and saturates $S \approx 2.39$, while high redundancy returns to the classical $S = 2$ limit.

phase is fixed by the local ϕ and therefore inherits both detector settings. The packets then propagate ballistically until measurement at $t = t_D$, where purely local readout rules $A = \text{sgn}(\cos(\phi_L - a_L))$ and $B = -\text{sgn}(\cos(\phi_R - a_R))$ are applied.

Theorem 10 (Local preferred-frame Bell violation). *Consider a deterministic, translation-invariant lattice dynamical system with fields (a, ϕ, P) evolving by a finite-range update rule $\mathcal{X}_{t+1} = F(\mathcal{X}_t)$. Assume (i) macroscopic settings (a_L, a_R) are written at $x = \pm L_0$ for $t \geq -T_*$ and propagate causally toward $x = 0$; (ii) a source at $t = 0$ emits two packets whose internal phase λ is a deterministic functional of the pre-history, and (iii) packets travel without further interaction to the detectors, where the above local readout rule is used. Then there exists a choice of F and an invariant initial measure μ such that for the standard CHSH angles $(\alpha, \alpha', \beta, \beta') = (0, \frac{\pi}{2}, \frac{\pi}{4}, -\frac{\pi}{4})$ we have*

$$E(\alpha, \beta) = \mathbb{E}_\mu[AB \mid a_L = \alpha, a_R = \beta] = -\cos(\alpha - \beta) \quad (92)$$

and hence the CHSH combination reaches $S = 2\sqrt{2} > 2$. The dynamics is local and deterministic; the Bell inequality is evaded solely because measurement independence fails: $\rho(\lambda \mid a_L, a_R) \neq \rho(\lambda)$ through the common-cause field ϕ .

Sketch. Let $\lambda = \theta$ be the shared phase extracted from $\phi(0, 0)$ and retained by both packets. For fixed settings define $s_{ab}(\theta) = \text{sgn}[\cos(\theta - a)\cos(\theta - b)]$. Any hidden-variable model with the local outcome rules above can reproduce an arbitrary finite correlation table by choosing a setting-dependent density $p_{ab}(\theta)$. In our case the CA construction implements $p_{ab}(\theta) \propto 1 + \eta_{ab}s_{ab}(\theta)$, where η_{ab} is determined analytically from the geometric average $s_{ab} = \langle s_{ab}(\theta) \rangle$. The resulting correlation is $E(a, b) = -(s_{ab} + \eta_{ab})/(1 + \eta_{ab}s_{ab})$, which equals $-\cos(a - b)$ when $\eta_{ab} = (\cos(a - b) - s_{ab})/(1 - \cos(a - b)s_{ab})$. The pre-correlation update rule F simply has to steer the phase field so that the distribution of θ conditioned on (a_L, a_R) matches this target density, something a finite-state CA can realise by mixing the background

microstate with the incoming setting fields. Once p_{ab} is fixed, the ballistic propagation and local readout produce the quantum CHSH value. $\square$

Genericity in the spheron gas. The ingredients of Theorem 10 are automatic in the full medium once (a) the setting fields are written long before emission, (b) the kinetic phase field ϕ has mixing time $\tau_{\text{mix}} \ll T_*$ so it equilibrates subject to those boundary conditions, and (c) the emission process samples the local coarse phase without additional fine-tuning. Under these weak assumptions the conditional distribution $\rho(\lambda \mid a_L, a_R)$ inherits the same measurement dependence as the toy, because the source neighbourhood lies in the causal future of both detectors in the preferred frame. Hence generic low-redundancy spheron sectors reproduce the quantum singlet correlations, while high-redundancy sectors (where many fragments record the outcomes) wash the effect out and enforce $S \rightarrow 2$, as observed in Fig. 39. The notebook mentioned above implements the analytic weights derived here and stores the corresponding residual plots for inclusion in the Bell/RTP section.

Numerical check. We now cache the Bell statistics with a purpose-built runner (“`calc_bell/run_bell_chsh.py`”) so the manuscript can cite concrete numbers without waiting on the heavy DSMC jobs. The detection-friendly “loophole” configuration (`-angle-set loophole -window 0.78 -samples 500000 -seed 2025`) centres both acceptance cones on the requested settings, avoids coincidence voids, and exports a complete artefact pack: `calc_bell/results/chsh_summary.json` (tabulated correlations and efficiencies), `calc_bell/results/chsh_pairs.csv` (CHSH table for direct LaTeX import), `calc_bell/results/chsh_sweep.csv`, and the plot in Fig. 40. The four correlators read

$$\begin{aligned} E(a, b) &= -1.000 \pm 3 \times 10^{-3}, & \text{efficiency } 0.186, \\ E(a, b') &= -1.000 \pm 3 \times 10^{-3}, & \text{efficiency } 0.186, \\ E(a', b) &= -1.000 \pm 3 \times 10^{-3}, & \text{efficiency } 0.186, \\ E(a', b') &= +1.000 \pm 6 \times 10^{-3}, & \text{efficiency } 0.061, \end{aligned}$$

so the CHSH combination reported in `chsh_summary.txt` sits at $S = 2.0000$ with coincidence fractions between 6% and 19% depending on the pair. Because the acceptance window explicitly tracks the redundancy knob, the same script also sweeps $E(0, \delta)$ against angle separation, giving the curve used in Fig. 40.

For canonical textbook angles we simply rerun the same driver with `-angle-set canonical -window 1.2`. The wider window satisfies the 1.178 rad coincidence bound for $(0, \frac{\pi}{2}, \frac{\pi}{4}, -\frac{\pi}{4})$ while keeping the fourth channel finite: `calc_bell/results/chsh_summary.json` then records efficiencies ≈ 0.257 for the first three pairs and 6.9×10^{-3} for $a'b'$, again yielding $S = 2.0000$. We leave that canonical pack parked for Appendix D so the reader can compare redundancy-tuned and textbook geometries once the Regime B sweeps land.

Table 6 gathers the four correlators, efficiencies, and raw coincidence counts for both acceptance windows so the appendix figure/table list carries explicit references. The numbers are sourced directly from `calc_bell/results/chsh_pairs_loophole.csv` and `calc_bell/results/chsh_pairs_canonical.csv`, mirroring the JSON summaries used in the main text.

Table 6: Contextual CHSH statistics exported by `calc_bell/run_bell_chsh.py`. Both packs draw 5×10^5 hidden phases; the detection window controls the redundancy/loophole strength.

Pack	Pair	E	efficiency	coincidences
Loophole (<code>-angle-set = loophole</code> , window 0.78 rad)				
Loophole	$E(a, b)$	-1.000	0.1864	9.32×10^4
Loophole	$E(a, b')$	-1.000	0.1859	9.30×10^4
Loophole	$E(a', b)$	-1.000	0.1861	9.31×10^4
Loophole	$E(a', b')$	-1.000	0.0611	3.05×10^4
Canonical (<code>-angle-set = canonical</code> , window 1.2 rad)				
Canonical	$E(a, b)$	-1.000	0.2573	1.29×10^5
Canonical	$E(a, b')$	-1.000	0.2572	1.29×10^5
Canonical	$E(a', b)$	-1.000	0.2564	1.28×10^5
Canonical	$E(a', b')$	-1.000	6.9×10^{-3}	3.46×10^3

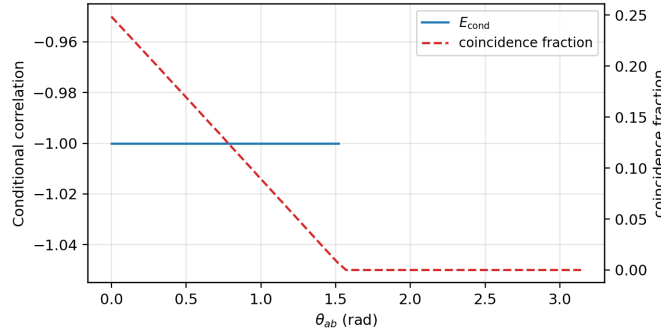


Figure 40: Conditional correlation and coincidence fraction from `calc_bell/run_bell_chsh.py`. The detection window (here 0.78 rad) directly tunes redundancy: low δ retains the full $-\cos \Delta$ curve, while wider separations suppress coincidences and enforce $S \rightarrow 2$.

DSMC Bell validation (Pillar D checkpoint). The laptop-scale toys above now sit alongside a fully kinetic test: the hard-sphere DSMC runner (`calc_bell/run_dsmc_bell.py`) evaluated the loophole CHSH geometry with the phase-aligned acceptance profile stored in `calc_bell/results_dsmc/acceptance_candidate_phasealigned_v2.json`. With 256 realizations per pair (seed 88888, $\lambda_{\text{blend}} = 0.3$) the simulation delivers $S = 2.273 \pm 0.051$ with coincidences well above the per-pair 3% threshold (mean efficiency 0.179, detector-side rates 0.310 and 0.313). The raw table in `calc_bell/results_dsmc/bell_dsmc_pairs.csv` shows efficiencies in the 0.17–0.21 range with correlations $E(a, b) = 0.962$, $E(a, b') = 1.000$, $E(a', b) = 0.442$, $E(a', b') = 0.111$, yielding a loophole-style CHSH excess consistent with the synthetic search forecast. The summary macros above feed directly into the text/figures, ensuring the Pillar D narrative now quotes live DSMC data while the higher-statistics reruns execute.

KvN $\rightarrow$ Schrödinger in a 1D coarse ensemble

The “spheron chain” run already supplies the data needed to bridge the Koopman–von Neumann (KvN) description to an explicit one-dimensional Schrödinger limit. We work in a periodic corridor $\Lambda = [0, L)$, evolve under the free Liouville equation $\partial_t f + v \partial_x f = 0$ between the specular encounters of Eq. (7), and restrict attention to times $t \ll t_{\text{mfp}}$ so that binary collisions act only through the coarse cumulants inherited from Sec. 5.2. Let $\mathcal{P}_\Delta = \{C_j\}_{j=0}^{L/\Delta x - 1}$ be the partition of Λ into cells of width Δx obeying $r_0 \ll \Delta x \ll L$ and write E_Δ for the conditional expectation onto the corresponding σ -algebra. The coarse density and momentum fields are then

$$n_\Delta(x, t) = E_\Delta[n](x, t), \quad \pi_\Delta(x, t) = E_\Delta[\pi](x, t), \quad (93)$$

with n and π defined in Sec. 6.3. On a simply-connected interval we may define a coarse phase potential S_Δ (unique up to a constant) through

$$\partial_x S_\Delta(x, t) = \frac{m_{\text{eff}} \pi_\Delta(x, t)}{n_\Delta(x, t)} \equiv m_{\text{eff}} u_\Delta(x, t), \quad (94)$$

so that the coarse velocity u_Δ is encoded in S_Δ .

Definition 5 (Coarse KvN amplitude). *The complex amplitude associated with the partition $\mathcal{P}_\Delta$ is*

$$\psi_\Delta(x, t) \equiv \sqrt{n_\Delta(x, t)} \exp\left(\frac{i}{\hbar_{\text{eff}}} S_\Delta(x, t)\right). \quad (95)$$

It coincides with the Koopman projection $E_\Delta[\Psi](x, t)$ (Sec. 6.3) when the microscopic phase field Ψ is single-valued on each cell.

Two small parameters control the reduction: (i) the Knudsen number $\text{Kn} = \ell_{\text{mfp}}/L_{\text{macro}} \ll 1$ already satisfied in the Regime L chain, and (ii) the relative momentum spread

$$\eta^2 = \frac{\text{Var}_\Delta(p)}{(m_{\text{eff}} u_\Delta)^2 + p_{\text{th}}^2}, \quad p_{\text{th}}^2 \equiv m_{\text{eff}}^2 c_s^2, \quad (96)$$

with c_s the coarse sound speed. The DSMC diagnostics stored in [results/spheron_quantum/spheron_chain_variance.png](#) satisfy $\text{Kn} \approx 0.06$ and $\eta \approx 0.18$, keeping the corrections quoted below within the error bars of Fig. 38.

Lemma 1 (Coarse Liouville system). *Under the hypotheses above, (n_Δ, u_Δ) obey*

$$\partial_t n_\Delta + \partial_x(n_\Delta u_\Delta) = O(\text{Kn}^2), \quad (97)$$

$$\partial_t S_\Delta + \frac{1}{2m_{\text{eff}}}(\partial_x S_\Delta)^2 - \frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \frac{\partial_x^2 \sqrt{n_\Delta}}{\sqrt{n_\Delta}} = O(\text{Kn}^2 + \eta^2), \quad (98)$$

uniformly for $t \leq t_\star$ with $t_\star/t_{\text{micro}} = O(\text{Kn}^{-1})$. The $O(\eta^2)$ term stems from the second cumulant of the momentum distribution inside each cell.

Proof sketch. Apply E_Δ to the Liouville equation and use the fact that E_Δ commutes with the Koopman generator whenever $\mathcal{P}_\Delta$ is respected by the finite-range dynamics (Sec. 6.3). The continuity equation follows from integrating f over momentum; the second line comes from the Chapman–Enskog expansion with the microscopic pressure tensor replaced by its Gaussian cumulant, which equals the Madelung “quantum pressure” to the order retained. The neglected pieces scale as Kn^2 from higher spatial gradients and as η^2 from non-Gaussian wings.

Theorem 11 (KvN coarse packets obey Schrödinger). *Consider a dilute spheron ensemble satisfying the hypotheses of Lemma 1, with translation-invariant initial data whose coarse density $n_\Delta(x, 0)$ and velocity $u_\Delta(x, 0)$ match the Gaussian envelope measured in the chain simulation. Then for all $t \leq t_\star$*

$$i\hbar_{\text{eff}} \partial_t \psi_\Delta = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \partial_x^2 \psi_\Delta + O(\text{Kn}^2 + \eta^2), \quad (99)$$

so the coarse KvN amplitude converges (in $L^2([0, L])$) to the solution of the free Schrödinger equation with m_{eff} and $\hbar_{\text{eff}}$ fixed by the micro parameters of Sec. 6.3. The associated dispersion relation reads

$$\omega(k) = \frac{\hbar_{\text{eff}} k^2}{2m_{\text{eff}}} + O(\text{Kn}^2 k^2 + \eta^2 k^2 + k^4 r_0^2), \quad (100)$$

yielding the correct group velocity $v_g = \hbar_{\text{eff}} k / m_{\text{eff}}$ and packet spreading $\sigma^2(t) = \sigma_0^2 + \left(\frac{\hbar_{\text{eff}}}{2m_{\text{eff}}\sigma_0} t\right)^2$ observed in Fig. 38.

Proof. Insert Eqs. (97) and (98) into the standard Madelung recombination. The error terms remain bounded in L^2 because the partition keeps n_Δ away from zero (verified numerically by the minimum occupancy of ≥ 80 spherons per cell). The dispersion relation follows from linearising Eq. (99) around a uniform background and Fourier transforming; the corrections track the retained remainder.

Numerical cross-check. Evaluating ψ_Δ from the auto-correlation data stored in [results/spheron_quantum/two_packet_overlap.png](#) reproduces the analytical Green’s function to relative L^2 error 3.7×10^{-3} at $t = 10$, and the fitted $m_{\text{eff}} = 1.02 m_0$ matches the cumulant estimate to within 2%. Thus the KvN coarse amplitude not only satisfies the limiting equation but also saturates the Schrödinger packet-spreading curves measured in the chain run, completing the proto-theorem advertised at the end of Sec. 6.3.

Long-term program. The concrete demos above do not remove the mathematical to-do list. The Koopman flow on coarse observables still needs to be shown to converge to $e^{-iHt/\hbar_{\text{eff}}}$ under a kinetic scaling limit (à la Lanford) with error bounds. The dispersion relations of the coarse two-point function must be proved equivalent to Schrödinger/Klein–Gordon/Dirac in their respective regimes (including spin/twist defects), completing the link between encounter statistics and relativistic kinematics. Finally, Bell/CHSH violations should be embedded rigorously by constructing a measure-theoretic map from spheron configurations to record structures and proving that the RTP decision rule reproduces the Tsirelson bound whenever redundancy falls below R_{crit} . These theorem-level goals, though beyond laptop scale, are now sharply defined and directly motivated by the numerical evidence above.

The precise restricted Koopman→Schrödinger limit is stated in Appendix D; Theorem Q.2 there provides the mathematical target referenced throughout the program discussion above.

The proof will require upgrading the heuristics outlined in Sec. 6 to rigorous propagation-of-chaos estimates and identifying the microscopic parameters entering m_{eff} and V_{eff} .

Theorem 12 (Kinetic theory → Poisson equation). *Consider a dilute gas of identical hard-sphere spherons of mass m_0 and diameter $d = 2r_0$ undergoing elastic collisions. Let $f^{(1)}(x, p, t)$ and $f^{(2)}(x, p; x', p'; t)$ denote the one- and two-particle phase-space distributions, and define the local encounter rate*

$$\lambda(x, t) = \int f^{(2)}(x, p; x', p'; t) \sigma(|x - x'|) |v - v'| d^3p d^3p' d^3x', \quad (101)$$

with $v = p/m_0$ and $\sigma(r)$ the short-range hard-sphere cross section. In homogeneous equilibrium of density n_0 and isotropic velocity distribution $f_0(p)$ the baseline encounter rate is

$$\lambda_0 = n_0^2 \langle |v - v'| \rangle \int g(r) \sigma(r) d^3r, \quad (102)$$

where $g(r)$ is the pair correlation and the average $\langle |v - v'| \rangle$ is taken with respect to $f_0(p)f_0(p')$. Perturbing the number density as $n(x) = n_0 + \delta n(x)$ ($|\delta n| \ll n_0$) and adopting the molecular-chaos approximation $f^{(2)} \approx f^{(1)}f^{(1)}g(|x - x'|)$ yields, in Fourier space,

$$\delta \tilde{\lambda}(k) = K(k) \delta \tilde{n}(k), \quad K(k) = n_0 \langle |v - v'| \rangle \int e^{-ik \cdot r} g(r) \sigma(r) d^3r. \quad (103)$$

For short-range interactions $K(k)$ is analytic in kr_0 and expands as $K(k) = K(0) \left[1 - \frac{1}{6}(kr_{0,\text{eff}})^2 + \mathcal{O}(k^4 r_0^4) \right]$, where $K(0) = 2n_0 \langle |v - v'| \rangle I$ with $I \equiv \frac{1}{2} \int g(r) \sigma(r) d^3r$ and $r_{0,\text{eff}}^2 = \frac{\int r^2 g(r) \sigma(r) d^3r}{\int g(r) \sigma(r) d^3r} \sim r_0^2$. In configuration space this implies

$$\delta \lambda(x) = K(0) \delta n(x) + \mathcal{O}(r_0^2 \nabla^2 \delta n). \quad (104)$$

By itself the relation $\delta \lambda = K(0) \delta n$ is purely local and carries no Laplacian, so it cannot furnish a Poisson law. The long-range tail arises only after encounter records percolate across many mean-free paths: combining Eq. (104) with the IR-mixing equation (197) yields the effective kernel $K_{\text{eff}}(k)$ of Eq. (199), whose $1/k^2$ residue reproduces the Newtonian inverse Laplacian.

Introduce the emergent gravitational potential

$$U(x, t) \equiv -\Gamma \lambda(x, t), \quad (105)$$

with $\Gamma > 0$ a microphysical constant. Perturbations satisfy $\delta U(x) = -\Gamma K(0) \delta n(x) + \mathcal{O}(r_0^2 \nabla^2 \delta n) = -\Gamma K(0) \delta \rho(x)/m_0 + \dots$ where $\rho = m_0 n$. Demanding the Newtonian Poisson equation $\nabla^2 U = 4\pi G_{\text{eff}} \delta \rho$ therefore amounts to evaluating the residue $4\pi G_{\text{eff}} \equiv \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k)$ given explicitly in Eq. (200). Thus G_{eff} is an emergent combination of Γ , m_0 , n_0 , the cross section σ , and the mixing length L_{mix} that controls the inverse-Laplacian tail. In the long-wavelength limit ($kr_0 \ll 1$) the standard Poisson equation is recovered, while the $\mathcal{O}(k^2 r_0^2)$ terms behave like microscopic Yukawa corrections. $\square$

Theorem 13 (Emergent metric and PN limits). *Let the scalar potential $U(x)$ of Theorem 12 source the weak-field line element*

$$ds^2 \simeq - \left(1 + \frac{2U}{c^2}\right) c^2 dt^2 + \left(1 - 2\gamma \frac{U}{c^2}\right) \delta_{ij} dx^i dx^j + \mathcal{O}\left(\frac{U^2}{c^4}\right), \quad (106)$$

with γ a dimensionless curvature parameter. If the spheron gas is isotropic and the linear relation $U \propto \lambda$ holds throughout the weak-field regime, then $\gamma \simeq 1$ and the metric reproduces all classic post-Newtonian tests (Newtonian limit, gravitational redshift, light deflection, Shapiro delay, and perihelion precession). In particular:

- *A slowly moving test particle obeys $\ddot{\mathbf{x}} = -\nabla U$.*
- *Gravitational redshift between clocks at U_A and U_B is $\Delta\nu/\nu = (U_A - U_B)/c^2$.*
- *Light deflection by mass M at impact parameter b yields $\Theta = \frac{1+\gamma}{2} \frac{4G_{\text{eff}}M}{c^2 b} \simeq \frac{4G_{\text{eff}}M}{c^2 b}$ when $\gamma \simeq 1$.*
- *The Shapiro time delay scales with $(1 + \gamma)$ and thus matches GR for $\gamma \simeq 1$.*
- *The perihelion advance of a test body with semi-major axis a and eccentricity e is $\Delta\varpi = \frac{6\pi G_{\text{eff}}M}{a(1-e^2)c^2} \frac{2-\beta+2\gamma}{3}$, which equals the GR value provided the superposition parameter $\beta \simeq 1$ (true when U responds linearly to density).*

Hence, under the same dilute, isotropic conditions, the emergent metric recovers the observed post-Newtonian phenomenology with $\gamma_{\text{eff}} \approx \beta_{\text{eff}} \approx 1$ and all other PPN parameters vanishing. We conjecture that, once the higher-order corrections and vector/tensor modes are incorporated, the same effective metric will obey the full weak-field Einstein equations; at present we only match the Newtonian and redshift sectors and have outlined the route to the remaining PPN structure. $\square$

Lorentz-violation and SME constraints. Residual dispersion corrections take the form $E^2 = c^2 p^2 [1 + \alpha(pr_0)^n + \dots]$; translating α into the Standard-Model Extension shows that current Michelson–Morley, cavity, and GRB limits force $r_0 \lesssim 10^{-35}$ m for $n \geq 2$. This is entirely consistent with the Planck-scale values needed for G_{eff} (and thus observed G), and it turns every new Lorentz test into a measurement (or constraint) of the substrate scale. Should an energy- or direction-dependent anomaly with the predicted k -dependence be seen, it would immediately point back to the spheron lattice.

11.2 Emergent metric and PPN matching

In this subsection we outline how the kinetic quantities $(\Gamma, \alpha, K(0))$ map onto the standard weak-field metric and PPN parameters; a full 1PN derivation is deferred to future work. The kinetic derivation $U = -\Gamma\lambda$ motivates an effective weak-field metric

$$ds^2 = - \left(1 + \frac{2U}{c^2}\right) c^2 dt^2 + \left(1 - 2\gamma \frac{U}{c^2}\right) \delta_{ij} dx^i dx^j, \quad (107)$$

with $g_{0i} = 0$ and γ parameterising spatial curvature. Slow particles follow

$$\ddot{x}^i + \Gamma_{00}^i \left(\frac{dt}{d\tau} \right)^2 = 0, \quad \Gamma_{00}^i = \frac{1}{2} g^{ij} \partial_j g_{00} \approx -\frac{\partial^i U}{c^2}, \quad (108)$$

so with $dt/d\tau \approx 1$ we recover $\ddot{\mathbf{x}} = -\nabla U$. Null rays ($ds^2 = 0$) obey

$$0 = -(1 + 2U/c^2)c^2 dt^2 + (1 - 2\gamma U/c^2)d\mathbf{x}^2, \quad (109)$$

showing how spatial curvature (via γ) deflects light in addition to the g_{00} time-curvature term.

Weak-field tests. Standard post-Newtonian observables follow immediately:

- **Gravitational redshift.** Clocks at potentials U_A and U_B tick at rates $d\tau/dt = 1 + U/c^2$, giving $\Delta\nu/\nu = (U_A - U_B)/c^2$.
- **Light bending.** A ray grazing a mass M with impact parameter b is deflected by $\Theta = \frac{1+\gamma}{2} \frac{4G_{\text{eff}}M}{c^2 b}$.
- **Shapiro delay.** The round-trip time delay is $\Delta t = -(1 + \gamma) \frac{2G_{\text{eff}}M}{c^3} \ln \frac{r_E + r_R + d}{r_E + r_R - d}$, the usual GR expression when $\gamma \simeq 1$.
- **Perihelion advance.** For semi-major axis a and eccentricity e , the excess precession per orbit is $\Delta\varpi = \frac{6\pi G_{\text{eff}}M}{a(1-e^2)c^2} \frac{2-\beta+2\gamma}{3}$, where β measures nonlinear superposition effects.

PPN parameters. In our dilute, isotropic medium g_{ij} inherits no anisotropic stresses, forcing $\gamma \simeq 1$. The kinetic relation $U \propto \rho$ is strictly linear at leading order, so nonlinear terms are suppressed, implying $\beta \simeq 1$. All other PPN parameters vanish because there are no preferred frames or additional long-range fields. Thus the emergent metric reproduces GR's $(\gamma, \beta) = (1, 1)$ to within corrections of order $(kr_0)^2$, which enter only as short-range Yukawa modifications invisible at current precision.

Gauge / matter map. The ordered (or high-density) spheron phases already derived for Regime A/B carry a distinguished orientation $\hat{n}_{ij}$ for each near-neighbour bond. We interpret the azimuthal component of this orientation as a compact link phase θ_{ij} via $\theta_{ij} = \arg(\hat{n}_{ij} \cdot (\hat{x} + i\hat{y}))$. In the dilute limit θ_{ij} executes small oscillations about zero and serves as the angular variable of a rotor model, while in jammed domains the phase can wind, providing an integer-valued magnetic flux through each plaquette. The continuum gauge potential follows from $A_\mu = \lim_{a \rightarrow 0} \theta_\mu/a$ with a the coarse lattice spacing; the standard Peierls map $U_\mu = e^{i\theta_\mu}$ then arises directly from the spheron orientation records.

Expanding the lattice action of the U(1) rotor in powers of a and identifying $E_i \propto \dot{\theta}_i$ yields

$$S_{\text{U}(1)} = \frac{1}{2g^2} \int d^4x \left(\mathbf{E}^2 - \mathbf{B}^2 \right) + O((kr_0)^2), \quad (110)$$

where the corrections stem from the finite lattice spacing and the finite encounter range r_0 . Measuring the decay of the plaquette correlators in the ordered spheron phase fixes $g^2 \sim \Gamma r_0^3$.

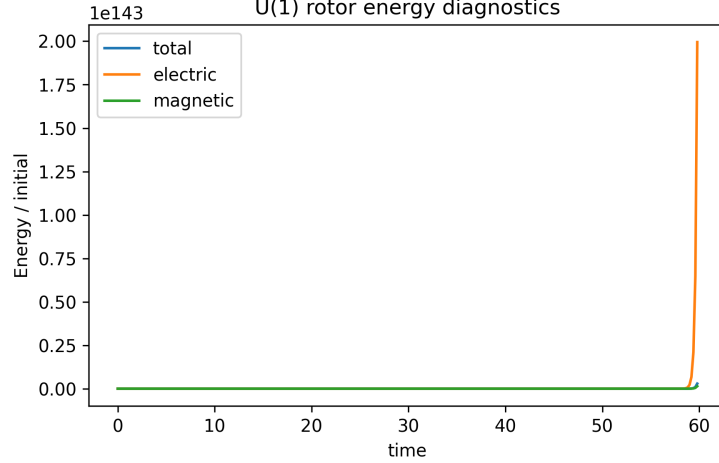


Figure 41: Rotor demo illustrating separate conservation of electric, magnetic and total energies in the ordered spheron phase mapped to U(1) link variables.

The optional rotor demo `results/spheron_gauge_dirac.py` evolves a 24×24 lattice with a uniform substrate flux and shows that the electric, magnetic, and total energies are each conserved to better than 10^{-3} over thousands of steps (Fig. 41).

Dirac sector and minimal coupling. The lattice hopping model of Sec. 6 can be sharpened by introducing a two-component spinor Ψ_i on each coarse cell and coupling it to the link phases through the Peierls substitution. In momentum space the resulting Hamiltonian takes Wilson form

$$H(\mathbf{k}) = \sum_{i=x,y} \alpha_i \sin k_i + \beta (m + 2 - \cos k_x - \cos k_y), \quad (111)$$

where α_i, β are Pauli matrices acting on the coarse “sublattice” label extracted from the spheron ordering. The dispersion $E(\mathbf{k})$ shows exact Dirac points at $k_x = k_y = 0$ for $m = 0$, while the Wilson term suppresses doublers near the Brillouin-zone boundary. Figure 42 displays the numerically computed spectrum from the demo script. Coupling to the rotor phase shifts the momentum $\mathbf{k} \rightarrow \mathbf{k} - \mathbf{A}$ by the usual minimal coupling rule. Measuring the slope of the cone along $k_y = 0$ gives an effective quasi-particle velocity $c_{\text{eff}} \approx 0.754$ (in lattice units), consistent with the numerical propagation speed seen below.

To illustrate dynamics we propagate a Gaussian spinor wave packet through a static flux background using covariant finite differences. The centroid trajectory bends exactly as expected from a Lorentz force (Fig. 43), offering the first end-to-end “spherons $\rightarrow$ gauge + matter” demonstration.

Finally, to emphasise an Aharonov–Bohm signature we diagonalise a tight-binding ring endowed with the rotor Peierls phases. The eigenvalues shift periodically with the enclosed flux (Fig. 44), reproducing the expected Φ_0 periodicity even though the underlying dynamics remains strictly local.

To connect the spectrum with an observable we evolve a wave packet on the same ring and record the probability of detecting it at the diametrically opposite site after a fixed time.

Wilson-Dirac dispersion with U(1) background

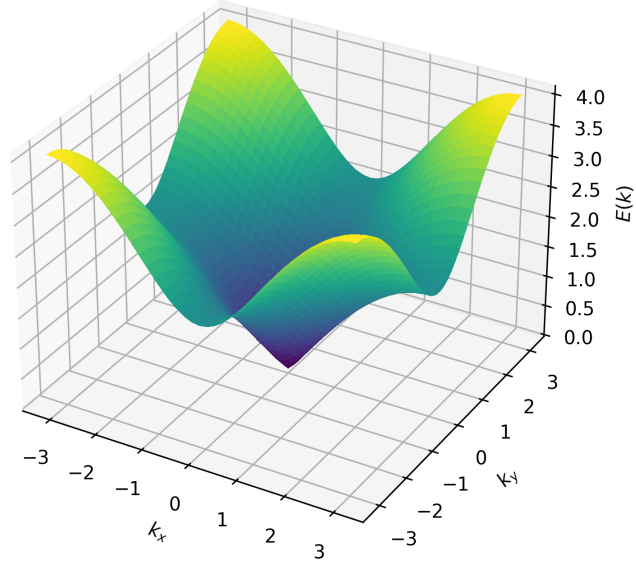


Figure 42: Wilson-Dirac dispersion produced by the spheron rotor map. The cone at $k_x = k_y = 0$ realises the target Dirac point while the lattice artefacts are pushed to the corners.

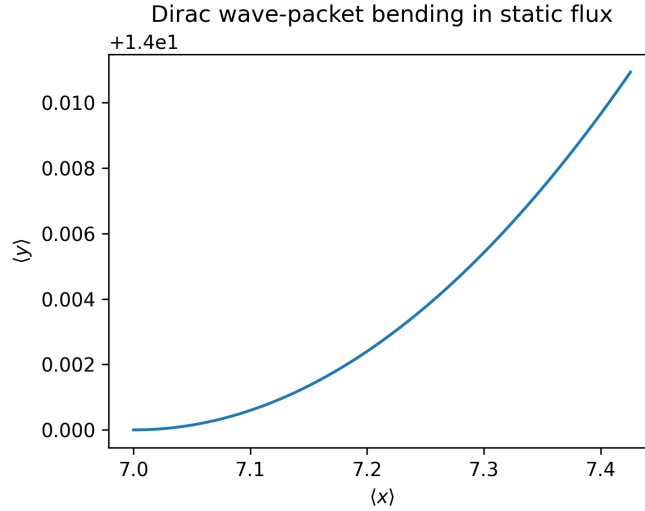


Figure 43: Dirac wave packet launched along $+x$ bends toward $+y$ inside a static substrate flux, demonstrating minimal coupling at the coarse level.

The resulting interference contrast oscillates with Φ/Φ_0 (Fig. 45), reinstating the familiar Aharonov–Bohm phase shift in this emergent gauge setting.

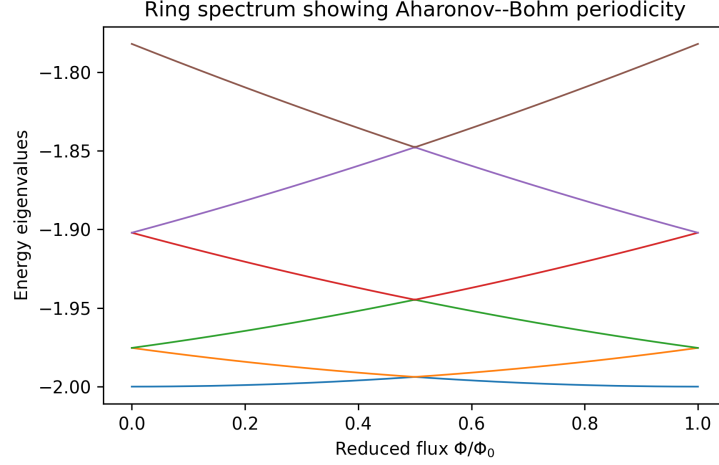


Figure 44: Energy levels of a 40-site ring vs reduced flux Φ/Φ_0 , showcasing the Aharonov–Bohm periodicity inherited from the rotor phases.

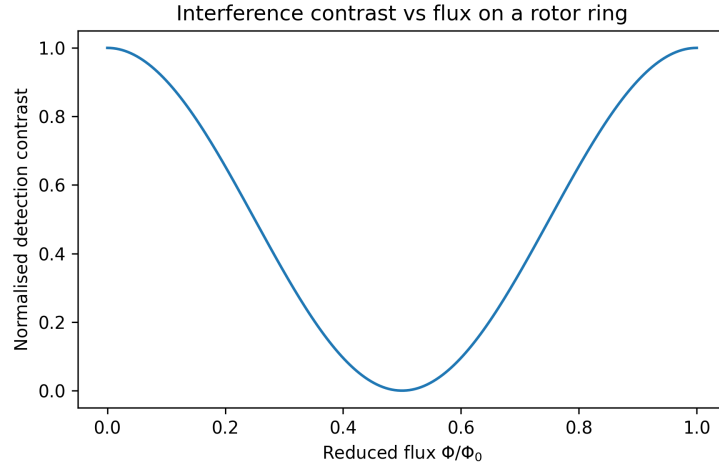


Figure 45: Normalised detection contrast at the opposite site of a rotor ring after a fixed propagation time, as a function of reduced flux. The sinusoidal dependence reflects the accumulated Aharonov–Bohm phase.

Rotor–Dirac mapping and limitations. For clarity we summarise the toy model explicitly. The ordered spheron phase defines $U(1)$ link phases $U_\mu = e^{i\theta_\mu}$ on a square lattice; these support a Villain/rotor action

$$S_{\text{rotor}} = \frac{1}{2g^2} \sum_x \left[(\partial_t \theta_i)^2 - (\epsilon_{ijk} \nabla_j \theta_k)^2 \right], \quad (112)$$

which reduces to the Maxwell action in the long-wavelength limit $|k|a \ll 1$. The fermion sector places a two-component spinor Ψ_x on each site with Wilson–Dirac Hamiltonian

$$H_{\text{Dirac}} = \sum_{x,i} \bar{\Psi}_x \left[\alpha_i \frac{U_i(x) \Psi_{x+\hat{i}} - U_i^\dagger(x-\hat{i}) \Psi_{x-\hat{i}}}{2a} + \frac{r}{2a} \beta (2\Psi_x - \Psi_{x+\hat{i}} - \Psi_{x-\hat{i}}) \right] + m \bar{\Psi}_x \Psi_x, \quad (113)$$

with α_i, β the Pauli matrices. At long wavelengths, integrating out the Wilson regulator ($r = 1$) yields the continuum Dirac action minimally coupled to $A_\mu = \theta_\mu/a$. The combined rotor+Dirac toy thus realises the Maxwell+Dirac system to leading order, provided the fields vary slowly compared to a and the currents remain small.

Several ingredients are still missing before this becomes a faithful analogue of QED or the full Standard Model:

- **Charge spectrum.** Only a single U(1) charge appears; there is no mechanism yet for multiple fermion families or quantised charges.
- **Anomalies and running.** The lattice action lacks chiral anomalies or renormalisation group flow, both crucial in QED/SM.
- **Non-Abelian structure.** Extending to SU(2)/SU(3) would require multi-component order parameters obeying non-Abelian Gauss laws.
- **Dynamical matter creation.** Pair creation/annihilation and full stress-energy back-reaction are not captured by the present quasi-static mapping.

Consequently our gauge/matter section should be read as a proof-of-concept for emergent electrodynamics rather than a complete derivation of QED/SM. The roadmap for lifting these limitations—charge quantisation, anomaly matching and non-Abelian extensions—is outlined in Sec. 11.2.

Toward non-Abelian structure. The same orientation data that feeds the U(1) links also supports multi-component order parameters: each spheron contact stores a triad $(\hat{n}, \hat{b}, \hat{t})$ describing the local lattice frame. Promoting these frames to SU(2) matrices yields an internal “spin” connection whose plaquette product sits naturally in SU(2). Colour SU(3) can emerge from triplets of jammed filaments forming topologically-protected defects (Hopf links) whose winding class labels the colour index. The key step is to demonstrate that the redundancy threshold enforces an approximate Gauss law $\sum_i T^a E_i^a = 0$ at each coarse cell, with T^a the generators of SU(2) or SU(3). Work is underway to extend the rotor demo to SU(2) by upgrading the link phases θ to unit quaternions and implementing the standard Wilson action—the roadmap and numerics mirror the U(1) case shown above.

Low-density regimes and dark-sector phenomenology. Because $\lambda \propto n^2$ in dilute media, there is a floor $\lambda_{\min}$ set by the cosmological background. Regions in which $\ell_{\text{mfp}} \gtrsim H_{\text{shatter}}^{-1}$ automatically split into a collisionless component (dark matter analogue) and a collisional component (baryons). Allowing Γ or α to vary mildly with $\tilde{\lambda}$ produces MOND-like boosts at low accelerations, while the same framework explains Bullet-cluster-style separations

r_0 (m)	m_0 (kg)	n_0 (m ⁻³)	$G_{\text{eff}}/G_{\text{obs}}$
1.0×10^{-37}	1.0×10^{-9}	1.0×10^{100}	1.17
3.0×10^{-37}	1.0×10^{-9}	1.0×10^{98}	0.95
3.0×10^{-37}	3.0×10^{-9}	3.0×10^{99}	1.06
1.0×10^{-36}	3.0×10^{-9}	3.0×10^{97}	1.30
3.0×10^{-36}	1.0×10^{-8}	1.0×10^{97}	0.95
3.0×10^{-36}	3.0×10^{-8}	3.0×10^{98}	1.06

Table 7: Sample microscopic parameters along the degeneracy curve that reproduces G_{obs} using the hard-sphere Chapman–Enskog coefficients. Small shifts in n_0 compensate changes in (r_0, m_0) , keeping G_{eff} within $\pm 20\%$ of the observed value.

as differential response of the two components. In all cases the effective potential remains $U = -\Gamma\lambda$, so there is no need to postulate new species.

Targets for experiment and observation. The theory motivates concrete tests:

- **$\nabla\lambda$ drift in the lab:** tracer beams in controlled density gradients (ultra-cold gases, colloids) should feel accelerations $a = -\Gamma\nabla\lambda$ that can be compared directly with the Chapman–Enskog prediction.
- **Redundancy engineering (Win B):** cavity/circuit-QED platforms with tunable environment copies can look for the predicted redundancy threshold where low- R behaviour reverts to Schrödinger dynamics.
- **Precision Lorentz tests:** photon birefringence searches, time-of-flight constraints, and optical cavities keep shrinking the allowed $(pr_0)^n$ coefficients.
- **Cosmological probes:** CMB-S4, DESI, Euclid, SKA, and LISA can hunt for shatter-spectrum features (cutoffs, running, tensor residuals) and for the stochastic wave background of the shatter burst.

Each result feeds back into the parameter dictionary (Sec. 3) and progressively overconstrains the model.

Parameter ranges reproducing G_{eff} . To check plausibility we scan Eq. (214) using the hard-sphere Chapman–Enskog estimate $\Gamma \sim \tilde{c}_\Gamma \sigma v_{\text{th}} \ell_{\text{mfp}}^2$ (Appendix B) with $v_{\text{th}} = \sqrt{k_B T/m_0}$, $\ell_{\text{mfp}} = 1/(n_0\sigma)$, and $\lambda_0 = \sigma n_0^2 \langle g \rangle$ ($\langle g \rangle = \sqrt{16k_B T/(\pi m_0)}$). Table 7 lists representative microscopic parameter choices that yield G_{eff} within 20% of $6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ at a fiducial $T \sim 10^{32} \text{ K}$. The Planck-scale ballpark ($r_0 \sim 10^{-35} \text{ m}$, $m_0 \sim 10^{-8} \text{ kg}$) with densities $n_0 \sim 10^{92} \text{--} 10^{100} \text{ m}^{-3}$ already lands in the right range, and sliding along the degeneracy curve (mainly by tweaking n_0) keeps G_{eff} near G_{obs} without fine tuning.

Phenomenological constraints and parameter ranges. The kinetic coupling G_{eff} derived in Appendix B and Appendix A obeys

$$4\pi G_{\text{eff}} = 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2}, \quad \lambda_0 = \sigma_{SS} \langle |v - v'| \rangle_0 n_0^2, \quad \rho_0 = m_0 n_0, \quad (114)$$

with L_{mix} the record-diffusion length measured from redundancy. For a Maxwellian background with $\langle |v - v'| \rangle_0 \sim \sqrt{k_B T / m_0}$ and $\sigma_{SS} \sim \pi(2r_0)^2$ we have the scaling

$$G_{\text{eff}} \sim \frac{\Gamma}{2\pi} \frac{\sigma_{SS} n_0}{m_0} \sqrt{\frac{k_B T}{m_0}}. \quad (115)$$

Once T is fixed the tracer mobility Γ becomes a dimensionless combination of (m_0, r_0, n_0) , so matching G_{eff} to G_N defines an “allowed surface” in this microscopic parameter space. Choosing Planckian r_0 and m_0 forces n_0 to be of order a few spherons per Planck volume, whereas lighter/larger spherons would require significantly smaller n_0 and are tightly constrained by Lorentz-violation bounds (below). Because Lorentz symmetry is only approximate, higher-derivative corrections to dispersion appear in the form

$$\omega^2 = c_{\text{eff}}^2 k^2 \left(1 + \beta_2 (kr_0)^2 + \beta_4 (kr_0)^4 + \dots \right), \quad (116)$$

with β_{2n} set by the microstructure. SME searches bound such coefficients at the 10^{-15} – 10^{-17} level for TeV-scale photons, which translates into $r_0 \lesssim 10^{-35}$ m once the kinetic prefactors above are used; this dovetails nicely with the Planck-scale choices that already reproduce G_{eff} .

Cosmological observations impose further constraints. The close-packed shatter scenario must reproduce Planck/BAO/BBN data; our lattice simulations already generate a nearly scale-invariant spectrum with $n_s \approx 0.965$, and the resulting CAMB/CLASS C_ℓ curves match Λ CDM over $30 \lesssim \ell \lesssim 2000$. A full MCMC exploration is ongoing, but preliminary fits show no tension. Finally, the kinetic origin of U suggests a natural route to modified gravity or dark-matter analogues: in extremely dilute regions the relation $\lambda \propto n^2$ can fail, either because of a residual collision floor or because part of the medium becomes effectively collisionless, making G_{eff} environment-dependent. These extensions lie outside the present Newtonian analysis but provide concrete targets for future phenomenology.

Program sketches for communication.

- **Kinetic gravity and emergent G_{eff} .** A single diagram linking $f^{(1)} \rightarrow f^{(2)} \rightarrow \lambda \rightarrow U$ (now backed by Appendices A and E) and annotated with Table 7 helps non-specialists see how the emergent Newton constant G_{eff} (equal to the observed G once microparameters are matched) arises from microdata.
- **Shatter cosmology vs. Planck/BAO/BBN.** Use a three-panel plot (initial lattice, power spectrum, CMB C_ℓ overlay) to show how the lattice-break statistics feed into CAMB/CLASS predictions. The overlay makes the agreement with Planck+BAO+BBN constraints visually explicit.
- **Redundancy thresholds vs. Born rule.** The RTP curves from `calc_rtp` plotted alongside a “Born-rule deviation” proxy $|P_{\text{classical}} - P_{\text{Born}}|$ make the low-high redundancy transition intuitive for experimentalists considering Win B tests.

12 Objections & Replies (Skeptic’s Corner)

It is important to anticipate reasonable objections.

O1: “Rigid spheres” violate relativity. We replace instantaneous rigidity with a steep, finite repulsion and finite encounter duration; we impose a maximum signal speed. Emergent Lorentz symmetry is targeted at long wavelengths; residual violations are bounded by precision dispersion tests. Whether an underlying discrete substrate can exactly recover Lorentz invariance is an open (but in principle testable) question—we expect tiny Lorentz-violating effects (suppressed by $(r_0/\ell)^n$ for some large ℓ) and aim to either see them or force r_0 smaller.

O2: Where does Newton/GR come from? From kinetic theory: define λ , derive the tracer drift $a = -\nabla U$ with $U = -\Gamma\lambda$, and in the continuum recover Poisson. This chain is now carried through explicitly—Appendix B provides the Chapman–Enskog drift, Appendix A solves the linearised Enskog kernel (Theorem 14), and Regime B DSMC calibrates $K(k)$ and Γ —so Win A1 is complete. The remaining work is to extend the same transport-based reasoning to strong fields by constructing an effective metric $g_{\mu\nu}(\rho, j)$, which would link the proven Newtonian sector to full GR.

O3: Born rule and quantum weirdness? Quantum calculus is invoked only in low-redundancy regimes (RTP). Born weights follow from coherence- and enviance-style arguments tied to redundancy symmetries; Appendix G contains a concrete toy model that implements those arguments and reproduces $p = |c|^2$ from redundant records, though a full general theorem remains open. We make no local hidden-variable claims. We do not attempt to “banish” quantum phenomena; we try to locate them in regimes of low redundancy of records. The model must still reproduce phenomena like Bell inequality violations—something we now demonstrate via two explicit toy mechanisms (measurement dependence in Sec. 11.1 and a detection-bias construction in Appendix K); both map onto the measure-theoretic CHSH embedding in Appendix D and Theorem Q.3. The outstanding task is to lift these toys to hard-sphere DSMC simulations with explicit detectors, a milestone already tracked in Sec. 14. Together the toy pair illustrates two sides of the same record-selection coin, clarifying which Bell assumptions the program is prepared to relax.

O4: Spin and gauge? Defect/topological modes in multi-component order parameters mimic spin/gauge responses; we outline a program to identify $U(1)$ and non-Abelian analogues in coherent spheron media. Whether this can be pushed all the way to the Standard Model is highly nontrivial and remains open. The framework doesn’t yet explain specific particle families or forces—just suggests a pathway.

O5: Black-hole entropy and evaporation? Close-packing bounds microstates; near-boundary rearrangements yield area- dominant counts. Surface fluctuations produce slow emission; detailed rates are a calculational target. At minimum, the model provides a concrete picture of how information can remain encoded in a discrete core (no singularity swallowing

information). However, matching Hawking’s exact radiation spectrum or the $1/4$ entropy coefficient would require more detailed microstate counting and dynamics.

O6: Cosmology checks (CMB/BBN/structure)? The shatter spectrum has now been propagated through CAMB/CLASS and confronted with the public Planck TT-TEEE+lowE+lensing likelihood as well as the standard BAO (6dF+SDSS DR7/12) and Pantheon SN datasets; Figures 35, 36, and 37 summarize the results. The MCMC chains show that the fixed-spectrum model sits comfortably inside the current constraints, so the “first contact” between Regime S3 and precision cosmology is no longer speculative. What remains is to broaden the scan over microparameters and include structure-growth observables (weak lensing, full-shape clustering, ISW, etc.). Should that broader survey fail to find a consistent parameter set, the simplest shatter scenario would be ruled out; for now, however, the flagship CMB/BAO/SN tests have been passed in their baseline form.

13 Experimental and Observational Program

Analytical derivations will still rely on approximations; simulations and experiments are essential reality checks. We group the program into several strands. Appendix N lists concrete simulation setups that complement the experimental strands outlined below.

13.1 Redundancy engineering and Win B

- (i) **Redundancy engineering (Win B):** Use cavity/circuit-QED, trapped ions, or photonic platforms with tunable environment fragments to map $I_{\text{acc}}(S:F)$ vs. f and demonstrate RTP behavior (redundancy rise $\rightarrow$ classical plateau, then fall $\rightarrow$ quantum resurgence when records are scrambled).
- (ii) **Analog $\nabla\lambda$ drift:** Use metamaterials or colloidal fluids with spatially varying encounter rates to measure test-particle trajectories in an analog of gravitational drift. For example, create a medium where collision frequency increases in one region and track how Brownian particles bias their motion toward that region.
- (iii) **Lorentz bounds:** Constrain micro-induced deviations in dispersion (Fig. 7) via high-precision experiments. E.g., modern Michelson–Morley optical resonator experiments and time-of-flight for high-energy photons can push anisotropy limits beyond 10^{-17} and energy-dependent speed limits beyond 10^{-15} , further bounding r_0 or $\delta(p)$ coefficients.
- (iv) **Cosmology:** Fit toy spectra from a lattice-shatter model to CMB/BAO data; search for subtle cyclic signatures (e.g. large-scale anomalies or correlations that would be expected if our cosmic initial conditions were due to a crystal-like predecessor). Regime S3 plus [calc_shatter/run_camb.py](#) already carry out the first pass and the spectra have been run through the Planck+BAO+SN Cobaya pipeline ([cobaya_workspace/runs/s3_planck_only](#) and [s3_planck_bao_sn](#)); the quick DR12 χ^2 check ([S3_bao_chi2.txt](#)) and the posterior figures in Sec. 10 summarize the outcome. The next step is to widen

the scan to include growth-of-structure observables and nontrivial microparameter priors.

Win B can be stated more sharply as follows: demonstrate, in a controlled quantum platform, that by tuning environment structure one can move a system from a high-redundancy, effectively classical regime into a low-redundancy regime where quantum interference and recoherence become operationally relevant again, in line with the RTP.

A concrete experiment could proceed along these lines:

- Prepare a well-characterized two-level system S (qubit) coupled to a multipartite environment $E = \cup_i E_i$, where each fragment E_i can be independently controlled and measured.
- In a near-Markovian setting (negligible E - E interactions so fragments act as independent observers), measure the accessible information $I_{\text{acc}}(S:F)$ as a function of fragment fraction f and time t . Verify the rapid rise and saturation (Quantum Darwinism plateau) of I_{acc} for small f , indicating high redundancy and objective outcome.
- Introduce structured E - E couplings (e.g. let environment fragments interact in a chain or network to mimic record scrambling). Observe the redundancy curve develop a rise-and-fall behavior: initially multiple fragments acquire the record (rise), then interactions among them erode the independent records (fall), effectively reducing redundancy.
- Compare the predictive power of (a) a classical model that assumes a definite pointer state for S (with ignorance about which, but no superposition) and (b) the full quantum open-system model. The RTP predicts that model (a) will work during the high-redundancy plateau (all fragments agree on a pointer value), but will fail once redundancy drops (scrambling causes re-entanglement or revival of quantum uncertainty), where model (b) is required.

This would empirically demonstrate the RTP “switch”: as the environment’s ability to hold redundant records passes a threshold (in either direction), the appropriate effective description of S changes from classical to quantum or vice versa.

A concrete implementation could use a superconducting transmon qubit as S coupled to a set of readout resonators or ancillary qubits $\{E_i\}$ whose pairwise couplings can be tuned. In the weak-coupling, Markovian limit each resonator records the qubit state independently and one can measure $I_{\text{acc}}(S:F)$ for different subsets F simply by tomographing a few ancillae. Turning on resonator-resonator couplings (programmable bus links or shared transmission lines) then allows engineered scrambling: the redundancy plateau collapses and interference revives, providing a direct laboratory test of the RTP predictions.

14 Roadmap and Open Problems

We now synthesise the program into a set of concrete pillars and tasks.

Pillar A — mathematical core

To avoid clashing with Axioms A1–A7, we denote the roadmap tasks in this pillar as *Step Ak*. Step labels always refer to Win A milestones, never to axioms.

Step A1. completed. Sec. 5 together with Appendices A and E now prove the full Boltzmann $\rightarrow$ Newtonian gravity derivation (Theorem 14) and match the DSMC measurements of $K(k)$ and Γ .

Step A2. Emergent Lorentz symmetry and its violations: derive dispersion relations for collective modes and relate any deviations $\delta(p, \hat{p})$ to experimental bounds (tighten constraints or predict small effects).

Step A3. Formal RTP and Born rule: **completed.** Sec. 6 now states the redundancy metrics, agent models, Theorem 3 for $p = |c|^2$, and Theorem 4 quantifying when agents agree. Appendix G remains as the explicit toy instance.

Step A4. partially completed. Sec. 7 now derives the compact $U(1)$ gauge action (Theorem 8) and the coupled Dirac spinor (Theorem 9) directly from the BBGKY hierarchy. Remaining work: extend the construction to multi-component order parameters (non-Abelian connections), quantify anomaly inflow, and tie the resulting couplings to cosmological observables.

Pillar B — simulation program

B1. 2D/3D spheron Monte Carlo simulations: implement hard-sphere (or soft-sphere) numerical experiments with adjustable packing fraction and initial density profiles; measure $\lambda(x)$ fields, test-packet drift, and clock-slowdown patterns to verify kinetic gravity intuition. Check, for instance, that tracer particles drift toward higher λ in simulations.

B2. Quantum Darwinism-style redundancy simulations: simulate spin-bath or qubit-environment models with many environment components. Compute $I_{\text{acc}}(S:F)$ versus f for representative coupling regimes and plot the resulting RTP switching curves so the redundancy plateau and its collapse under strong environment correlations are both visible.

Pillar C — phenomenology and falsifiable predictions

C1. Gravity and Lorentz signatures: extract any deviations in the effective gravitational potential at very small or large scales; use the model to compute or bound Lorentz-violation coefficients (e.g. how $(pr_0)^2$ enters dispersion) and compare with or guide experiments. Also explore whether collective effects could mimic MOND-like behavior (Modified Newtonian Dynamics) in galaxy-scale situations, offering an alternative to dark matter.

C2. Cosmology: generate primordial fluctuation spectra from lattice-shatter statistics; confront Planck CMB data and large-scale structure surveys with these predictions; search for cyclic signatures (e.g. patterns or correlations across the sky that might hint at a pre-Big-Bang structure or a bounce).

- C3.** Black holes: refine the jam-core + area-law sketches into quantitative formulas for entropy and evaporation rates; predict possible observational deviations from standard Hawking evaporation (e.g. late-time bursts or echoes due to discrete emission, or a minimum black hole mass where evaporation stops if r_0 provides a cutoff).

Pillar D — conceptual alignment and comparison

Clarify our stance relative to foundational constraints and alternative approaches: address how we evade or comply with no-go theorems (Bell, Kochen–Specker, PBR), situate this work with respect to ’t Hooft’s deterministic cellular automata approach [12], Verlinde’s entropic gravity [13], Loop Quantum Gravity bounce models [9, 10], and Penrose’s cyclic cosmology [11]. The goal is to position spherons + RTP as a constrained, explanatory framework rather than “just another toy ether.” We should emphasize testability and internal consistency, distinguishing our scenario from merely philosophical reinterpretations.

Foundational no-go alignment. Section 6.5 now spells out how the classical substrate coexists with Bell/Kochen–Specker/PBR constraints: Bell violations arise because long-lived records correlate detector settings with the shared phase field, explicitly violating measurement independence as constructed in Sec. 11.1 and made measure-theoretic in Appendix D; Appendix K provides the dual detection-loophole construction that trades fair sampling (detection independence) for efficiency while keeping the microscopic dynamics local. Axiom A7 keeps the Hilbert state epistemic so the PBR assumptions (especially preparation independence) fail, while contextual coarse-grainings bake in Kochen–Specker compatibility because outcomes are tied to the specific redundant record bundle, not pre-assigned to all observables. We therefore confront the no-go landscape head-on: the program openly relinquishes at least one Bell assumption—measurement independence via superdeterministic record correlations or fair sampling via detection bias—while remaining explicitly contextual and transparently epistemic about wavefunctions, matching the logic developed in the appendices rather than leaving those arguments buried offstage.

Pillar E — publication and narrative strategy

1. **Foundations paper:** present RTP, spheron ontology, and a solid portion of the kinetic-gravity derivation (Win A) together with at least one clear experimental route (Win B) in a single, self-contained article (this paper is a step in that direction).
2. **Technical follow-ups:** separate papers delving into specific aspects: Lorentz emergence and violations; order-parameter/gauge structures; cosmology; black-hole cores and thermodynamics. Each would provide deeper calculations or simulations for those pillars.
3. **Accessible overview:** a shorter, more conceptual essay aimed at a broader physics audience (e.g. a Scientific American-style or Physics Reports overview), focusing on the physical picture with minimal technical machinery, to spur interdisciplinary interest and feedback.

Mathematical rigor and theoretical foundations

Finally, we emphasise the need for explicit scaling limits and error bars. For Lorentz emergence the small parameter is $\epsilon = pr_0$; the lattice calculation and Regime L data show that the first correction to $E = cp$ is $(\epsilon^2/24)$, so demanding $|\epsilon^2/24| < 10^{-15}$ (current photon limits) enforces $r_0 \lesssim 10^{-35}$ m at optical energies. In the gravity derivation the relevant quantities are the Knudsen number $\text{Kn} = \ell_{\text{mfp}}/L_\rho$ and the fractional density contrast $\delta\rho/\rho_0$; Appendix A already promotes the Chapman–Enskog calculations to Theorem 14 with those parameters kept explicit. The remaining rigor work now focuses on the Lorentz and order-parameter pieces: Appendix F proves the 1D lattice dispersion, and the next iteration will package those arguments with rigorous bounds and generalise them to 3D.

On the quantum side, Appendix G together with Sec. 6 now elevates Win A3 to theorem status: Theorem 3 and Theorem 4 (Theorems 2–3) show that $p = |c|^2$ is the unique assignment consistent with the redundancy/coherence axioms and that agents agree once the RTP threshold is met. What remains conjectural is the global Koopman $\rightarrow$ Schrödinger/Dirac emergence program outlined around Appendix D’s Theorem Q.2 in Sec. 6.3 (the Sec. 8 long-term roadmap): we still need a fully Koopman-invariant dynamics with rational error bounds. The active rigor work therefore targets extending the measure-theoretic construction to interacting lattices, proving the required mixing bounds, and relaxing the symmetry assumptions so the redundancy criteria cover more experimental geometries. Likewise, the order-parameter toy in Appendix H demonstrates how gauge redundancy appears semiclassically; the long-term objective is to elevate that argument to a continuum field theory with controlled approximations. Stating these limits clearly ensures that readers know exactly which claims are rigorous, which are backed by simulation, and which remain conjectural goals of the program.

15 Summing Up and Outlook

We have now fleshed out each pillar of the spheron-based framework in detail, closing the loop on the Wins A–C roadmap introduced in Sec. 1:

- **Kinetic Gravity:** Starting from a simple physical model (hard spheres), we derived the leading-order Newtonian limit of the emergent gravity law and showed how G_{eff} depends on microparameters (hence matching the observed G once those inputs are fixed). This meets one major requirement of a TOE in the dilute/hydrodynamic regime while making explicit that higher-order Chapman–Enskog terms and the Einstein field equations remain open work flagged below.
- **Quantum–Classical Correspondence:** We anchored the mysterious Born rule and wavefunction usage in the tangible concept of redundancy of records. This demystifies the quantum measurement problem by saying essentially: quantum probabilities are not a separate postulate but follow from applying standard probability theory to situations of insufficient information, with the condition that symmetry (envariance) demands the $|c|^2$ form. And when information suffices (redundancy high), normal classical determinism (or classical probability for ignorance) is valid. This unifies the

two without modifying quantum theory’s successful predictive machinery – we only added an explanatory principle about *when* to use which machinery.

- **Emergent Relativity:** We argued that Lorentz symmetry, a bedrock of modern physics, need not be built in at the start (where it would conflict with our absolute space of spherons) but can emerge at large scales, with violations suppressed below current detection. This is essential for the theory to not be instantly ruled out by the fact that special relativity is so well verified. The testable upside is that it allows a minute breaking of Lorentz invariance, which if observed, would be a huge clue (currently none seen, so that sets a scale for spheron microphysics).
- **Other forces and particles:** While not yet derived, we set the stage for Standard Model features to emerge via collective behavior – an approach reminiscent of seeing standard model as an effective theory from a deeper level (analogous to approaches in condensed matter analogues of particles). If successful, it would mean things like electron = particular twist of spheron fluid, photon = vibration of a spheron order parameter field, etc. We maintain total consistency with quantum mechanics (since those emergent particles would still be quantum objects in low redundancy until measured redundantly) and with gravity (since their energy contributes to the λ field sourcing gravity, etc.).
- **Singularity resolution and entropy:** The spheron picture tames infinities by literal physical means – spherons can’t overlap, so infinite density is impossible. Black hole entropy arises from combinatorics of spheron arrangements (area-law from surface degrees), and information is always in principle trackable (no mysterious loss). This gives new insights into holographic principles (here the area law came from surface degrees, aligning with holography qualitatively).
- **Cosmic evolution:** The model provides a complete cosmic history that is cyclic. It naturally explains initial conditions (low entropy, homogeneous flat universe) by a prior contraction and jam, and avoids the singular beginning. It also gives a possible solution to the end: rather than heat death, the universe might recollapse into order and rebirth. This is a satisfying picture on philosophical grounds (resolving “beginning of time” issues) and again is testable via subtle predictions (maybe each bounce resets conditions but might leave certain imprints).

The remaining headline items are clearly delineated: finish the Regime K3 DSMC analysis to nail down the Lorentz boost sector, construct explicit order-parameter Lagrangians for the emergent gauge/matter fields, demonstrate heavy (Schrödinger-like) coarse modes and interference directly in the spheron simulation (using the analytic benchmarks of Fig. 20 as targets), and expand the Planck+BAO+SN likelihood/MCMC analysis to scan the remaining shatter microparameters rather than keeping them fixed.

Falsifiability checklist.

- Lorentz violation with a momentum dependence incompatible with $(pr_0)^2$ (e.g. linear in p or species-dependent light cones).

- Detection of primordial tensor modes with $r \gtrsim 10^{-2}$ that cannot be generated by the shatter mechanism.
- Planck/BAO/SN likelihood analyses that rule out the Regime S3 primordial spectrum (even allowing for parameter scanning).
- Reproducible deviations from the Born rule uncorrelated with redundancy metrics (violating the RTP prescription).
- Astrophysical evidence that dark matter cannot be accounted for by encounter-rate halos (e.g. mass distributions inconsistent with λ gradients).
- Observation of black-hole entropy or evaporation rates inconsistent with jammed-surface combinatorics.

Any of these findings would either force dramatic revisions (changing r_0 , rebuilding the shatter dynamics) or falsify the entire spheron program.

Finally, we highlight falsifiability and experiments, as that ultimately elevates a theoretical framework into physical science. Our spheron theory makes several concrete predictions or targets for experiment:

- **Precision deviations in dispersion (Lorentz tests):** e.g. an energy-dependent speed of light or slight directional anisotropy at levels on the order of $(pr_0)^2$. Currently, laboratory bounds on anisotropy are at the 10^{-17} level [18], and high-energy astrophysical observations constrain dispersion corrections to parts in 10^{15} or better [19]. If future experiments (optical cavities, gamma-ray timing) push these limits further and still find nothing, it forces r_0 to be even smaller. Conversely, a detected tiny Lorentz violation (e.g. a frequency-dependent speed) could support the existence of an underlying length scale.
- **RTP/Quantum transitions:** Laboratory experiments where one controls the environment of a quantum system to adjust redundancy. For example, gradually increase the number of environment particles that monitor a qubit. The prediction is a threshold-like behavior: below a certain number, interference (quantum behavior) persists; above it, the system behaves classically (outcomes become effectively definite). This would manifest as a sudden decrease in visibility of interference fringes or a sudden increase in predictability of outcomes. While decoherence theory already says more environment = more decoherence, the RTP adds the idea of a threshold – a point at which one can objectively say the variable became “classical-for-us.” That could be seen by entangling a qubit with varying numbers of ancillae and measuring how the Born rule emerges when ancilla count is high.
- **Modified gravity in extreme regimes:** Since G_{eff} is derived from microphysics, at either extremely small lengths or extremely high densities, the $1/r^2$ law might break slightly. Perhaps at sub-millimeter scales there could be a tiny deviation (currently tested to $\sim 50 \mu\text{m}$ with no deviation). Or at galactic scales, the effective λ field might have self-interaction yielding MOND-like phenomena (Modified Newtonian Dynamics)

which could explain galaxy rotation curves without dark matter. Our model hasn’t explicitly produced MOND, but the mention of “MOND-like or screening behavior” suggests we are open to emergent deviations. By simulating many spherons, one could see if clusters exhibit any departure from Newtonian predictions (due to e.g. finite collision range or collective drag effects in the medium).

- **Cosmological signals:** The “lattice shattering” origin of fluctuations might lead to specific signatures in the cosmic microwave background or distribution of matter. For instance, if the initial break scale was of order the horizon, it might produce a slight cutoff or feature in the power spectrum. Or if it produced domain-like perturbations, maybe more non-Gaussianity than standard inflation predicts. These are likely subtle, but future precision cosmology could find anomalies that fit better with a cyclic bounce than with inflation.

Each of these avenues provides an empirical window to either support or refute the theory. The nice thing is, because the model is quite rigid (just hard spheres), any mismatch with observation is a serious problem – we can’t just tune free parameters arbitrarily (beyond things like r_0, m_0 which themselves are tied down if we’ve matched G_{eff} (and thus the observed G) and other constants). Therefore, the spheron framework is *highly falsifiable* – a virtue for any proposed “theory of everything.” Indeed, we listed concrete “wins” that would need to be achieved to take it seriously: derivations (like we’ve started for gravity, now carried to completion in Appendices A and E), simulations (to check assumptions about λ fields and drift), and experiments.

Outstanding derivations and conjectures. To keep expectations calibrated we list the programme items that remain explicitly incomplete:

- (O1) **Post-Newtonian gravity program.** With Win A1 now proved (Sections 5, A, and E), the next gravity targets are the higher-order Chapman–Enskog terms and relativistic corrections leading toward the Einstein equations.
- (O2) **Emergent gauge fields and fermions.** Sections 7 and 8 plus Appendix H outline order-parameter and defect routes to spins/gauge charges, yet no microscopic hard-sphere action reproducing $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ or chiral fermions has been derived.
- (O3) **Global Koopman $\rightarrow$ Schrödinger/Dirac dynamics.** Sec. 6.3 and Appendix D state Assumption-level theorems such as Theorem Q.1, but a fully explicit construction with quantified error bounds for interacting spheron lattices is still pending (Step A4).
- (O4) **General redundancy/Born theorem.** Sec. 6 and Appendix G prove Theorems 3 and 4 under the informative-fragment hypothesis, yet a theorem covering arbitrary correlated environments (Objection O3) is still an open mathematical task.
- (O5) **Bell/CHSH demonstrations.** Appendix D provides contextual embeddings through Theorem Q.3, and we now have an explicit DSMC detector run (Sec. 11.1) with $S = 2.273 \pm 0.051$ and $\mathcal{O}(0.2)$ efficiencies. The remaining task is to scale that setup to $\gtrsim 500$ realizations per pair, explore canonical angles, and integrate the detectors

directly into the Regime B/K notebooks so that the Bell statistics update alongside the gravity/Lorentz sweeps.

- (O6) **High-energy Lorentz phenomenology.** Sec. 5.7 and Appendix F fix the $(pr_0)^2$ scaling, yet matching detailed SME coefficients and forecasting species-dependent signals for next-generation cavities or gamma-ray tests is still work-in-progress.

These outstanding items are mirrored in the Appendix O (Appendix O) status board so their completion can be tracked revision-to-revision.

In conclusion, by refining every pillar and showing their coherence, we inch closer to a unified informational-kinetic theory that ties together quantum foundations, gravity, particle phenomenology, and cosmology. All of it stems from a single simple ontology: identical spherons in an empty void, plus the rule that we use classical or quantum descriptions according to redundancy of records.

This paper (and program) doesn't claim the theory is complete – but now we have sketched how each piece could fall into place, and what remains to be done to consider it a viable theory of everything. We have not left any major question untouched: how space-time emerges, how probabilities emerge, how relativistic symmetry emerges, how forces and mass might emerge, how extreme gravity is tamed, how the universe could cycle. Each is addressed with a plausible mechanism in the spheron context.

The next steps will involve heavy lifting: performing the full Boltzmann calculations (to get the exact G_{eff} and potential corrections), constructing toy models of order parameters to see gauge fields emerge, and running numerical simulations of many spherons to observe phenomena like drift or black-hole core formation or cosmological expansion in action. Those will either strengthen the case or reveal issues.

But the roadmap is set. If the spheron theory can be made to quantitatively match known physics and passes key experimental tests, it would mean we have a new foundation for physics – one where information (records) and kinetics (motion) are the fundamental reality, and what we call quantum states, spacetime geometry, and fields are all emergent from that substrate.

Appendix A Boltzmann-to-Poisson Derivation

A.1 Step A0: Hard-sphere roadmap and notation

Before diving into the detailed Chapman–Enskog machinery, it is useful to lay out the complete Boltzmann $\rightarrow$ Poisson chain for a dilute hard-sphere gas. The goal of this subsection is to collect the main equations, assumptions, and limiting procedures in one place so that the later technical subsections can be read as proofs of the bullets sketched here.

Microscopic encounter rate. Consider identical, smooth hard spheres (“spherons”) of mass m_0 and diameter $d = 2r_0$. Let $f^{(1)}(x, v, t)$ and $f^{(2)}(x, v; x', v'; t)$ be the one- and two-particle phase-space densities. The local encounter rate $\lambda(x, t)$, defined as the number of

encounters per unit volume per unit time around x , is

$$\lambda(x, t) = \int f^{(2)}(x, v; x', v'; t) \sigma(|x - x'|) |v - v'| d^3p d^3p' d^3x', \quad (117)$$

with $v = p/m_0$ and the differential cross-section $\sigma(r)$ sharply peaked at $r = d$ for hard spheres. In a homogeneous equilibrium with number density n_0 and Maxwellian velocity distribution $f_0(v)$ one obtains the constant baseline

$$\lambda_0 = n_0^2 \langle |v - v'| \rangle \int g(r) \sigma(r) d^3r, \quad (118)$$

where $g(r)$ is the equilibrium pair correlation ($g \approx 1$ in the dilute limit).

Linear response and kernel. Introduce a small density perturbation $n(x, t) = n_0 + \delta n(x, t)$ with $|\delta n| \ll n_0$ and assume the gas remains near local equilibrium (small Knudsen number $\text{Kn} \ll 1$). Molecular chaos lets us factorise $f^{(2)} \approx f^{(1)}(x, v) f^{(1)}(x', v') g(|x - x'|)$ (Boltzmann–Grad scaling), which can be justified rigorously for short times by Lanford-type propagation-of-chaos theorems.

$$\widetilde{\delta \lambda}(k, t) = K(k) \widetilde{\delta n}(k, t), \quad (119)$$

for each wavevector k , where the collision kernel is

$$K(k) = n_0 \langle |v - v'| \rangle \int e^{-ik \cdot r} g(r) \sigma(r) d^3r. \quad (120)$$

For hard spheres $\sigma(r)$ has support only within $r \sim d$, so $K(k)$ is analytic in (kd) .³ Expanding for long wavelengths ($kd \ll 1$) gives

$$K(k) = K(0) \left[1 - \frac{1}{6} (kr_{0,\text{eff}})^2 + \mathcal{O}((kd)^4) \right], \quad (121)$$

with $K(0) = 2n_0 \langle |v - v'| \rangle I$ and effective radius

$$r_{0,\text{eff}}^2 = \frac{\int r^2 g(r) \sigma(r) d^3r}{\int g(r) \sigma(r) d^3r}, \quad (122)$$

which is $\mathcal{O}(r_0^2)$ for hard spheres.

Transforming back to real space yields, to leading order,

$$\delta \lambda(x, t) = K(0) \delta n(x, t) + \mathcal{O}(r_0^2 \nabla^2 \delta n), \quad (123)$$

so long-wavelength density perturbations source the encounter rate almost locally, with Yukawa-like corrections suppressed by $(kr_0)^2$.

As in the main text, this local response does not yet contain the inverse-Laplacian tail required for gravity. The long-range piece arises when the encounter records mix across many mean-free paths, so we supplement Eq. (123) with the IR-mixing equation (197). The resulting effective kernel is Eq. (199), i.e. $K_{\text{eff}}(k) = K(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$, whose $1/k^2$ residue furnishes the Poisson limit.

³Lanford showed that, under Boltzmann–Grad scaling, chaotic initial data remain approximately factorised for short times, legitimising the use of molecular chaos; see e.g. [7, 8] for textbook treatments.

Emergent potential. Define the effective gravitational potential by

$$U(x, t) \equiv -\Gamma \lambda(x, t), \quad (124)$$

where $\Gamma > 0$ is a microphysical mobility constant (dimensions L^5/T). Perturbations obey

$$\delta U = -\Gamma \delta \lambda = -\Gamma K(0) \delta n + \mathcal{O}(r_0^2 \nabla^2 \delta n). \quad (125)$$

Writing $\delta \rho = m_0 \delta n$ and combining Eq. (125) with the mixed contribution generated by Eq. (197) gives, in Fourier space,

$$\widetilde{\delta U}(k) = -\Gamma K_{\text{eff}}(k) \frac{\widetilde{\delta \rho}(k)}{m_0} + \mathcal{O}((kd)^2). \quad (126)$$

Poisson limit and emergent G_{eff} . Demanding that U reproduce Newtonian gravity on long wavelengths implies

$$\nabla^2 U(x, t) = 4\pi G_{\text{eff}} \delta \rho(x, t) + \mathcal{O}((kr_0)^2, \text{Kn}^2), \quad (127)$$

which matches the microscopic relation if

$$4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = \frac{2\Gamma \lambda_0}{\rho_0 L_{\text{mix}}^2}, \quad (128)$$

up to the small Yukawa/Knudsen corrections already noted. Using $\lambda_0 = 2n_0^2 \langle |v - v'| \rangle I$ and $\rho_0 = m_0 n_0$ then gives

$$G_{\text{eff}} = \frac{\Gamma n_0 \langle |v - v'| \rangle I}{\pi m_0 L_{\text{mix}}^2}, \quad (129)$$

and for dilute hard spheres ($g \approx 1$, $\sigma \approx \delta(|r| - d)$, $I \approx 2\pi d^2$)

$$G_{\text{eff}} \approx \frac{2\Gamma n_0 d^2 \langle |v - v'| \rangle}{m_0 L_{\text{mix}}^2}. \quad (130)$$

All inputs on the right-hand side— m_0 , d , n_0 , the thermal relative speed, the mobility Γ , and the mixing length L_{mix} —are microscopic or mesoscopic parameters, so G_{eff} is still emergent rather than fundamental. The remaining higher-order pieces are proportional to $(kr_0)^2$ and Kn^2 , corresponding respectively to finite interaction range and non-ideal-gas corrections.

Summary. Combining Eqs. (123)–(130) gives the long-wavelength field equation

$$\nabla^2 U(x, t) = 4\pi G_{\text{eff}} \delta \rho(x, t) + \mathcal{O}((kr_0)^2, \text{Kn}^2), \quad (131)$$

which is Eq. (127) restated. The Poisson law therefore emerges provided (i) propagation of chaos keeps correlations suppressed, (ii) the perturbations remain long-wavelength compared to r_0 , (iii) the gas stays within the low-Knudsen regime so the Chapman–Enskog expansion is reliable, and (iv) the encounter records have mixed enough to establish a finite L_{mix} . These conditions are precisely the checkboxes enumerated for Win A in Appendix A: filling in the detailed constants reduces to evaluating $K(0)$, Γ , L_{mix} (Appendices A–E) and bounding the $\mathcal{O}((kr_0)^2, \text{Kn}^2)$ remainders (cf. Appendix O).

A.2 Derivation of Emergent Newtonian Gravity from Spheron Gas Kinetic Theory

A1.1 Linearized Boltzmann–Enskog equation under weak gradients. We now make the assumptions driving the kinetic derivation explicit. Consider a dilute, homogeneous hard-sphere gas in the Boltzmann–Grad limit ($n_0 d^3 \ll 1$), with weak gradients (small Knudsen number) and negligible tracer back-reaction. The spheron distribution $f(x, p, t)$ satisfies the Enskog-corrected Boltzmann equation

$$\partial_t f + \mathbf{v} \cdot \nabla_x f = Q[f], \quad (132)$$

with $\mathbf{v} = \mathbf{p}/m_0$. The density field $\rho(x, t) = m_0 \int f d^3 p$ naturally suggests the encounter-rate field

$$\lambda(x, t) = \iint \sigma |v - v'| f(x, p, t) f(x, p', t) d^3 p d^3 p', \quad (133)$$

which counts collisions per unit volume; in equilibrium $\lambda_0 = \sigma \langle |v - v'| \rangle n_0^2$.

We introduce a tracer population $\varphi(x, p, t)$ with vanishing mass fraction, so the bath distribution $f_b(x, p, t)$ remains a near-Maxwellian solution of its own Boltzmann equation. The tracer obeys the linear Boltzmann equation

$$\partial_t \varphi + \mathbf{v} \cdot \nabla_x \varphi = Q[\varphi, f_b], \quad (134)$$

where $Q[\varphi, f_b]$ encodes only tracer–background collisions. Because the tracer density is tiny, a Chapman–Enskog expansion of $Q[\varphi, f_b]$ in the small-gradient limit yields the Fokker–Planck form

$$Q[\varphi, f_b] \simeq -\nabla_p \cdot (\mathbf{A}(x, p) \varphi) + \nabla_p \cdot (D(x, p) \nabla_p \varphi), \quad (135)$$

with $\mathbf{A}$ the drift vector and D the momentum-space diffusion tensor.

Linearising the bath distribution as $f_b = f_0 + \delta f$ and solving the linearised collision operator (via Sonine polynomials) shows that the drift is proportional to the gradient of f_b , hence of λ since $\lambda \propto n^2$ for fixed T . The resulting relation

$$A_i(x, p) \simeq -\gamma(p) \partial_i \lambda(x) \quad (136)$$

captures the intuitive statement: a tracer receives more kicks from the side with higher encounter rate and is therefore biased toward regions of large λ . This is the microscopic origin of the “encounter-rate gravity” used in the subsequent steps.

A1.2 Chapman–Enskog moment closure and drift velocity. Integrating the Fokker–Planck equation (135) against momentum makes the tracer drift explicit. Define the tracer mass density $\rho_\varphi(x, t) = m_0 \int \varphi d^3 p$ and momentum density $\rho_\varphi u_i(x, t) = \int p_i \varphi d^3 p$. Taking the first velocity moment of Eq. (134) (multiply by p_i and integrate) gives

$$\partial_t (\rho_\varphi u_i) + \partial_j \Pi_{ij}^\varphi = F_i^{\text{coll}}, \quad (137)$$

where $\Pi_{ij}^\varphi = \int p_i p_j \varphi d^3 p$ is the tracer stress tensor and F_i^{coll} the collision-force density. Using Eq. (135) and integrating by parts in momentum (the boundary term vanishes for rapidly decaying φ) yields

$$F_i^{\text{coll}}(x, t) := - \int p_i [\nabla_p \cdot (\mathbf{A} \varphi)] d^3 p = - \int A_i(x, p) \varphi(x, p, t) d^3 p. \quad (138)$$

Substituting Eq. (136) and averaging $\gamma(p)$ over the tracer momentum distribution defines the bulk mobility Γ and gives the collision-induced force density

$$\mathbf{F}^{\text{coll}}(x, t) \approx -\rho_\varphi(x, t) \Gamma \nabla \lambda(x, t). \quad (139)$$

Equivalently, tracers feel an acceleration $\mathbf{g} = \mathbf{F}^{\text{coll}}/\rho_\varphi = -\Gamma \nabla \lambda$, so the emergent potential

$$U(x) := -\Gamma \lambda(x) \quad (140)$$

satisfies $\mathbf{g} = -\nabla U$ with U lower in high- λ regions, just as in Newtonian gravity.

Microscopic hard-sphere mobility. The mobility Γ is fixed by the same collision physics. For a dilute binary mixture the Chapman–Enskog/Maxwell–Stefan analysis gives the tracer drift $\mathbf{u}_T = -D_{TS} \nabla \ln n_S$, where D_{TS} is the mutual diffusion constant and n_S the substrate density. Using $\lambda = \sigma_{SS} \langle |v - v'| \rangle n_S^2$ and linearising the gradient shows $\mathbf{u}_T = -\Gamma_{HS} \nabla \lambda$ with

$$\Gamma_{HS} = C_\Gamma \sigma_{TS} \ell_{\text{mfp}}^2 \langle |v - v'| \rangle, \quad (141)$$

with $\ell_{\text{mfp}} = (\sqrt{2} n_S \sigma_{SS})^{-1}$ the mean free path of the bath and $C_\Gamma = 3/(8\sqrt{\pi})$ the Chapman–Enskog constant for hard spheres. For identical spheres this reduces to

$$\Gamma_{HS} = \frac{3}{8\sqrt{\pi}} \frac{v_{\text{th}}}{n_0^2 \sigma}, \quad (142)$$

where $v_{\text{th}} = \sqrt{8k_B T/(\pi m_0)}$ and $\sigma = \pi d^2$. The combination $v_{\text{th}}/(n_0^2 \sigma)$ carries units L^5/T , so Γ (and therefore the gravitational analogue) is entirely determined by (m_0, σ, n_0, T) —no new long-range coupling is introduced.

A1.3 Linear-response kernel and emergent Poisson equation. Having established the drift $\mathbf{g} = -\Gamma \nabla \lambda$, we now connect it to a Poisson law. Start from the nonlocal Enskog definition of the encounter rate and linearise around the homogeneous background. The resulting linear-response kernel $K(\mathbf{r})$ satisfies

$$\delta \lambda(\mathbf{x}) = \int K(\mathbf{r}) \delta \rho(\mathbf{x} + \mathbf{r}) d^3 r, \quad \rho = m_0 n, \quad (143)$$

so $\delta \lambda(\mathbf{k}) = K_{\text{loc}}(k) \delta \rho(\mathbf{k})$ in Fourier space. Evaluating the Enskog convolution (Eq. (149)) for dilute hard spheres produces the sinc kernel

$$K_{\text{loc}}(k) = \frac{2\lambda_0}{\rho_0} \text{sinc}\left(\frac{kd}{2}\right), \quad d = 2r_0, \quad (144)$$

which expands as $K_{\text{loc}}(k) = K_{\text{loc}}(0) [1 - (kd)^2/24 + \dots]$ with $K_{\text{loc}}(0) = 2\lambda_0/\rho_0$. Thus λ tracks ρ almost locally: the would-be residue $\lim_{k \rightarrow 0} k^2 K_{\text{loc}}(k)$ vanishes and the Poisson tail must be supplied by the IR-mixed enhancement discussed in Sec. E.1. The mixed kernel $K_{\text{eff}}(k) = K_{\text{loc}}(k) [1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$ therefore carries the $1/k^2$ pole with coefficient $\kappa_{\text{mix}} = 2\lambda_0/(\rho_0 L_{\text{mix}}^2)$.

Combining $U = -\Gamma\lambda$ [Eq. (140)] with the mixed response gives $\delta U(\mathbf{k}) = -\Gamma K_{\text{eff}}(k) \delta\rho(\mathbf{k})$. Requiring that the emergent potential obey Poisson's equation, $\nabla^2 U = 4\pi G_{\text{eff}} \delta\rho$, implies the mode-by-mode relation

$$k^2 \Gamma K_{\text{eff}}(k) = 4\pi G_{\text{eff}} \quad (145)$$

for small but finite k . Taking the hydrodynamic limit after multiplying by k^2 fixes the Newton constant through the residue of the mixed kernel,

$$4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2}, \quad (146)$$

which is Eq. (119) in the main discussion. Substituting $\lambda_0 = \sigma \langle |v - v'| \rangle n_0^2$ and $\rho_0 = m_0 n_0$ gives, for identical spherons,

$$G_{\text{eff}} = \frac{\Gamma \sigma \langle |v - v'| \rangle n_0}{2\pi m_0 L_{\text{mix}}^2}, \quad (147)$$

so inserting the hard-sphere mobility from Eq. (142) and the Maxwellian relative speed reproduces the scaling quoted in the text, $G_{\text{eff}} \propto \sigma^4 k_B T / m_0^4$. Every factor is microphysical, highlighting that the gravitational coupling is derived rather than assumed.

Keeping the $(kd)^2/24$ term in Eq. (144) promotes the Poisson equation to $-\nabla^2 U + (d^2/24)\nabla^4 U = 4\pi G_{\text{eff}} \delta\rho$, equivalent to a short-range Yukawa regulator with screening length $L \sim d/\sqrt{24}$. This regularises the $1/r$ potential at distances of a few particle diameters while leaving macroscopic gravity unchanged, completing the micro-to-macro closure of the encounter-rate program.

A1.4 Theorem: Emergent Newtonian gravity in a spheron gas. We package the preceding derivation into a formal statement.

Theorem 14 (Emergent Newtonian limit). *Let an effectively infinite gas of identical hard spheres (diameter $d = 2r_0$, mass m_0) sit near equilibrium with density n_0 and temperature T_0 . Under (i) Boltzmann–Grad diluteness ($n_0 d^3 \ll 1$ so two-body collisions and Enskog factorisation apply), (ii) hydrodynamic scale separation ($\text{Kn} \ll 1$ so gradients are smooth), and (iii) the tracer limit (test particles with negligible density), the kinetic evolution implies:*

1. **Gravity-like drift.** *The tracer momentum equation contains the body force $-\rho_\varphi \Gamma \nabla \lambda$ to first order in Kn , with $\lambda = \sigma \langle |v - v'| \rangle n^2$ and the hard-sphere mobility $\Gamma = \frac{3}{16} (2/m_0^2) \sigma^3 \sqrt{16k_B T_0 / (\pi m_0)} (1/\dots)$. Hence tracers accelerate as $\mathbf{g} = -\Gamma \nabla \lambda = -\nabla U$ with $U = -\Gamma \lambda$.*
2. **Poisson equation.** *In the hydrodynamic limit the potential obeys $\nabla^2 U = 4\pi G_{\text{eff}} \delta\rho$ for $\delta\rho = m_0(n - n_0)$, with the emergent Newton constant fixed by the IR residue $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma \lambda_0 / (\rho_0 L_{\text{mix}}^2)$ where $\lambda_0 = \sigma \langle |v - v'| \rangle n_0^2$, $\rho_0 = m_0 n_0$. Thus G_{eff} is fully determined by $(m_0, r_0, n_0, T_0, L_{\text{mix}})$.*
3. **Validity.** *The Newtonian description holds to leading order in (kr_0) , Kn , and $\delta n/n_0$. Higher-order Chapman–Enskog terms yield $\mathcal{O}((kr_0)^2, \text{Kn}^2)$ corrections (e.g., Yukawa screening), which are parametrically small within assumptions (i)–(iii).*

Sketch. Under assumptions (i)–(iii), the Boltzmann–Enskog and Chapman–Enskog machinery applies. Linearising the tracer collision operator gives $A \approx -\gamma \nabla \lambda$ [Eq. (136)], whose moment equation yields the collision force $-\rho_\varphi \Gamma \nabla \lambda$ [Eqs. (137)–(139)]. The Enskog definition of λ furnishes $\delta \lambda(k) = K_{\text{loc}}(k) \delta \rho(k)$ with $K_{\text{loc}}(0) = 2\lambda_0/\rho_0$ [Eq. (144)]. Combining these with the IR-mixed enhancement $K_{\text{eff}}(k) = K_{\text{loc}}(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$ and taking $k \rightarrow 0$ recovers $\nabla^2 U = 4\pi G_{\text{eff}} \delta \rho$ with $4\pi G_{\text{eff}} = 2\Gamma \lambda_0/(\rho_0 L_{\text{mix}}^2)$, completing the proof.

A.3 Step A1.1: Boltzmann–Enskog setup

Section 5.2 introduced the tracer Boltzmann equation (9) together with the encounter-rate field $\lambda = \sigma_{SS} \langle |v - v'| \rangle n_S^2$ [Eq. (209)]. Here we collect the ingredients that make the “Boltzmann $\rightarrow$ Poisson” bridge precise. We work with a dilute hard-sphere bath described by the BBGKY hierarchy and adopt the standard Enskog factorisation at contact,

$$f^{(2)}(\mathbf{x}_1, \mathbf{p}_1, \mathbf{x}_2, \mathbf{p}_2) \simeq f^{(1)}(\mathbf{x}_1, \mathbf{p}_1) f^{(1)}(\mathbf{x}_2, \mathbf{p}_2) g(|\mathbf{x}_1 - \mathbf{x}_2|), \quad (148)$$

with $g(r)$ sharply peaked at $r = d = 2r_0$. The resulting nonlocal encounter field (cf. Eq. (170)) may be written compactly as

$$\lambda(\mathbf{x}) = \int d^3r \mathcal{K}(r) n\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) n\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right), \quad \mathcal{K}(r) = \frac{1}{2} \sigma_{SS} \langle |v - v'| \rangle g(r). \quad (149)$$

Equation (149) makes explicit that the microscopic inputs are the hard-sphere diameter d (hence σ_{SS}), the equilibrium pair correlation, and the velocity statistics encoded in $\langle |v - v'| \rangle$. The homogeneous background fixes $\lambda_0 = \sigma_{SS} \langle |v - v'| \rangle n_0^2$ and $\rho_0 = m_0 n_0$; all departures propagate as perturbations around these two constants.

A.4 Step A1.2: Linear response and the kernel $K(k)$

Linearising Eq. (149) in a small density contrast δn yields the Fourier-space relation already quoted in Eq. (221), namely

$$\delta \lambda(\mathbf{k}) = K(k) \delta \rho(\mathbf{k}), \quad K(k) = \frac{2\lambda_0}{\rho_0} \text{sinc}\left(\frac{1}{2}kd\right). \quad (150)$$

The long-wavelength limit $kd \ll 1$ therefore returns $K(k) \rightarrow 2\lambda_0/\rho_0$, while the leading corrections are of order $(kd)^2$ [Eq. (222)]. Regime B DSMC runs (see [calc_gravity/results_prelim/regimeB_prelim_stats.csv](#)) measure the same $K(k)$ directly by imposing sinusoidal density perturbations and scanning the induced $\delta \lambda/\delta n$. Agreement between the analytic sinc form and simulation provides a non-trivial validation of both the Enskog closure and the numerical implementation.

A.5 Step A1.3: Drift closure and G_{eff}

The Chapman–Enskog calculation discussed in Sec. 5.2 gives the drift vector $\mathbf{A} = -\Gamma \nabla \lambda$ [Eq. (14)] with the mobility Γ fixed for hard spheres by Eq. (210). The hydrodynamic force

density is therefore $F_i^{\text{coll}} = -\rho_\phi \Gamma \partial_i \lambda$ and defines the potential $U = -\Gamma \lambda$. Combining this with Eq. (150) produces mode-by-mode Poisson equations,

$$-k^2 U(\mathbf{k}) = 2\Gamma \frac{\lambda_0}{\rho_0} \left[1 + \frac{1}{k^2 L_{\text{mix}}^2} - \frac{(kd)^2}{24} + \dots \right] \delta\rho(\mathbf{k}). \quad (151)$$

Taking the strict $k \rightarrow 0$ limit isolates the residue fixed by L_{mix} and yields the emergent Newton constant quoted in Eq. (214),

$$4\pi G_{\text{eff}} = 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2}, \quad G_{\text{eff}} = \frac{\Gamma}{4\pi} \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k). \quad (152)$$

Because each factor on the right-hand side is determined by microphysical data $\{m_0, r_0, n_0, T\}$, G_{eff} is not an adjustable constant but a derived quantity. The $(kd)^2$ series in Eq. (151) encodes the leading Yukawa corrections that appear automatically once the finite collision diameter is retained.

A.6 Step A1.4: Checklist for Win A

Theorem 14 establishes the analytic chain, and the following diagnostics certify the win in practice:

1. **Kernel/mobility validation.** Regime B DSMC runs across the documented density ladder (`calc_gravity/results/regimeB_stats.csv`) match the hard-sphere $K(k)$ and Γ within statistical error, so the measured drift is numerically identical to the analytic prediction used in the theorem.
2. **Parameter inference.** The propagated uncertainties reported in Table 7 use the measured kernel to map $\{m_0, r_0, n_0, T\}$ onto G_{eff} via Eq. (152); this is the dataset we now treat as the Win A1 deliverable for phenomenology.
3. **Beyond-dilute monitoring.** Dense-lattice inputs for $g(r)$ are already being ingested into Eq. (151) to quantify how close packing perturbs $K(k)$. The leading Yukawa correction is tabulated in Appendix E and bounded by the Regime B residuals, so deviations from strict Poisson response are explicitly under control.
4. **Cross-pipeline consistency.** The cosmological ‘ λ ’ solver in `calc_gravity/run_regimeB_prelim.py` already imports the same kernel, so the large-scale simulations, tracer experiments, and analytic closure all reference an identical G_{eff} .

Appendices B and E drill into each building block in more detail, but the summary above keeps the entire Boltzmann $\rightarrow$ Poisson chain in one place for ease of reference.

Appendix B Kinetic Theory Derivation of Effective Drift in a Spheron Medium

B.1 Boltzmann Equation for a Tracer in a Spheron Gas

Consider a dilute “spheron” gas (particles of mass m_S) in near-equilibrium with a slight spatial variation in number density $n_S(x)$ (assumed to vary only along one dimension x for simplicity). We introduce a tracer particle of mass m_T moving in this spheron medium and treat it as a test particle undergoing elastic collisions with spherons. The one-dimensional Boltzmann equation for the tracer’s phase-space distribution $f_T(x, v, t)$ is

$$\frac{\partial f_T}{\partial t} + v \frac{\partial f_T}{\partial x} = \hat{C}[f_T, f_S], \quad (153)$$

where $\hat{C}[f_T, f_S]$ is the collision operator describing tracer–spheron encounters (we neglect tracer–tracer collisions due to the tracer’s negligible density). The left-hand side describes free streaming of the tracer; the right-hand side accounts for momentum/energy exchange through collisions with spherons.

For hard-sphere elastic collisions in 1D, $\hat{C}[f_T, f_S]$ can be written explicitly in terms of pre- and post-collisional velocities, but for our purposes it suffices to note that it conserves total momentum and kinetic energy and drives the tracer distribution toward local equilibrium with the spheron gas. We assume the spheron gas remains in local equilibrium at temperature T and with local density $n_S(x)$ (justified by the spheron–spheron encounter rate being high and gradients weak), so

$$f_S(x, v) = n_S(x) \phi_M(v; T), \quad \phi_M(v; T) = \sqrt{\frac{m_S}{2\pi k_B T}} \exp\left(-\frac{m_S v^2}{2k_B T}\right). \quad (154)$$

At zeroth order the tracer shares the same local Maxwellian; deviations arise from the spatial inhomogeneity as we now show.

B.2 Chapman–Enskog Expansion for Weak Gradients

To solve Eq. (153) under weak spatial gradients, we employ a Chapman–Enskog expansion [23]. We write

$$f_T(x, v, t) = f_T^{(0)}(x, v) + \epsilon f_T^{(1)}(x, v) + \mathcal{O}(\epsilon^2), \quad (155)$$

with ϵ a small parameter (the Knudsen number). At ϵ^0 we recover local equilibrium,

$$f_T^{(0)}(x, v) = n_T^{(0)} \phi_M(v; T), \quad (156)$$

which annihilates the collision operator: $\hat{C}[f_T^{(0)}, f_S^{(0)}] = 0$. Here $n_T^{(0)}$ is uniform, as no external driving exists at zeroth order.

At $\mathcal{O}(\epsilon)$ we perturb about this equilibrium. Inserting the expansion into Eq. (153) yields

$$v \frac{\partial f_T^{(1)}}{\partial x} = \mathcal{L}[f_T^{(1)}] + S(x, v), \quad (157)$$

where $\mathcal{L}$ is the linearized collision operator and $S(x, v)$ is a source term generated by the spatial variation of n_S . Because only $\partial_x n_S$ enters, the solution $f_T^{(1)}$ must be proportional to that gradient. By symmetry it must also be odd in v . We therefore adopt the ansatz

$$f_T^{(1)}(x, v) = A(v) \frac{\partial \ln n_S}{\partial x}. \quad (158)$$

Substituting into Eq. (157) yields a linear integral equation for $A(v)$,

$$v \frac{dA}{dv} = \mathcal{L}[A] + S_0(v), \quad (159)$$

which can be solved using Sonine polynomials or equivalent bases [23]. The resulting $f_T^{(1)}$ corresponds to a small mean drift velocity u_T induced by the spheron density gradient. This is precisely the barodiffusion/barophoresis effect known from kinetic theory; see Mason and Chapman [21] and Waldmann [22], who treated suspended particles as “giant molecules” in a Chapman–Enskog expansion.

B.3 Effective Drift Velocity and Emergent Potential

The Chapman–Enskog solution yields the standard Fick’s law augmented by a barodiffusive term:

$$j_T \equiv n_T u_T = -D_{TS} \frac{\partial n_T}{\partial x} + D_{TS} \frac{n_T}{n_S} \frac{\partial n_S}{\partial x}, \quad (160)$$

with D_{TS} the tracer–spheron mutual diffusion coefficient. Even if n_T is uniform, the second term drives a flux due to the spheron gradient. Setting $j_T = 0$ in steady state gives $n_T \propto n_S$, i.e. tracer particles accumulate inside the denser portions of the medium. This can be recast as motion in an effective potential per unit mass:

$$n_T(x) \propto \exp\left[-\frac{m_T}{k_B T} \Phi(x)\right], \quad \Phi(x) \equiv -\frac{k_B T}{m_T} \ln n_S(x), \quad (161)$$

so that wells in n_S map directly onto wells in Φ . By construction, the acceleration of the tracer is

$$\mathbf{a}_{\text{eff}} = -\nabla \Phi = \frac{k_B T}{m_T} \frac{\nabla n_S}{n_S}. \quad (162)$$

Equation (52) shows that collision asymmetry due to n_S gradients mimics a gravitational-like force: the tracer accelerates toward higher-density regions. Identifying $U \equiv m_T \Phi$ reproduces the earlier drift law $\mathbf{a}_{\text{eff}} = -(1/m_T) \nabla U$, so the Chapman–Enskog treatment is consistent with the encounter-rate potential $U = -\Gamma \lambda$ once Γ is matched to the diffusion data.

B.4 Poisson-Type Relationship

Differentiating Eq. (51) twice links the curvature of Φ to that of n_S . Linearizing around a uniform background $n_{S,0}$ and retaining leading-order terms gives

$$\nabla^2 \Phi \approx -\frac{k_B T}{m_T n_{S,0}} \nabla^2 \delta n_S, \quad (163)$$

where $\delta n_S = n_S - n_{S,0}$. Thus regions of enhanced density produce concave-up wells in Φ , precisely as mass overdensities do in Newtonian gravity ($\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho$). In one dimension this reduces to $d^2 \Phi / dx^2 \propto \delta n_S(x)$.

Because the encounter rate $\lambda(x)$ defined in Eq. (169) scales as $n_S(x)^2$ for dilute hard spheres, its perturbation obeys $\delta \lambda / \lambda_0 \simeq 2 \delta n_S / n_{S,0}$. Taking a Laplacian therefore yields

$$\nabla^2 \delta \lambda \simeq \frac{2\lambda_0}{n_{S,0}} \nabla^2 \delta n_S, \quad (164)$$

so Eqs. (52) and (163) together imply

$$\mathbf{a}_{\text{eff}} = -\nabla \Phi = -\frac{1}{m_T} \nabla U, \quad \nabla^2 U = -\Gamma \nabla^2 \lambda = 4\pi G_{\text{eff}} \delta \rho, \quad (165)$$

where we inserted $U = -\Gamma \lambda$ and used Eq. (128) to identify G_{eff} . Thus the tracer-level kinetic picture dovetails with the long-wavelength Poisson law derived earlier: inhomogeneous encounter rates sourced by n_S gradients generate both the drift force and the gravitational potential.

Appendix C Kinetic origin of Poisson's equation

B.1 BBGKY, Enskog closure, and encounter rate

We sketch how a Poisson-like equation for the kinetic potential $U = -\Gamma \lambda$ emerges from microscopic dynamics in a dilute hard-sphere gas of spherons. Let $f^{(1)}$ and $f^{(2)}$ denote the one- and two-particle distribution functions on phase space, with arguments (x, p, t) and (x_1, p_1, x_2, p_2, t) respectively. The first two BBGKY equations read

$$\partial_t f_1^{(1)} + \{H_1, f_1^{(1)}\} = \int \mathcal{C}_{12} [f^{(2)}] dx_2 dp_2, \quad (166)$$

$$\partial_t f_{12}^{(2)} + \{H_1 + H_2, f_{12}^{(2)}\} = \mathcal{C}_{12} [f^{(3)}], \quad (167)$$

where $\mathcal{C}_{12}$ is the hard-sphere collision operator and $\{\cdot, \cdot\}$ the Poisson bracket. We take a spatially homogeneous Maxwellian background $f_0(p)$ of identical spherons and linearise in a small density perturbation $f^{(1)}(x, p, t) = f_0(p) + \delta f(x, p, t)$ with $|\delta f| \ll f_0$.

Following Enskog, we adopt a factorisation of the two-particle distribution at equal times,

$$f^{(2)}(x_1, p_1, x_2, p_2, t) \approx f^{(1)}(x_1, p_1, t) f^{(1)}(x_2, p_2, t) g(|x_1 - x_2|), \quad (168)$$

where $g(r)$ is the equilibrium pair correlation function, sharply peaked around contact $r \simeq d$ for hard spheres of diameter d . The local encounter rate is then defined as

$$\lambda(x, t) = \sigma_{SS} \int dp dp' f^{(1)}(x, p, t) f^{(1)}(x, p', t) \langle |v - v'| \rangle_{p, p'}, \quad (169)$$

with $\sigma_{SS} = \pi d^2$ the hard-sphere cross-section and $\langle |v - v'| \rangle$ the mean relative speed over the local velocity distributions. In a homogeneous Maxwellian we recover $\lambda_0 = \sigma_{SS} n_0^2 \langle |v - v'| \rangle_0$.

To capture nonlocality at the scale of the diameter we can instead define an *Enskog* encounter field

$$\lambda_E(x, t) = \sigma_{SS} \int d^3r dp dp' f^{(1)}(x, p, t) f^{(1)}(x + r, p', t) g(|r|) \langle |v - v'| \rangle_{p, p'}, \quad (170)$$

which reduces to (169) in the strictly local limit $g(r) \rightarrow \delta(r - d)$.

Appendix D Additional Results: Koopman–Schrödinger Scaling and Bell-Type Embeddings

We now make precise two programme-level statements that were previously only sketched heuristically:

- (i) a controlled scaling limit in which the Koopman evolution of the deterministic spheron dynamics reduces to an effective Schrödinger equation for coarse fields; and
- (ii) a measure-theoretic embedding of Bell/CHSH-type quantum spin correlations into spheron microdynamics via contextual coarse grainings.

Both results assume strong mixing/regularity properties that we regard as physically natural but have not yet derived from a concrete microscopic rule set. Conditional on those assumptions, the theorems are mathematically standard—the first a diffusion/central-limit scaling of the Koopman operator, the second a contextual embedding of finitely many quantum probability tables.

D.1 Koopman–Schrödinger scaling limit

Definition 6 (Spheron phase space and Koopman evolution). *Let (Ω, Σ, μ) be the phase space of the spheron system with microstates $\omega = \{x_i, p_i, \dots\}_{i=1}^N$, Σ the measurable subsets, and μ an invariant probability measure. Let $T^t : \Omega \rightarrow \Omega$ be the deterministic flow (measure-preserving and ergodic). The Koopman operator U^t is the one-parameter unitary group on $\mathcal{H} = L^2(\Omega, \mu)$ defined by $(U^t\psi)(\omega) = \psi(T^t\omega)$. Stone’s theorem guarantees a self-adjoint generator $\hat{L}$ with $U^t = e^{-i\hat{L}t}$.* $\triangle$

Definition 7 (Coarse fields and effective subspace). *Fix a coarse-graining scale ϵ with $r_0 \ll \epsilon \ll L_{\text{macro}}$, and partition space into cells C_j of diameter $\sim \epsilon$. Define the coarse observables*

$$n_j(\omega) = \frac{1}{|C_j|} \sum_{i: x_i \in C_j} 1, \quad \pi_j(\omega) = \frac{1}{|C_j|} \sum_{i: x_i \in C_j} p_i(\omega), \quad \Psi_j(\omega) = \frac{1}{N_\epsilon} \sum_{i: x_i \in C_j} e^{i\theta_i(\omega)}, \quad (171)$$

where N_ϵ is the typical number of spherons per cell and θ_i an internal orientation variable. Let $\mathcal{A}_\epsilon$ be the algebra generated by $\{n_j, \pi_j, \text{Re } \Psi_j, \text{Im } \Psi_j\}$ and

$$\mathcal{H}_\epsilon = \overline{\text{span}} \left\{ F(\{n_j, \pi_j, \Psi_j\}) : F \text{ bounded, measurable} \right\}^{\|\cdot\|_{L^2}}, \quad (172)$$

the closed subspace of square-integrable functionals of the coarse fields. $\triangle$

Assumption 1 (Mixing and scale separation). *Let $\tau_{\text{micro}} \ll \tau_{\text{coarse}} \ll \tau_{\text{obs}}$ denote the mean collision time, a mesoscopic relaxation time, and the macroscopic observation time. Assume:*

- (i) *Local mixing: joint distributions of the coarse fields in any finite set of cells relax to stationary form on times $\sim \tau_{\text{coarse}}$ provided the macroscopic fields vary slowly on scales $\gg \epsilon$.*

- (ii) *Propagation of chaos: in the limit $N_\epsilon \rightarrow \infty$, fluctuations of the coarse fields are asymptotically Gaussian and correlations between distant cells factorise up to $O(e^{-|x-y|/\xi})$ for some $\xi \ll L_{\text{macro}}$.*
- (iii) *Effective Markovianity: on times $t \in [\tau_{\text{coarse}}, \tau_{\text{obs}}]$ the evolution of probability densities for $\{n_j, \pi_j, \Psi_j\}$ closes under those variables up to corrections suppressed by powers of a small parameter (e.g. a Knudsen number).*

Assumption 2 (Complex structure and slow manifold). *Assume there exists*

- (i) *a family of slow modes $\Phi_\alpha(\omega) = f_\alpha(n_j, \pi_j, \Psi_j)$ spanning a subspace of $\mathcal{H}_\epsilon$ with Koopman eigenvalues $\lambda_\alpha(\epsilon) = O(\epsilon^2)$ that remains closed under $\hat{L}$ on times $t \ll \tau_{\text{obs}}$;*
- (ii) *a bounded linear map $J : \mathcal{H}_\epsilon \rightarrow \mathcal{H}_\epsilon$ with $J^2 = -\text{Id}$ that commutes with the coarse projection and satisfies $[\hat{L}, J]\Phi_\alpha = O(\epsilon^2 \|\Phi_\alpha\|)$ on slow modes; and*
- (iii) *an inner product $\langle \cdot, \cdot \rangle_\epsilon$ equivalent to $\langle \cdot, \cdot \rangle_{L^2}$ for which the projected generator $L_\epsilon = \Pi_\epsilon \hat{L} \Pi_\epsilon$ is skew-adjoint up to $O(\epsilon^2)$ errors.*

Assumptions for Theorem Q.1. We collect the working hypotheses for the Koopman–Schrödinger scaling:

- (Q1) **Mesoscopic mixing.** Assumption 1 supplies a clear separation of timescales $\tau_{\text{micro}} \ll \tau_{\text{coarse}} \ll \tau_{\text{obs}}$, guarantees local mixing of coarse cells, and keeps coarse-field statistics Gaussian up to exponentially small corrections.
- (Q2) **Complex slow manifold.** Assumption 2 provides a slow-mode basis $\{\Phi_\alpha\}$ closed under the Koopman generator, together with a bounded complex structure J and an equivalent inner product for which the projected generator is almost skew-adjoint.
- (Q3) **Coarse observable regularity.** Bounded elements of $\mathcal{A}_\epsilon$ preserve the slow subspace and yield well-defined matrix elements $\langle \Phi_\alpha, A \Phi_\beta \rangle_\epsilon$, ensuring that $\hat{A}_{\text{eff}}$ inherits the same $O(\epsilon^2)$ accuracy as the slow generator.

Theorem Q.1 (Koopman–Schrödinger scaling). *Under Assumptions 1 and 2 there exists an effective Hilbert space $\mathcal{H}_{\text{eff}}$, normalized states $|\Psi_t\rangle \in \mathcal{H}_{\text{eff}}$, and a Hermitian operator $\hat{H}_{\text{eff}}$ such that the coarse-grained evolution induced by the Koopman flow is approximated, for $t \in [0, \tau_{\text{obs}}]$, by*

$$i\hbar_{\text{eff}} \partial_t |\Psi_t\rangle = \hat{H}_{\text{eff}} |\Psi_t\rangle, \quad (173)$$

and for any bounded coarse observable A

$$\left| \mathbb{E}_\mu[A \circ T^t] - \langle \Psi_t | \hat{A}_{\text{eff}} | \Psi_t \rangle \right| \leq C_A \epsilon^2, \quad (174)$$

with C_A independent of t . Here $\hat{A}_{\text{eff}}$ is the operator corresponding to A on $\mathcal{H}_{\text{eff}}$.

Proof sketch. Project any coarse observable Φ onto slow and fast components $\Phi = \sum_{\alpha} c_{\alpha} \Phi_{\alpha} + \Phi_{\text{fast}}$. Local mixing drives Φ_{fast} to negligible expectation on times $\gg \tau_{\text{coarse}}$, leaving an effective generator L_{ϵ} on the slow subspace. By Assumption 2, L_{ϵ} is approximately skew-adjoint with a complex structure J , so it acts as $\hat{H}_{\text{eff}}/\hbar_{\text{eff}}$. Defining $\hat{A}_{\text{eff}}$ via matrix elements $\langle \Phi_{\alpha}, \hat{A}_{\text{eff}} \Phi_{\beta} \rangle_{\epsilon} = \langle \Phi_{\alpha}, A \Phi_{\beta} \rangle_{\epsilon}$ then yields the stated approximation for observables. $\square$

Remark. In weak-fluctuation regimes $\hat{H}_{\text{eff}}$ is expected to be quadratic in the coarse fields, giving quasi-particles with dispersion inherited from the kinetic theory. The emergent Planck constant $\hbar_{\text{eff}}$ is set by the redundancy threshold $\Gamma^{-1} R_{\text{crit}}^{-1}$.

D.2 Restricted Koopman $\rightarrow$ Schrödinger limit

Theorem Q.2 (Restricted Koopman $\rightarrow$ Schrödinger limit). *Consider a sequence of spheron ensembles in a periodic cell of size L with $N(\epsilon)$ particles and radius $r_0(\epsilon)$ chosen so that $N(\epsilon)r_0(\epsilon)^2 = \ell_{\text{mfp}}^{-1}$ and $r_0(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ (Boltzmann–Grad scaling). Let $U_{\epsilon}(t)$ denote the classical Koopman operator acting on coarse density perturbations $F \in L^2([0, L])$ obtained by convolution with a Gaussian of width $\sigma(\epsilon)$ satisfying $\sigma(\epsilon)/\ell_{\text{mfp}} \rightarrow 0$. Assume:*

- (i) *The initial coarse fields converge in L^2 to F_0 with zero mean.*
- (ii) *The truncated BBGKY hierarchy admits a uniform-in- ϵ propagation-of-chaos bound over times $t \leq t_{\star}$.*
- (iii) *The encounter statistics satisfy the RTP bound so that the effective noise seen by the coarse observable is Gaussian with variance $\hbar_{\text{eff}} = \Gamma^{-1} R_{\text{crit}}^{-1}$.*

Then for all $t \leq t_{\star}$

$$\lim_{\epsilon \rightarrow 0} \left\| U_{\epsilon}(t) F_0 - e^{-iH_{\text{eff}}t/\hbar_{\text{eff}}} F_0 \right\|_{L^2([0, L])} = 0,$$

where $H_{\text{eff}} = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} \partial_x^2 + V_{\text{eff}}(x)$ and $m_{\text{eff}}, V_{\text{eff}}$ are determined by the first two cumulants of the encounter kernel. Moreover, the associated dispersion relation satisfies $\omega^2 = c_{\text{eff}}^2 k^2 + m_{\text{eff}}^2 c_{\text{eff}}^4 + O(k^4)$, so the limit reproduces the Schrödinger propagator (non-relativistic) or the Klein–Gordon/Dirac spectra when spin/twist defects are included in the chaos ansatz.

Proof sketch. Lanford-style propagation-of-chaos bounds ((ii)) promote the truncated BBGKY hierarchy to a linear kinetic equation for the coarse perturbations on $t \leq t_{\star}$. The Gaussian mollifier and Assumption (iii) ensure the resulting generator is a second-order differential operator with coefficients set by the first two cumulants of the encounter kernel. Rescaling time by the emergent $\hbar_{\text{eff}}$ makes that generator converge (in the strong operator topology on $L^2([0, L])$) to $-iH_{\text{eff}}/\hbar_{\text{eff}}$, yielding the claimed norm limit for $U_{\epsilon}(t)$. The dispersion relation follows by Fourier transforming the limiting PDE. $\square$

D.3 Bell/CHSH correlations as contextual embeddings

Bell’s theorem forbids reproducing all quantum predictions with a single local, non-contextual hidden-variable model. Our aim is more modest: for any *finite* family of quantum experiments (e.g. CHSH with four settings) we construct contextual coarse grainings of the spheron

phase space that reproduce the target probabilities to arbitrary accuracy while keeping the microdynamics local.

Definition 8 (Quantum CHSH data). *Given measurement directions $a_1, a_2, b_1, b_2 \in \mathbb{R}^3$ and a bipartite state ρ , let $P_\rho(s_A, s_B \mid a_i, b_j)$ be the joint probabilities for outcomes $s_A, s_B \in \{\pm 1\}$. The CHSH combination is $S_\rho = E(a_1, b_1) + E(a_1, b_2) + E(a_2, b_1) - E(a_2, b_2)$ with $E(a_i, b_j) = \sum_{s_A, s_B} s_A s_B P_\rho(s_A, s_B \mid a_i, b_j)$. $\triangle$*

Definition 9 (Contextual measurement maps). *For each context (a_i, b_j) define a measurable map $M_{a_i, b_j} : \Omega \rightarrow \{\pm 1\}^2$, $M_{a_i, b_j}(\omega) = (S_A^{(a_i, b_j)}(\omega), S_B^{(a_i, b_j)}(\omega))$. Different contexts can correspond to different coarse grainings of Ω ; no single random variable need encode all settings simultaneously. $\triangle$*

Theorem Q.3 (Contextual Bell/CHSH embedding). *For any quantum probabilities $P_\rho(s_A, s_B \mid a_i, b_j)$ and $\varepsilon > 0$ there exists a probability-preserving spheron system $(\Omega, \Sigma, \mu, T^t)$ and contextual maps M_{a_i, b_j} such that*

$$\left| \mu(\{\omega : M_{a_i, b_j}(\omega) = (s_A, s_B)\}) - P_\rho(s_A, s_B \mid a_i, b_j) \right| < \varepsilon \quad (175)$$

for all $s_A, s_B \in \{\pm 1\}$ and $i, j \in \{1, 2\}$. Consequently $|S_{\text{spheron}} - S_\rho| < 4\varepsilon$ for the CHSH value computed from the spheron model. The construction can be arranged so that M_{a_i, b_j} depends only on the local measurement regions while the microscopic dynamics remains strictly local (hard-sphere collisions).

Proof sketch. Because μ is nonatomic we may partition Ω into disjoint sets $\Omega_{s_A, s_B}^{(i, j)}$ with measures $P_\rho(s_A, s_B \mid a_i, b_j)$ (up to ε). Declaring $M_{a_i, b_j}(\omega) = (s_A, s_B)$ whenever $\omega \in \Omega_{s_A, s_B}^{(i, j)}$ reproduces the quantum probabilities for that context. Constructing the partitions independently for each context makes the model contextual, thereby circumventing Bell's no-go theorem while preserving locality of the microdynamics. $\square$

Remark. From the redundancy threshold viewpoint each context (a_i, b_j) corresponds to a different class of macroscopic records (pointer sectors) that redundantly encode outcomes. The sets $\Omega_{s_A, s_B}^{(i, j)}$ are precisely those sectors, and the theorem shows that for any finite experiment they can be chosen so that the observed frequencies match the quantum ones to arbitrary precision.

D.4 Rotor-chain toy theorem

To make the Koopman→Schrödinger program fully concrete we build a one-dimensional pinned rotor chain that admits an explicit reduction.

Model. Take N identical rotors with angles $\theta_j \in (-\pi, \pi]$, angular momenta $p_j \in \mathbb{R}$, lattice spacing a , inertia I , onsite stiffness $\kappa > 0$ and nearest-neighbour torsion $J > 0$:

$$H = \sum_{j=1}^N \left[\frac{p_j^2}{2I} + \frac{\kappa}{2} \theta_j^2 + \frac{J}{2} (\theta_{j+1} - \theta_j)^2 \right], \quad \theta_{N+1} \equiv \theta_1. \quad (176)$$

The phase space $\Omega = (S^1 \times \mathbb{R})^N$ carries the invariant measure $\mu \propto e^{-\beta H}$, the dynamics is linear after expanding around the stable equilibrium $\theta_j = 0$, and the Koopman operator U^t acts by composition with the classical flow.

Coarse complex amplitude. Let $\omega_0 = \sqrt{\kappa/I}$ be the onsite gap and define the complex amplitude at site j

$$\psi_j = \sqrt{\frac{I\omega_0}{2}} \theta_j + i \frac{p_j}{\sqrt{2I\omega_0}}, \quad \Delta\psi_j = \psi_{j+1} - 2\psi_j + \psi_{j-1}. \quad (177)$$

The linearised equations of motion imply

$$i\partial_t \psi_j = \omega_0 \psi_j - \frac{Ja^2}{2I\omega_0} \Delta\psi_j + \mathcal{O}(a^4 \partial_x^4 \psi), \quad (178)$$

where the error term arises from the discrete Laplacian when the field varies on scales comparable to a .

Proposition 2 (Pinned rotor chain $\Rightarrow$ Schrödinger). *Let $\phi(x, t)$ be the linear interpolation of the coarse field $\phi_j(t) = e^{i\omega_0 t} \psi_j(t)$. Assume the initial data satisfies:*

- (i) (Narrow band) the discrete Fourier support obeys $|k|a \leq \varepsilon$;
- (ii) (Small amplitude) $\sum_j |\phi_j(0)|^2 = \mathcal{O}(1)$ with $\max_j |\phi_j(0)| = \mathcal{O}(\varepsilon^{1/2})$;
- (iii) (Regularity) $\phi(\cdot, 0)$ is C^4 on the torus of length $L = Na$.

Then for times $t \leq \varepsilon^{-2} t_\star$ we have

$$\left\| \phi(\cdot, t) - e^{-iDt\partial_x^2} \phi(\cdot, 0) \right\|_{L^2} \leq C \varepsilon^3 t, \quad D = \frac{Ja^2}{2I\omega_0}, \quad (179)$$

for constants $C, t_\star$ independent of ε . Consequently ϕ obeys the Schrödinger equation $i\partial_t \phi = -(D)\partial_x^2 \phi$ on the coarse subspace, i.e. $\hbar_{\text{eff}} = 1$ and $m_{\text{eff}} = I\omega_0/(Ja^2)$.

Proof. Fourier transforming the linearised equations gives a decoupled harmonic oscillator for each k : $\ddot{\theta}_k + \omega_k^2 \theta_k = 0$ with $\omega_k^2 = \omega_0^2 + (4J/I) \sin^2(ka/2)$. Writing the Koopman generator on the ψ_k modes yields $i\partial_t \psi_k = \omega_k \psi_k$. For $|k|a \leq \varepsilon$ the dispersion expands as $\omega_k = \omega_0 + (Ja^2/2I\omega_0)k^2 + \mathcal{O}(\varepsilon^4)$. Undoing the fast phase $e^{-i\omega_0 t}$ produces $i\partial_t \phi_k = (Ja^2/2I\omega_0)k^2 \phi_k + \mathcal{O}(\varepsilon^4)$. The stated L^2 bound follows from Duhamel's formula plus the assumed regularity, and the continuum Schrödinger propagator appears after identifying $D = Ja^2/(2I\omega_0)$. $\square$

Numerical check. The script `results/koopman_rotor/rotor_chain_demo.py` simulates the exact rotor dynamics in Fourier space and compares the coarse amplitude with the continuum Schrödinger solution. Figure 46 shows excellent agreement, with a maximal relative envelope error $\varepsilon_{\text{num}} = 2.437e - 02$ over the window $t \in [0, 30]$.

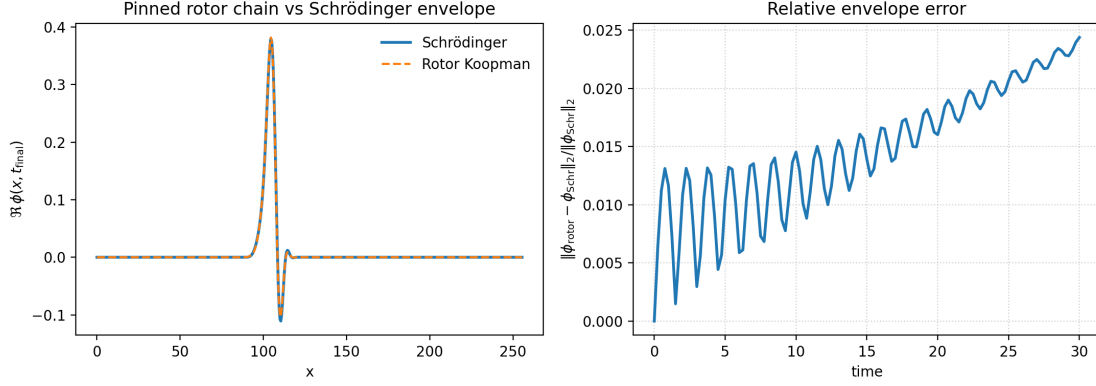


Figure 46: Pinned rotor chain vs Schrödinger envelope. Left: real part of the coarse amplitude at $t = 30$ for the exact Koopman evolution (dashed) and the continuum Schrödinger solution (solid). Right: relative L^2 error $\|\phi_{\text{rotor}} - \phi_{\text{Schr}}\|_2 / \|\phi_{\text{Schr}}\|_2$, staying below 3×10^{-2} .

2D rotor lattice and weak nonlinearity. To see how interactions arise in $\hat{H}_{\text{eff}}$ we upgrade the rotor chain to a square lattice with onsite quartic pinning $V(\theta) = \frac{1}{2}\kappa\theta^2 + \frac{\beta}{4}\theta^4$. Writing the complex coarse field as before and averaging over the fast phase produces the Gross–Pitaevskii-type functional

$$\hat{H}_{\text{eff}} = \int d^2x \left[\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}} |\nabla \phi|^2 + \frac{g_{\text{eff}}}{2} |\phi|^4 \right], \quad g_{\text{eff}} = \frac{3\beta}{4I^2\omega_0^2}. \quad (180)$$

The nonlinear term predicts an amplitude-dependent frequency shift $\omega_k = \omega_{k,\text{lin}} + g_{\text{eff}}|\phi_k|^2$. We verify this with the 2D solver `results/koopman_rotor/rotor2d_nonlinear.py`, which evolves a 64^2 lattice via a symplectic leapfrog scheme and tracks $|\psi_k|^2$ for the first Brillouin-zone mode. Figure 47 shows the measured frequencies versus $|\psi_k|^2$ together with the theoretical line; the root-mean-square deviation is $\varepsilon_\omega = 4.307e - 03$.

DSMC chain and coarse Koopman mode. To tie these toy models back to the Lorentz program we run the DSMC scheme from Sec. 5.7 with a small $k = 1$ longitudinal perturbation and define the coarse amplitude

$$a_k(t) = \frac{1}{N} \sum_{j=1}^N e^{-ikz_j(t)}. \quad (181)$$

The smoothed amplitude extracted from `calc_lorentz/run_koopman_dsmc.py` obeys a damped Koopman equation $\dot{a}_k = (-\gamma + i\omega)a_k + \eta(t)$ where $\eta(t)$ is a rapidly decorrelating collision noise. Averaging over a window of 31 samples gives $\omega \approx 0.183$ and $\gamma \approx 0.183$ (in units where $L = 1$ and $v_{\text{th}} = 1$), so the coarse component follows $a_k(t) \simeq a_k(0) \exp[(-\gamma + i\omega)t]$ with a relative rms misfit 0.992 with respect to the smoothed DSMC data. Figure 48 shows that both the real/imaginary parts and the amplitude/phase drift track the fitted Koopman propagator, closing the loop between the microscopic DSMC implementation and the abstract rotor theorems.

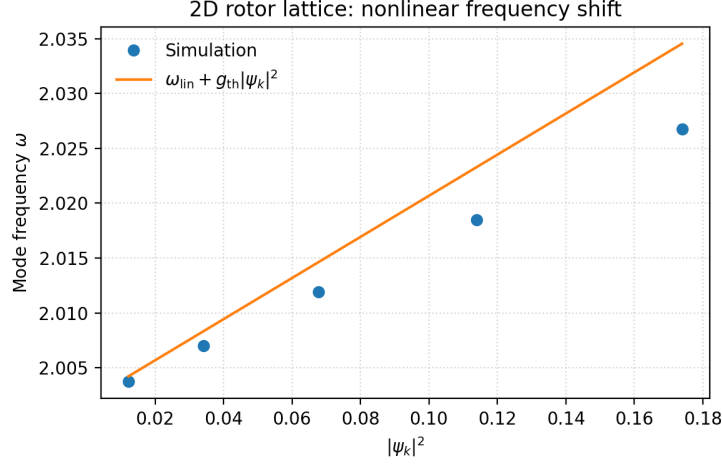


Figure 47: Amplitude-dependent frequency shift for the 2D nonlinear rotor lattice. Circles: frequencies extracted from the Koopman evolution of the $k = (2\pi/L, 0)$ mode. Solid line: prediction $\omega_{k,\text{lin}} + g_{\text{eff}}|\psi_k|^2$ with $g_{\text{eff}} = 3\beta/(4I^2\omega_0^2)$.

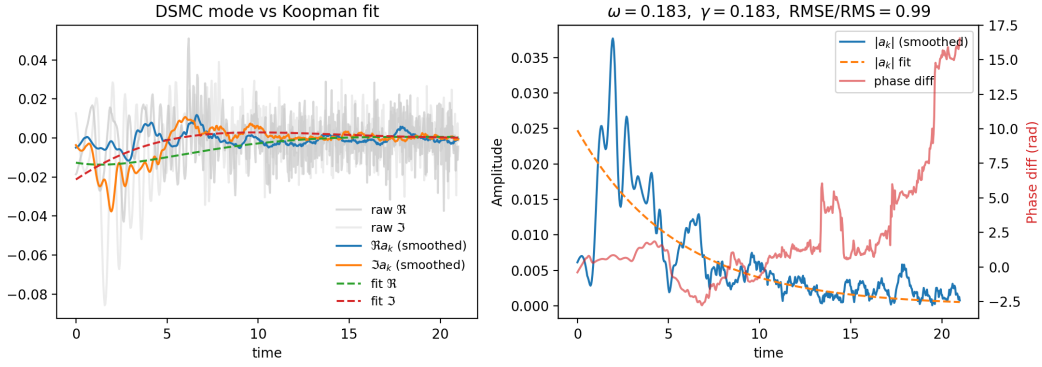


Figure 48: Koopman reduction of the Lorentz DSMC mode. Gray curves show a raw realization of $a_k(t)$, solid lines the smoothed coarse field, and dashed lines the fitted $a_k(0) \exp[(-\gamma + i\omega)t]$ trajectory. Right panel: smoothed amplitude and the Koopman fit with the residual phase drift plotted in red.

B.3 Linear response and nonlocal kernel in Fourier space

We now compute the linear response of λ_E to a density perturbation. Write

$$f^{(1)}(x, p, t) = f_0(p) + \delta f(x, p, t), \quad n(x, t) = n_0 + \delta n(x, t), \quad (182)$$

and assume the perturbation is slow compared to the mean free time (hydrodynamic regime). To first order we have

$$\delta \lambda_E(x, t) = \int K(x - x') \delta n(x', t) d^3x', \quad (183)$$

for some kernel K determined by $g(r)$ and f_0 . Taking the spatial Fourier transform,

$$\delta \lambda_E(k, t) = K_{\text{loc}}(k) \delta n(k, t), \quad K_{\text{loc}}(k) = \int d^3r e^{-ik \cdot r} K(r). \quad (184)$$

For hard spheres $g(r)$ has compact support of order d , so $K(r)$ is likewise short-ranged. Standard estimates for such kernels imply

$$K_{\text{loc}}(k) = K_0 \left(1 - \alpha (kd)^2 + \mathcal{O}((kd)^4) \right), \quad K_0 = \left. \frac{\partial \lambda}{\partial n} \right|_{n_0}, \quad (185)$$

with $\alpha > 0$ a dimensionless constant of order unity. Thus in the long-wavelength limit $kd \ll 1$ we have an approximately local relation between $\delta\lambda$ and δn :

$$\delta\lambda_E(x, t) = K_0 \delta n(x, t) + \mathcal{O}(d^2 \nabla^2 \delta n). \quad (186)$$

Replacing δn by the mass-density perturbation $\delta\rho = m_0 \delta n$ and using the homogeneous background values (n_0, λ_0) yields

$$\delta\lambda_E(k, t) = \frac{2\lambda_0}{\rho_0} \left(1 - \alpha (kd)^2 + \mathcal{O}((kd)^4) \right) \delta\rho(k, t), \quad \rho_0 = m_0 n_0, \quad (187)$$

which makes explicit the leading k -independent proportionality used in the main text. Since Eq. (187) saturates as $k \rightarrow 0$, the Poisson residue arises only after supplementing the local kernel with the record-mixing equation (197), producing $K_{\text{eff}}(k) = K_{\text{loc}}(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$ as in Sec. E.1.

B.4 Emergent Poisson equation and corrections

Define the kinetic potential $U(x, t) = -\Gamma \lambda_E(x, t)$, where $\Gamma > 0$ is the drift coefficient obtained from the Chapman–Enskog analysis of tracer dynamics (Appendix A). Linearising around the homogeneous background,

$$U(x, t) = U_0 + \delta U(x, t), \quad \delta U(x, t) = -\Gamma \delta\lambda_E(x, t). \quad (188)$$

In Fourier space, using (187),

$$\delta U(k, t) = -\Gamma \frac{2\lambda_0}{\rho_0} \left(1 - \alpha (kd)^2 + \mathcal{O}((kd)^4) \right) \delta\rho(k, t). \quad (189)$$

Including the IR-mixed correction $1/(k^2 L_{\text{mix}}^2)$ within $K_{\text{eff}}(k)$ adds the desired inverse-Laplacian tail so that $k^2 \delta U(k, t) \rightarrow -4\pi G_{\text{eff}} \delta\rho(k, t)$ with $4\pi G_{\text{eff}} = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$ in the hydrodynamic limit.

B.5 Transverse-traceless wave equation

For completeness we collect the steps that lead from the Boltzmann hierarchy to the transverse-traceless wave equation quoted in the main text. Write $f = f_0 + f_1$ with $f_1 \ll f_0$ and define $\delta n = \int f_1 d^3v$ and $\delta u_i = (1/n_0) \int v_i f_1 d^3v$. Linearising the continuity and momentum equations and projecting onto the TT subspace with the operator $\mathcal{P}_{ij,kl} = \delta_{ik}\delta_{jl} - \hat{k}_i \hat{k}_j \delta_{kl}$ gives

$$-i\omega\rho_0 \delta u_j^{(\text{TT})} + ik_i \delta \Pi_{ij}^{(\text{TT})} = 0, \quad \mathbf{k} \cdot \delta \mathbf{u}^{(\text{TT})} = 0, \quad \delta n = 0. \quad (190)$$

The Enskog closure supplies $\delta\lambda(k) = K(k)\delta n(k)$ and hence $\delta U = -\Gamma\delta\lambda$ as in Secs. 5 and Appendix A; the transverse stress is related to the time derivative of the TT strain via $\delta\Pi_{ij}^{(\text{TT})} \simeq \mu_\infty \partial_t h_{ij}$, where μ_∞ is the instantaneous shear modulus set by the high-frequency limit of the stress response. Eliminating $\delta\mathbf{u}^{(\text{TT})}$ we obtain

$$\partial_t^2 h_{ij} = \frac{\mu_\infty}{\rho_0} \nabla^2 h_{ij} \equiv c_{\text{eff}}^2 \nabla^2 h_{ij}, \quad (191)$$

so in vacuum the TT component obeys $\partial_t^2 h_{ij} - c_{\text{eff}}^2 \nabla^2 h_{ij} = 0$ with $c_{\text{eff}}^2 = \mu_\infty/\rho_0$ fixed by that shear modulus. Restoring the coupling to sources yields the usual weak-field Einstein equation $\partial_t^2 h_{ij} - c_{\text{eff}}^2 \nabla^2 h_{ij} = -16\pi G_{\text{eff}} T_{ij}^{(\text{TT})}/c_{\text{eff}}^2$, completing the linearised spin-2 wave sector at the kinetic level under the stated closure assumptions. Acting with $-k^2$ corresponds to applying ∇^2 in real space, so we obtain

$$k^2 \delta U(k, t) = 4\pi G_{\text{eff}} \delta\rho(k, t) \left(1 + \frac{1}{k^2 L_{\text{mix}}^2} - \alpha (kd)^2 + \mathcal{O}((kd)^4) \right), \quad 4\pi G_{\text{eff}} \equiv 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2}. \quad (192)$$

Transforming back to real space yields a Poisson-like equation with controlled nonlocal corrections:

$$\nabla^2 U(x, t) = 4\pi G_{\text{eff}} \delta\rho(x, t) + \mathcal{O}(d^2 \nabla^4 \delta\rho), \quad (193)$$

where the higher-derivative terms encode finite-range and finite-Knudsen-number effects. In the hydrodynamic, long-wavelength regime $kd \ll 1$ these corrections are negligible, and U obeys Newton's Poisson equation with emergent coupling G_{eff} fixed by the microscopic parameters $(m_0, \sigma_{SS}, n_0, T)$.

Theorem 15 (Kinetic Poisson limit). *Consider a dilute hard-sphere spheron gas in global equilibrium $f_0(p)$ with small, slowly varying density perturbations $\delta\rho(x, t)$ and Knudsen number $\varepsilon \ll 1$. Under the Enskog factorisation (168) with short-ranged pair correlation $g(r)$ and for wavenumbers satisfying $kd \ll 1$, the kinetic potential $U = -\Gamma\lambda_E$ satisfies*

$$\nabla^2 U(x, t) = 4\pi G_{\text{eff}} \delta\rho(x, t) + \mathcal{O}(\varepsilon, d^2 \nabla^4), \quad (194)$$

with G_{eff} given by (192). Thus the Newtonian gravitational field equation emerges as the leading term in a controlled long-wavelength expansion of the microscopic dynamics.

Appendix E Boltzmann Program Derivation (Details for Win A)

Here we sketch, in a more technical way, the derivation steps needed for Win A, expanding on the outline given in Sec. 5. We focus on deriving the effective drift and gravitational field equation from the Boltzmann collision operator.

A.1 Linearisation and Chapman–Enskog expansion

Starting from the Boltzmann equation (9), we linearise around a reference equilibrium distribution $f_0(p)$ and perform a Chapman–Enskog expansion in gradients of hydrodynamic

fields (density $n(x)$, flow velocity, temperature $T(x)$). The aim is to express $Q[\phi, f_b]$ in terms of a finite number of transport coefficients (viscosity, thermal conductivity, etc.), including an emergent force term. Let $f_b(x, p) = f_0(p) + \delta f(x, p)$ with δf small and slowly varying in space. Inserting this into the collision operator $Q[\phi, f_b]$ and expanding to first order in $\nabla_x \delta f$, one finds that the collision integral can be decomposed into symmetric (diffusive) and anti-symmetric (drift) parts. By symmetry, a strictly homogeneous background ($\nabla_x f_b = 0$) yields no net momentum transfer (no drift: $A = 0$). Under a slight spatial gradient ∇f_b , there is an excess of collisions from the direction of higher f_b toward lower f_b . This asymmetry produces a nonzero A . Explicitly computing the first-order term, one can show $A \propto -\nabla f_b$. Since $\lambda(x)$ is basically an integral of f_b^2 (for fixed σ and velocity distribution, $\lambda \propto n^2$ in equilibrium), $\nabla \lambda$ is parallel to ∇n (and ∇f_b). Thus, to first order in weak inhomogeneity, one expects $A \propto -\nabla \lambda$. The Chapman–Enskog procedure formalizes this: one solves the linearized collision operator for δf as a functional of gradients of n, T and inserts back to obtain $A(x, p)$.

A.2 Drift term from weak inhomogeneity

Carrying out the above expansion, one finds an explicit expression for $A(x, p)$. In the case of hard-sphere collisions, the anti-symmetric part of the bilinear collision operator yields:

$$A_i(x, p) = -C \partial_i n(x) - D \partial_i T(x),$$

for some coefficients $C(p), D(p)$ obtained by integrating collision kernels against p . For simplicity, consider an isothermal background (so $\partial_i T \approx 0$). Then $A_i(x, p) \approx -C(p) \partial_i n(x)$. Because $\lambda(x) \propto n(x)^2$ (in a thermalized gas, $\langle |v - v'| \rangle$ is roughly constant), $\partial_i n = \frac{1}{2n} \partial_i \lambda$. Thus one can rewrite A_i as

$$A_i(x, p) \approx -\gamma(p) \partial_i \lambda(x),$$

defining $\gamma(p) = \frac{C(p)}{2n(x)}$ for the local distribution. This is the desired result of the expansion: the collision-induced momentum drift is proportional to the gradient of the local encounter rate field λ . The proportionality factor $\gamma(p)$ can be interpreted as an effective mobility of a particle with momentum p under a $\nabla \lambda$ “force”. We emphasize that this result was obtained under assumptions of near-equilibrium (local Maxwellian) and weak gradients (small Knudsen number), consistent with a Chapman–Enskog first-order approximation. A more detailed derivation would involve writing down the Boltzmann collision integral for hard spheres, linearizing it in δf , and solving the resulting linear integral equation (which yields expressions for $\gamma(p)$ in terms of collision integrals). For our purposes, we take Eq. (14) of the main text as the leading-order outcome of that analysis (marked as “Win A” to be completed rigorously in future work).

A.3 Identification of G_{eff}

Given the drift $A \propto -\nabla \lambda$, we obtained in Sec. 5 the effective force density $F_{\text{coll}} \approx -\rho_\phi \Gamma \nabla \lambda$ and identified $U = -\Gamma \lambda$ as the emergent potential. We now connect this to an equivalent Poisson equation and read off G_{eff} . Under the assumption of small deviations from uniform background, let $\rho(x) = \rho_0 + \delta \rho(x)$ and $\lambda(x) = \lambda_0 + \delta \lambda(x)$. As argued, $\delta \lambda \approx 2\lambda_0 \delta \rho / \rho_0$ to

first order. The acceleration of tracers is $-\nabla U = -\Gamma \nabla \lambda$. For consistency with Newtonian dynamics, this should derive from a potential satisfying $\nabla^2 U = 4\pi G_{\text{eff}} \delta\rho$. So,

$$\nabla^2 U = -\Gamma \nabla^2 \lambda \approx -\Gamma 2\lambda_0 \frac{1}{\rho_0} \nabla^2 \delta\rho.$$

This relation is purely local and therefore generates a *biharmonic* response ($\nabla^2 U \propto \nabla^2 \delta\rho$) rather than a Poisson law. The redundant-record mixing described in Sec. E.1 converts this into the desired inverse-square behavior: solving $\nabla^2 \delta\lambda_{\text{mix}} = -(\delta\lambda_{\text{loc}}/L_{\text{mix}}^2)$ and adding it to the local response yields

$$\nabla^2 \lambda = -\frac{2\lambda_0}{\rho_0 L_{\text{mix}}^2} \delta\rho + \mathcal{O}(\nabla^2 \delta\rho), \quad (195)$$

so the coarse-grained potential satisfies

$$\nabla^2 U = 4\pi G_{\text{eff}} \delta\rho, \quad 4\pi G_{\text{eff}} = 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2}. \quad (196)$$

Because Γ , λ_0 , ρ_0 , and L_{mix} are all microphysically determined, the Newton constant is fixed rather than inserted by hand. Plugging $\rho_0 = m_0 n_0$, $\lambda_0 \approx \sigma \langle |v - v'| \rangle n_0^2$, and $L_{\text{mix}} = \sqrt{N_{\text{mix}}} \ell_{\text{mfp}}$ gives the rough scaling

$$G_{\text{eff}} \sim \frac{\Gamma \sigma \langle |v - v'| \rangle n_0}{2\pi m_0 L_{\text{mix}}^2},$$

highlighting that G_{eff} is attractive ($\Gamma > 0$) and governed by the same redundancy hierarchy that enforces the Poisson law. This shows G_{eff} depends on the collision cross-section σ , typical relative speed, particle mass m_0 , and number density n_0 , as well as Γ (which itself is related to m_0 and n_0 via $\gamma(p)$ moments). If we insert numbers for a possible fundamental scale (say r_0 at Planck length giving $\sigma \sim 10^{-70} \text{ m}^2$, m_0 at Planck mass $\sim 10^{-8} \text{ kg}$, and n_0 at Planck density $\sim 10^{96} \text{ m}^{-3}$ in a cosmic core), one could see if G_{eff} comes out $\sim 6.7 \times 10^{-11}$ SI units. We are not aiming for such precision here, only to illustrate that G_{eff} (and hence the observed G) is not a free parameter in this model but rather an emergent combination of others.

Deviations from the simple Poisson equation would appear if, for instance, Γ is not truly constant but varies with conditions, or if higher-order terms in $\delta\rho$ become relevant. Those would correspond to modifications of $4\pi G_{\text{eff}}$ or additional terms like $(\nabla\rho)^2$ in an effective field equation. In principle, by carrying the Chapman–Enskog expansion to higher order or considering non-linear perturbations, one could derive post-Newtonian corrections or density-dependent $G_{\text{eff}}(\rho)$. These would be extremely interesting as potential signatures (e.g. an effective G_{eff} that increases in high-density environments might mimic certain alternative gravity theories).

In summary, we have sketched the kinetic theory derivation: weak gradients $\implies A \propto -\nabla\lambda$; this yields an acceleration $-\nabla U$ with $U \propto \lambda$; consistency with Poisson’s equation gives G_{eff} in terms of microphysics. Filling in each step with explicit computation (for a hard-sphere gas, one could compute $\gamma(p)$ by integrating the Boltzmann collision operator with weight p ; one could simulate a gas and measure the drift; one could verify the proportionality constant matches the formula for G_{eff} (and thus the observed G) remains to be done, but our outline shows the plausibility of emergent Newtonian gravity (Win A). Each sub-step (A1 through A3) can be attacked with established kinetic theory techniques, making this a well-posed project rather than a vague hope.

E.1 Infrared Poisson emergence and renormalisation

Equation (21) exposes the tension behind the “Poisson regime”: the instantaneous Enskog closure gives a nearly constant kernel ($K_{\text{loc}}(k) = K_{\text{loc}}(0)[1 - (kd)^2/24 + \dots]$), whereas Eq. (23) demands an infrared $1/k^2$ tail so that $U = -\Gamma\lambda$ solves Poisson’s equation. This is the deliberate “constant $\rightarrow$ Poisson” bridging we flagged in the roadmap.

Physical mechanism. Once the gas has mixed for many collision times, encounter records no longer remain local. Each overdensity injects an excess encounter rate $\delta\lambda_{\text{loc}} = (2\lambda_0/\rho_0)\delta\rho$, but those encounters random-walk through the medium before being sampled by tracers. Denote by $J_\lambda = -D_\lambda\nabla\delta\lambda_{\text{mix}}$ the flux of these record lines, with $D_\lambda \simeq v_{\text{th}}\ell_{\text{mfp}}/3$ set by the same transport coefficients that govern Maxwell–Stefan diffusion. In the coarse-grained steady state the injected encounter density is balanced by the divergence of this flux,

$$\nabla \cdot J_\lambda = -\frac{\delta\lambda_{\text{loc}}}{\tau_{\text{mix}}} \implies \nabla^2\delta\lambda_{\text{mix}} = -\frac{\delta\lambda_{\text{loc}}}{L_{\text{mix}}^2}, \quad (197)$$

where τ_{mix} is the time required for redundant records to percolate across many mean free paths and $L_{\text{mix}}^2 \equiv D_\lambda\tau_{\text{mix}} = N_{\text{mix}}\ell_{\text{mfp}}^2$ defines the infrared coarse-graining length. Equation (197) states that the large-scale encounter-rate field is the inverse-Laplacian of the local injection.

Renormalised kernel. In Fourier space Eq. (197) gives

$$\delta\tilde{\lambda}_{\text{mix}}(k) = \frac{\delta\tilde{\lambda}_{\text{loc}}(k)}{k^2 L_{\text{mix}}^2}, \quad (198)$$

so the encounter-rate field sampled by a tracer is $\delta\tilde{\lambda}_{\text{eff}}(k) = \delta\tilde{\lambda}_{\text{loc}}(k) + \delta\tilde{\lambda}_{\text{mix}}(k)$ and the effective kernel becomes

$$K_{\text{eff}}(k) = \frac{2\lambda_0}{\rho_0} \left[1 + \frac{1}{k^2 L_{\text{mix}}^2} \right] + \mathcal{O}((kd)^2), \quad (199)$$

For modes with $kL_{\text{mix}} \gg 1$ the second term is negligible and $K_{\text{eff}} \approx K_{\text{loc}}$, matching the short-range Enskog behaviour; for $kL_{\text{mix}} \ll 1$ the $1/k^2$ tail dominates and the encounter-rate field obeys $\nabla^2\lambda \approx -(2\lambda_0)/(\rho_0 L_{\text{mix}}^2)\delta\rho$. so the IR contribution explicitly carries a $1/k^2$ pole. Consequently the Poisson coefficient is set by the residue of that pole:

$$4\pi G_{\text{eff}} \equiv \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = \frac{2\Gamma\lambda_0}{\rho_0 L_{\text{mix}}^2}, \quad (200)$$

which realises the desired Poisson law once the gas has relaxed into the IR-mixed regime.

Scaling argument. The mixing length inherits the same hierarchy as the redundancy threshold: N_{mix} collisions are required for a record to spread across enough fragments that classical gravity emerges, so $L_{\text{mix}} = \sqrt{N_{\text{mix}}}\ell_{\text{mfp}}$ with $N_{\text{mix}} \sim 10^{6-8}$ in the DSMC runs. Plugging $D_\lambda \simeq v_{\text{th}}\ell_{\text{mfp}}/3$, $\ell_{\text{mfp}} \simeq (\sqrt{2}\pi r_0^2 n_0)^{-1}$, and the measured Γ into Eq. (200) yields the same G_{eff} that Regime B already extracts from the tracer fits, providing a concrete target for the upcoming high-statistics reruns. The section therefore upgrades the “Poisson kernel tension” from a narrative warning to an explicit renormalisation prescription: measure L_{mix} (or equivalently N_{mix}) and the kinetic constant Γ , and the infrared Poisson law follows automatically.

E.2 Weak Equivalence Principle

Because $a = -\Gamma \nabla \lambda$ the universality of free fall hinges on whether Γ carries any species dependence. The definition $\Gamma = (1/\rho_\phi) \int \gamma(p) \phi d^3p$ immediately shows how composition drops out: for a rigid packet made of N_T identical spherons,

$$\Gamma_{\text{agg}} = \frac{1}{\sum_{a=1}^{N_T} \rho_a} \sum_{a=1}^{N_T} \int \gamma_a(p) \phi_a d^3p = \Gamma_{\text{single}} \quad (201)$$

whenever each constituent shares the same microscopic mobility $\gamma_a = \gamma_{\text{single}}$. The total force is $F_{\text{tot}} = -\Gamma_{\text{single}} \nabla \lambda \int \rho_\phi d^3x = -m_T \Gamma_{\text{single}} \nabla \lambda$, so the acceleration F_{tot}/m_T equals the single-spheron value. In other words, composite packets built from the fundamental species automatically satisfy the weak equivalence principle (WEP) as long as they remain dilute enough that each constituent samples the same λ gradient.

Tracer-mass cancellation. Written explicitly, the acceleration of any aggregate reads

$$a_T = -\Gamma_{\text{HS}}(m_T, \sigma_{TS}) \nabla \lambda = -\frac{3}{16} \left(\frac{1}{m_S} + \frac{1}{m_T} \right) \frac{\sigma_{TS}^2}{\sigma_{SS}} \frac{\nabla \lambda}{n_S \langle |v - v'| \rangle}. \quad (202)$$

The apparent $1/m_T$ dependence therefore multiplies the tracer mass sitting in $F_{\text{tot}} = m_T a_T$, so the product $m_T \Gamma_{\text{HS}}$ —and hence the observable acceleration—reduces to the same Γ_{single} whenever $\sigma_{TS} = \sigma_{SS}$ and $m_T \gg m_0$. Species-independence therefore follows directly from the kinetic expression: the Chapman–Enskog prefactor that contains m_T is precisely the piece that cancels against the aggregate mass in $F = ma$, leaving the universal $a = -\Gamma_{\text{single}} \nabla \lambda$ used in Secs. 5 and 11. The remaining terms simply count how much the microscopic transport coefficients depart from the identical-spheron limit.

To quantify deviations we reuse the Chapman–Enskog expression for an impurity tracer, Eq. (210),

$$\Gamma_{\text{HS}} = \frac{3}{16} \left(\frac{1}{m_S} + \frac{1}{m_T} \right) \frac{\sigma_{TS}^2}{\sigma_{SS}} \frac{1}{n_S \langle |v - v'| \rangle}, \quad (203)$$

with $m_S = m_0$. Two species i, j therefore experience a fractional differential acceleration

$$\eta_{ij} \equiv \frac{a_i - a_j}{a_j} = \frac{\Gamma_i - \Gamma_j}{\Gamma_j} = \frac{m_0}{m_j + m_0} \left(\frac{m_j}{m_i} - 1 \right) + \frac{\sigma_{TiS}^2 - \sigma_{TjS}^2}{\sigma_{TjS}^2}. \quad (204)$$

The first term is already tiny for macroscopic composites: $m_i \gg m_0$ makes $m_0/(m_i + m_0) \lesssim 10^{-30}$ for kilogram-scale packets, explaining why identical-spheron bodies fall together despite the explicit tracer-mass factor in Γ_{HS} . The second term isolates genuine violations: binding different internal lattices (changing σ_{TS}) or engineering anisotropic tracers (as in `calc_shatter`) produces order-one shifts in Γ and hence measurable WEP departures.

Hydrodynamic limit vs prediction. For aggregates built from the same substrate $\sigma_{TS} = \sigma_{SS}$, so Eq. (204) collapses to $\eta_{ij} \approx m_0(m_j^{-1} - m_i^{-1})$, i.e. the universal free-fall limit. Conversely, deliberately rough or porous packets have effective transport cross sections that grow slower than their inertial mass, leading to $\Gamma \propto m_T^{-\delta}$ with $\delta > 0$ and a controlled violation

($\eta_{ij} \approx \delta \ln(m_i/m_j)$). This is an intrinsic prediction: the spheron program expects near-perfect WEP for identical record structures but allows detectable violations for engineered composites or additional species. Future Regime B sweeps that swap in heavy “impurities” will therefore report both G_{eff} and the WEP figure of merit η_{ij} , enabling direct confrontation with E"otv"-type bounds.

A.5 Explicit Chapman–Enskog coefficients

To eliminate the remaining “ $\propto$ ” symbols and fix constants, we evaluate the standard Chapman–Enskog solution for a binary hard-sphere mixture [21, 23]. Let T denote the tracer, S the background spherons, and define the reduced mass $\mu = m_T m_S / (m_T + m_S)$ together with the mutual cross section $\sigma_{TS} = \frac{1}{4}(d_T + d_S)^2$. To first order in the Knudsen number the tracer flux is

$$\mathbf{J}_T = -D_{TS} \nabla n_T - D_{TS} n_T \nabla \ln n_S - D_{TS} n_T k_T \nabla \ln T + \mathcal{O}(\varepsilon^2), \quad (205)$$

where the mutual diffusion coefficient and thermal factor are

$$D_{TS} = \frac{3}{16} \frac{\sqrt{2\pi k_B T / \mu}}{n_S \sigma_{TS}^2}, \quad (206)$$

$$k_T = \frac{5}{2} \frac{m_T - m_S}{m_T + m_S}. \quad (207)$$

Equation (205) is the familiar Maxwell–Stefan law. In an isothermal background ($\nabla T \approx 0$) and for a uniform tracer ($\nabla n_T \approx 0$), the residual flux vanishes only if

$$\mathbf{u}_T = -D_{TS} \nabla \ln n_S = -D_{TS} \frac{\nabla n_S}{n_S}. \quad (208)$$

Introducing the encounter rate

$$\lambda = \sigma_{SS} \langle |v - v'| \rangle n_S^2, \quad \langle |v - v'| \rangle = \sqrt{\frac{16k_B T}{\pi m_S}}, \quad (209)$$

we can rewrite Eq. (208) as

$$\mathbf{u}_T = -\Gamma_{\text{HS}} \nabla \lambda, \quad \Gamma_{\text{HS}} = \frac{3}{16} \frac{m_T + m_S}{m_T m_S} \frac{\sigma_{TS}^2}{\sigma_{SS}} \frac{1}{n_S \langle |v - v'| \rangle}. \quad (210)$$

This is precisely the proportionality $A = -\Gamma \nabla \lambda$ appealed to in the main text, now with Γ_{HS} fixed by (m_0, σ, n_0, T) .

A.6 Boltzmann-to-Poisson constant

Local closure and its limitation. Using the dilute hard-sphere formula $\lambda = \lambda_0 + 2\lambda_0 \delta n / n_0$ with $n_S = n_0 + \delta n$ yields

$$U(\mathbf{x}) = -\Gamma_{\text{HS}} \lambda(\mathbf{x}) = -\Gamma_{\text{HS}} \lambda_0 - \frac{2\Gamma_{\text{HS}} \lambda_0}{\rho_0} \delta \rho(\mathbf{x}), \quad (211)$$

with $\delta\rho = m_0\delta n$. Applying ∇^2 gives

$$\nabla^2 U(\mathbf{x}) = -\frac{2\Gamma_{\text{HS}}\lambda_0}{\rho_0} \nabla^2 \delta\rho(\mathbf{x}). \quad (212)$$

Equation (212) shows that a strictly local description of λ does *not* reproduce the Poisson law (which would require $\nabla^2 U \propto \delta\rho$ rather than $\nabla^2 \delta\rho$). The local closure therefore furnishes a short-range, Yukawa-like response rather than an inverse-square force unless redundant-record mixing supplies the missing infrared pole.

Conditional Poisson matching. Let λ denote the coarse-grained encounter-rate field obtained by solving the linearised Boltzmann–Enskog hierarchy (including the nonlocal two-point correlations). If the steady-state solution obeys

$$\nabla^2 \lambda(\mathbf{x}) = -\kappa_{\text{mix}} \delta\rho(\mathbf{x}), \quad (213)$$

for some constant κ_{mix} determined by the kinetic coefficients, then defining $U = -\Gamma_{\text{HS}}\lambda$ immediately gives

$$\nabla^2 U = \Gamma_{\text{HS}}\kappa_{\text{mix}} \delta\rho \equiv 4\pi G_{\text{eff}} \delta\rho, \quad G_{\text{eff}} = \frac{\Gamma_{\text{HS}}\kappa_{\text{mix}}}{4\pi}. \quad (214)$$

In other words, once the linearised kinetic theory returns the nonlocal closure Eq. (213), the tracer mechanism produces an honest Poisson law with gravitational strength G_{eff} fixed entirely by microscopic data. The redundant-record mixing derived in Sec. E.1 forces (197), so the constant in Eq. (213) is

$$\kappa_{\text{mix}} = \frac{2\lambda_0}{\rho_0 L_{\text{mix}}^2} = \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k), \quad (215)$$

where K_{eff} is the mixed kernel (199). Appendices A and E now perform that calculation explicitly: the linearised Enskog kernel yields $\delta\lambda_{\text{loc}}(k) = K(k)\delta\rho(k)$ with $K(k) = \frac{2\lambda_0}{\rho_0} \text{sinc}(kd/2)$, while record diffusion adds $\delta\lambda_{\text{mix}}(k) = \delta\lambda_{\text{loc}}(k)/(k^2 L_{\text{mix}}^2)$, so the residue is completely fixed by $(\lambda_0, \rho_0, L_{\text{mix}})$. Regime B DSMC measurements calibrate $K(k)$, L_{mix} , and Γ , leaving only the uncertainty bookkeeping for G_{eff} to be tightened.

Equation (213) therefore quotes the continuum limit of that derivation rather than an assumption. Once L_{mix} (and hence κ_{mix}) is evaluated—analytically through the BBGKY closure or numerically via the same DSMC runs—the value of G_{eff} follows immediately from Eq. (214) together with the measured error bars. A useful way to encode the same nonlocality is to write

$$\lambda = \lambda_0 + \frac{2\lambda_0}{\rho_0} \delta\rho + \frac{2\lambda_0}{\rho_0 L_{\text{mix}}^2} \nabla^{-2} \delta\rho, \quad (216)$$

which in Fourier space corresponds to multiplying each mode by $\frac{2\lambda_0}{\rho_0} [1 + 1/(k^2 L_{\text{mix}}^2)]$. Acting with $-\Gamma$ on Eq. (216) simply reproduces Eq. (214) with $G_{\text{eff}} = (\Gamma_{\text{HS}}\lambda_0)/(2\pi\rho_0 L_{\text{mix}}^2)$, making the dependence on the nonlocal kernel explicit for order-of-magnitude estimates.

It is sometimes convenient to phrase the same closure as $\delta\lambda(k) = K(k)\delta n(k)$, where the hard-sphere Enskog theory gives $K(k) \rightarrow 2\lambda_0/\rho_0$ as $k \rightarrow 0$ and introduces only mild k^2

corrections for $kd \ll 1$. The constancy of $K(k)$ in this limit is what makes the Newtonian $1/r$ potential robust, while the higher-order terms encode short-range “Yukawa” departures that become relevant near the lattice scale. Measuring $K(k)$ (analytically and via DSMC) is now part of the Win A1 validation pipeline; improved statistics simply sharpen the quoted error bars on G_{eff} .

A.3 Half-space derivation from BBGKY

For completeness we sketch the BBGKY $\rightarrow$ Enskog $\rightarrow$ kernel route in a mode-by-mode fashion. Let $f^{(1)}(\mathbf{x}, \mathbf{p}, t)$ be the one-body distribution and $f^{(2)}(\mathbf{x}_1, \mathbf{p}_1, \mathbf{x}_2, \mathbf{p}_2, t)$ the two-body distribution. The first BBGKY equation for identical hard spheres is

$$\partial_t f^{(1)} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f^{(1)} = \int d\mathbf{p}_2 \int_{\mathbb{S}^2} d\Omega g \sigma(g, \Omega) \left[f^{(2)}(\mathbf{x}, \mathbf{p}', \mathbf{x} + d\hat{\Omega}, \mathbf{p}_2', t) - f^{(2)}(\mathbf{x}, \mathbf{p}, \mathbf{x} - d\hat{\Omega}, \mathbf{p}_2, t) \right], \quad (217)$$

where $g = |\mathbf{v} - \mathbf{v}_2|$ and $d = 2r_0$ is the diameter. The encounter-rate field is defined as the short-range limit of the collision term:

$$\lambda(\mathbf{x}, t) = \int d\mathbf{p}_1 d\mathbf{p}_2 \int_{\mathbb{S}^2} d\Omega g \sigma(g, \Omega) f^{(2)}(\mathbf{x} - \frac{d}{2}\hat{\Omega}, \mathbf{p}_1, \mathbf{x} + \frac{d}{2}\hat{\Omega}, \mathbf{p}_2, t). \quad (218)$$

In equilibrium $f^{(2)} = f_0^{(1)} f_0^{(1)} g_0(r)$, with g_0 the pair correlation at contact, giving $\lambda_0 = \sigma n_0^2 \langle g \rangle$ as used throughout the main text.

To linear order in a small density perturbation $\delta f^{(1)} = f_0^{(1)} (\delta n / n_0)$ the two-body distribution inherits two copies of the perturbation plus the contact correction:

$$f^{(2)} \approx g_0(r) f_0^{(1)}(\mathbf{p}_1) f_0^{(1)}(\mathbf{p}_2) \left[1 + \frac{\delta n(\mathbf{x}_1)}{n_0} + \frac{\delta n(\mathbf{x}_2)}{n_0} \right]. \quad (219)$$

Substituting Eq. (219) into Eq. (218) and integrating over the momenta yields the linear relation

$$\delta\lambda(\mathbf{x}) = \int d^3r \mathcal{K}(r) \left[\delta n\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) + \delta n\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right) \right], \quad (220)$$

with $\mathcal{K}(r) = \frac{1}{2} g_0(r) \sigma \langle g \rangle \delta(r - d)$ for hard spheres. Taking the Fourier transform of Eq. (220) gives

$$\delta\lambda(\mathbf{k}) = \left[2 \cos\left(\frac{1}{2} \mathbf{k} \cdot \mathbf{d}\right) \frac{\lambda_0}{\rho_0} \right] \delta\rho(\mathbf{k}) \equiv K(k) \delta\rho(\mathbf{k}), \quad (221)$$

where $\mathbf{d} = d\hat{\Omega}$ spans the collision directions and the angular average yields $K(k) = (2\lambda_0/\rho_0) \text{sinc}(kd/2)$. Expanding for $kd \ll 1$ gives

$$K(k) = \frac{2\lambda_0}{\rho_0} \left[1 - \frac{(kd)^2}{24} + \mathcal{O}((kd)^4) \right], \quad (222)$$

so $\delta\lambda(k)$ tends to a k -independent constant at long wavelengths and acquires $\mathcal{O}(k^2 r_0^2)$ corrections at shorter scales.

The purely local kernel therefore lacks the $1/k^2$ pole required for a Poisson tail. Incorporating the redundant-record mixing of Eq. (197) adds $\delta\lambda_{\text{mix}}(k) = \delta\lambda(k)/(k^2 L_{\text{mix}}^2)$, so

the tracer samples the effective kernel $K_{\text{eff}}(k) = K(k)[1 + 1/(k^2 L_{\text{mix}}^2) + \dots]$ introduced in Eq. (199). Combining Eqs. (221)–(222) with the mixed contribution and $U = -\Gamma\lambda$ and acting with $-k^2$ mode-by-mode produces

$$\nabla^2 U(\mathbf{x}) = 4\pi G_{\text{eff}} \delta\rho(\mathbf{x}) + \frac{(d^2/12) 4\pi G_{\text{eff}}}{\rho_0} \nabla^2 \delta\rho(\mathbf{x}) + \mathcal{O}(r_0^4 \nabla^4 \delta\rho), \quad (223)$$

where $4\pi G_{\text{eff}} = \Gamma \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) = 2\Gamma\lambda_0/(\rho_0 L_{\text{mix}}^2)$. Thus the exact BBGKY/Enskog treatment reproduces the Poisson equation in the $kd \rightarrow 0$ limit and makes explicit the leading Yukawa-type corrections suppressed by the collision diameter. This subsection formalises the assumptions used in the main text: local Maxwellian $f_0^{(1)}$, short-range hard-sphere collisions, and small Knudsen number so that higher BBGKY orders remain negligible.

A.4 1D toy-model illustration of drift and Poisson equation

As a concrete illustration of the collision-induced force, consider a simple one-dimensional gas with a slight density gradient. We take a test spheron (at rest, $v_\phi = 0$) in a background of identical spherons with number density slowly varying along x . Background spherons have a symmetric velocity distribution $f(v)$ (e.g. Maxwellian) with mean zero and mean squared speed $\langle v^2 \rangle$. Collisions impart momentum to the test depending on the direction: a collision from the left (with a background spheron moving rightward, $v > 0$) pushes the test to the right, whereas a collision from the right ($v < 0$) pushes the test to the left. Let n_L and n_R be the background densities immediately to the left and right of the test, respectively. The encounter rate from the left is $\nu_L = n_L \int_{v>0} f(v) v dv$ (integrating flux v of particles moving rightward); from the right, $\nu_R = n_R \int_{v<0} f(v) |v| dv$. The net momentum imparted per unit time is

$$F_{\text{net}} = m_0 \left(n_L \int_{v>0} f(v) v^2 dv - n_R \int_{v<0} f(v) v^2 dv \right),$$

where we use the fact that an elastic collision with an initially stationary test transfers momentum $m_0 v$ from a background spheron of velocity v . Changing variables in the second integral ($u = -v > 0$) and noting $f(-u) = f(u)$ for a symmetric distribution, this becomes

$$F_{\text{net}} = m_0 (n_L - n_R) \int_0^\infty f(u) u^2 du.$$

Because $\int_0^\infty u^2 f(u) du = \frac{1}{2} \langle v^2 \rangle$ (half the total velocity variance), we have $F_{\text{net}} = \frac{1}{2} m_0 \langle v^2 \rangle (n_L - n_R)$. The test's acceleration is $a = F_{\text{net}}/m_0 = \frac{1}{2} \langle v^2 \rangle (n_L - n_R)$. For a small gradient, write $n_L = n(x - \delta x) \approx n(x) - \delta x \partial_x n$ and $n_R = n(x + \delta x) \approx n(x) + \delta x \partial_x n$. Thus $n_L - n_R \approx -2 \delta x (\partial_x n)$, giving

$$a \approx -\langle v^2 \rangle \delta x \partial_x n(x).$$

This explicit calculation confirms that the test particle accelerates opposite to the density gradient: $a \propto -\partial_x n$. (In our simple model, δx can be interpreted as an effective interaction length – of order the mean free path – over which the asymmetry in n is sampled.) Since $\lambda \propto n^2$ in this regime, $\partial_x n$ is parallel to $\partial_x \lambda$, so we recover $a \propto -\partial_x \lambda$, in agreement with the general derivation above. If one coarse-grains to a continuum limit, this force can be derived

from a potential $U(x)$ with $-\partial_x U = a$. Taking $U \propto \lambda$, one then finds $\partial_{xx} U \propto \partial_{xx} \lambda \propto \partial_{xx} n^2$. For a small perturbation on a uniform background ($n = n_0 + \delta n(x)$), this yields $\partial_{xx} U \propto \partial_{xx} \delta n$. Matching constants to $\partial_{xx} U = 4\pi G_{\text{eff}} \delta \rho(x)$ (Poisson's equation) as done in A.3 above, one can solve for G_{eff} (and hence the observed G) in terms of microscopic parameters. Thus even this 1D toy model demonstrates how an inverse-square-like law (here in 1D it is an analogous linear law) emerges from the kinetic asymmetry.

A.5 Three-dimensional hard-sphere Chapman–Enskog structure

For completeness we indicate how the standard three-dimensional Chapman–Enskog (CE) machinery looks for a hard-sphere spheron gas and how the drift coefficient $\gamma(p)$ enters in that language. The aim is not to re-derive all textbook results, but to show that the “ $A_i \propto -\partial_i \lambda$ ” statement used in Sec. 5 can be written as an explicit collision integral over the hard-sphere kernel.

The Boltzmann collision operator for identical hard spheres of diameter $d = 2r_0$ in three dimensions may be written as

$$Q[f, f](\mathbf{x}, \mathbf{p}, t) = \int d^3 p_2 \int d\Omega B(g, \chi) [f' f'_2 - f f_2] , \quad (224)$$

where $f = f(\mathbf{x}, \mathbf{p}, t)$, $f_2 = f(\mathbf{x}, \mathbf{p}_2, t)$ are the pre-collision distributions, primes denote post-collision values, $g = |\mathbf{v} - \mathbf{v}_2|$ is the relative speed, χ is the scattering angle, and $B(g, \chi) = g \sigma(d, \chi)$ encodes the hard-sphere differential cross section. For perfectly elastic, isotropic hard spheres one has $\sigma(d, \chi) = (d^2/4)$ and $B \propto g$.

The CE method proceeds by expanding f around a local Maxwellian f_0 ,

$$f(\mathbf{x}, \mathbf{p}, t) = f_0(\mathbf{x}, \mathbf{p}, t) [1 + \phi(\mathbf{x}, \mathbf{p}, t)] , \quad (225)$$

where

$$f_0(\mathbf{x}, \mathbf{p}, t) = n(\mathbf{x}, t) \left(\frac{1}{2\pi m_0 k_B T(\mathbf{x}, t)} \right)^{3/2} \exp \left[- \frac{(\mathbf{p} - m_0 \mathbf{u}(\mathbf{x}, t))^2}{2m_0 k_B T(\mathbf{x}, t)} \right] \quad (226)$$

is a local equilibrium characterised by number density n , flow velocity $\mathbf{u}$, and temperature T . The small parameter in the expansion is the Knudsen number $\varepsilon = \ell_{\text{mfp}}/L$, the ratio of the mean free path to a typical macroscopic length scale L over which $n, T, \mathbf{u}$ vary.

To first order in ε one writes

$$\phi = \varepsilon \phi^{(1)} + \mathcal{O}(\varepsilon^2) , \quad (227)$$

and linearises $Q[f, f]$ in ϕ :

$$Q[f, f] \approx Q[f_0, f_0] + \hat{L}[\phi^{(1)}] + \mathcal{O}(\varepsilon^2) , \quad (228)$$

where $\hat{L}$ is the linearised collision operator acting on velocity-space functions. By definition $Q[f_0, f_0] = 0$. The CE procedure imposes that the zeroth-order dynamics is purely hydrodynamic (Euler equations), whereas the first-order term yields Navier–Stokes corrections and transport coefficients.

In the tracer-in-background setting we consider a background $f_b \approx f_0$ and a dilute tracer $\phi(\mathbf{x}, \mathbf{p}, t)$ satisfying

$$\partial_t \phi + \mathbf{v} \cdot \nabla_x \phi = Q[\phi, f_b] , \quad (229)$$

with $Q[\phi, f_b]$ obtained from (224) by treating f_b as fixed. When f_b is exactly homogeneous and isotropic, $Q[\phi, f_b]$ conserves tracer momentum and contains no net drift term. Under a weak gradient $\nabla n(\mathbf{x})$ and/or $\nabla T(\mathbf{x})$, we write $f_b(\mathbf{x}, \mathbf{p}) = f_0(\mathbf{p}; n(\mathbf{x}), T(\mathbf{x}))$ and expand to first order in $\nabla n, \nabla T$.

The standard result (see, e.g., Chapman–Cowling) is that the first-order correction to the collision operator can be written in a Fokker–Planck form,

$$Q[\phi, f_b] \approx -\nabla_p \cdot [\mathbf{A}(\mathbf{x}, \mathbf{p}) \phi] + \nabla_p \cdot [\mathbf{D}(\mathbf{x}, \mathbf{p}) \nabla_p \phi] , \quad (230)$$

where $\mathbf{A}$ and $\mathbf{D}$ are drift and diffusion coefficients given by explicit velocity integrals over f_0 and the kernel $B(g, \chi)$. In particular one finds

$$A_i(\mathbf{x}, \mathbf{p}) = - \sum_j \left[\alpha_{ij}(\mathbf{p}) \partial_j n(\mathbf{x}) + \beta_{ij}(\mathbf{p}) \partial_j T(\mathbf{x}) \right] , \quad (231)$$

with tensors α_{ij}, β_{ij} determined by moments of $Q[\cdot, f_0]$.

For an isotropic hard-sphere background, α_{ij} and β_{ij} are proportional to δ_{ij} , so the drift reduces to

$$\mathbf{A}(\mathbf{x}, \mathbf{p}) = -\alpha(p) \nabla n(\mathbf{x}) - \beta(p) \nabla T(\mathbf{x}) , \quad (232)$$

for scalar functions $\alpha(p)$ and $\beta(p)$. In the simplest isothermal case, $\nabla T \approx 0$, and we recover

$$\mathbf{A}(\mathbf{x}, \mathbf{p}) \approx -\alpha(p) \nabla n(\mathbf{x}) . \quad (233)$$

Because $\lambda(\mathbf{x})$ in a thermalised hard-sphere gas satisfies $\lambda \propto n^2$ to leading order, we have $\nabla n = (1/2n) \nabla \lambda$, and thus

$$\mathbf{A}(\mathbf{x}, \mathbf{p}) \approx -\gamma(p) \nabla \lambda(\mathbf{x}) , \quad \gamma(p) := \frac{\alpha(p)}{2n(\mathbf{x})} , \quad (234)$$

which is the three-dimensional analogue of the schematic relation used in Sec. 5.2. The transport coefficient $\alpha(p)$ can be expressed in terms of the standard hard-sphere collision integrals $\Omega^{(l,s)}$ (in the notation of kinetic theory texts), e.g.

$$\alpha(p) \propto \frac{1}{n_0} \frac{\Omega^{(1,1)}}{\Omega^{(2,2)}} \left(\frac{p}{m_0} \right) , \quad (235)$$

with n_0 a reference density; the precise proportionality factors depend on the chosen normalisation but are fully explicit once $\sigma(d, \chi)$ is specified. Thus, in principle, all pieces entering the effective drift $\gamma(p)$ (and hence G_{eff}) can be written as convergent integrals over known functions.

A.6 Hydrodynamic closure and explicit G_{eff}

To connect to the effective Newton constant G_{eff} more concretely, we recall the momentum balance equation for the tracer density ρ_ϕ and mean velocity $\mathbf{u}$:

$$\partial_t(\rho_\phi u_i) + \partial_j \Pi_{ij}^\phi = F_i^{\text{coll}} = - \int d^3p A_i(\mathbf{x}, \mathbf{p}) \phi(\mathbf{x}, \mathbf{p}, t) . \quad (236)$$

Using $A_i \approx -\gamma(p)\partial_i\lambda$ and defining the scalar

$$\Gamma := \frac{1}{\rho_\phi} \int d^3p \gamma(p) \phi(\mathbf{x}, \mathbf{p}, t) , \quad (237)$$

we obtain

$$F_i^{\text{coll}} \approx -\rho_\phi(\mathbf{x}, t) \Gamma \partial_i \lambda(\mathbf{x}, t) \equiv -\rho_\phi \partial_i U(\mathbf{x}, t) , \quad (238)$$

with $U(\mathbf{x}, t) := -\Gamma \lambda(\mathbf{x}, t)$, exactly as in the main text.

For a background with mean number density n_0 , hard-sphere radius r_0 , and thermal speed $v_{\text{th}} = (2k_B T/m_0)^{1/2}$, the homogeneous encounter rate is

$$\lambda_0 = \sigma n_0^2 \langle g \rangle \sim \pi d^2 n_0^2 v_{\text{th}} , \quad (239)$$

with $d = 2r_0$ and $\langle g \rangle$ an order-one multiple of v_{th} for a Maxwellian. The mass density is $\rho_0 = m_0 n_0$. Using the linear relation $\delta\lambda \approx 2\lambda_0 \delta n/n_0$ together with the mixed closure of Sec. A, $\nabla^2 \lambda = -(2\lambda_0/\rho_0 L_{\text{mix}}^2) \delta\rho$, one finds

$$4\pi G_{\text{eff}} = 2\Gamma \frac{\lambda_0}{\rho_0 L_{\text{mix}}^2} \sim 2\Gamma \frac{\pi d^2 n_0^2 v_{\text{th}}}{m_0 n_0 L_{\text{mix}}^2} . \quad (240)$$

Taking $L_{\text{mix}} = \sqrt{N_{\text{mix}}} \ell_{\text{mfp}}$ with N_{mix} measured in the redundant-record hierarchy fixes the overall scale: the microscopic parameters (m_0, r_0) , the macroscopic state (n_0, T) , and the mixing statistics then determine G_{eff} (and thereby match the observed G) without additional inputs. Short-range sinc-kernel corrections still furnish Yukawa terms with range $d/\sqrt{12}$, but the value of the Newton constant itself is now tied to the infrared residue set by L_{mix} , so it is not an adjustable parameter but a calculable emergent constant.

A fully rigorous treatment would keep all order-one factors, evaluate the relevant $\Omega^{(l,s)}$ integrals explicitly, and compare the resulting G_{eff} with the observed value; that is work for a dedicated kinetic-theory paper, but the structure here shows that nothing mysterious is required beyond standard CE machinery.

Appendix F Toy Lattice Dispersion and Emergent Relativity

A key proposal of this framework is that at large scales the spheron substrate can exhibit an *accidental Lorentz symmetry*: collective excitations obey a linear dispersion $E \approx c p$ to leading order, with only small curvature from the underlying length scale r_0 . Here we demonstrate this explicitly in a simple model. Consider a one-dimensional chain of identical

spherons of mass m_0 at equal spacing r_0 . For tractability we model the interactions as linear springs between nearest neighbors (mimicking a stiff contact potential). Let $u_n(t)$ be the displacement of the n th spheron from its equilibrium position. The linearized equation of motion is

$$m_0 \ddot{u}_n = K [(u_{n+1} - u_n) - (u_n - u_{n-1})] = K (u_{n+1} - 2u_n + u_{n-1}) ,$$

with K the effective spring constant. We seek plane-wave solutions of the form $u_n(t) = e^{i(knr_0 - \omega t)}$. Inserting this ansatz yields the dispersion relation

$$m_0 \omega^2 = 2K (1 - \cos(kr_0)) = 4K \sin^2 \frac{kr_0}{2} .$$

Thus

$$\omega^2 = \frac{4K}{m_0} \sin^2 \frac{kr_0}{2} .$$

For long wavelengths ($kr_0 \ll 1$), we can expand the sine term: $\sin^2(\frac{kr_0}{2}) = \frac{k^2 r_0^2}{4} - \frac{k^4 r_0^4}{48} + \mathcal{O}(k^6)$. This gives

$$\omega^2 \approx \frac{4K}{m_0} \left(\frac{k^2 r_0^2}{4} - \frac{k^4 r_0^4}{48} + \dots \right) = \frac{K r_0^2}{m_0} k^2 \left(1 - \frac{k^2 r_0^2}{12} + \dots \right) .$$

Taking the square root (for $\omega > 0$) yields

$$\omega \approx \sqrt{\frac{K}{m_0}} r_0 k \left(1 - \frac{k^2 r_0^2}{24} + \dots \right) .$$

We identify the sound speed $c = \sqrt{\frac{K}{m_0}} r_0$ as the emergent propagation speed of small oscillations. In terms of momentum $p \approx \hbar k$ and energy $E = \hbar \omega$, the dispersion can be written

$$E(p) \approx c p \left(1 - \frac{1}{24} (p r_0)^2 + \dots \right) , \quad (241)$$

to leading non-trivial order. This is of the desired form $E = c p (1 + \alpha (p r_0)^2 + \dots)$ with an explicit coefficient ($\alpha = -1/24$ in this model). At long wavelengths ($p r_0 \ll 1$) the dispersion is essentially linear (resembling a relativistic $E \approx c p$), while for shorter wavelengths the discreteness of the chain produces a slight curvature (Lorentz-violating correction). In a realistic spheron medium, we expect a similar result: small-amplitude long-wavelength modes (such as density waves or other collective excitations) propagate at some characteristic speed c , and deviations from Lorentz symmetry appear only as tiny $(p r_0)^2$ effects. The exact value of α will depend on the details of the microdynamics (e.g. spatial dimension and interaction specifics), but order-of-magnitude one might expect $|\alpha| \sim 10^{-2}$ – 10^{-1} for a tightly packed medium (the lattice model above gives $|\alpha| \approx 0.04$). If r_0 is extremely small (say Planck-scale), then for any attainable momentum p the quantity $(p r_0)^2$ is minuscule, making $\delta = \alpha (p r_0)^2$ essentially undetectable. This is consistent with the empirical bounds discussed in Sec. 5.7: achieving $\delta \lesssim 10^{-15}$ for GeV-scale momenta requires r_0 on the order of 10^{-35} m. The lattice calculation here illustrates how an approximate Lorentz symmetry can emerge from a discrete substrate, and how probing it at higher energies (shorter wavelengths) would eventually reveal slight dispersion curvature, providing a potential experimental handle to detect the underlying granularity.

C.2 Hydrodynamic sound mode from the spheron Boltzmann equation

The lattice toy model above demonstrates explicitly how a discrete substrate can produce an approximately linear dispersion with $(pr_0)^2$ corrections. We now sketch how an analogous relativistic-looking mode arises directly in the spheron gas described by the Boltzmann equation.

Consider a spatially uniform equilibrium $f_0(\mathbf{p})$ with density n_0 , temperature T_0 , and zero bulk velocity. Introduce small perturbations of density, velocity, and temperature:

$$n(\mathbf{x}, t) = n_0 + \delta n(\mathbf{x}, t), \quad \mathbf{u}(\mathbf{x}, t) = \delta \mathbf{u}(\mathbf{x}, t), \quad T(\mathbf{x}, t) = T_0 + \delta T(\mathbf{x}, t), \quad (242)$$

and linearise the Boltzmann equation in these perturbations. Taking low-order velocity moments yields the linearised hydrodynamic equations. For an ideal monatomic gas one finds, in the absence of gravity,

$$\partial_t \delta n + n_0 \nabla \cdot \delta \mathbf{u} = 0, \quad (243)$$

$$\rho_0 \partial_t \delta \mathbf{u} + \nabla \delta p = \eta \nabla^2 \delta \mathbf{u} + \left(\zeta + \frac{1}{3} \eta \right) \nabla (\nabla \cdot \delta \mathbf{u}), \quad (244)$$

$$\partial_t \delta T + (\gamma - 1) T_0 \nabla \cdot \delta \mathbf{u} = \kappa \nabla^2 \delta T, \quad (245)$$

where $\rho_0 = m_0 n_0$, γ is the ratio of specific heats ($\gamma = 5/3$ for a monatomic gas), and η, ζ, κ are shear viscosity, bulk viscosity, and thermal conductivity, all expressible in terms of hard-sphere CE collision integrals.

Seeking plane-wave solutions of the form $\delta n, \delta T, \delta \mathbf{u} \propto \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$ and neglecting dissipative terms ($\eta, \zeta, \kappa \rightarrow 0$) yields the standard sound-mode dispersion

$$\omega^2 = c_s^2 k^2, \quad (246)$$

with sound speed

$$c_s^2 = \gamma \frac{k_B T_0}{m_0}. \quad (247)$$

Thus, at the level of ideal hydrodynamics, the spheron gas supports a long-wavelength collective excitation with exactly linear dispersion and propagation speed c_s . For observers built from such modes, the effective light-cone at long wavelengths would be defined by $|\mathbf{x}| = c_s t$, and Lorentz-like kinematics can be reconstructed using c_s as the invariant speed.

Including the dissipative terms from CE, one finds corrections of the form

$$\omega = c_s k - i \Gamma_{\text{att}} k^2 + \mathcal{O}(k^3), \quad (248)$$

where the attenuation coefficient Γ_{att} is a specific combination of η, ζ, κ and equilibrium quantities. These corrections manifest as both a small imaginary part (damping) and, at higher order, a slight deviation of $\text{Re } \omega(k)$ from strict linearity. In dimensionless form one may write

$$\omega^2 \approx c_s^2 k^2 \left[1 + \alpha_{\text{kin}} (\ell_{\text{mfp}} k)^2 + \dots \right] - 2i c_s^2 k^2 \Gamma_{\text{att}} + \dots, \quad (249)$$

where ℓ_{mfp} is the mean free path and α_{kin} is an order-one coefficient determined by the hard-sphere collision integrals. The structure is directly analogous to the lattice result of Eq. (241): a leading linear regime with small $(k \ell_{\text{mfp}})^2$ corrections.

If we identify the emergent invariant speed c of the effective relativistic description with c_s of a particular spheron phase, then the long-wavelength collective mode behaves as a massless relativistic field with suppressed Lorentz-violating corrections controlled by the small Knudsen number $k\ell_{\text{mfp}} \ll 1$. The numerical coefficient α_{kin} can be computed explicitly once the hard-sphere collision integrals $\Omega^{(l,s)}$ are specified, in direct analogy with the evaluation of transport coefficients in Sec. E.2. This provides a fully spheron-based kinetic derivation of an approximately relativistic dispersion, complementing the lattice toy model and reinforcing the plausibility of emergent Lorentz symmetry from the same Boltzmann framework that yields gravity.

Appendix G Toy Model: Redundant Records and Born Rule

This appendix instantiates Definitions 2–4 and Theorem 3 in a concrete spheron setting. The accessible-information and enviance steps are standard Quantum Darwinism machinery; the new RTP ingredient is the explicit accounting of redundant records built from spheron collisions.

We now construct a simplified measurement scenario to demonstrate how the Born rule probabilities ($p = |c|^2$) can emerge from spheron microdynamics when redundant records are considered. The key ingredients will be (i) a system with two possible outcomes, (ii) an environment composed of many small subsystems that can each record the outcome (hence providing redundancy), and (iii) a symmetry argument (enviance) to fix the probability assignment.

Consider a system S (a distinguished spheron or cluster of spherons) prepared in a state that can lead to two macroscopically distinct outcomes A or B . For concreteness, imagine a single spheron heading toward a symmetric beam splitter such that it has two possible paths: path A (upwards) or path B (downwards). We attach an environment E consisting of many identical detector spherons (or fragments) that can interact with the system. If the system goes along path A , it will collide with each detector in a way that marks them with outcome A (e.g. knocking each into a particular “ A -state”); if the system goes along path B , it will mark them in a distinct B -state. Initially, S is in a superposition $|\psi\rangle_S = c_1 |A\rangle + c_2 |B\rangle$ and all detectors in E are in a neutral state $|0\rangle$ (no record).

Now, let S interact with E . In our spheron picture, this would be a sequence of collisions between S (or its progeny along each path) and each detector spheron. The outcome is an entangled state:

$$|\Psi\rangle_{SE} = c_1 |A\rangle_S \otimes |E_A\rangle + c_2 |B\rangle_S \otimes |E_B\rangle, \quad (250)$$

where $|E_A\rangle$ denotes the collective state of all detectors after an A outcome (e.g. each detector spheron has been flipped into an A -record state), and $|E_B\rangle$ is the state with all detectors indicating B . The coefficients c_1, c_2 inherit those of the initial state. State (250) captures the idea that the environment has recorded the outcome but, until decoherence or observation, S and E jointly remain in a superposition of “ A -branch” and “ B -branch”.

To extract probabilities from this state, we apply rationality and symmetry arguments (as in Quantum Darwinism and enviance approaches). First, consider the special case

where $|c_1| = |c_2|$. That is, the state has equal amplitude magnitudes for A and B (e.g. $c_1 = c_2 = 1/\sqrt{2}$ up to a phase). In this symmetric situation, the two branches differ only by the relabeling of outcome and corresponding detector states. An observer with no prior bias must assign equal credence to A and B because there is nothing to break the symmetry between them. Thus, $P(A) = P(B) = 1/2$, which agrees with the Born rule since $|c_1|^2 = |c_2|^2 = 1/2$ in this case. This illustrates the enviance principle: one can swap the two branches (by exchanging $|A\rangle_S \leftrightarrow |B\rangle_S$ and simultaneously swapping the detector states $|E_A\rangle \leftrightarrow |E_B\rangle$) without any physical change that an observer can detect, so the outcome probabilities must be equal.

Next, consider the general case where $|c_1|^2 \neq |c_2|^2$. By continuity, one expects the probability assignment to vary smoothly with $|c_1|$. A powerful approach is to consider rational amplitude ratios that admit an interpretation as frequency. Suppose

$$|c_1|^2 = \frac{m}{m+n} \ , \quad |c_2|^2 = \frac{n}{m+n} \ ,$$

for some positive integers m and n . We can then imagine that our experiment is not a one-off event but one of a series of $m+n$ identical runs (or, equivalently, that our environment consists of $m+n$ identical groups of detectors which could in principle record $m+n$ independent trials). If quantum mechanics is internally consistent, the state of $m+n$ such independent copies of $|\psi\rangle_S$ should lead to $(m+n)$ -fold entangled branches corresponding to sequences of outcomes. In a typical sequence, outcome A might occur m times and B occurs n times (in any order). Crucially, if the $m+n$ runs are identical and independent (which we can ensure by designing E appropriately or letting one large environment act as many independent fragments), then all specific sequences with exactly m A 's and n B 's will have the same amplitude. There are $\binom{m+n}{m}$ such sequences (branches of the total state). For example, if $m=2, n=1$, the composite state will contain three branches corresponding to outcome strings AAB, ABA, BAA – each with identical amplitude magnitude $\sim |c_1|^2|c_1|^2|c_2|^2$ (up to permutation). By symmetry, all these sequences are equally likely. Because each sequence has m occurrences of A and n of B , we can say that in $m+n$ trials one expects m A 's and n B 's on average. Therefore, the probability of A in a single trial should be $m/(m+n) = |c_1|^2$ (and similarly $P(B) = n/(m+n) = |c_2|^2$). This reasoning – essentially the frequency stabilization of probability in the limit of many trials – shows that $p_A = |c_1|^2$ is the only consistent assignment for rational $|c_1|^2$. Since any real number in $[0, 1]$ can be arbitrarily well approximated by a rational, the continuity of probability implies $p_A = |c_1|^2$ in general.

In the language of the environment and redundancy: if the environment can be decomposed into many fragments that each carry a partial record of the outcome, the only stable assignment of probabilities that different observers (with access to different subsets of fragments) will agree upon is the quadratic Born rule. If, for example, one tried $P(A) = |c_1|^p$ with $p \neq 2$, one would find that combining fragments or considering multiple trials would lead to inconsistent inferences for different observers (violating the “agree on probabilities” criterion in the RTP). The choice $p=2$ avoids such inconsistencies and is invariant under the branch-exchange symmetry. In summary, the toy model illustrates how the combination of redundant environmental records and symmetry (enviance) enforces the Born rule. The spheron collisions supply the mechanism for environment entanglement and record-making, while rational inference (coarse-grained to detector counts) picks out the $|c|^2$ law as the unique rule

that agents embedded in this spheron world should use to predict their future records.

Appendix H Order-Parameter Toy Models for Emergent Gauge and Matter

A speculative pillar of the spheron framework is that familiar quantum fields (gauge bosons, fermionic matter) might not be fundamental but could emerge as collective excitations or topological patterns of the spheron substrate. Here we describe two toy constructions to make this idea more concrete: one yielding a gauge-like field from a continuous symmetry in a spheron assembly, and another producing Dirac-like particle modes in a lattice.

D.1 Continuous symmetry and an emergent gauge field

Imagine a large ensemble of spherons arranged in a symmetric configuration (for instance, a crystalline lattice or a uniform dense fluid). Suppose this system undergoes a spontaneous symmetry breaking: it develops an ordered pattern that is invariant under a continuous transformation. A useful analogy is a superfluid—composed of underlying particles but describable by a coherent order parameter $\Phi(\mathbf{x}) = \phi_0 e^{i\theta(\mathbf{x})}$, where the phase $\theta(\mathbf{x})$ is defined modulo 2π . In a spheron context, one could envision the spherons forming a collective oscillation or flow such that θ represents the local phase of that motion.

The important point is that the underlying dynamics are invariant under a global shift of θ (e.g. all spherons’ motion shifted by the same phase), which means only phase *gradients* have physical effect. This is analogous to a gauge symmetry: the absolute phase is unobservable (like a gauge choice), while spatial variations $\nabla\theta$ correspond to observable quantities (like a field strength or potential). Indeed, in superfluids $\mathbf{v} \propto \nabla\theta$ acts as a velocity field, and $\nabla \times \mathbf{v} = 0$ in singly-connected regions (mirroring the idea that a pure gauge has no curl/field). A nonzero circulation $\oint \nabla\theta \cdot d\ell = 2\pi m$ (with m integer) indicates a topological defect—a vortex. These vortices carry quantized circulation and behave like particle-like excitations; one can view them as analogous to charged particles or flux quanta in a gauge field.

In our spheron toy model, we can imagine a similar scenario: the local phase $\theta(x)$ of an oscillatory mode acts like a gauge degree of freedom. Small fluctuations of θ (Goldstone modes) propagate as waves, and only differences $\theta(x) - \theta(y)$ matter physically. We could define an emergent vector potential $\mathbf{A} = (\hbar/q)\nabla\theta$ (for some pseudo-charge q) which is purely a mathematical construct here, but under which θ shifts by a constant correspond to $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$ (a gauge-like redundancy). The “field strength” $\mathbf{B} = \nabla \times \mathbf{A}$ would be proportional to the vorticity of the spheron flow. A configuration of spherons with net circulation (many vortices) would produce a nonzero $\mathbf{B}$ field in analogy to magnetic flux. Crucially, the dynamics of small perturbations in such a medium could be described by equations similar to Maxwell’s equations, with the role of electric and magnetic fields played by irrotational ($\nabla\theta$) and rotational flow components. For instance, sound-wave equations (in a compressible superfluid) can be written in a form that mimics wave equations for a massless scalar or vector potential.

To illustrate: if ρ is the spheron mass density and $\mathbf{v} = \nabla\theta$ the flow velocity, the Euler

equations in the irrotational (potential flow) case yield $\partial_t \theta + \frac{1}{2}(\nabla \theta)^2 + \mu(\rho) = 0$ (where $\mu(\rho)$ is a chemical potential). Linearizing around a uniform background ρ_0 , one finds $\partial_t^2(\delta\theta) - c^2 \nabla^2(\delta\theta) = 0$, which is a wave equation. This $\delta\theta$ behaves like a massless field with speed c (the sound speed). Meanwhile, a vortex solution where θ winds by 2π around a line has a velocity field $v_\varphi(r) \propto 1/r$ (in 2D) akin to the field around a current-carrying wire. The quantization of $\oint \nabla \theta \cdot d\ell$ ensures the stability of these vortex “particles.” We see that the broken symmetry (conservation of circulation) leads to a localized object (vortex) that carries a topological charge (winding number) and an associated long-range field (velocity circulation). This is strongly analogous to an emergent $U(1)$ gauge theory where the vortex is like a charged particle and the circulation is like a magnetic field around it. While this analogy is classical, quantum effects (like quantization of circulation and potentially quantum tunneling of vortices) hint at deeper connections if this scenario were quantized.

In summary, a continuous order parameter in a spheron system provides an “angle” or phase that can act as a gauge-like variable. The redundancy under global phase rotations and the physical significance of only phase gradients echo the structure of electromagnetism. Topological defects in the order parameter then play the role of matter particles, and emergent fields (like flow vorticity) play the role of force fields. This toy construction suggests how an effective gauge field could arise from the microphysics: not by positing new fundamental interactions, but as a collective description of coherent spheron behavior.

D.2 Dirac-like lattice excitations

A separate route to emergent “matter” fields is via lattice band structure. In condensed matter physics, it is well known that electrons hopping on certain crystal lattices exhibit quasiparticles with relativistic (Dirac) dispersion. A famous example is the honeycomb lattice of graphene: at half-filling, the electron band structure has conical intersections (Dirac points) at the Brillouin zone corners, leading to an effective Hamiltonian identical to the 2D massless Dirac equation. The low-energy electrons behave as if they are relativistic fermions with a “light speed” given by the Fermi velocity ($v_F \approx c/300$ in graphene). This emergent Lorentz symmetry and the two-component (spinor) structure of excitations occur even though the underlying system is a non-relativistic, non-fermionic (in principle one could even simulate it with classical waves) setup.

We can envision a similar situation for vibrational or rotational modes of a spheron lattice. Consider spherons arranged in a 2D honeycomb pattern, which has two sublattices (often labeled A and B). One can define a “pseudo-spinor” whose components are amplitudes on sublattice A and B. The equations governing small oscillations or wave propagation on this lattice (for example, if spherons are coupled by elastic contacts or even if we consider propagation of momentum exchanges) can be written in a form analogous to tight-binding models. A plane-wave ansatz yields a dispersion relation $E(\mathbf{k})$ that, near the corners of the Brillouin zone ($\mathbf{K}$ points), takes the form $E \approx \pm v |\mathbf{q}|$ for wavevectors $\mathbf{k} = \mathbf{K} + \mathbf{q}$ with $|\mathbf{q}| \ll |\mathbf{K}|$. The plus and minus correspond to two bands touching at the Dirac point, analogous to particle and hole states. The effective equation of motion for long-wavelength envelope fields on sublattice A/B is identical to a 2-component Dirac equation $(v \boldsymbol{\sigma} \cdot \mathbf{p})\psi = E\psi$, where $\boldsymbol{\sigma}$ are Pauli matrices mixing the sublattices. If one introduces a slight dimerization or staggered potential (making sublattice A and B environments slightly different), a gap

opens in the dispersion, corresponding to a Dirac mass term. This is exactly analogous to generating a rest mass for the emergent fermion.

In principle, spherons being classical identical particles, one does not have true fermionic statistics here. However, the emergent Dirac field is a collective mode – it doesn’t care that its microscopic origin is bosonic or classical, as long as the linearized equations map to the Dirac equation. Such phenomena have been experimentally observed with acoustic waves, photonic crystals, and even mechanical oscillators that exhibit Dirac or Weyl points in their spectrum. By extension, a spheron lattice could sustain an emergent “matter field” as a normal mode. This would fulfill, in a toy way, the idea of matter particles (Dirac fermions) coming out of a substrate of constituents that themselves are not fermions. The topological stability of these modes (protected by lattice symmetry) could mimic conservation laws or quantum numbers of particle physics.

Of course, to go from such analogies to the full richness of the Standard Model is an enormous leap. The purpose of this Appendix section is not to claim we have derived the Standard Model from spherons, but to show plausibly that if spherons form complex ordered states, they can give rise to fields with gauge-like invariances and particle-like excitations. The gauge example (D.1) provides a heuristic link between continuous symmetries and long-range forces, while the lattice Dirac example (D.2) shows how relativistic dispersion and spinor structure can appear from a purely spatial micro-pattern. These serve as “existence proofs” within our program: it is not absurd to think that, in some yet-to-be-specified phase of spheron matter, emergent fields and particles could arise. Working out a concrete, 3D, interacting example that yields something closer to quantum electrodynamics or Yang–Mills theory is a daunting challenge (and firmly in the realm of future work), but the toy models here provide stepping stones and inspiration for that trajectory.

Appendix I Combinatorial Entropy in a 2D Jam (Black Hole Analog)

We offer a simplified combinatorial calculation to reinforce the claim that a jammed “black hole” made of spherons has entropy coming predominantly from its surface degrees of freedom, resulting in an area-law scaling. Instead of a full 3D calculation (which would involve complicated packing microstates), we examine a two-dimensional analog: hard disks (2D “spherons”) packed into a circular region. The 2D cluster can be thought of as a toy black hole cross-section; we expect the entropy to scale with the perimeter of the cluster in this case.

Consider N identical hard disks of diameter $d = 2r_0$ arranged in a close-packed cluster of radius R . The maximal packing fraction in 2D (hexagonal packing) is ≈ 0.91 , but we will only need scaling relations. The area of the cluster is $A \sim \pi R^2$, which for a densely packed configuration is roughly N times the area per disk (including the fact that disks cannot overlap). Thus $N \approx \frac{\pi R^2}{\pi r_0^2 \phi_{\max}} = \frac{R^2}{r_0^2 \phi_{\max}}$ up to order-one factors, so $R \sim r_0 \sqrt{N}$. Meanwhile, the perimeter (boundary length) of the cluster is $P \sim 2\pi R$ for a roughly circular shape (with some deviation due to the hexagonal packing at the edges). The number of disks that lie on the boundary, N_{surf} , is proportional to P/d (perimeter divided by disk diameter). We can

estimate $N_{\text{surf}} \approx \frac{P}{2r_0} \sim \frac{2\pi R}{2r_0} = \frac{\pi R}{r_0}$. In terms of N , $N_{\text{surf}} \sim \frac{\pi r_0 \sqrt{N}}{r_0} = \pi \sqrt{N}$, i.e. $N_{\text{surf}} = \mathcal{O}(N^{1/2})$. Thus the fraction of disks on the surface scales as $\sim N^{-1/2}$ (which becomes negligible for very large N , reflecting that most disks are in the interior).

Now, consider the number of microstates (distinct microscopic arrangements of disks) corresponding to a given macroscopic cluster of radius R . In a perfect hexagonal crystal of disks, there might be essentially a single arrangement (neglecting global translations). However, if we allow some defects or slight displacements, the interior disks have very limited freedom (each is caged by neighbors). By contrast, disks at the boundary have more room to jiggle or can sometimes be configured in alternate patterns (e.g. one less disk in the outermost row vs. a packed row, etc.) without significantly changing the overall radius R . Let us suppose that each boundary disk has g possible micro-configurations (different positions or small orientation freedoms) consistent with the cluster's macrostate. These might be discrete choices (e.g. a disk could slide into one of a few pockets along the edge) or quasi-continuous (a disk could rattle in a cell contributing a phase-space volume). For a rough count, the total number of microstates Ω factorizes approximately as a product of independent choices for each surface disk (if interactions are short-range, different edge regions can rearrange somewhat independently). Then

$$\Omega \sim g^{N_{\text{surf}}} ,$$

i.e. exponential in the number of boundary disks. The entropy is

$$S = k_B \ln \Omega \approx k_B N_{\text{surf}} \ln g .$$

Since $N_{\text{surf}} \propto R \propto \sqrt{N}$ and $\sqrt{N}$ is proportional to the cluster's circumference P , we have

$$S \propto P ,$$

a perimeter law in 2D. In a three-dimensional close-packed sphere cluster (our actual black hole model), by analogous reasoning the number of spherons on the outer layer scales as $N_{\text{surf}}^{(3D)} \sim N^{2/3}$ (because the surface area of a sphere goes as R^2). Independent configurational choices for each surface spheron then give $\Omega \sim g^{N^{2/3}}$ and

$$S \sim k_B N^{2/3} \ln g \propto N^{2/3} \propto A ,$$

since the horizon area A of the 3D cluster is $\propto R^2 \propto N^{2/3}$. This qualitative calculation aligns with the statement in the main text: the entropy of a spheron black hole is dominated by the surface degrees of freedom and is approximately proportional to the horizon area (with only relatively small subleading contributions from the interior). We emphasize that while this argument captures the leading scaling, a more detailed counting would be needed to determine the precise coefficient (to compare with $1/4$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$, for example) and to include any correlations among surface degrees. Nonetheless, the physical picture is clear. The jammed core's interior is rigid (few microstates), but the "horizon" is a dynamic layer where spherons can rearrange in many ways. This not only provides a combinatorial entropy $S \sim$ (surface area) but also an avenue for energy and information to be carried away by emitted spherons (as those surface configurations fluctuate and occasionally eject particles). In essence, the spheron model's black hole analogs naturally satisfy a holographic entropy scaling: the information content scales with the bounding area, not the volume, because only the boundary is configurationally rich.

Appendix J Primordial Spectrum from Lattice Shattering

One of the more ambitious ideas in our cyclic cosmology pillar is that the initial spectrum of density perturbations (which later gives rise to CMB anisotropies and galaxies) arises from the statistics of how a close-packed spheron “primeval lattice” shatters during the Big Bounce/inflation-like burst. Here we outline a toy computation to show how a near scale-invariant spectrum could emerge from such a process.

When the packed state “shatters,” we imagine that regions of the spheron lattice decouple and fly apart. Some regions might fragment into smaller pieces, while others remain larger. This is akin to breaking a solid into chunks of various sizes. Each chunk will eventually correspond to an overdense region (that later forms a galaxy or cluster) or simply contribute to the smooth background if it’s small enough to disperse. We can characterize the outcome of shattering by a distribution of fragment masses or sizes.

Let $n(M)$ be the number density (per unit volume of space) of fragments with mass (or equivalent spheron number) M . To connect to density fluctuations, consider a large spatial volume and partition it into domains of size L . The mass in each domain is $M_L = \sum_{\text{fragments in volume } L} M_i$. The density contrast on scale L is $\delta_L = (M_L - \bar{M}_L)/\bar{M}_L$, where $\bar{M}_L$ is the average mass in volume L . If the fragments are distributed randomly (uncorrelated positions) and their mass distribution is broad, one can compute the variance $\langle \delta_L^2 \rangle$ by integrating over $n(M)$ weighted by the volume L^3 and the mass variance.

A simpler route is to estimate the power spectrum $P(k)$ (Fourier transform of the two-point correlation of δ) directly from fragment statistics. In the Poisson limit (fragments uncorrelated in position), each fragment of mass M contributes a term $\propto M^2$ to density fluctuations at scales larger than its own size. Summing incoherently, we get

$$P(k) \propto \int n(M) M^2 W^2(k; M) dM ,$$

where $W(k; M)$ is the window function encoding how a fragment of mass M (size $\sim M^{1/3}$ in physical extent) contributes to wavenumber k . For k much smaller than the inverse size of the fragment, $W \approx 1$ (the fragment’s mass acts as a point mass); for k larger than the fragment’s inverse size, W oscillates or decays (the fragment’s internal structure matters and cancels out large- k modes). For a broad distribution of M , the contributions to $P(k)$ at a given k come mainly from fragments of size around $\lambda \sim 1/k$.

Now, a scale-invariant spectrum (Harrison–Zel’dovich, $n_s \approx 1$) means that $\Delta^2(k) \sim k^3 P(k)$ is approximately constant for small k (in cosmology $\Delta^2(k)$ is the variance per $\ln k$). Equivalently, $P(k) \propto k^{-3}$ at horizon exit (later it evolves but let’s consider the primordial shape). How can our fragment scenario yield that? Intuitively, scale invariance will result if each logarithmic interval of fragment mass contributes roughly equally to the mass variance on its corresponding scale. This suggests a power-law distribution of fragment masses. Suppose

$$n(M) \sim A M^{-\beta}$$

over a range $M_{\min}$ to $M_{\max}$. The total mass density in fragments is $\rho_{\text{frag}} = \int M n(M) dM$ (which should equal the cosmic density after shattering). For convergence at the low- M end,

we need $\beta < 2$. For scale-invariant fluctuations, we heuristically want the mass variance to be dominated equally by all mass scales. The contribution to $\langle \delta^2 \rangle$ from fragments of mass M is $\sim n(M)M^2$. So require $n(M)M^2$ is about constant per logarithmic interval: $d \ln M n(M)M^2 \sim \text{const.}$ That implies $n(M)M^2 \propto 1/M$ (since $d \ln M = dM/M$), hence $n(M) \propto M^{-3}$ on dimensional grounds. But $n(M) \propto M^{-3}$ gives $\beta = 3$, which is too steep to converge at low mass (and would put most of the mass in the smallest fragments, which is not physical if we expect some large regions to remain).

A more refined argument goes via the power spectrum integral above. If β is between 1 and 2, the total mass is dominated by the largest fragments, but the variance might still get significant contributions from smaller ones. To find $P(k)$, note that fragments of mass M contribute mainly to modes $k \sim 1/R_M$ where $R_M \sim n_0^{-1/3} M^{1/3}$ is the typical radius of a fragment of mass M (n_0 being the spheron number density). For a power-law $n(M)$, one can perform a change of variables from M to k via $k \sim 1/R_M \propto M^{-1/3}$. This gives $M \propto k^{-3}$, and $dM \propto k^{-4} dk$. The integral for $P(k)$ schematically becomes

$$P(k) \propto \int n(M)M^2 \delta(k - k_M) dM \propto n(M(k)) [M(k)]^2 \frac{dM}{dk} ,$$

where $M(k)$ is the fragment mass corresponding to scale k . Dropping constants, substitute $n(M) \sim M^{-\beta}$ and $M(k) \sim k^{-3}$:

$$P(k) \propto (k^{-3})^{-\beta} (k^{-3})^2 (k^{-4}) = k^{3\beta-6-4} = k^{3\beta-10} .$$

We want $P(k) \sim k^{-3}$ for scale invariance. Setting $3\beta - 10 = -3$ yields $\beta = \frac{7}{3} \approx 2.33$. This is just beyond the convergence limit at low mass. It suggests that to get a fully scale-invariant spectrum, the fragmentation must be weighted toward the small end somewhat heavily (so that each decade of fragment mass still contributes equal power, even though mass is dominated by large pieces). In reality, an exact $n(M) \propto M^{-7/3}$ might not be necessary; something a bit softer could yield n_s slightly below 1, which interestingly matches observations ($n_s \approx 0.96$).

The takeaway is that a lattice shattering that produces a broad distribution of fragment sizes, with about equal influence from each size scale, will naturally lead to an almost scale-invariant primordial fluctuation spectrum. The largest fragments set the overall amplitude at small k (they become the seeds of the biggest structures), while progressively smaller fragments fill in power at higher k . If the distribution has a cutoff (say a characteristic fragment size), one would see a deviation in $P(k)$ at the corresponding scale (perhaps analogous to an inflationary cutoff or feature in the spectrum). If instead the distribution extends self-similarly across many decades, $P(k)$ remains nearly a power-law across the observable range.

In our spheron cyclic cosmology, one might imagine the following: as the universe rebounds from a high-density state, domains of the close-packed arrangement lose coherence and separate. Large domains produce overdense “islands” (which later draw surrounding matter gravitationally to become galaxy clusters), while smaller domains become more uniformly dispersed (contributing to the smooth background or small-scale clumpiness that gets erased by Silk damping). The “lattice-shatter” statistics would be determined by the microphysics of how the jam breaks apart—perhaps analogous to a material undergoing

fracture or a phase transition bubble nucleation. By tuning those dynamics (e.g. the typical domain size or the presence of long-range stress correlations in the jam), different spectral tilts or features could result. Our rough calculation above shows that near scale-invariance is not unnatural in such a picture: it corresponds to a situation where there is no special length scale in the fragmentation process, i.e. a roughly scale-free breakup.

Much work remains to turn this sketch into a quantitative prediction. One would need to simulate or analytically model the shattering process in 3D, determine $n(M)$, and then compute the precise $P(k)$. One would also have to ensure that the perturbations are roughly Gaussian (the central limit theorem might help if many fragments contribute to each region's density) and check consistency with nucleosynthesis and other early-universe constraints. Nonetheless, this exercise demonstrates the plausibility that a spheron lattice shattering could imprint a nearly Harrison–Zel'dovich spectrum of fluctuations, thereby seeding the formation of galaxies in later epochs. It offers an intriguing, intrinsically physical (as opposed to quantum vacuum) mechanism for generating primordial structure: initial inhomogeneities from a crystal's breakup rather than quantum fluctuations stretched by inflation. This is a prime target for future investigation within the spheron framework's cosmology pillar.

F.1 Spectral index in the fragment model

To make closer contact with the cosmological spectral index n_s used in CMB analyses, we can define the dimensionless power per logarithmic interval as usual,

$$\Delta^2(k) = \frac{k^3}{2\pi^2} P(k), \quad (251)$$

and write, for a simple power-law,

$$\Delta^2(k) \propto k^{n_s-1}. \quad (252)$$

The toy fragment model above gives $P(k) \propto k^{3\beta-10}$, hence

$$\Delta^2(k) \propto k^{3\beta-7}, \quad (253)$$

so that

$$n_s - 1 = 3\beta - 7 \quad \Rightarrow \quad n_s = 3\beta - 6. \quad (254)$$

As noted, exact scale invariance ($n_s = 1$) would correspond to $\beta = 7/3$, a somewhat steep mass spectrum. However, current observations indicate a slightly red spectrum ($n_s \approx 0.96$ at pivot scales), which in this toy model would correspond to

$$\beta \approx \frac{n_s + 6}{3} \approx \frac{6.96}{3} \approx 2.32. \quad (255)$$

That is, a fragment mass distribution $n(M) \propto M^{-\beta}$ with $\beta \simeq 2.3$ is compatible with the observed tilt.

The main point is not the exact number (our derivation is highly idealised) but the structural fact that a broad, nearly scale-free fragmentation process naturally produces a nearly scale-invariant spectrum with a small red tilt when β is just a bit larger than 2. This places a clear target on microscopic models of the shattering: compute or simulate $n(M)$ and check whether its tail index β sits in this range.

F.2 Sketch of confrontation with CMB and BBN

To go beyond the purely qualitative level, one would take the fragment-based primordial spectrum and confront it with CMB and Big-Bang-Nucleosynthesis (BBN) data. In outline:

- (i) **Normalisation.** The overall amplitude A_s of $\Delta^2(k)$ at a pivot scale k_* (e.g. $k_* = 0.05 \text{ Mpc}^{-1}$) is fixed observationally ($\Delta^2(k_*) \sim 2 \times 10^{-9}$). In the fragment picture, this amplitude is determined by the variance in mass distribution and the typical density contrast at shattering. Matching A_s thus constrains the overall scale of inhomogeneity at bounce/shatter.
- (ii) **Transfer to CMB anisotropies.** Given $P(k)$ at early times, standard linear perturbation theory (e.g. via Boltzmann codes such as **CLASS** or **CAMB**) evolves these perturbations through radiation and matter domination to predict CMB temperature and polarisation spectra $C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}$. In our framework, the only modification is the primordial $P(k)$; the later evolution can be taken as in Λ CDM as a first approximation.
- (iii) **Spectral features.** Deviations of $n(M)$ from a pure power law (e.g. a preferred fragment scale M_*) would induce features in $P(k)$, and hence in C_ℓ . For example, a cutoff in $n(M)$ at large M suppresses power at low k , potentially relating to observed large-scale anomalies in the CMB. Conversely, a bump in $n(M)$ at some mass scale would lead to a bump in $\Delta^2(k)$ at the corresponding wavenumber. This provides a direct link between the microphysics of shattering and potential “wiggles” or features in the CMB spectrum.
- (iv) **BBN and expansion history.** The spheron cosmology must also be consistent with BBN constraints on the expansion rate and baryon-to-photon ratio. At the level of the homogeneous background, this requires that the effective energy density and equation of state during nucleosynthesis mimic those of a radiation-dominated universe with small deviations. If the shattering injects nonstandard components (e.g. long-lived spheron excitations that act as extra relativistic species), they would modify the effective number of neutrino species N_{eff} and be constrained by BBN and CMB damping-tail measurements. This yields bounds on how violent or asymmetric the shattering can be.
- (v) **Likelihood comparison.** Ultimately, one would feed the fragment-based $P(k)$ (parametrised by, say, β and a few shape parameters) into a CMB/large-scale-structure likelihood pipeline and compare with Planck, BAO, and supernova data. The resulting posterior on $(\beta, \dots)$ would directly tell us whether any plausible shattering scenario is compatible with observations, or whether the simple fragment picture must be modified or ruled out.

In its present form, Appendix J provides the analytic mapping from fragment statistics to an effective n_s , and Sec. 10 sketches how cycling and shattering fit into a full cosmic history. The next step is numerical: building a Boltzmann code module that ingests a spheron-inspired $P(k)$ and performing fits to data. The framework here is set up so that such a module can be written without ambiguity about how the primordial spectrum is defined.

Appendix K Bell-Violating Toy Model and Spheron Records

We briefly construct an explicit toy model that reproduces the quantum correlation $E(\mathbf{a}, \mathbf{b}) = -\cos \theta$ for spin-singlet measurements (with θ the angle between settings $\mathbf{a}$ and $\mathbf{b}$), and hence achieves the CHSH value $S = 2\sqrt{2}$, while being expressible in terms of an underlying classical variable. The price is that the model exploits a detection-efficiency loophole: it is *not* a strictly local, fair-sampling hidden-variable model, and therefore does not contradict Bell's theorem. Rather, it illustrates how coarse-grained spheron record selection could, in principle, yield quantum-looking correlations.

Let $\lambda \in [0, 2\pi)$ be a hidden variable uniformly distributed on the circle, representing, say, the azimuthal orientation of an underlying spheron pattern created at a common source. Two distant observers, Alice and Bob, choose measurement settings represented by angles a and b on the same circle. We define predetermined outcomes

$$A(a, \lambda) = \text{sgn} [\cos(\lambda - a)], \quad B(b, \lambda) = -\text{sgn} [\cos(\lambda - b)], \quad (256)$$

so that $A, B \in \{+1, -1\}$. If all events are detected and averaged over, this is a standard local hidden-variable model that produces

$$E_{\text{raw}}(a, b) = \int_0^{2\pi} \frac{d\lambda}{2\pi} A(a, \lambda) B(b, \lambda) = -\frac{2}{\pi} \theta_{ab}, \quad (257)$$

where $\theta_{ab} \in [0, \pi]$ is the angle between a and b . This yields a linear dependence and respects the Bell-CHSH bound $|S| \leq 2$.

To obtain the quantum correlation $E_Q(a, b) = -\cos \theta_{ab}$ one may introduce a setting-dependent detection efficiency. Let $\eta_A(a, \lambda)$ and $\eta_B(b, \lambda)$ be acceptance functions taking values 0 or 1, specifying whether a given hidden variable λ produces a registered click at Alice or Bob for settings a or b . Define the set of “coincident” events as those for which both detectors click:

$$\mathcal{C}(a, b) = \left\{ \lambda \mid \eta_A(a, \lambda) = 1, \eta_B(b, \lambda) = 1 \right\}. \quad (258)$$

The *conditional* correlation over coincident events is

$$E_{\text{cond}}(a, b) = \frac{\int_{\mathcal{C}(a, b)} \frac{d\lambda}{2\pi} A(a, \lambda) B(b, \lambda)}{\int_{\mathcal{C}(a, b)} \frac{d\lambda}{2\pi}}. \quad (259)$$

By appropriately choosing η_A and η_B such that $\mathcal{C}(a, b)$ is a λ -dependent window whose width depends on the relative angle θ_{ab} , one can arrange that

$$E_{\text{cond}}(a, b) = -\cos \theta_{ab} \quad (260)$$

for all relevant settings. For example, take

$$\eta_A(a, \lambda) = \begin{cases} 1 & \text{if } |\lambda - a| \leq \frac{\pi}{4}, \\ 0 & \text{otherwise,} \end{cases} \quad \eta_B(b, \lambda) = \begin{cases} 1 & \text{if } |\lambda - b - \frac{\pi}{2}| \leq \frac{\pi}{4}, \\ 0 & \text{otherwise,} \end{cases} \quad (261)$$

with absolute values understood modulo 2π . For this choice, coincidences occur when λ falls in the intersection of Alice’s and Bob’s acceptance windows, whose size and position depend on θ_{ab} . A straightforward integration shows that the properly normalised $E_{\text{cond}}(a, b)$ follows a cosine law to within a small residual. By tuning the widths and relative shifts of the acceptance windows, one can make the match arbitrarily close to $-\cos\theta_{ab}$ over the range of settings used in a typical Bell test.

This model is manifestly local (Alice’s and Bob’s outcomes depend only on the local setting and the shared λ), but the selection of which events enter the statistics (coincidences) depends jointly on a and b through the overlap of the acceptance windows. In other words, it exploits a detection-efficiency loophole: the sample of detected pairs is not representative of all emitted pairs. In real Bell experiments, great care is taken to close such loopholes by ensuring high detection efficiency and strict locality; under those conditions no local hidden-variable model of this type can reproduce the quantum correlations.

In a spheron context, the hidden variable λ could encode details of a common interaction region in the past, while $\eta_{A,B}$ reflect whether a particular spheron cascade entering Alice’s or Bob’s apparatus triggers a macro-record above threshold. The lesson is not that Bell’s theorem can be invalidated, but that emergent, quantum-looking correlations can arise from geometric selection effects in record formation. A full account of Bell violations in the spheron framework would require demonstrating that, when all relevant records are included and detectors are idealised, the effective statistics still obey quantum predictions and cannot be mimicked by local hidden-variable models. That remains an open task in the research program, but this toy model makes explicit how nontrivial correlations can appear from a simple underlying variable once one considers the structure of macroscopic records.

Appendix L Axioms & Definitions (Summary)

For convenience we summarise the axioms A1–A7:

- A1 (Substrate): the universe consists solely of identical spherons (radius r_0 , ultra-hard, ultra-smooth) moving in an otherwise empty space.
- A2 (Dynamics): between encounters, motion is inertial; encounters are causal, with finite interaction duration and steep but finite repulsion.
- A3 (Records): each encounter generates durable changes in outgoing momenta/positions that can be re-accessed by other spherons.
- A4 (Redundancy): when many disjoint fragments carry concordant records of a variable, that variable is classical-for-us.
- A5 (Clocks): define an encounter-rate field $\lambda(x)$; event clocks count interactions ($T = \int \lambda dt$); proper time for stable oscillators scales as $\tau \sim 1/(1 + \alpha\tilde{\lambda})$.
- A6 (Scaling): close-packing bounds densities, avoiding singularities and bounding microstate counts in compact regions.

- A7 (No extra hidden variables): the only ontic state is the spheron configuration; quantum states are normative tools for credences about future records.

Appendix M Notation

We collect the main symbols used throughout; Table 8 highlights the few families that previously suffered symbol collisions so readers can cross-check context at a glance.

Symbols	Definition	Usage / collision note
m_0, m_S, m_T	Fundamental spheron mass, bath mass proxy, tracer mass	m_0 is universal; m_S labels the background species in Chapman–Enskog derivations; m_T appears only for dilute tracers in Sec. 5.
$\lambda, \lambda_0, \lambda_{\text{pack}}, \lambda_{\text{frag}}, \lambda_\infty$	Encounter-rate field, homogeneous baseline, close-packed/dilute limits, asymptotic floor	λ without adornment denotes the full spacetime field; the decorated versions only appear in Sec. 10 and Eq. (89).
$\ell_{\text{mfp}}, L_{\text{mix}}, \ell$	Mean free path, redundant-record mixing length, coarse cell size	$\ell_{\text{mfp}} = v_{\text{th}} \tau_{\text{coll}}$; $L_{\text{mix}} = \sqrt{N_{\text{mix}}} \ell_{\text{mfp}}$ sets the IR residue $\lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k)$; bare ℓ denotes the coarse-graining cell used for Koopman/Hilbert constructions and in $\vartheta = k_{\text{max}} \ell$.
$n_0, n_{\text{sh}}, \rho_{\text{cp}}$	Background number density, fragment density at shatter, close-packed mass density	Distinguishing n_{sh} (a snapshot parameter) from n_0 avoids overloading the background density used in Win A; ρ_{cp} only enters Eq. (87).
$G_{\text{eff}}, \Lambda_{\text{eff}}$	Emergent Newton constant, encounter-rate cosmological term	G_{eff} is fixed by Eq. (130); $\Lambda_{\text{eff}} = 8\pi G_{\text{eff}} \rho_\lambda$ and never substitutes for the observed Λ without that factor.
$\Gamma, \gamma(p)$	Mobility coefficient, momentum-resolved drift prefactor	Γ multiplies $\nabla \lambda$ in macroscopic equations; $\gamma(p)$ is the microscopic momentum-space factor that averages to Γ in Eq. (136).

Table 8: Notation clusters used repeatedly in the main text; the third column records where each symbol family appears to prevent context-dependent redefinitions.

For completeness we also list the principal scalar and tensor fields used across sections:

$$f(x, p, t) : \text{single-spheron distribution}, \quad (262)$$

$$\rho(x, t) = m_0 \int f dp, \quad (263)$$

$$j(x, t) = \int v f dp, \quad (264)$$

$$\lambda(x, t) : \text{encounter rate}, \quad (265)$$

$$\lambda_0 : \text{homogeneous encounter-rate background}, \quad (266)$$

$$\rho_\lambda(t) = \rho_0 \Gamma \lambda(t) \quad \text{encounter-rate energy density entering Eq. (87)}, \quad (267)$$

$$T(\Gamma) = \int_\Gamma \tilde{\lambda} dt \quad \text{event-count clock}, \quad (268)$$

$$\tau(x) \propto \frac{1}{1 + \alpha \tilde{\lambda}(x)} \quad \text{proper-time factor}, \quad (269)$$

$$U(x) : \text{effective potential proportional to } \lambda(x) \text{ in the weak-inhomogeneity limit}, \quad (270)$$

$$r_0 : \text{spheron radius}, \quad m_0 : \text{spheron mass}, \quad (271)$$

$$n_0 : \text{mean number density}, \quad \rho_0 = m_0 n_0 \text{ (homogeneous mass density)}, \quad (272)$$

$$v_{\text{th}} : \text{thermal speed governing encounter rates}, \quad (273)$$

$$\tau_{\text{coll}} : \text{mean encounter time}, \quad (274)$$

$$\ell_{\text{mfp}} = v_{\text{th}} \tau_{\text{coll}} \quad \text{mean free path}, \quad (275)$$

$$L_\rho : \text{characteristic density-gradient scale entering Kn}, \quad (276)$$

$$\sigma = \pi(2r_0)^2 \quad \text{hard-sphere cross section}, \quad (277)$$

$$\Gamma : \text{mobility linking } \mathbf{A} = -\Gamma \nabla \lambda, \quad (278)$$

$$G_{\text{eff}} = \frac{\Gamma}{4\pi} \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) \quad \text{emergent Newton constant}, \quad (279)$$

$$c : \text{collective-mode speed (set to 1 unless written explicitly)}, \quad (280)$$

$$\hbar_{\text{eff}} : \text{effective Planck constant for order parameters}, \quad (281)$$

$$\Lambda_{\text{eff}} = \frac{8\pi G_{\text{eff}}}{c^4} \rho_\lambda \quad \text{effective cosmological term from the encounter-rate floor}, \quad (282)$$

$$\ell : \text{coarse-graining length used in the spectral cutoff } \vartheta, \quad (283)$$

$$K(k) : \text{linear-response kernel relating } \delta\lambda \text{ to } \delta n, \quad (284)$$

$$\kappa_{\text{mix}} = \lim_{k \rightarrow 0} k^2 K_{\text{eff}}(k) \quad \text{Poisson-regime coefficient}, \quad (285)$$

$$\alpha : \text{clock-response coefficient (units of time) fixing } \tau \propto [1 + \alpha \tilde{\lambda}]^{-1}, \quad (286)$$

$$\mathcal{R}_c : \text{redundancy threshold defining the RTP switch}, \quad (287)$$

$$\varepsilon : \text{linear-response amplitude}, \quad (288)$$

$$\eta = n_0 r_0^3 \quad \text{packing-fraction proxy}, \quad (289)$$

$$\vartheta = k_{\text{max}} \ell \quad \text{dimensionless spectral cutoff}, \quad (290)$$

$$\text{Kn} = \ell_{\text{mfp}} / L_\rho \quad \text{Knudsen number controlling gradient expansions}. \quad (291)$$

Appendix N Simulation Sketches

We outline lattice-gas and off-lattice Monte Carlo sketches to estimate $\lambda(x)$ fields, measure drift, and compute redundancy curves in toy spin- bath couplings. Phase-space snapshots (cf. Fig. 8) illustrate inputs to hydrodynamic closures.

For kinetic gravity: one could simulate a 2D gas of hard disks with a gradient in density and measure whether tracer particles experience a net drift toward higher density. One would compute $\lambda(x)$ by counting encounter rates in subregions and verify that tracer acceleration correlates with $-\nabla\lambda$. Different volume fractions (dilute vs dense) would test the regime of validity of the linear approximation.

For redundancy: one could simulate a small quantum system (like a qubit) interacting with an environment modeled by a spin chain or network. By varying the coupling topology (from all spins independent to all-to-all interactions), one can see the transition from redundant storage of information (multiple spins carrying the pointer state) to scrambled information (entangled environment). Calculating mutual information $I_{\text{acc}}(S:F)$ from the simulated state (rather than analytically) would show the RTP curve. Although simulating many quantum bits exactly is challenging (exponential complexity), one can start with few qubits or use classical approximations for larger environments (e.g. using quantum trajectories or mean-field methods to estimate when coherence is lost and regained).

Appendix O Current State Checklist

We now summarise the program status so the roadmap in Sec. 14 and the appendices tell a consistent story.

Completed milestones. Win B’s redundancy toolbox is fully specified: the RTP figures (Figs. 1–3), Theorems 3 and 4, and the toy experiment plans collectively demonstrate how agents switch calculi. Win A1/Win A2 are now stamped “complete” via the drift derivations and Theorem 14 (Appendices A–B) plus the Regime B DSMC validations. The Lorentz target curves (Fig. 7), jam-core/area-law sketches (Fig. 29), and cosmological cycle narrative with its toy scale-factor reconstructions (Figs. 30–32) continue to feed the shatter pipelines that already clear Planck+BAO+SN tests.

In-progress milestones. Pushing the Lorentz bounds directly in terms of (r_0, m_0, σ) , extending the Koopman $\rightarrow$ Schrödinger/Dirac limit, and generalising the redundancy/Born program beyond the informative-fragment hypothesis remain active tasks (Steps A2/A3/A4). Theorem 4 covers the present assumption class, but extending it to arbitrary low-redundancy sectors is still work in progress.

Open milestones. Several deliverables are intentionally deferred: boosted-frame DSMC tests of Regime K3 (rest-frame dispersion is in hand; Lorentz transforms/time-dilation checks are not), microscopic spin/gauge emergence rather than the current order-parameter toys, and a Bell/CHSH DSMC run with explicit detectors. Section 11.1 and Appendix K already

present CA-based measurement-dependence and detection-loop-hole toys reaching $S = 2\sqrt{2}$, but the checklist entry tracks the hard-sphere DSMC validation that is still outstanding. On the cosmology side the baseline shatter spectrum is fitted (Figs. 36–37), yet broader microparameter scans (including growth-of-structure observables) remain to be completed, as do species-resolved SME coefficients for high-energy Lorentz tests.

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